Homological Algebra

— *EYNTKA* Series —

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Part I

Homological Algebra

Throughout this book, the reader has encountered many functors, some notable examples being $\mathrm{Hom}_R(-, N)$, $\mathrm{Hom}_R(M, -)$ and $M \otimes_R -$ among many others. Each of these functors give some information about the objects and the morphisms passing through the functor. As we saw for the Hom and $M \otimes_R -$ functors, certain appropriate objects preserve exactness, while most preserve it only partially. Perhaps a crowning achievement of mid 20th century mathematics is to be able to quantify this lack of exactness, and rather spectacularly use the information given by this quantification to extract a (sometimes copious) amount of information about the objects of study. The great effectiveness with which homological algebra can produce interesting information may seem surprising at first, for example, using homological algebra the notion of *dimension* as it is understood in geometry is readily translatable to algebra, or that we may measure how far away a module is from being a vector-space (details are in K-theory)[1] How this is the case is perhaps best motivated by going through learning the subject to see for oneself, but it may be helpful for some readers to have some of my (the authors) intuition presented to stand on some ground before proceeding.

If we have two objects M_1 and M_2, we may wish to consider a way of constructing a new object M given M_1 and M_2. Ideally, we may form a short exact sequence like so:

$$0 \to M_1 \to M \to M_2 \to 0$$

More generally, we may have a collection of objects $\{M_i\}_{i \in I}$ which we use to construct an object via a long exact sequence. However, it is often not the case that such an exact sequence is too strong, that is, it is not the case that $\mathrm{im}(\varphi_i) = \ker(\varphi_{i+1})$. Even if such a construction exists, we often then want to "transform" this information via a functor which shall then lose us exactness (one can think of a functor as asking some sort of "question" whose output is the answer). In this case, it has been eerily common that the image of φ_i is contained in the kernel of φ_{i+1}:

$$\mathrm{im}(\varphi_i) \subseteq \ker(\varphi_{i+1}) \tag{1}$$

Conceptually, this can be thought as the object "being obstructed" from fully being an extension. This new type of sequence is called a *cochain complex*. The reader with a good memory may recall [7, Exact Chain Complex] where it was given that another name for a long exact sequence is an exact cochain complex. In this way, the condition given by equation 1 is a generalization of the exact condition. The theoretical heart of homological algebra can be thought of how to extract "quantitatively" how a cochain complex *fails* to be an exact cochain complex. For example, in [7, Extension Problem: Prelude to Homological Algebra] it was have eluded that through the use of homology theory, the Ext functor gives us all the ways two modules M_1, M_2 may be extended. In this case, the failure of $\mathrm{Hom}(M, -)$ or $\mathrm{Hom}(-, N)$ preserving injectivity or surjectivity (equivalently, the "obstruction" to these functors being exact) gave rise to the Ext functor and the study of its homology.

The term "algebraic object" has been is a bit ambiguous. What type of algebraic objects can be used to in the theory of homological algebra? As it turns out, not all algebraic objects satisfy the right conditions for the theory of homological algebra (for example, **Grp** and **Fld** will be shown to be inappropriate for a theory of homology). However, there are non-algebraic objects for which a homological theory can be induced (famously, topological spaces and smooth manifolds admit a theory of homological algebra). This spurred the need to generalize the theory of homological algebra to the categorical level, where homological theory is well-defined on *abelian categories*. We thus shall start homological algebra with an overview of abelian categories. For those who are already familiar

[1] This is overviewed in chapter 3

with some homological algebra, they may have studied homological algebra within the category of R-**Mod**, which is a concrete abelian category. Doing so will in fact not loose any generality due to Freyd-Mitchell Theorem (theorem 1.1.15) which states that any [small] abelian category A admits a fully faithful exact functor to a category R-**Mod** for a suitable ring R. Working with the concrete category R-**Mod** brings it benefits as it may be easier to conceptualize the manipulation of elements, and so the reader may ask for the benefit in sticking to working with general abelian categories. As the author, I believe the benefits comes in being comfortable with more advanced category theory. Future algebraists are going to encounter more complicated problems involving category theory (higher order categories, topoi's, and stacks, are all examples the reader may see in the not too distant future). Hence, introducing a categorical perspective will familiarize the reader for more complicated categorical notions in their not too distant future[2].

The reader may wonder why a cochain complex is not called a chain complex. This is because we can work in the dual category, where all arrows are flipped. In the dual category, we shall get chain complexes. Homological algebra in the dual category is theoretically the same, however we shall see that cohomology is often computationally easier to work in, permitting more useful algebraic structures we can use to study. It is for this reason that most mathematicians work with cohomology rather than homology, even though every theorem we shall show for homology translates directly to cohomology[3]. In this book, we shall start with homology, and switch to cohomology when starting to work with more computational parts of the theory to get the reader accustomed to both modes of thinking.

Goal

Chapter 1 shall introduce the theory of abelian categories which shall be the types of categories in which homological algebra is well-defined.

The main result Chapter 2 of the book is trying to convey is what information from R-modules give rise to certain properties in its respective chain complex(es), and conversely what information from it's chain complex(es) gives us information about modules. A (co)chain complex need not be exact, and (co)homology shall be the algebraic object quantifying this failure of exactness. We shall see that given an object M within an abelian category, there are in general an infinite number of ways we may associate a [co]chain complex to it. Thus, there will have to be a way to either associate a canonical [co]chain complex (in a natural functorial manner), or we assign [co]chain complexes to an object up to appropriate notion of isomorphism. The latter shall be the approach in this book, where we shall see that we will want to consider [co]chain complexes up to something called *quasi-isomorphism* of [co]chain complexes. We shall then see what cohomological information we may deduce given different choices of functors and how they transform [co]chain complexes.

After having developed the ground work for extracting cohomological information, in Chapter 3 we shall put this knowledge to good use and apply cohomology theory to G-invariant functors and see what this this can tell us about groups. This shall also finally address what was foreshadowed in [7, Foreshadowing] where it was said the units of a Galois field form a trivial 1st homology group. We shall define the notion of homological dimension which shall line up with the results presented in the section on commutative algebra. Finally, we present an interesting technical result to check

[2] assuming, naturally, that the reader holds great interest in algebra and shall keep pursuing branches within the field

[3] there is even an equivalent algebraic structure for the additional algebraic structure that is induced in cohomology, however it is less useful

flatness of a module over a local ring that builds on the same intuitions the Nakayama lemma ([4, Nakayama's Lemma]) presented.

Finally, this Part concludes with the introduction to spectral sequences in chapter 4, which are the quintessential tool for many concrete computations in homological algebra and any serious work in many disciplines of algebra and geometry. The main idea is that most object of study can be broken down into a filtration of simpler objects whose cohomology is more computed. The spectral sequence will put piece together this cohomological information.

Connection to Singular Homology/Cohomology

For readers familiar with algebraic topology and the theory of (co)homology, the connection between the above abstraction and the homology theory of topological spaces comes from noticing that singular (co)homology on a manifold can be interpreted as taking the projective resolution $C_n(X) = \mathrm{Hom}_{\mathbf{Top}}(\Delta^n, X)$. In particular, given M, we can apply $C_*(-)$ to it and get a projective (even free) resolution from which we extract homological information. This should also motivate the use of $\mathrm{Hom}_R(M, -)$ and $\mathrm{Hom}_R(-, N)$ as the "right" functors to be considering when generalizing classical algebraic topology to the algebraic setting. As the tensor product it adjoint ([7, hom-Tensor Adjoint]) it is another natural candidate.

0.0.1 Foreshadowing: Yoneda's Lemma The Hom is very useful as it is "represented" by an element (either in the domain or codomain). Even more is true: it can be used to represent other functors. In [2, The Yoneda Lemma] this is studied *representable functors* which are functors where $F(-) \Rightarrow \mathrm{Hom}_C(A, -)$ for natural transformation $\Rightarrow$, category C, object $A \in C$, and $F : C \to \mathbf{Set}$. The Yoneda lemma will give a way of representing objects using Hom.

Historical Origins

While modern homological algebra is phrased in the language of categories, its machinery evolved from two distinct 19th-century sources which converged in the mid-20th century.

The algebraic lineage begins around 1890 with David Hilbert's Invariant Theory. To prove the Basis Theorem, Hilbert studied the relations ("syzygies") between generators of an ideal. He constructed a chain of free modules to resolve these relations:

$$0 \to F_n \to \cdots \to F_1 \to F_0 \to M \to 0$$

Hilbert proved the *Syzygy Theorem*, showing that for polynomial rings, this chain of relations eventually terminates in an exact sequence of free modules. However, this theory encountered a significant obstruction when applied to rings with singularities (such as coordinate rings of singular curves). In these non-regular settings, Hilbert's resolutions do not terminate but continue infinitely. This failure shifted the mathematical focus from simply computing invariants to essentially classifying the severity of a singularity by the infinite length of its resolution.

Parallel to this, in the 1930s, number theorists like Emmy Noether, Richard Brauer, and Helmut Hasse were attempting to classify division algebras (generalizations of the quaternions) over number fields. They classified these algebras using "Crossed Products," which relied on complex systems of functions called “Factor Sets” ($c : G \times G \to K^\times$). While they successfully defined the *Brauer Group* of a field to organize these algebras, the calculations were notoriously opaque. The “Factor Sets” satisfied a messy algebraic identity that seemed arbitrary, acting as a computational blocker that obscured the underlying structure of the fields they were studying.

Simultaneously, from 1895 onwards, the topological roots of the subject were forming as Henri Poincaré classified manifolds. He decomposed spaces into simplices and represented their boundaries using incidence matrices. He defined Betti numbers as the ranks of these matrices. However, Poincaré’s numerical approach had a blind spot: rank obliterates “torsion.” A twisted shape like the projective plane appeared numerically identical to a trivial one (both have rank 0 in dimension 1).

In 1925, Emmy Noether revolutionized this view. She realized that the matrices topologists were diagonalizing were actually presentations of abelian groups. By shifting focus from the numerical rank to the *Homology Group* itself ($\ker \partial / \operatorname{im} \partial$), she captured the torsion information, effectively “algebraizing” the topology. She showed that the matrix was not just a calculator for a number, but a presentation of a structure.

The convergence of these fields occurred in the 1940s when algebraists attempted to classify *Group Extensions*. Saunders Mac Lane and Samuel Eilenberg attempted to classify these extensions and rediscovered the same “Factor Sets” used by Brauer and Hasse. They discovered that the algebraic identity these functions satisfied was *identical* to the boundary formula for a 2-simplex in topology; essentially they realized they were calculating $H^2(G, N)$ (see [6] for this computation in the EYNTKA series). This was the turning point: it proved that “cohomology” was not just a property of geometric shapes, but a fundamental algebraic structure that arose whenever one tried to “extend” one object by another, whether in topology, group theory, or number theory.

The grand unification arrived in 1956 when Cartan and Eilenberg realized that the topological concept of “boundaries” and the algebraic concept of “extensions” were instances of the same phenomenon: the *failure of exactness*. They observed that while Hilbert’s resolution is exact, applying a functor (like the tensor product or Hom) usually destroys that exactness. They realized that the “homology” groups were simply the mathematical terms required to repair this broken exactness. This birth of the *Derived Functor* (Tor and Ext) transformed the field from the study of static objects to the study of how functors distort structure. To build this machinery, they faced a technical blocker: Hilbert’s reliance on “Free Modules.” While free modules worked for Homology (Tor), they failed for the dual theory of Cohomology (Ext), as the dual of a free module is rarely free. To achieve the necessary symmetry, they replaced the specific condition of “Freeness” with the categorical properties of *Projectivity* and *Injectivity*. This allowed them to define derived functors for any ring, regardless of whether a basis existed.

With this machinery established, the period from 1955 to 1980 saw these newly developed tools being used to solving the original obstructions in geometry and number theory. In geometry, Jean-Pierre Serre proved that a local ring is regular (non-singular) if and only if it has finite global homological dimension (theorem 3.2.8). This transformed the “problem” of infinite resolutions into a precise detection tool: the non-terminating resolution was effectively the algebraic signature of a geometric singularity. Alexander Grothendieck later introduced *Local Cohomology* ($H^i_{\mathfrak{m}}$) to quantify the depth and severity of these singularities, eventually leading to Intersection Homology which restored Poincaré Duality to singular spaces.

In number theory, the mysterious “Factor Sets” of the 1930s were identified as elements of the second Galois Cohomology group, $H^2(\mathrm{Gal}(L/K), L^\times)$. This realization allowed Emil Artin and John Tate to rewrite Class Field Theory entirely in the language of cohomology. The obscure identities of the past were revealed to be standard cohomological relations, and the *Tate Cohomology* groups ($\hat{H}^i$) were developed specifically to handle the arithmetic of finite groups, solving the classification problems that had stalled the previous generation (see [10] for the EYNTKA series exposition of modern class field theory).

1

Abelian Categories

This chapter focuses on the category in which we may define "cohomological theories" as not all categories are well-suited for this endeavor

1.1 Definitions and Properties

We have extensively worked with categories which have a multitudes of different properties. Some important terms to recall are the following:

- *zero-object*: an object 0 in a category C $0 \in \mathrm{Obj}(C)$ that is both the initial and final object. Such an object is also called a terminal object. Similarly, a zero morphism is a morphism $f : X \to Y$ such that for any other morphisms $g, h : X \to Y$, $fg = fh$ and $hf = gf$.
- *Products* and *Coproducts*: Given two objects $a, b \in \mathrm{Obj}(C)$, if there exists an object $c \in \mathrm{Obj}(C)$ that is the limit (resp. colimit) of the diagram:

  $$\overset{A}{\bullet} \qquad \overset{B}{\bullet}$$

  then c is the product (resp. coproduct) of a and b, and is usually written as $a \times b$ (resp. $a \amalg b$)
- *kernels* and *cokernels*: Given a morphism $f : a \to b$, the kernel (resp. cokernel) of f is an object k and a morphism $\iota : k \to a$ such that k is the limit (resp. colimit) of the diagarm:

  $$\overset{a}{\bullet} \xrightarrow{f} \overset{b}{\bullet}$$

  with the added restriction that the map $\varphi : k \to y$ must be the zero morphism (namely, by commutativity, $f\iota = 0_{kb}$ must be the zero morphism). In terms of diagrams, we get that that for any $\iota' : K \to A$ such that $f\pi\iota' = 0$, that there exists a morphism φ st the following diagram commutes:

$$\begin{tikzcd}
K' \arrow[r, "\iota'"'] \arrow[rr, bend left, "0"] \arrow[dr, dashed, "\varphi"'] & A \arrow[r, "\varphi"'] & B \\
& K \arrow[u, "\iota"'] &
\end{tikzcd}$$

and simmilarly for cokernels:

$$\begin{tikzcd}
& K' \arrow[d, "\pi'"] \arrow[dr, dashed, "\psi"] & \\
A \arrow[r, "\varphi"] \arrow[rr, bend right, "0"'] & B \arrow[r, "\pi"] & K
\end{tikzcd}$$

Kernels and cokernels do not necessarily exist in a category, or not all morphisms have kernels (ex. the category of finitely generated R-modules for any ring R need has morphisms without kernels). Furthermore, monomorphism need not be kernels (not all subgroups are normal) and epimorphisms need not be cokernels (recall $\mathbb{Z} \to \mathbb{Q}$ is an epimorphism in **Ring**)

Not all categories we have worked with have these properties. For example, **Fld** doesn't have products (what would be the characteristic of the product of $\mathbb{F}_p$ and $\mathbb{F}_q$ for $p \neq q$?), some categories don't have zero morphisms (set being an example), some have monomorphisms/epimorphisms that do not correspond to kernels (as showed above). We first limit our attention to categories which have these nice properties:

Definition 1.1.1: Additive Category

Let $\mathbf{A}$ be a category. Then $\mathbf{A}$ is said to be *additive* if it has a zero-object, has both finite products and coproducts, and each $\mathrm{Hom}_A(X, Y)$ can be given an abelian group structure such that the composition of maps:

$$\mathrm{Hom}_A(X, Y) \times \mathrm{Hom}_A(Y, Z) \to \mathrm{Hom}_A(X, Z)$$

is bilinear. A functor between additive categories is *additive* if it is a group homomorphism on hom-sets.

A functor between two additive categories is called an *additive functor* if the maps between the hom-sets is a homomorphism.

The term "additive" comes froms the fact that we may interpret the additive category as a monoidal category (a category which we may add a functor $\otimes : C \times C \to C$ that is given the property of a monoid operation, for an additive category this would be the direct product), where the hom-sets are "enriched" to also be abelian groups. Another way of looking at additive categories is in which the algebra of matrices is well-defined.

Example 1.1.2: Additive Categories

1. The category of R-modules is an example of an additive category.

2. **Grp** is not additive, for $\mathrm{Hom}_{\mathbf{Grp}}(A, B)$ doesn't have an abelian group strucuture satisfying the above condition.

3. The category of matrices over a ring is an additive category.

4. The category of finitely generated R-modules is an additive category

A key realisation is that in the absence of kernels and cokernels as objects monomorphisms and epimorphisms take their place:

Lemma 1.1.3: Additive Categories: Monomorphisms and Kernels I

Let A be an additive category. Then kernels are monomorphisms and cokernels are epimorphisms

***Proof*:**
Let $k\colon K \to A$ be a kernel of $\varphi\colon A \to B$, and let $f, g\colon X \to K$ satisfy $k \circ f = k \circ g$. Because the hom-sets of an additive category are abelian groups and composition is bilinear, this says $k \circ (f - g) = 0$.

Now apply the universal property of the kernel to the zero morphism $0\colon X \to A$. It satisfies $\varphi \circ 0 = 0$, so there is a *unique* morphism $u\colon X \to K$ with $k \circ u = 0$. Both $f - g$ and the zero morphism $X \to K$ have this property, so uniqueness forces $f - g = 0$, that is $f = g$. Hence k is a monomorphism.

The statement for cokernels is the same argument in the opposite category: a cokernel of φ in A is a kernel of φ in A^{op}, an additive category, and a monomorphism there is an epimorphism in A, completing the proof.

Note that kernels and cokernels need not exist in an additive category, hence the converse of each accession need not be the case (in a vacuous way). If a morphism φ has a kernel then we may identify exactly what this kernel is:

Lemma 1.1.4: Additive Categories: Monomorphisms and Kernels II

Let A be an additive category and $\varphi : X \to Y$ a morphism. Then:

1. If φ has a kernel, then φ is a monomorphism if and only if $0 \to X$ is its kernel

2. If φ has a cokernel, then φ is an epimorphism if and only if $Y \to 0$ is its cokernel

***Proof*:**

1. Suppose first that the zero morphism $0 \to X$ is a kernel of φ, and let $f, g\colon Z \to X$ satisfy $\varphi \circ f = \varphi \circ g$. Then $\varphi \circ (f - g) = 0$, so $f - g$ factors through the kernel: there is $u\colon Z \to 0$ with $f - g = 0 \circ u = 0$. Hence $f = g$ and φ is a monomorphism.

   Conversely suppose φ is a monomorphism and let $k\colon K \to X$ be a kernel of φ. From

$\varphi \circ k = 0 = \varphi \circ 0$ and φ monic it follows that $k = 0$. But k is itself a monomorphism by lemma 1.1.3, and $k \circ \mathrm{id}_K = 0 = k \circ 0$, so $\mathrm{id}_K = 0$. An object whose identity is the zero morphism is a zero object, so $K \cong 0$ and the kernel is $0 \to X$.

2. This is the statement of (1) in the opposite category, where kernels and monomorphisms become cokernels and epimorphisms.

This completes the proof.

As mentioned, it is not guaranteed that kernels and cokernels exist. We shall shortly axiomatically add them to our category by adding them into the definition of a new category that supersedes additive categories, but first we show a few more important results about abelian categories. Namely, in an additive category, products and coproducts are equivalent:

Lemma 1.1.5: Additive Categories: Products and Coproducts

Let A be an additive category and $X, Y \in \mathrm{Obj}(A)$. Then if $X \times Y$ exists, so does $X \amalg Y$, and if $X \amalg Y$ exists, so does $X \times Y$. Furthermore,

$$X \times Y \cong X \amalg Y$$

***Proof*:**
Suppose $P = X \times Y$ exists, with projections $p_X : P \to X$ and $p_Y : P \to Y$. Let $i_X : X \to P$ be the unique morphism with $p_X i_X = \mathrm{id}_X$ and $p_Y i_X = 0$, and let $i_Y : Y \to P$ be defined symmetrically; both exist by the universal property of the product.

Step 1: The key identity $i_X p_X + i_Y p_Y = \mathrm{id}_P$. Two morphisms out of P into P agree as soon as their composites with p_X and with p_Y agree, again by the universal property of the product. Compute

$$p_X(i_X p_X + i_Y p_Y) = (p_X i_X)p_X + (p_X i_Y)p_Y = \mathrm{id}_X \circ p_X + 0 \circ p_Y = p_X$$

using bilinearity of composition, and symmetrically $p_Y(i_X p_X + i_Y p_Y) = p_Y$. Since id_P has the same two composites, the identity holds.

Step 2: (P, i_X, i_Y) is a coproduct. Let $f : X \to Z$ and $g : Y \to Z$ be given and set $h = f p_X + g p_Y : P \to Z$. Then

$$h i_X = f(p_X i_X) + g(p_Y i_X) = f, \qquad h i_Y = f(p_X i_Y) + g(p_Y i_Y) = g$$

so h is a morphism with the required restrictions. For uniqueness, suppose h' also satisfies $h' i_X = f$ and $h' i_Y = g$. Then by Step 1,

$$h' = h' \mathrm{id}_P = h'(i_X p_X + i_Y p_Y) = (h' i_X)p_X + (h' i_Y)p_Y = f p_X + g p_Y = h$$

Hence P with i_X, i_Y satisfies the universal property of $X \amalg Y$, and in particular $X \times Y \cong X \amalg Y$.

If instead $X \amalg Y$ is assumed to exist, the same argument in the opposite category produces the product, completing the proof.

Hence, additive categories behave very much like abelian groups, however as we pointed out kernels and cokernels need not exist. We thus define a new category that forces these conditions:

Definition 1.1.6: Abelian Category

[Abelian Category] Let A be an additive category. Then A is an *abelian* category if

1. kernels and cokernels exist in A;
2. every monomorphism is the kernel of some morphism, and every epimorphism is the cokernel of some morphism.

The second condition is not a consequence of the first, and it is the one that does the work. An additive category satisfying only (1) is called *pre-abelian*, and there the canonical map from the coimage to the image of a morphism (theorem 1.1.9) need not be an isomorphism; the category of torsion-free abelian groups is pre-abelian but not abelian, since $\mathbb{Z} \xrightarrow{2} \mathbb{Z}$ is there both a monomorphism and an epimorphism without being an isomorphism. Condition (2) rules this out, and is what makes lemma 1.1.8 true.

Note that lemma 1.1.4 is a weaker statement, not this one: it characterizes a monomorphism by having *zero* kernel, which says nothing about whether the monomorphism is itself a kernel.

Since each kernel is associated to a monomorphism, we may intuitively think of the association:

$$\text{kernel} \;\Leftrightarrow\; \text{submodule}$$

and similarly:

$$\text{cokernel} \;\Leftrightarrow\; \text{quotient module}$$

In many ways, an abelian category should be thought of as the minimum number of conditions necessary to define exact sequence[1]. To see this, we require the following:

Lemma 1.1.7: Abelian Category: Kernels And Cokernels

Let A be an abelian category. Then every kernel is the kernel of its cokernel, and every cokernel is the cokernel of its kernel.

***Proof*:**
Let $k\colon K \to A$ be a kernel, say of $\varphi\colon A \to B$, and let $c\colon A \to C$ be a cokernel of k. The claim is that k is a kernel of c.

First, $c \circ k = 0$ holds by the definition of a cokernel. Now let $t\colon T \to A$ satisfy $c \circ t = 0$; a unique factorization of t through k is needed. Since $\varphi \circ k = 0$ and c is the cokernel of k, the morphism φ factors as $\varphi = \bar{\varphi} \circ c$ for some $\bar{\varphi}\colon C \to B$. Therefore

$$\varphi \circ t = \bar{\varphi} \circ c \circ t = 0$$

and since k is the kernel of φ, there is a unique $u\colon T \to K$ with $k \circ u = t$. Uniqueness is automatic because k is a monomorphism by lemma 1.1.3. So k satisfies the universal property of $\ker c$.

[1] Those familiar with some algebraic topology may find it insightful that an abelian category is the the category with ninimum number of condition so that the snake lemma (theorem 2.1.15) holds

The dual statement, that every cokernel is the cokernel of its kernel, is this argument in the opposite category, completing the proof.

Lemma 1.1.8: Abelian Categories: Isomorphisms

[Abelian Categories, Isomorphisms] Let A be an abelian category and $\varphi : A \to B$ a morphism in A. Then if φ is a monomorphism and an epimorphism, it is an isomorphism

Note that this does not hold in additive categories, nor even in pre-abelian ones: condition (2) of definition 1.1.6 is exactly what the proof below consumes.

***Proof*:**
Since φ is an epimorphism, condition (2) of definition 1.1.6 makes it the cokernel of some morphism $\psi\colon C \to A$. By the definition of a cokernel $\varphi \circ \psi = 0$, so ψ factors through $\ker \varphi$ by the universal property of the kernel.

Now φ is also a monomorphism, so $\ker \varphi = 0$ by lemma 1.1.4. The factorization of the previous paragraph then runs through the zero object, so $\psi = 0$ outright and φ is a cokernel of the zero morphism $0 \to A$. But the identity $\mathrm{id}_A\colon A \to A$ is also a cokernel of $0 \to A$: any $f\colon A \to X$ with $f \circ 0 = 0$, which is every f, factors uniquely through id_A. Cokernels are unique up to unique isomorphism, so there is an isomorphism $A \to B$ commuting with φ and id_A, which forces φ to be that isomorphism, as we sought to show.

Theorem 1.1.9: Abelian Category: Canonical Decomposition

[Abelian Category, Canonical Decomposition] Let A be an abelian category and $\varphi : A \to B$ a morphism. Write $C = \mathrm{coker}(\ker \varphi)$ and $K = \ker(\mathrm{coker}\varphi)$. Then φ decomposes in the following cannonical way:

$$\begin{tikzcd} A \arrow[r, two heads] \arrow[rrr, bend left, "\varphi"] & C \arrow[r, "\tilde{\varphi}"'] & K \arrow[r, tail] & B \end{tikzcd}$$

where the first morphism is an epimorphism, the last a monomorphism, and the middle morphism $\tilde{\varphi}$ is an *isomorphism*.

***Proof*:**
Write $k\colon N \to A$ for $\ker \varphi$ and $p\colon A \to C$ for its cokernel, and $q\colon B \to Q$ for $\mathrm{coker}\varphi$ and $\iota\colon K \to B$ for its kernel. By lemma 1.1.3 the morphism p is an epimorphism and ι is a monomorphism.

Step 1: φ factors through ι. Since q is the cokernel of φ, $q \circ \varphi = 0$, and since ι is the kernel of q there is a unique $\varphi'\colon A \to K$ with $\varphi = \iota \circ \varphi'$.

Step 2: φ' factors through p. From $\iota \circ \varphi' \circ k = \varphi \circ k = 0$ and ι monic it follows that $\varphi' \circ k = 0$. As p is the cokernel of k, there is a unique $\tilde{\varphi}\colon C \to K$ with $\varphi' = \tilde{\varphi} \circ p$. Combining, $\varphi = \iota \circ \tilde{\varphi} \circ p$, which is the displayed factorization.

Step 3: ι is the smallest subobject through which φ factors. Let $m\colon V \to B$ be any monomorphism with $\varphi = m \circ \psi$ for some ψ. By condition (2) of definition 1.1.6 the monomorphism m is a kernel, so by lemma 1.1.7 it is the kernel of its own cokernel; write $d\colon B \to D$ for coker$m$, so that $m = \ker d$. Then

$$d \circ \varphi = d \circ m \circ \psi = 0$$

so d factors through the cokernel of φ, say $d = \bar{d} \circ q$. Since $\iota = \ker q$ gives $q \circ \iota = 0$, it follows that $d \circ \iota = \bar{d} \circ q \circ \iota = 0$, and therefore ι factors through $\ker d = m$.

Step 4: φ' is an epimorphism. Let $c\colon K \to W$ be the cokernel of φ' and $n\colon V \to K$ its kernel. From $c \circ \varphi' = 0$ the morphism φ' factors as $\varphi' = n \circ \psi$, whence $\varphi = \iota \circ n \circ \psi$. The composite $\iota \circ n$ is a monomorphism, being a composite of two (each is a kernel, so lemma 1.1.3 applies), so Step 3 gives a t with $\iota = \iota \circ n \circ t$. As ι is monic this yields $n \circ t = \mathrm{id}_K$, so n is a split epimorphism as well as a monomorphism, hence an isomorphism by lemma 1.1.8. Then $c \circ n = 0$ with n invertible forces $c = 0$; but c is a cokernel, hence an epimorphism by lemma 1.1.3, and $\mathrm{id}_W \circ c = 0 = 0 \circ c$ then gives $\mathrm{id}_W = 0$, so $W \cong 0$. A morphism with zero cokernel is an epimorphism, so φ' is one.

Step 5: $\tilde{\varphi}$ is an isomorphism. By Step 4 the composite $\tilde{\varphi} \circ p = \varphi'$ is an epimorphism, and if $u \circ \tilde{\varphi} = 0$ then $u \circ \tilde{\varphi} \circ p = 0$ and hence $u = 0$; so $\tilde{\varphi}$ is an epimorphism. Dually, running Steps 1 to 4 in the opposite category shows that the induced morphism $C \to B$, which is $\iota \circ \tilde{\varphi}$, is a monomorphism; and a composite $g \circ h$ can be a monomorphism only if h is, so $\tilde{\varphi}$ is a monomorphism. Being both, it is an isomorphism by lemma 1.1.8, completing the proof.

From the above, we may think of ker(coker) : $K \to B$ as representing the "image" of φ:

Lemma 1.1.10: Abelian Category: Images

Let $\varphi : A \to B$ be a morphism in an abelian category, and let $\iota : K \to B$ be the kernel of the cokernel of φ. Then:

1. ι is a monomorphism
2. φ factors through ι
3. ι is initial with these properties

***Proof*:**
Part (1) is lemma 1.1.3, since ι is a kernel. Part (2) is Step 1 of the proof of theorem 1.1.9: writing $q = \mathrm{coker}\varphi$, the identity $q \circ \varphi = 0$ makes φ factor uniquely through $\iota = \ker q$.

Part (3) is the minimality established in Step 3 of that proof: if $m\colon V \to B$ is any monomorphism through which φ factors, then ι factors through m. Concretely, m is a kernel by condition (2) of definition 1.1.6, hence the kernel of its own cokernel d by lemma 1.1.7; then $d \circ \varphi = 0$ makes d factor through q, so $d \circ \iota = 0$ and ι factors through $\ker d = m$, as we sought to show.

Definition 1.1.11: Abelian Category: Image

Let $\varphi : A \to B$ be a morphism in an abelian category. Then the *iamge* of φ is $\ker(\operatorname{coker}(\varphi))$ and is denoted $\operatorname{im}(\varphi)$. Similarly, its co-image is $\operatorname{coker}(\ker(\varphi))$ and is dentoed $\operatorname{coim}(\varphi)$.

With our intuition from working with R-modules, given a morphism $\varphi : A \to B$ with image $\iota : K \to B$, we may imagine there is an induced epimorphism $\overline{\varphi} : A \to K$. This is indeed the case:

Proposition 1.1.12: Abelian Category: sub object and quotient objects

Let $\varphi : A \to B$ be a morphism in an abelian category with image $\operatorname{im}\varphi \colon K \to B$ and coimage $\operatorname{coim}\varphi \colon A \to C$. Then the induced morphism $A \to K$ is an epimorphism and the induced morphism $C \to B$ is a monomorphism.

***Proof*:**
The induced morphism $A \to K$ is the morphism φ' of theorem 1.1.9, shown to be an epimorphism in Step 4 of that proof. The induced morphism $C \to B$ is the composite $\iota \circ \tilde{\varphi}$, a composite of a monomorphism with an isomorphism, hence a monomorphism, completing the proof.

From the cannonical decomposition, we may say that the image and coimage are "isomorphic" (which is not strictly true since these are morphisms, however by the Freyd-Mitchell Theorem this is a safe way of looking at it).

Theorem 1.1.13: Abelian Category: Exactness

Consider a sequence of objects in an abelian cateogy

$$\cdots \to A \xrightarrow{\varphi} B \xrightarrow{\psi} C \to \cdots$$

Then this sequence is *exact* if:

1. $\psi \circ \varphi = 0$
2. $\operatorname{coker}\varphi \circ \ker \psi = 0$

Example 1.1.14: Categorical Exactness

1. Aluffi p.577

Freyd-Mitchell Theorem

As mentioned, R-mod is an abelian category, and as we shall develop a bit later it the categories R-mod "represent" all abelian categories.

Theorem 1.1.15: Freyd-Mitchell Theorem

Let A be a small abelian category. Then there is a fully faithful exact functor $A \to R$-Mod for a suitable ring R. Such a functor reflects exactness: a sequence in A is exact if and only if its image in R-Mod is.

***Proof Key ideas*:**
The proof constructs R as the endomorphism ring of a suitable injective cogenerator in the category of left exact functors $A^{\text{op}} \to \mathbf{Ab}$, and occupies a chapter's worth of category theory that this book does not otherwise need; it is given in full in *An Introduction to Homological Algebra* [12], §1.6. The idea is that A embeds into its category of left exact presheaves by Yoneda, that this larger category is a Grothendieck category with enough injectives, and that a single injective cogenerator there converts the whole category into modules over its endomorphism ring.

The clause about reflecting exactness is immediate from the rest and is what the theorem is used for: a fully faithful exact functor reflects kernels and cokernels, so a diagram chase valid in R-Mod transfers back to A. This is what licenses the element-based arguments in lemma 2.1.17 and throughout chapter 4.

1.2 Localization of Categories

The constructions of this book repeatedly produce morphisms that ought to be invertible but are not. The model for what to do about it is already familiar from commutative algebra: a ring R and a multiplicative set S of elements one wishes to invert are replaced by $S^{-1}R$, together with a map $R \to S^{-1}R$ that is universal among ring maps sending S into the units. Nothing in that description mentions elements of a ring rather than morphisms of a category, and the same universal property can be imposed one level up.

The payoff comes in chapter 2, where the class to be inverted is the class of quasi-isomorphisms and the resulting category is the derived category. The construction is recorded here, before it is needed, because it is a statement about categories and not about complexes.

Definition 1.2.1: Localization of a Category

Let $\mathcal{C}$ be a category and W a class of morphisms of $\mathcal{C}$. A *localization* of $\mathcal{C}$ at W is a category $\mathcal{C}[W^{-1}]$ together with a functor $Q\colon \mathcal{C} \to \mathcal{C}[W^{-1}]$ such that

1. $Q(w)$ is an isomorphism for every $w \in W$;
2. for every category $\mathcal{D}$ and every functor $F\colon \mathcal{C} \to \mathcal{D}$ carrying each element of W to an isomorphism, there is a functor $\overline{F}\colon \mathcal{C}[W^{-1}] \to \mathcal{D}$ with $\overline{F} \circ Q = F$, and $\overline{F}$ is unique with this property.

As with every universal property, the definition determines the answer up to canonical isomorphism as soon as an answer exists: two localizations receive unique comparison functors from one another whose composites must be the identity, by the uniqueness clause applied to Q itself. Existence is the substantive point, and it is settled by a construction that simply writes down the morphisms that formal inversion forces to exist.

Theorem 1.2.2: Existence of the Localization

Let $\mathcal{C}$ be a category and W a class of morphisms. Then a localization $Q\colon \mathcal{C} \to \mathcal{C}[W^{-1}]$ exists.

***Proof*:**
Step 1: The construction. Let $\mathcal{C}[W^{-1}]$ have the same objects as $\mathcal{C}$. For objects X and Y, consider finite *zigzags* from X to Y: alternating strings of morphisms of $\mathcal{C}$

$$X = X_0 \;\mapstochar\relbar\; X_1 \;\mapstochar\relbar\; \cdots \;\mapstochar\relbar\; X_n = Y$$

in which each link $X_i \mapstochar\relbar X_{i+1}$ is either a morphism $X_i \to X_{i+1}$ of $\mathcal{C}$, called a forward link, or a morphism $X_{i+1} \to X_i$ that belongs to W, called a backward link. Enlarging W to contain all identities and to be closed under composition changes neither the class of functors inverting it nor, therefore, the localization, so assume this without loss of generality; it is what makes the second move below well formed. Impose on the collection of zigzags the smallest equivalence relation generated by: replacing two consecutive forward links by their composite; replacing two consecutive backward links by their composite; deleting a forward or backward link that is an identity; and deleting a pair of adjacent links w forward followed by w backward, or w backward followed by w forward, for $w \in W$. Let $\mathrm{Hom}_{\mathcal{C}[W^{-1}]}(X, Y)$ be the collection of equivalence classes, with composition given by concatenation, which is well defined because the generating relations are local to the links they touch. Concatenation is associative and the empty zigzag is an identity, so this is a category. Let Q send an object to itself and a morphism to the one-link forward zigzag.

Step 2: Q inverts W. For $w \in W$ the backward one-link zigzag on w is a morphism in the opposite direction, and concatenating it with the forward one-link zigzag on w in either order produces a pair of adjacent links which the last generating relation deletes, leaving the empty zigzag. So $Q(w)$ is an isomorphism.

Step 3: The universal property. Let $F\colon \mathcal{C} \to \mathcal{D}$ carry W into isomorphisms. Define $\overline{F}$ to be F on objects, and on a zigzag to be the composite in $\mathcal{D}$ obtained by applying F to each forward link and $F(-)^{-1}$ to each backward link. Each generating relation is respected: composites go to composites by functoriality, identities to identities, and an adjacent pair w, w of opposite orientation goes to $F(w)F(w)^{-1}$ or $F(w)^{-1}F(w)$, both identities. So $\overline{F}$ is well defined on equivalence classes, and it is a functor because concatenation goes to composition. By construction $\overline{F} \circ Q = F$.

For uniqueness, suppose $G \circ Q = F$ for a functor G. Every morphism of $\mathcal{C}[W^{-1}]$ is a concatenation of one-link zigzags, each of which is $Q(f)$ or $Q(w)^{-1}$; a functor is determined by its values on these, and $G(Q(f)) = F(f)$ forces $G(Q(w)^{-1}) = G(Q(w))^{-1} = F(w)^{-1}$. So G agrees with $\overline{F}$ on every morphism, completing the proof.

1.2.3 Warning: Localization and Set Theory The construction above builds each $\mathrm{Hom}_{\mathcal{C}[W^{-1}]}(X, Y)$ out of *all* zigzags, and for a category $\mathcal{C}$ that is not small there is no guarantee that the result is a set rather than a proper class. The property at risk is *local smallness*: the localization of a locally small category need not be locally small.

Two remedies are standard. The first is to assume $\mathcal{C}$ small, which is of no use here since $\mathrm{Ch}(\mathbf{A})$ never is. The second is to exhibit a calculus of fractions as in theorem 1.2.5, which replaces arbitrary zigzags by roofs; note that this alone does not bound anything, since the roofs over a fixed object are indexed by a proper class. What is needed in addition is that the W-morphisms into each object admit a *set* that is cofinal among them, and a multiplicative system with that property is what makes the localization locally small.

Neither remedy applies to $\mathrm{Ch}(\mathbf{A})$ localized at quasi-isomorphisms, because the quasi-isomorphisms there are *not* a multiplicative system. Both conditions (2) and (3) of definition 1.2.4 fail: taking $C^\bullet = (\mathbb{Z} \xrightarrow{\mathrm{id}} \mathbb{Z})$ in degrees $-1, 0$, which is acyclic, and $h\colon \mathbb{Z}[0] \to C^\bullet$ the chain map that is the identity in degree 0, every chain map $t\colon X^\bullet \to \mathbb{Z}[0]$ has only a degree-zero component, so $h \circ t = 0$ forces $t = 0$, which is not a quasi-isomorphism. The standard route is instead to pass to the homotopy category first: quasi-isomorphisms *are* a multiplicative system in $K(\mathbf{A})$, and $D(\mathbf{A}) = K(\mathbf{A})[\mathrm{qis}^{-1}]$, which is how section 2.5 presents it.

The zigzag description settles existence and is useless for computation: a morphism is an equivalence class of arbitrarily long strings, and deciding when two strings agree is not something one does by hand. Under a mild hypothesis on W every zigzag collapses to a single roof, and then morphisms become genuinely manipulable.

Definition 1.2.4: Multiplicative System

A class W of morphisms in a category $\mathcal{C}$ is a *multiplicative system* if

1. W contains every identity and is closed under composition;
2. (Ore condition) for every $f\colon Z \to B$ in $\mathcal{C}$ and every $t\colon V \to B$ in W there are $t'\colon Z' \to Z$ in W and $f'\colon Z' \to V$ in $\mathcal{C}$ with $f \circ t' = t \circ f'$;
3. for every pair $f, g\colon X \to Y$ and every $s\colon Y \to Y'$ in W with $s \circ f = s \circ g$, there is $t\colon X' \to X$ in W with $f \circ t = g \circ t$.

Condition (2) is what lets two roofs be composed, and condition (3) is what makes the resulting notion of equality between roofs transitive. Both are visible in the ring case: for a commutative ring every S is a multiplicative system, and the conditions are the familiar manipulations that put two fractions over a common denominator.

Theorem 1.2.5: Calculus of Fractions

Let W be a multiplicative system in $\mathcal{C}$. Then:

1. every morphism $A \to B$ in $\mathcal{C}[W^{-1}]$ is represented by a *roof*

$$\begin{tikzcd} & Z \arrow[dl, "s"'] \arrow[dr, "f"] & \\ A & & B \end{tikzcd}$$

   with $s \in W$, standing for $Q(f) \circ Q(s)^{-1}$;

2. if two roofs (s, f) and (s', f') are *commonly dominated*, meaning there are $u\colon Z'' \to Z$ and $u'\colon Z'' \to Z'$ with $su = s'u' \in W$ and $fu = f'u'$, then they represent the same morphism;

3. conversely, two roofs representing the same morphism are commonly dominated, so that roofs modulo common domination compute $\mathrm{Hom}_{\mathcal{C}[W^{-1}]}(A, B)$.

***Proof*:**

Step 1: Roofs compose. Given roofs $A \xleftarrow{s} Z \xrightarrow{f} B$ and $B \xleftarrow{t} V \xrightarrow{g} C$, apply the Ore condition to $f\colon Z \to B$ and $t \in W$ to obtain $t'\colon Z' \to Z$ in W and $f'\colon Z' \to V$ with $ft' = tf'$. Then $A \xleftarrow{st'} Z' \xrightarrow{gf'} C$ is a roof, since $st' \in W$ by closure under composition, and in $\mathcal{C}[W^{-1}]$ it represents

$$Q(g)Q(f')Q(t')^{-1}Q(s)^{-1} = Q(g)Q(t)^{-1}Q(f)Q(s)^{-1}$$

the identity following by rearranging $Q(f)Q(t') = Q(t)Q(f')$, both $Q(t)$ and $Q(t')$ being invertible. This is the composite of the two given morphisms.

Step 2: Part (2). Suppose (s, f) and (s', f') are commonly dominated by u, u'. Since $su = s'u'$ lies in W and $Q(s)$ is invertible, $Q(u)$ is invertible too, and then

$$Q(f)Q(s)^{-1} = Q(fu)Q(su)^{-1} = Q(f'u')Q(s'u')^{-1} = Q(f')Q(s')^{-1}$$

so the two roofs represent the same morphism.

Step 3: Common domination is an equivalence relation. Reflexivity and symmetry are immediate. For transitivity, suppose Z_1 dominates (s, f) and (s', f') through u_1, v_1 with $\sigma_1 := su_1 = s'v_1 \in W$, and Z_2 dominates (s', f') and (s'', f'') through u_2, v_2 with $\sigma_2 := s'u_2 = s''v_2 \in W$. Apply the Ore condition to $\sigma_1\colon Z_1 \to A$ and $\sigma_2 \in W$ to get $t'\colon Z_3 \to Z_1$ in W and $w\colon Z_3 \to Z_2$ with $\sigma_1 t' = \sigma_2 w$. Then

$$s'(v_1 t') = \sigma_1 t' = \sigma_2 w = s'(u_2 w)$$

so condition (3) of definition 1.2.4, applied to the parallel pair $v_1 t', u_2 w\colon Z_3 \rightrightarrows Z'$ and to $s' \in W$, yields $t\colon Z_4 \to Z_3$ in W with $v_1 t' t = u_2 w t$. The roof with middle leg $\sigma_1 t' t \in W$ and legs $u_1 t' t$ and $v_2 w t$ then dominates both (s, f) and (s'', f''): the middle leg lies in W by closure, and the two outer compatibilities follow because $fu_1 = f'v_1$ and $f''v_2 = f'u_2$ both factor through f'.

Step 4: Part (1). Every morphism of $\mathcal{C}[W^{-1}]$ is a zigzag, and it suffices to show every zigzag is equivalent to one of the form backward-then-forward. Assign to a zigzag the number of pairs $i < j$ with link i forward and link j backward. Composing two adjacent links of the same orientation strictly decreases the length and does not increase this number; and given an adjacent pair $Z \xrightarrow{f} B \xleftarrow{t} V$ with $t \in W$, the Ore condition supplies $Z \xleftarrow{t'} Z' \xrightarrow{f'} V$ with $t' \in W$ and $ft' = tf'$, which represents the same morphism because $Q(t)^{-1}Q(f) = Q(f')Q(t')^{-1}$, and which strictly decreases the count. The count cannot decrease forever, so the process terminates at a zigzag with no forward link preceding a backward one; composing along each run leaves a single roof.

Step 5: Part (3). What remains is the converse of Step 2. The standard route builds the category whose morphisms are roofs modulo common domination, with composition as in Step 1, and checks that this is well defined on classes, associative and unital, and that it satisfies the universal property of definition 1.2.1; comparing with $\mathcal{C}[W^{-1}]$ through that universal property then forces the two to agree, which is exactly the converse. Those verifications are bookkeeping of the same kind as Steps 1 and 3, carrying no further idea, and they are carried out in full in *An Introduction to Homological Algebra* [12]. This completes the proof.

The construction now applies verbatim to the situation this book treats, and it is recorded as a corollary so that later chapters may cite existence rather than assume it.

Corollary 1.2.6: Existence of the Derived Category

Let $\mathbf{A}$ be an abelian category and let qis denote the class of quasi-isomorphisms in $\mathrm{Ch}(\mathbf{A})$. Then the localization

$$D(\mathbf{A}) := \mathrm{Ch}(\mathbf{A})[\mathrm{qis}^{-1}]$$

exists, and the canonical functor $Q\colon \mathrm{Ch}(\mathbf{A}) \to D(\mathbf{A})$ is universal among functors carrying quasi-isomorphisms to isomorphisms.

***Proof*:**
Apply theorem 1.2.2 with $\mathcal{C} = \mathrm{Ch}(\mathbf{A})$ and $W = \mathrm{qis}$; no hypothesis on W is needed for existence. The universal property is clause (2) of definition 1.2.1.

The construction says nothing about local smallness, and by box 1.2.3 neither remedy applies here, so $D(\mathbf{A})$ is produced only as a possibly non-locally-small category. The route that repairs this, and the one section 2.5 follows, is to localize $K(\mathbf{A})$ rather than $\mathrm{Ch}(\mathbf{A})$, where the quasi-isomorphisms do form a multiplicative system and theorem 1.2.5 applies, as we sought to show.

2

Complexes and (co)Homology

Let $\mathbf{A}$ be a fixed abelian category, which by theorem 1.1.15 may without loss of generality it can be considered ${}_R\mathrm{Mod}$. In this chapter, we shall focus on studying (co)chains of such modules, that is sequence of modules that respect equation (1):

$$\mathrm{im}(d^n) \subseteq \ker(d^{n+1})$$

The collection of all chains shall be denoted $\mathrm{Ch}(\mathbf{A})$. We shall see that $\mathrm{Ch}(\mathbf{A})$ is an abelian category, the notion of extensions (and snakes lemma) is well defined, which will be explored. In the right categories, it is relatively easy to construct a concrete complex given the category has enough objects, for example $M \in \mathrm{Obj}(R\text{-}\mathbf{Mod})$ always has a complex known as a *free resolution* (see [7, Free Resolution]). There will be many different ways of creating complexes given an object, giving us different types of resolutions. We shall see that when a category has enough projective objects (resp. Injective objects), then "up to homotopy equivalence", projective resolutions shall be *initial* (resp. final) in the category of complexes $\mathrm{Ch}(A)$. Thus, we shall take the time to develop the homology theory in categories with enough projective or injective objects. We shall see that under appropriate circumstances, taking (co)homology of a projective resolution of an object can give us some important information about said object.

Homology and cohomology are duals of each other, which means on the theoretical level they are the same. As cohomology is the more computationally useful tool, we shall be focusing on cohomology on this chapter, with the idea that the reader shall be comfortable being able to switch indices when necessary. Taking the cohomology of a complex shall be shown to be an additive functorial process, meaning for a given morphism of complexes $\alpha^\bullet : A^\bullet \to B^\bullet$, we shall get a morphism each cohomology $H^i(\alpha) : H^i(A^\bullet) \to H^i(B^\bullet)$ which will be shown can be to form a new complex $H^\bullet(\alpha) : H^\bullet(A^\bullet) \to H^\bullet(B^\bullet)$.

The goal of this chapter is to show how to extract information given a complexes (co)homology. Given this, it is important to know what information a morphism of complexes preserves in cohomology, that is given a morphism of complexes, $\alpha^\bullet : A^\bullet \to B^\bullet$, when is $H^\bullet(\alpha) : H^\bullet(A^\bullet) \to H^\bullet(B^\bullet)$ an isomorphism? This shall lead us to the notion of *quasi-isomorphisms*, and their "representation" through mapping cones. As it turns out, quasi-isomorphisms do not behave nicely in general. What we shall then seek out is a stricter concept known as a *homotopy equivalence* that will still be a quasi-isomorphism, but will be invertible in the appropriate category.

2.1 Basic Definition and Properties

Starting with defining the generalization of an exact sequence:

Definition 2.1.1: Chain Complex

A *chain complex* $(C_\bullet, d_\bullet)$ is a sequence of objects and morphisms:

$$\cdots \xleftarrow{d_i} M_i \xleftarrow{d_{i+1}} M_{i+1} \xleftarrow{d_{i+2}} \cdots$$

with the indices running through $\mathbb{Z}$, such that

$$d_i \circ d_{i+1} = 0$$

and each d_i is called a *boundary map*. Similarly, a *cochain complex* $(C^\bullet, d^\bullet)$ is a sequence of objects

$$\cdots \xrightarrow{d^i} M^i \xrightarrow{d_{i-1}} M^{i-1} \xrightarrow{d^{i+2}} \cdots$$

with the indices running through $\mathbb{Z}$, such that

$$d^i \circ d^{i-1} = 0$$

where $d^\bullet$ are is called the *differential*[a].

[a]Due to it's origin's in differential geometry and algebraic topology

Chains and cochains can be interchanged by taking

$$M_i = M^{-i}$$

Working with one or the other makes essentially no difference for homological algebra. In certain cases, some structures present themselves more in the cohomological case (ex. a natural ring structure called the *cohomology ring* shall become important to us), and so we opt to work with the cohomology, even though at this stage this can be considered to be an arbitrary choice. The condition that $d^i \circ d^{i-1} = 0$ is the same as $\mathrm{im} d^{i-1} \subseteq \ker d^i$. If the complex is exact, then $\mathrm{im} d^{i-1} = \ker d^i$ for all i. We also often write $C^\bullet$ instead of $(C^\bullet, d^\bullet)$ if the mappings $d^\bullet$ are clear from context or implied. In some textbooks, even the bullet is dropped and M represents the complex. Sometime, for homology, the notation $\partial_\bullet$ shall be used instead of $d_\bullet$ due to it's connection to being the boundary map for topological spaces when creating complexes out of topological spaces[1].The condition that $d^i \circ d^{i-1} = 0$ (resp. $d_i \circ d_{i+1} = 0$) is often abbreviated to $d \circ d = 0$.

Due to historical origins in algebraic topology, the kernel and image of the differential

[1]Similarly, $d^\bullet$ stands for the *differential* due to it's link to the derivative, which would be explored in a class in algebraic topology, or see EYNTKA Algebraic Topology

Definition 2.1.2: Cycles and Boundaries

Let $C_\bullet$ (resp. $C^\bullet$) be a chain complex (resp. cochain complex). Then:

1. $\ker(d_n)$ is called the n-*cycles* of $C_\bullet$ and $\mathrm{im}(d_{n+1})$ is called the n-boundary of $C_\bullet$. The n-cycle is represented as $Z_n = Z_n(C_\bullet)$ and the n-*boundary* is represented as $B_n = B_n(C_\bullet)$

2. $\ker(d^n)$ is called the n-*cocycles* of $C^\bullet$ and $\mathrm{im}(d^{n-1})$ is called the n-*coboundary* of $C^\bullet$. The n-cocycle is represented as $Z^n = Z_n(C_\bullet)$ and the n-coboundary is represented as $B_n = B_n(C_\bullet)$

The name cycles and boundaries has it's origin in algebraic topology:

Example 2.1.3: Cycles and Boundaries
(Put that image from p.67 from alg top notes)

The collection of all complexes forms a Category:

Definition 2.1.4: Category Of Complexes

Let $\mathbf{A}$ be an abelian category. Then $\mathrm{Ch}^\bullet(\mathbf{A})$ is a new category called the *category of complexes of A* where

1. $\mathrm{Obj}(\mathrm{Ch}^\bullet(\mathbf{A})) = \{\text{cochain complexes in } \mathbf{A}\}$

2. For two complexes $C^\bullet$ and $N^\bullet$, a morphism $\alpha^\bullet \in \mathrm{Hom}_{\mathrm{Ch}(\mathbf{A})}(C^\bullet, N^\bullet)$ consists of morphisms st the following diagram commute:

$$\begin{tikzcd}
\cdots \arrow[r] & M^{i-1} \arrow[r, "d^{i-1}_{C^\bullet}"] \arrow[d, "\alpha^{i-1}"] & M^i \arrow[r, "d^i_{C^\bullet}"] \arrow[d, "\alpha^{i-1}"] & M^{i+1} \arrow[r] \arrow[d, "\alpha^{i-1}"] & \cdots \\
\cdots \arrow[r] & N^{i-1} \arrow[r, "d^{i-1}_{N^\bullet}"] & N^i \arrow[r, "d^i_{N^\bullet}"] & N^{i+1} \arrow[r] & \cdots
\end{tikzcd}$$

Similarly for $\mathrm{Ch}_\bullet(\mathbf{A})$ with chain complexes.

If unambiguous, the bullet is dropped giving $\mathrm{Ch}(\mathbf{A})$. It should be checked that $\mathrm{Ch}(A)$ is indeed a category. There are many variations of $\mathrm{Ch}(A)$ which will be important:

1. $C^+(A)$ is the full subcategory of $\mathrm{Ch}(A)$ of complexes which are bounded from bellow, so that $L^i = 0$ for $i \ll 0$

2. $C^-(A)$ is the full subcategory of $\mathrm{Ch}(A)$ of complexes which are bounded from above, so that $L^i = 0$ for $i \gg 0$

3. $C^{\geq 0}(A)$ (resp. $C^{\leq 0}$) for which $L^i = 0$ for $i \leq 0$ (resp. $i \geq 0$)

4. P, I are full subcategories of A consisting of projective and injective objects in A respectively, and hence we have $\mathrm{Ch}(P)$, $\mathrm{Ch}(I)$, and all its variants

There are also a few simple functors we may immediately define:

1. Take $\iota_r : \mathbf{A} \to \mathrm{Ch}(\mathbf{A})$ that sends an object A form $\mathbf{A}$ to a complex that is zero every and A in the rth degree. We shall commonly identify A with the complex given by ι_0.
2. Take $s_r : \mathrm{Ch}(\mathbf{A}) \to \mathrm{Ch}(\mathbf{A})$ which shifts over the complex, namely
$$C^\bullet \mapsto M[r]^\bullet$$
where $M[r]^i = M^{i+r}$ and $d^i_{M[r]^\bullet} = (-1)^r d^{i+r}_{C^\bullet}$ (the $(-1)^r$ shall be explained shortly).
3. the truncation functor shall truncate the complex in the natural way.

More importantly, $\mathrm{Ch}(A)$ is an abelian category, allowing the notion of exactness between complexes to be well-defined:

Lemma 2.1.5: Complex Category Is Abelian

Let A be an abelian category and $\mathrm{Ch}(A)$ the category of complexes of A. Then $\mathrm{Ch}(A)$ is an abelian category

***Proof*:**
Left as an exercise, check that:

1. morphisms between two complexes form an abelian group, namely since $\alpha^i + \beta^i$ will be well-defined
2. Finite products and coporducts exist in $\mathrm{Ch}(A)$ as a consequence of them existing in A
3. kernels and cokernls will come from the commutativity of the diagram defining morphisms of complexes

Since it is an abelian category, we shall be able to consider sequences of complexes, and in particular exact sequences of complexes:
$$\cdots \to L^\bullet \to C^\bullet \to N^\bullet \to \cdots$$
which give a commutative diagram like so:

$$\begin{tikzcd}
 & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & \\
\cdots \arrow[r] & A_{n+2} \arrow[r] \arrow[d, "\iota"] & A_{n+1} \arrow[r] \arrow[d, "\iota"] & A_n \arrow[r] \arrow[d, "\iota"] & A_{n-1} \arrow[r] \arrow[d, "\iota"] & \cdots \\
\cdots \arrow[r] & B_{n+2} \arrow[r] \arrow[d, "\pi"] & B_{n+1} \arrow[r] \arrow[d, "\pi"] & B_n \arrow[r] \arrow[d, "\pi"] & B_{n-1} \arrow[r] \arrow[d, "\pi"] & \cdots \\
\cdots \arrow[r] & C_{n+2} \arrow[r] \arrow[d] & C_{n+1} \arrow[r] \arrow[d] & C_n \arrow[r] \arrow[d] & C_{n-1} \arrow[r] \arrow[d] & \cdots \\
 & 0 & 0 & 0 & 0 &
\end{tikzcd} \tag{2.1}$$

Example 2.1.6: Complexes

1. Given any object M, we may always create the trivial complex

$$\cdots 0 \to 0 \to M \to 0 \to 0 \to \cdots$$

   Verify that this is indeed a complex.

2. As mentioned, In R-**Mod**, the free resolution is an example of a complex.

3. Any short exact sequence can be thought of an (exact) complex if we add 0's that extend arbitrarily to the left and right.

4. Let $C^n = \mathbb{Z}/8\mathbb{Z}$ for $n \geq 0$ and $C^n = 0$ for $n < 0$. Let each d^n be defined as:

$$\overline{x} \mapsto \overline{4x}$$

   Show that this is a cochain complex.

5. Let $\{V_i, W_i\}_{i\in\mathbb{Z}}$ be a collection of vector spaces. Define:

$$A_n = V_n \oplus W_n \oplus V_{n-1}$$

   Combining the projection and inclusion maps $A_n \to V_{n-1} \to A_{n-1}$, show that $A_\bullet$ forms a chain complex.

6. Let $C^\bullet$ be a cochain complex, choose an object $A \in \mathbf{A}$, and apply $\mathrm{Hom}_A(-, A)$ to each object. Show that this is a cochain complex.

Since $\mathrm{Ch}(\mathbf{A})$ is an abelian category, kernels and cokernels exist and match with monomorphisms/epimorphisms, hence giving us the following:

Definition 2.1.7: Sub-complexes And Quotient Complexes

Let $C^\bullet$ be a (co)chain complex. Then a (co)chain complex $N^\bullet$ is called a *subcomplex* if each N_n is a sub-object of C_n and the differential on B_n is the restriction of the differential in C_n (categorically, the inclusion map $\iota : N^\bullet \to C^\bullet$ is a morphism of complexes).

Furthermore, C^n/B^n is a *quotient-complex*. Furthermore, by the canonical decomposition of morphisms in abelian categories, for any complex-morphism $\alpha^\bullet$, $\{\ker(\alpha^\bullet)\}$ and $\{\mathrm{coker}(\alpha^\bullet)\}$ is a sub-complex and quotient complex respectively.

2.1.1 Homology and Cohomology

Chain complexes need not be exact. The cohomology of a complex measures how far away a complex is from being exact, that is how far away a complex is form representing a sequence of extensions

Definition 2.1.8: Homology and Cohomology Of A Complex

Let $(C^\bullet, d^\bullet)$ be a complex. Then the ith homology and cohomology of the complex at M^i is, respectively:

$$H_i(C^\bullet) = \frac{\ker d_i}{\mathrm{im} d_{i+1}} \qquad H^i(C^\bullet) := \frac{\ker d^i}{\mathrm{im} d^{i-1}}$$

Though homology and cohomology are similar, it is not necessarily the case that $H^i(C^\bullet) = H_{n-i}(C^\bullet)$ for some appropriate n around which the chain can "pivot". We shall see such examples soon, and later also show under what conditions we may compare homology and cohomology[2]

Example 2.1.9: Cohomology Of Complex

Let's compute the cohomology of sequences in example 2.1.6.

1. The homology and cohomology at the 0th entry is:

$$\frac{\ker(d_0)}{\mathrm{im}(d_1)} = \frac{\ker(d^1)}{\mathrm{im}(d^0)} = M$$

   which shows that at this position, the chain is "off" by M in being exact.

2. As we have defined exact free resolutions, they are exact, meaning $\mathrm{im}(d^i) = \ker(d^{i-1})$, hence their homology and cohomology is always 0
3. Any exact sequence has trivial homology and cohomology
4. This sequence is not exact, and yet it's homoogy/cohomology is trivial (as the reader should verify). Such a sequence is called *acyclic*.
5. The reader should calculate this cohomology as an exercise
6. This cohomology shall be calculated later in this chapter.

Lemma 2.1.10: Cohomology Functor

For every $i \in \mathbb{Z}$, there exists a functor:

$$H^i : \mathrm{Ch}(A) \to A \qquad C^\bullet \mapsto H^i(C^\bullet)$$

which is additive and covariant.

This shows that taking cohomology is indeed a way of analyzing resolutions, for it preserves enough of the structures of modules and complexes. Notice it is not necessarily an exact functor, hence it will not in general preserve extensions.

[2]The universal coefficient theorem and Poincaré duality being two famous examples the reader may have heard of if they took algebraic topology

***Proof*:**
Let $\alpha^\bullet : C^\bullet \to N^\bullet$ be a morphism of complexes and take $H^i(C^\bullet)$ and $H^i(N^\bullet)$. Let $x \in H^i(H^\bullet)$ so that $x = k + \mathrm{im}(d_M^{i-1})$ for d^{i-1}. Since $d^i(x) = 0$, by commutativity it must be that $\alpha^i(x) \in \ker(d_N^i)$. Furthermore, by commutativity we see that $\overline{\alpha^i}$ is zero on the image of $\mathrm{im}(d_M^{i-1})$, hence $H^i(\alpha^\bullet)$ is well-defined.

This functor is certainly covariant, and is additive since $\mathrm{Hom}_A(H^i(C^\bullet), H^i(N^\bullet))$ is an abelian group.

Notice that each H^i takes d^i to the zero map. Hence, we may consider the following functor:

$$H^\bullet : \mathrm{Ch}(\mathbf{A}) \to \mathrm{Ch}(\mathbf{A})$$

by taking $H^i(C^\bullet)$ for each i and connecting them by the zero-morphism. This is the cohomology of this complex then equals that of $C^\bullet$, namely:

$$H^\bullet(H^\bullet(C^\bullet)) = H^\bullet(C^\bullet)$$

Lemma 2.1.11: Cohomology And Products

Let $\{A_\alpha^\bullet\}$ be a collection of complexes in $\mathbf{A}$, and suppose the coproduct (respectively the product) indexed by α exists and is an exact functor. Then

$$H^n\Big(\bigoplus_\alpha A_\alpha^\bullet\Big) = \bigoplus_\alpha H^n(A_\alpha^\bullet) \qquad H^n\Big(\prod_\alpha A_\alpha^\bullet\Big) = \prod_\alpha H^n(A_\alpha^\bullet)$$

The exactness hypothesis is automatic for *finite* index sets, where the two agree by lemma 1.1.5, and it holds for arbitrary index sets in R-Mod.

***Proof*:**
Treat the coproduct; the product is the same argument in the opposite category. The complex $\bigoplus_\alpha A_\alpha^\bullet$ has n-th term $\bigoplus_\alpha A_\alpha^n$ and differential $\bigoplus_\alpha d_\alpha^n$.

An exact functor preserves kernels and images, so

$$\ker\Big(\bigoplus_\alpha d_\alpha^n\Big) = \bigoplus_\alpha \ker d_\alpha^n, \qquad \mathrm{im}\Big(\bigoplus_\alpha d_\alpha^{n-1}\Big) = \bigoplus_\alpha \mathrm{im} d_\alpha^{n-1}$$

and it preserves the quotient of the second by the first, being exact. Hence

$$H^n\Big(\bigoplus_\alpha A_\alpha^\bullet\Big) = \frac{\bigoplus_\alpha \ker d_\alpha^n}{\bigoplus_\alpha \mathrm{im} d_\alpha^{n-1}} = \bigoplus_\alpha \frac{\ker d_\alpha^n}{\mathrm{im} d_\alpha^{n-1}} = \bigoplus_\alpha H^n(A_\alpha^\bullet)$$

For a finite index set the coproduct is a finite biproduct, which every additive functor preserves, so no hypothesis is needed; in R-Mod both $\bigoplus$ and $\prod$ are exact and kernels and images are computed componentwise, giving the statement for arbitrary index sets. This completes the proof.

Now, given a morphism of complexes $\alpha^\bullet : C^\bullet \to N^\bullet$, we may ask what (co)homological information such a morphism preserves. In other words, given a morphism $\alpha^\bullet$, when is $H^\bullet(\alpha^\bullet)$ an isomorphism? We give such morphisms a name:

Definition 2.1.12: Quasi-Isomorphism

Let $\alpha^\bullet : C^\bullet \to N^\bullet$ be a morphism of complexes. Then if $H^\bullet(\alpha^\bullet) : H^\bullet(C^\bullet) \to H^\bullet(N^\bullet)$ is an isomorphism, $\alpha^\bullet$ is called a *quasi-isomorphism*

One may wonder if the fact that the (co)homology is isomorphism is strong enough of a condition for the chains to be isomorphic. This turns out to be rather far from the case:

Example 2.1.13: Quasi-Isomorphism

1. The following counter example shall come up with many different minor variations, and so the reader should keep this in mind. Take the following morphism of complexes:

$$\begin{tikzcd}
\cdots \arrow[r] & 0 \arrow[r] \arrow[d] & \mathbb{Z} \arrow[r, "\cdot 2"] \arrow[d] & \mathbb{Z} \arrow[r] \arrow[d, "\pi"] & 0 \arrow[r] \arrow[d] & \cdots \\
\cdots \arrow[r] & 0 \arrow[r] & 0 \arrow[r] & \mathbb{Z}/2\mathbb{Z} \arrow[r] & 0 \arrow[r] & \cdots
\end{tikzcd}$$

   This is a quasi-isomorphism, as the reader should check, however it does not have an inverse. More generally, it is not even an equivalence relation, failing to be symmetric (show there is no morphism the other way around that is a quasi-isomorphism).

2. The natural maps $H^\bullet(C^\bullet) \to C^\bullet$ and $C^\bullet \to H^\bullet(C^\bullet)$ are quasi-isomorphisms, given the property of $H^\bullet(H^\bullet(C^\bullet)) = H^\bullet(C^\bullet)$ discussed earlier.

3. Let $\iota(A) \in \mathrm{Obj}(\mathrm{Ch}(\mathbf{A}))$ be the complex with A at the 0th position and zero's everywhere else. Say that $C^\bullet$ is a resolution such that $C^\bullet \to \iota(A)$ is a quasi-isomorphism. Show that this implies $C^\bullet$ is exact everywhere except possibly at $H^0(M^0)$

The lack of symmetry means that quasi-isomorphisms do not form an equivalence relation, but a *weak equivalence relation* (a relation that is reflective and transitive, but not symmetric). This is rather a pity, since if it did form an equivalence relation, then the equivalence class of morphisms up to weak isomorphism would hold all the necessary (co)homological information. Alas, this was not mean to be. Thus, in the pursuit of finding (co)homological invariants, it productive to ask how (co)homological information can be detected between (co)chain complexes. One idea is to map $\mathrm{Ch}(\mathbf{A})$ into another category $D(\mathbf{A})$ in which quasi-isomorphisms become isomorphisms. If quasi-isomorphisms formed equivalence relationships, we could have formed a congruence category which would satisfy this condition, however as just mentioned this cannot be the case.

Such a functor has already been encountered: the (co)homology functor is already one such example. It is more fruitful to as if this can be done in a *univerasl* way, that is if there is a category $D(\mathbf{A})$ where for any category D for which there exists an additive functor $\mathscr{F} : \mathrm{Ch}(\mathbf{A}) \to D$ where every quasi-isomorphism in $\mathrm{Ch}(\mathbf{A})$ becomes an isomorphism in D, there exists a unique functor such that the following diagram commutes:

$$\begin{tikzcd} \mathrm{Ch}(\mathbf{A}) \arrow[r, "\mathscr{F}"] \arrow[d] & D \\ D(\mathbf{A}) \arrow[ur, "\exists!"'] & \end{tikzcd} \tag{2.2}$$

This category does indeed exist, by corollary 1.2.6, and is called the *derived category* (hence the $D(-)$ notation). This category shall further be explored in the coming sections. In some nice categories $\mathbf{A}$, quasi-isomorphisms will line up with the nicer notion of *homotopy-equivalence*, in which case the derived category will be eminently understandable. We shall explore these all in greater detail section 2.2.

Though quasi-isomorphisms may not be ideal in computing (co)homological invariants, they are still important as the defining concept for (co)homological information. For example, the following shows the relationship between quasi-isomorphism and exactness:

Lemma 2.1.14: Quasi-isomorphisms And Exactness

Let $C^\bullet$ be a cochain complex. Then the following are equivalent:

1. $C^\bullet$ is exact
2. $C^\bullet$ is acyclic, that is $H^\bullet(C^\bullet) = 0$
3. The morphism of complexes $0 \to C^\bullet$ is a quasi-isomorphism

***Proof*:**
(1) $\Leftrightarrow$ (2): exactness of $C^\bullet$ at degree n means $\ker d^n = \mathrm{im} d^{n-1}$, and $H^n(C^\bullet) = \ker d^n / \mathrm{im} d^{n-1}$ by definition 2.1.8. A quotient object is zero exactly when the two are equal, so $C^\bullet$ is exact at every degree if and only if $H^n(C^\bullet) = 0$ for every n.

(2) $\Leftrightarrow$ (3): the zero complex has $H^n(0) = 0$ for every n, and the unique morphism $0 \to C^\bullet$ induces the unique morphism $0 \to H^n(C^\bullet)$ in each degree. That morphism is an isomorphism precisely when $H^n(C^\bullet) = 0$. So $0 \to C^\bullet$ is a quasi-isomorphism, meaning it induces isomorphisms in every degree, if and only if $H^\bullet(C^\bullet) = 0$, completing the proof.

2.1.2 Snake Lemma

We often have multiple ways of constructions a resolution for an object M. We shall often extend resolutions using smaller resolutions. We then get cohomology of this new extended resolution by simply applying the cohomology functor. However, is is this functor exact? More generally, let's say we have a short exact sequence of complexes, then after applying cohomology do we get a short exact sequence of homology complexes? The answer is no: if

$$0 \to L^\bullet \to C^\bullet \to N^\bullet \to 0$$

is a short exact sequence, then we get the (not necessarily exact) sequence:

$$H^i(L^\bullet) \to H^i(C^\bullet) \to H^i(N^\bullet)$$

In terms of morphisms, if we have:

$$\alpha^\bullet : C^\bullet \to N^\bullet$$

Then the morphism after applying the cohomology functor $H^\bullet(\alpha^\bullet) : H^\bullet(C^\bullet) \to H^\bullet(N^\bullet)$ need not be exact, i.e. it doesn't preserve kernels and cokernels. Can we "fix" this in some way by recovering some exact sequence from this and thus allowing our analysis of building more complicated resolutions for M to be independent of cohomology. The answer is yes, Namely, if we take the cohomology of complexes, we may "snake" like in the following diagram:

$$\begin{tikzcd}
 & 0 \arrow[d, dotted] & 0 \arrow[d, dotted] & 0 \arrow[d, dotted] & 0 \arrow[d, dotted] & \\
\cdots \arrow[r, dotted] & H_{n+2}(A) \arrow[r, dotted] \arrow[d, "\iota"] & H_{n+1}(A) \arrow[r, dotted] \arrow[d, "\iota"] & H_n(A) \arrow[r, dotted] \arrow[d, "\iota"] & H_{n-1}(A) \arrow[r, dotted] \arrow[d, "\iota"] & \cdots \\
\cdots \arrow[r, dotted] & H_{n+2}(B) \arrow[r, dotted] \arrow[d, "\pi"] & H_{n+1}(B) \arrow[r, dotted] \arrow[d, "\pi"] & H_n(B) \arrow[r, dotted] \arrow[d, "\pi"] & H_{n-1}(B) \arrow[r, dotted] \arrow[d, "\pi"] & \cdots \\
\cdots \arrow[r, dotted] & H_{n+2}(C) \arrow[r, dotted] \arrow[d, dotted] \arrow[uur] & H_{n+1}(C) \arrow[r, dotted] \arrow[d, dotted] \arrow[uur] & H_n(C) \arrow[r, dotted] \arrow[d, dotted] \arrow[uur] & H_{n-1}(C) \arrow[r, dotted] \arrow[d, dotted] & \cdots \\
 & 0 & 0 & 0 & 0 &
\end{tikzcd}$$

Theorem 2.1.15: Snake Lemma

Given a short exact sequence of complexes $0 \to A \to B \to C \to 0$, we may form a long exact sequence of modules using cohomology:

$$\cdots \to H^n(A) \xrightarrow{i^*} H^n(B) \xrightarrow{\pi^*} H^n(C) \xrightarrow{\partial} H^{n+1}(A) \xrightarrow{i^*} H^{n+1}(B) \xrightarrow{\pi^*} H^{n+1}(C) \to \cdots$$

Equivalently, in homological indexing, where the differential lowers degree and H_n is written for the homology in degree n, the same statement reads

$$\cdots \to H_n(A) \xrightarrow{i_*} H_n(B) \xrightarrow{\pi_*} H_n(C) \xrightarrow{\partial} H_{n-1}(A) \xrightarrow{i_*} H_{n-1}(B) \to \cdots$$

and it is this form that the proof below constructs. The two are the same statement under $H_n = H^{-n}$, and both are used in this book.

The following proof can be found (in the Homological variant) in EYNTKA Algebraic Topology

***Proof*:**
We first define the map $\partial : H_n(C) \to H_{n-1}(A)$. We should keep in mind the following diagram:

$$\begin{tikzcd}
 & A_{n-1} \arrow[d, "\iota"] \\
B_n \arrow[r, "\partial"] \arrow[d, "\pi"] & B_{n-1} \\
C_n &
\end{tikzcd}$$

Let $c \in C_n$ be a cycle so that $c \in \ker \partial_n$. Since π is surjective, $c = \pi(b)$ for some $b \in B_n$. Then by definition $\partial : B_n \to B_{n-1}$ so $\partial(b) \in B_{n-1}$. Then by commutativity $j(\partial(b)) = \partial(j(b)) = \partial(c) = 0$ (since c is a cycle), thus $\partial b \in \ker(\pi)$. Since $\ker j = \mathrm{im} i$, $\partial b = i(a)$ for some $a \in A_{n-1}$. Since $i(\partial a) = \partial i(a) = \partial\partial b = 0$ since i is injective, $\partial a = 0$. Thus, define $\partial : H_n(C) \to H_{n-1}(A)$ by $[c] \mapsto [a]$. This is well defined since:

1. The element a is uniquely determined by ∂b since i is injective
2. for differnet choice b' for b, we get $j(b') = j(b)$, os $b' - b \in \ker(j)$ which is equal to $\mathrm{im}(i)$. Thus, $b' - b = i(a')$ for some a', ehcen $b' = b + i(a')$. But then this means we just chanage a to another representative (a homologous element) $a + \partial a'$, namely since
$$i(a + \partial a') = i(a) + i(\partial a') = \partial b + \partial i(a') = \partial(b = i(a'))$$
3. A difference choice of c within its homology class would have hte form $c = \partial c'$. To see this, since $c' = j(b')$ for some b',
$$c + \partial c' = c\partial j(b') = c + j(\partial b') = j(b + \partial b')$$
thus b is replaced by $b = \partial b'$, which gives ∂b and thus a is unchanged.

Finally, $\partial : H_n(C) \to H_{n-1}(A)$ is a homomorphism, for if $\partial[c_1] = [a_1]$ and $\partial[c_2] = [a_2]$ via elements b_1, b_2, then
$$j(b_1 + b_2) = j(b_1) + j(b_2) = c_1 = c_2$$
and
$$i(a_1 + a_2) = i(a_1) + i(a_2) = \partial b_1 + \partial b_2 = \partial(b_1 + b_2)$$
hence
$$\partial([c_1] + [c_2]) = [a_1] + [a_2]$$
giving us the map ∂ is indeed a homomorphism. What's left to show is the following inclusions:

1. $\mathrm{im} i_* \subseteq \ker j_*$. Since $ji = 0$, $j_* i_* = 0$
2. $\mathrm{im} j_* \subseteq \ker \partial$. By definition of ∂, $\partial b = 0$ so $\partial j_* = 0$.
3. $\ker j_* \subseteq \mathrm{im} i_*$. Let $b \in B_n$ with boundary represent a homology class in $\ker j_*$, so $j(b) = \partial c'$ for some $c' \in C_{n+1}$. Since j is surjectiv,e $c' = j(b')$ for some $b' \in B_{n+1}$. Then since $\partial j(b') = \partial c' = j(b)$,
$$j(b - \partial b') = j(b) - j(\partial b') = j(b) - \partial j(b') = 0$$

Thus, $b - \partial b' = i(a)$ for some $a \in A - n$. Since $i(\partial a) = \partial i(a) = \partial(b - \partial b') = \partial b = 0$ and i is injective, a is a cycle. Thus:

$$\iota_*[a] = [b - \partial b'] = [b]$$

showing that i_* maps onto $\ker j_*$

4. $\ker \partial \subseteq \mathrm{im} j_*$. By definition of ∂, if c represents a homology class in $\ker \partial$, then $a = \partial a'$ for some $a' \in A_n$. The element $b - i(a')$ is a cycle since $\partial(b - i(a')) = \partial b - \partial i(a') = \partial b - i(\partial a') = \partial b - i(a) = 0$. Since

$$j(b - i(a')) = j(b) - ji(a') = j(b) = c$$

we get that $j_*[b - i(a')] = [c]$

5. $\ker i_* \subseteq \mathrm{im}\partial$. Given a cycle $a \in A_{n-1}$ such that $i(a) = \partial b$ for seom $b \in B_n<$ then $j(b)$ is a cycle since $\partial j(b) = j(\partial b) = ji(a) = 0$, hence $\partial[j(b)] = [a]$

Doing this shall completing the proof.

Example 2.1.16: Uses Of Snake Lemma

1. *The six-term kernel-cokernel sequence.* The classical statement that carries the name "snake" is a special case of theorem 2.1.15. Suppose given a commutative diagram with exact rows

$$\begin{tikzcd}
0 \arrow[r] & A^0 \arrow[r] \arrow[d, "f"] & B^0 \arrow[r] \arrow[d, "g"] & C^0 \arrow[r] \arrow[d, "h"] & 0 \\
0 \arrow[r] & A^1 \arrow[r] & B^1 \arrow[r] & C^1 \arrow[r] & 0
\end{tikzcd}$$

Read each column as a complex concentrated in degrees 0 and 1, so that $A^\bullet = (A^0 \xrightarrow{f} A^1)$ and likewise for $B^\bullet$ and $C^\bullet$. Exactness of the two rows says precisely that $0 \to A^\bullet \to B^\bullet \to C^\bullet \to 0$ is a short exact sequence of complexes, and for a two-term complex $H^0 = \ker$ and $H^1 = \mathrm{coker}$. Since H^n vanishes outside degrees 0 and 1, the long exact sequence of theorem 2.1.15 has only six nonzero terms:

$$0 \to \ker f \to \ker g \to \ker h \xrightarrow{\partial} \mathrm{coker} f \to \mathrm{coker} g \to \mathrm{coker} h \to 0$$

The connecting morphism ∂ is the "snake" that crosses from the kernel row to the cokernel row.

2. *A computation.* Apply the six-term sequence to multiplication by a fixed integer $n \geq 1$ on the short exact sequence $0 \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{Q}/\mathbb{Z} \to 0$. Multiplication by n is injective on $\mathbb{Z}$ and bijective on $\mathbb{Q}$, and $\mathbb{Q}/\mathbb{Z}$ is divisible, so

$$\ker f = 0, \quad \ker g = 0, \quad \ker h = \tfrac{1}{n}\mathbb{Z}/\mathbb{Z} \cong \mathbb{Z}/n\mathbb{Z}$$

$$\mathrm{coker} f = \mathbb{Z}/n\mathbb{Z}, \quad \mathrm{coker} g = 0, \quad \mathrm{coker} h = 0$$

The six-term sequence collapses to $0 \to \mathbb{Z}/n\mathbb{Z} \xrightarrow{\partial} \mathbb{Z}/n\mathbb{Z} \to 0$, forcing ∂ to be an isomorphism. The explicit description confirms it: an element of $\ker h$ is the class of some $q \in \mathbb{Q}$ with $nq \in \mathbb{Z}$, and ∂ sends it to the class of nq in $\mathbb{Z}/n\mathbb{Z}$; the generator $1/n$ goes to the generator 1.

3. *Where the machine is used later.* Every long exact sequence in this book is an instance of theorem 2.1.15. The long exact sequence of derived functors (theorem 2.3.10) applies it to a short exact sequence of resolutions, the exact triangle of a mapping cone (proposition 2.1.19) applies it to the cone sequence, and the exact couple that generates a spectral sequence (theorem 4.2.2) is nothing but all the long exact sequences of a filtration assembled at once. The topological applications, the Mayer-Vietoris sequence and the long exact sequence of a pair, are developed where they are used, in *Algebraic Topology* [9].

The most-used consequence of the snake lemma is a criterion for recognizing an isomorphism in the middle of a diagram from isomorphisms on either side. It is stated here because it is needed whenever two exact sequences are compared, and in particular in the comparison theorem for spectral sequences (theorem 4.2.9).

Lemma 2.1.17: Five Lemma

Consider a commutative diagram in an abelian category with exact rows

$$\begin{tikzcd}
A_1 \arrow[r] \arrow[d, "f_1"] & A_2 \arrow[r] \arrow[d, "f_2"] & A_3 \arrow[r] \arrow[d, "f_3"] & A_4 \arrow[r] \arrow[d, "f_4"] & A_5 \arrow[d, "f_5"] \\
B_1 \arrow[r] & B_2 \arrow[r] & B_3 \arrow[r] & B_4 \arrow[r] & B_5
\end{tikzcd}$$

If f_2 and f_4 are isomorphisms, f_1 is an epimorphism, and f_5 is a monomorphism, then f_3 is an isomorphism.
More precisely, f_3 is a monomorphism as soon as f_2 and f_4 are monomorphisms and f_1 is an epimorphism, and f_3 is an epimorphism as soon as f_2 and f_4 are epimorphisms and f_5 is a monomorphism.

***Proof*:**
By theorem 1.1.15 the diagram may be chased with elements. Label the horizontal maps $\alpha_i\colon A_i \to A_{i+1}$ and $\beta_i\colon B_i \to B_{i+1}$.

Step 1: f_3 is a monomorphism. Assume f_2, f_4 monomorphisms and f_1 an epimorphism, and let $a \in A_3$ with $f_3(a) = 0$. Then $f_4(\alpha_3(a)) = \beta_3(f_3(a)) = 0$, and f_4 is a monomorphism, so $\alpha_3(a) = 0$. By exactness of the top row at A_3 there is $b \in A_2$ with $\alpha_2(b) = a$. Now $\beta_2(f_2(b)) = f_3(\alpha_2(b)) = f_3(a) = 0$, so by exactness of the bottom row at B_2 there is $c \in B_1$ with $\beta_1(c) = f_2(b)$. Since f_1 is an epimorphism, $c = f_1(e)$ for some $e \in A_1$. Then

$$f_2(\alpha_1(e)) = \beta_1(f_1(e)) = \beta_1(c) = f_2(b)$$

and f_2 is a monomorphism, so $\alpha_1(e) = b$. Therefore $a = \alpha_2(b) = \alpha_2(\alpha_1(e)) = 0$, the last equality by exactness of the top row at A_2.

Step 2: f_3 is an epimorphism. Assume f_2, f_4 epimorphisms and f_5 a monomorphism, and let $b \in B_3$. Since f_4 is an epimorphism, $\beta_3(b) = f_4(a)$ for some $a \in A_4$. Then

$$f_5(\alpha_4(a)) = \beta_4(f_4(a)) = \beta_4(\beta_3(b)) = 0$$

by exactness of the bottom row at B_4, and f_5 is a monomorphism, so $\alpha_4(a) = 0$. By exactness of the top row at A_4 there is $c \in A_3$ with $\alpha_3(c) = a$. The element $b - f_3(c)$ satisfies

$$\beta_3(b - f_3(c)) = \beta_3(b) - f_4(\alpha_3(c)) = f_4(a) - f_4(a) = 0$$

so by exactness of the bottom row at B_3 there is $e \in B_2$ with $\beta_2(e) = b - f_3(c)$. Since f_2 is an epimorphism, $e = f_2(g)$ for some $g \in A_2$, and then

$$f_3\big(\alpha_2(g) + c\big) = \beta_2(f_2(g)) + f_3(c) = \big(b - f_3(c)\big) + f_3(c) = b$$

so b lies in the image of f_3.

Under the hypotheses of the first statement both steps apply, so f_3 is both a monomorphism and an epimorphism, hence an isomorphism by lemma 1.1.8, completing the proof.

There is another way of looking at chain complexes which shall be important in elucidating triangle categories in section ref:HERE.

Mapping Cones

Using a particular total complex, we can detect when a quasi-isomorphism is an isomorphism. This result requires the snake lemma to prove, hence the need to put this section after the snake lemma.

Definition 2.1.18: Mapping Cone

Let $\alpha^\bullet : L^\bullet \to M^\bullet$ be a morphism of complexes. Then the *mapping cone* is the complex denoted by $MC(\alpha^\bullet)$ defined by:

$$MC(\alpha)^i := L[1]^i \oplus M^i = L^{i+1} \oplus M^i$$

where the morphism $d^i_{MC(\alpha)} : MC(\alpha)^i \to MC(\alpha)^{i+1}$ are given by:

$$d^i_{MC(\alpha)}(l, m) := (-d^{i+1}_L(l), \alpha^{i+1}(l) + d^i_M(m))$$

The term mapping cone comes from the topological notion of a mapping cone of a function. Visually, the mapping cone can be seen as the following commutative diagram:

$$\begin{tikzcd}
\cdots \arrow[r] \arrow[ddr] & L^{i+1} \arrow[r, "-d_L^{i+1}"] \arrow[ddr, "\alpha^{i+1}"] & L^{i+2} \arrow[r] \arrow[ddr] & \cdots \\
 & \oplus & \oplus & \\
\cdots \arrow[r] & M^i \arrow[r, "d_M^i"'] & M^{i+1} \arrow[r] & \cdots
\end{tikzcd}$$

Given $MC(\alpha)^\bullet$, we may define two natural chain maps $C^\bullet \to MC(\alpha)^\bullet$ and $MC(\alpha)^\bullet \to L[1]^\bullet$ induced by the natural exact morphisms $M^i \to L^{i+1} \oplus M^i \to L^{i+1}$ which implies that:

$$0 \to C^\bullet \to MC(\alpha)^\bullet \to L[1]^\bullet \to 0$$

is exact.

Proposition 2.1.19: Mapping Cone And Exact Triangles

There is an exact triangle

$$\begin{tikzcd} & H^\bullet(C^\bullet) \arrow[dr] & \\ H^\bullet(L[1]^\bullet) \arrow[ur, "+1\ \delta"] & & H^\bullet(MC(\alpha)^\bullet) \arrow[ll] \end{tikzcd}$$

where δ is the morphism induced by cohomology

***Proof*:**
The short exact sequence of complexes displayed above,

$$0 \to C^\bullet \to MC(\alpha)^\bullet \to L[1]^\bullet \to 0$$

is exact, so theorem 2.1.15 produces a long exact sequence in cohomology

$$\cdots \to H^n(C^\bullet) \to H^n(MC(\alpha)^\bullet) \to H^n(L[1]^\bullet) \xrightarrow{\delta} H^{n+1}(C^\bullet) \to \cdots$$

Exactness at each of the three kinds of term is precisely exactness of the displayed triangle at each of its three corners, the degree shift $H^n(L[1]^\bullet) = H^{n+1}(L^\bullet)$ being what the label $+1$ records.

The connecting morphism is the one induced by α, with no intervening sign; the projection $MC(\alpha)^\bullet \to L[1]^\bullet$ is a chain map precisely because the shift carries the differential $-d_L$, which is what cancels against the sign in the cone. To see this, unwind definition 2.1.18 on representatives: a class in $H^n(L[1]^\bullet)$ is represented by some $l \in L^{n+1}$ with $d_L l = 0$, and its preimage in $MC(\alpha)^n = L^{n+1} \oplus C^n$ may be taken to be $(l, 0)$. Applying the differential of the cone,

$$d_{MC(\alpha)}(l, 0) = \big(-d_L l,\ \alpha^{n+1}(l)\big) = \big(0,\ \alpha^{n+1}(l)\big)$$

which is the image of $\alpha^{n+1}(l) \in C^{n+1}$ under the inclusion. Hence $\delta[l] = [\alpha^{n+1}(l)]$, that is $\delta = H^{n+1}(\alpha)$, completing the proof.

2.1.20 Remark Not all categories have this triangle property, categories which will have this will be called *triangulated categories*

Proposition 2.1.21: Quasi-isomorphism And Isomorphism

Let $\alpha^\bullet : C^\bullet \to N^\bullet$. Then $H^\bullet(\alpha^\bullet)$ is an isomorphism if and only if $MC(\alpha)^\bullet$ is an exact complex.

***Proof*:**
By proposition 2.1.19, applied with $\alpha\colon C^\bullet \to N^\bullet$, the cone sits in a long exact sequence

$$\cdots \to H^n\big(MC(\alpha)^\bullet\big) \to H^n\big(C[1]^\bullet\big) \xrightarrow{H^{n+1}(\alpha)} H^{n+1}(N^\bullet) \to H^{n+1}\big(MC(\alpha)^\bullet\big) \to \cdots$$

the connecting morphism having been identified there as the map induced by α, and $H^n(C[1]^\bullet) = H^{n+1}(C^\bullet)$.

Suppose $MC(\alpha)^\bullet$ is exact, so that $H^\bullet(MC(\alpha)^\bullet) = 0$ by lemma 2.1.14. Deleting the vanishing terms leaves

$$0 \to H^{n+1}(C^\bullet) \xrightarrow{H^{n+1}(\alpha)} H^{n+1}(N^\bullet) \to 0$$

exact for every n, so each $H^{n+1}(\alpha)$ is an isomorphism.

Conversely suppose every $H^n(\alpha)$ is an isomorphism. Fix n and read the sequence around $H^n(MC(\alpha)^\bullet)$:

$$H^{n-1}(C[1]^\bullet) \xrightarrow{H^n(\alpha)} H^n(N^\bullet) \to H^n\big(MC(\alpha)^\bullet\big) \to H^n(C[1]^\bullet) \xrightarrow{H^{n+1}(\alpha)} H^{n+1}(N^\bullet)$$

The left-hand map is surjective and the right-hand map is injective, both being isomorphisms. Exactness at $H^n(N^\bullet)$ then makes the morphism $H^n(N^\bullet) \to H^n(MC(\alpha)^\bullet)$ zero, and exactness at $H^n(C[1]^\bullet)$ makes the morphism $H^n(MC(\alpha)^\bullet) \to H^n(C[1]^\bullet)$ zero. Exactness at $H^n(MC(\alpha)^\bullet)$ therefore gives

$$H^n\big(MC(\alpha)^\bullet\big) = \ker\big(\text{zero morphism}\big) = \operatorname{im}\big(\text{zero morphism}\big) = 0$$

As n was arbitrary the cone is acyclic, hence exact by lemma 2.1.14, completing the proof.

2.1.3 Chain Homotopies and Homotopic Categories

The notion of homotopies originated in topology as a weaker alternative to continuous functions[3]. The topological notion of homotopy settled on the level of continuous functions, where $f, g : X \to Y$ are homotopic (denoted $f \simeq g$) if there exists a continuous function $F : X \times [0,1] \to Y$ where $F(\cdot, t) = f_t$ is continuous, $f_0 = f$ and $f_1 = g$. If we represent X, Y as one dimensional lines, then we may visualize a homotopy as:

[3] every homeomorphic is a homotopy equivalence, however not every homotopy equivalence is a homeomorphism, hence homotopy can be seen as weaker than continuity

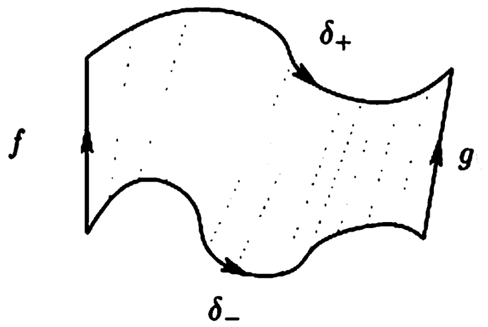

Figure 2.1: homotopy

In the development of homotopy theory, it was found that the (co)homology and homotopy of topological spaces (or their representations as chain complexes) is homology-invariant, hence homotopies are examples of quasi-isomorphisms. The proofs of these results turn out to be independent of the underlying topological structures and is purely algebraic, and hence the notion of homotopy has found a generalization as a purely algebraic notion between two (co)chain complexes.

Due to its origin in topology, the intuitions behind the following definitions will proceed more naturally with a more geometric build-up. Since homotopy may be interpreted purely algebraically, this geometrica build-up will be followed by a more algebraic interpretation. First, the reader may take for granted that given a topological space X, we may induce a chain $C_\bullet$, which is a topological object[4]. Given two continuous maps $f, g : X \to Y$, we may push forward the chain like so:

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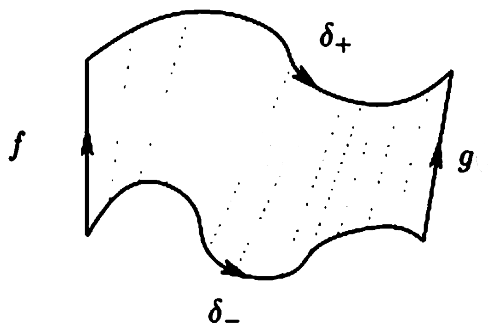


Figure 2.2: homotopy on chain

The space $h(C_\bullet)$ is a topological space, hence we may ask for it's border, $\partial h(C_\bullet)$. By inspecting the image, we see that:

$$\partial h(C_\bullet) = g(C_\bullet) - \delta_+ - f(C_\bullet) + \delta_- = g(C_\bullet) - f(C_\bullet) - h(\partial C_\bullet)$$

re-ordering, we get:

$$g(C_\bullet) - f(C_\bullet) = \partial h(C_\bullet) + h(\partial C_\bullet)$$

[4]See EYTNKA algebraic topology or Hatcher as for more details

Abstracting notation, we get:

$$g - f = \partial h + h\partial \tag{2.3}$$

Now, recall that in a chain complex, the image of the function ∂ is called the boundary, and that by definition of homology the boundary becomes trivial. In a topological setting, this vocabulary reflects the concrete of the boundary map ∂: it maps a chain to it's boundary. What this means for us is that when taking homology on both sides, we get:

$$H^k(g|_{C_\bullet} - f|_{C_\bullet}) = \overline{0}$$

meaning the two maps are equivalent up to homology! In other words, this "geometric intuition" shows that a homotopy is an example of a quasi-isomorphism. Even better, homotopies shall form an equivalence relation, hence they are an example of a quasi-isomorphism that shall capture some (co)homological information (and in the case where all quasi-isomorphisms are homotopies, they could fully capture the (co)homological information!)

Though this is the origins of chain homotopies being (co)homology-invariant, since they may be interpreted in pure algebraic terms, it is worthwhile taking a moment to have an algebraic build-up as well. Algebraically, homotopies can be seen as forming equivalence classes of (co)chain complexes up to *obstruction from being split-exact.* Recall that a split exact sequence is a short exact sequence that is also split at each map, that is there exists a section morphism $s^n : C^n \to C^{n-1}$ such that $d^{n-1} \circ s^n \circ d^{n-1} = d$, or if we drop the indices and collect all the section maps:

$$d = dsd$$

In other words, it is possible to "inverse" the map d via the map s in such a way that it "cancels" the effect of d (for we may re-apply the map d after sd and get the same result). The simplest source of split-exact (co)chains are exact (co)chains of vector-spaces, for any vector-space maps split[5]. Given a cochain, we may write:

$$C^n = Z^n \oplus B' \qquad (B^n)' \cong C^n/Z^n = d(C^n) = B^{n+1}$$

$$Z^n = B^n \oplus (H^n)' \qquad (H^n)' = Z^n/B^n = H^n(C)$$

Hence, $C^n \cong Z^n \oplus B^{n+1}$. This allows for the definition of the section map via the composition:

$$C^n \hookrightarrow Z^n \hookrightarrow B^n \cong d(C^n) \cong B^{n+1} \subseteq C^{n+1}$$

We may thus define the maps $d \circ s$ and $s \circ d$ where $d \circ s(C^n) = B^n$ and $s \circ d(C^n)(B^n)' \cong B^{n+1}$. Adding these maps together, we get a map $ds + sd : C^n \to C^n$ whose kernel is the homology:

$$\ker(ds + sd)_n = H^n(C^\bullet)$$

The kernel (and cokernel) of the chain map $ds + sd$ is naturally $H^\bullet(C^\bullet)$. Working over more general R-modules (and more generally in an arbitrary abelian category), we loose the natural section maps, for example $0 \to \mathbb{Z} \twoheadrightarrow \mathbb{Z}/2\mathbb{Z} \to 0$ does note have a section map for each map. Instead, we may define arbitrary maps $s^n : C^n \to D^{n-1}$ where we may put them in the following (*not* necessarily commutative) diagram:

$$\begin{tikzcd}
\cdots \arrow[r] & C^{n-1} \arrow[r, "d"] \arrow[dl] & C^n \arrow[r, "d"] \arrow[dl, "s^n"] & C^{n+1} \arrow[r] \arrow[dl, "s^{n+1}"] & \cdots \arrow[dl] \\
\cdots \arrow[r] & D^{n-1} \arrow[r, "d"] & D^n \arrow[r, "d"] & D^{n+1} \arrow[r] & \cdots
\end{tikzcd}$$

[5] More generally, any maps of projective modules split

Given any such collection of morphisms, we may define $f^n = s^{n+1} \circ d^n + d^{n-1} \circ s^n$ (or, if dropping indices and abbreviate, $f = sd + ds$). Notice that even though the maps $s^\bullet$ do not commute, $f^\bullet$ *will* create a commutative diagram, in particular it is a chain map. To see this, notice that (after dropping indices):

$$df = d(ds + sd) = dsd = (ds + sd)d = fd$$

showing it is indeed a chain map:

$$\begin{tikzcd}
\cdots \arrow[r] \arrow[dl, dotted] & C^{n-1} \arrow[r, "d"] \arrow[d, "f^{n-1}"] \arrow[dl, dotted] & C^n \arrow[r, "d"] \arrow[d, "f^n"] \arrow[dl, dotted, "s^n"] & C^{n+1} \arrow[r] \arrow[d, "f^{n+1}"] \arrow[dl, dotted, "s^{n+1}"] & \cdots \arrow[dl, dotted] \\
\cdots \arrow[r] & D^{n-1} \arrow[r, "d"] & D^n \arrow[r, "d"] & D^{n+1} \arrow[r] & \cdots
\end{tikzcd}$$

Thus, given any collection of morphisms $s^\bullet$, we may naturally define a chain map using the differentials. Conversely, is every morphism of complexes representable as $ds + sd$. This is not in general the case; maps being representable like so are given a name:

Definition 2.1.22: Null-homotopic Maps

Let $f^\bullet : C^\bullet \to D^\bullet$ be a morphism of complexes. Then f is called *null-homotopic* if there exists a collection $\{s^i\}_{i\in\mathbb{Z}}$ of morphisms $s^i : C^i \to D^{i-1}$ such that

$$f^\bullet = ds + sd$$

The maps $s^\bullet$ are called the *chain contraction* of $f^\bullet$

To show that not all maps are null-homotopic, consider the following lemma:

Lemma 2.1.23: Null-homotopic And Split Exact Complexes

Let $C^\bullet$ be a complex. Then $C^\bullet$ is split exact if and only if the identity map is null-homotopic.

***Proof*:**

Throughout, split exact means exact together with a family $s^n : C^n \to C^{n-1}$ satisfying $dsd = d$, and null-homotopic means $\mathrm{id} = ds + sd$ for some such family.

Step 1: Null-homotopic implies split exact. Suppose $\mathrm{id} = dt + td$. For exactness, let $c \in \ker d^n$; then

$$c = dt(c) + td(c) = dt(c)$$

so $c \in \mathrm{im} d^{n-1}$, and the reverse inclusion $\mathrm{im} d^{n-1} \subseteq \ker d^n$ holds because $d \circ d = 0$. For the splitting, compute

$$dtd = d(td) = d(\mathrm{id} - dt) = d - d^2 t = d$$

so t is a splitting family.

Step 2: Split exact implies null-homotopic. Suppose $C^\bullet$ is exact and $dsd = d$. Set

$$\varphi = ds + sd$$

Then $d\varphi = dsd = d$ and $\varphi d = dsd = d$, both using $d \circ d = 0$; in particular φ commutes with d.

Put $\psi = \mathrm{id} - \varphi$. From the two identities just proved, $d\psi = d - d = 0$ and $\psi d = d - d = 0$. The first says $\mathrm{im}\psi \subseteq \ker d$, and exactness gives $\ker d = \mathrm{im} d$, so $\mathrm{im}\psi \subseteq \mathrm{im} d$; the second says ψ vanishes on $\mathrm{im} d$. Composing, $\psi \circ \psi = 0$.

Therefore $\varphi = \mathrm{id} - \psi$ is invertible, with inverse $\mathrm{id} + \psi$, and φ^{-1} commutes with d because φ does. Set $t = \varphi^{-1}s$. Then

$$dt + td = d\varphi^{-1}s + \varphi^{-1}sd = \varphi^{-1}(ds + sd) = \varphi^{-1}\varphi = \mathrm{id}$$

so the identity is null-homotopic with contraction t, completing the proof.

The name "null-homotopic" comes from how the (co)homology of such maps are trivial:

Proposition 2.1.24: Null-homotopic And Cohomology

Let $f^\bullet : C^\bullet \to D^\bullet$ be null-homotopic. Then $H^\bullet(f)$ is the zero map

***Proof*:**
Since f is null-homotopic, let $f = ds + sd$. Take a n-cycle representative x of an equivalence class of $H^n(C^\bullet)$. Then:

$$f(x) = d(s(x))$$

Hence, $f(x)$ is an n-boundary in D^n, hence $\overline{f(x)} = \overline{0}$ in $H^n(D^\bullet)$, as we sought to show. (this is an elementary result, but a similar proof is given in Aluffi p.612 if needed)

Thus, maps that are null-homotopic induce zero (co)homology maps![6] Thus, we may think of maps that are null-homotopic as being the "zero" maps. This leads us to the following definition:

Definition 2.1.25: Chain-homotopy

Let $f^\bullet, g^\bullet : C^\bullet \to D^\bullet$ be two morphisms of complexes. Then we say that these two maps are *chain homotopic* or just *homotopic* if there exists maps $h^i : C^i \to D^{i-1}$ such that

$$g^i - f^i = d^{i-1}_{D^\bullet} \circ h^i + h^{i+1} \circ d^i_{C^\bullet}$$

or, dropping indices:

$$g^\bullet - f^\bullet = dh + hd$$

If $f^\bullet, g^\bullet$ are homotopic, we write $f^\bullet \sim g^\bullet$

The s notation has been replaced with h notation to emphasize now the "homotopy" aspect of it rather than the "section" property that was evident in the case of vector spaces. If we were working with chains, then we would have

$$g_\bullet - f_\bullet = \partial h + h\partial$$

which, the reader may recognize as equation (2.3). If we were working with chains, then $h_i : C_i \to D_{i+1}$ would shift up a degree, which would be interpreted in the geometrical setting as moving *up* a dimension, in the same way that homotopies are a "dimension more" than the original spaces X, Y.

[6]TBD How much can we say about the converse, if a map induces the zero morphism, then is it null-homotopic?

Furthermore, the reader should verify the "up to homotopy" *is* an equivalence relation, as the reader should verify!

Importantly, two chain-homotopic maps induce the same (co)homology:

Proposition 2.1.26: Chain-Homotopy and Cohomology

Let $f^\bullet, g^\bullet : C^\bullet \to D^\bullet$ be chain homotopic. Then

$$H^\bullet(f^\bullet) = H^\bullet(g^\bullet)$$

***Proof*:**
Since $f^\bullet$ and $g^\bullet$ are chain-homotopic, by proposition 2.1.24 for each element in $H^k(C^\bullet)$ (since it is an additive functor):

$$H^k(f^\bullet)(\overline{x}) - H^k(g^\bullet)(\overline{x}) = 0 \qquad \Rightarrow \qquad H^k(f^\bullet)(\overline{x}) = H^k(g^\bullet)(\overline{x})$$

as we sought to show.

The above result shows promise in using chain-homotopies to hold (co)homological information. The following definition is the important connection idea:

Definition 2.1.27: Homotopy Equivalent Chains

Let $f^\bullet : C^\bullet \to D^\bullet$ be a morphism of complexes. Then $f^\bullet$ is a *homotopy equivalence* if there exists a morphisms of complexes $g^\bullet : D^\bullet \to C^\bullet$ such that

$$f^\bullet \circ g^\bullet \sim 1_{D^\bullet} \qquad \text{and} \qquad g^\bullet \circ f^\bullet \sim 1_{C^\bullet}$$

If $f^\bullet : C^\bullet \to D^\bullet$ is a homotopy equivalence, then $C^\bullet$ and $D^\bullet$ are said to be *homotopy equivalent*

Proposition 2.1.28: Homotopy Equivalence Ans Cohomology

Let $f^\bullet : C^\bullet \to D^\bullet$ be a homotopy equivalence. Then $f^\bullet$ is a quasi-isomorphism, that is $H^\bullet(f^\bullet)$ is an isomorphism

***Proof*:**
Since $f^\bullet$ is a homotopy equivalence, we have $g^\bullet$ that is it's homotopic inverse. Then the maps $f^\bullet \circ g^\bullet$ and $g^\bullet \circ f^\bullet$ induce inverse morphisms of cohomology by proposition 2.1.26, and hence give an isomorphism, as we sought to show.

As demonstrated in example 2.1.13 not all quasi-isomorphisms are invertible even up to homotopy equivalence, hence quasi-isomorphisms are more general. As mentioned earlier, in nice enough categories these two concepts shall coincide.

Now that homotopy equivalences is established, can be asked if given an (additive) functor $\mathscr{F} : \mathbf{A} \to \mathbf{B}$, if there are two homotopic complexes in $\mathrm{Ch}(\mathbf{A})$, will the (co)homology be isomorphic in $\mathbf{B}$? In other words, when trying to define the derived category that was mentioned earlier, can this category be defined at least up to homotopy? The answer is yes! The proof of this result essentially follows from the following lemma:

Lemma 2.1.29: Homotopy Equivalence And Additive Functors

Let $\mathscr{F} : \mathbf{A} \to \mathbf{B}$ be an additive functor between two abelian categories, $C(\mathscr{F})$ the induced functor on chains. Then if $f^\bullet \sim g^\bullet$ in $\mathrm{Ch}(\mathbf{A})$, $C(\mathscr{F})(f^\bullet) \sim \mathrm{Ch}(\mathbf{B})(g^\bullet)$ in $\mathrm{Ch}(\mathbf{B})$. Furthermore, if $C^\bullet$ and $D^\bullet$ are homotopy equivalence, $C(\mathscr{F})(C^\bullet)$ and $C(\mathscr{F})(D^\bullet)$ are homotopy equivalent

***Proof*:**
Given the first assertion, the second is an immediate consequence, hence we prove the first. Let $f^\bullet \sim g^\bullet$ so that :

$$f^i - g^i = d^{i-1}_{D^\bullet} \circ h^i + h^{i+1} \circ d^i_{C^\bullet}$$

Since $\mathscr{F}$ is an additive functor:

$$\mathscr{F}(f^i) - \mathscr{F}(g^i) = \mathscr{F}\left(d^{i-1}_{D^\bullet}\right) \circ \mathscr{F}(h^i) + \mathscr{F}\left(h^{i+1}\right) \circ \mathscr{F}\left(d^i_{C^\bullet}\right)$$

hence, the morphisms $\mathscr{F}(h^i)$ gives a homotopy between $\mathscr{F}(f^\bullet)$ and $\mathscr{F}(g^\bullet)$, as we sought to show.

Note this does *not* hold for quasi-isomorphisms (exercise 4.15 in Aluffi), needing the stronger condition of exact functors.

Theorem 2.1.30: Derived Category And Homotopy Equivalence

Let $\mathscr{F} : \mathbf{A} \to \mathbf{B}$ be an additive functor between two abelian categories. Then if $C^\bullet$ and $D^\bullet$ are homotopy-equivalent, then:

$$H^\bullet(\mathscr{F}(C^\bullet)) \cong H^\bullet(\mathscr{F}(D^\bullet))$$

that is, their cohomology is isomorphic.

***Proof*:**
By lemma 2.1.29, an additive functor carries homotopy-equivalent complexes to homotopy-equivalent complexes, so $\mathscr{F}(C^\bullet)$ and $\mathscr{F}(D^\bullet)$ are homotopy equivalent in $\mathrm{Ch}(\mathbf{B})$. By proposition 2.1.28 a homotopy equivalence is a quasi-isomorphism, so the induced map on cohomology is an isomorphism, giving $H^\bullet(\mathscr{F}(C^\bullet)) \cong H^\bullet(\mathscr{F}(D^\bullet))$, as we sought to show.

Hence, up to homotopy, the choice of cochain shall give the same cohomology information!

Homotopic Categories

With chain homotopies being a nicer sub-class of quasi-isomorphisms, their exploration is worth further pursuing. They are in fact almost the solution to the universal problem given in equation (2.2). Let us take for granted that a category $D(A)$ exists that satisfies the universal property. What necessary conditions can be deduced about $D(A)$?

Lemma 2.1.31: Derived Category And Additive Functors

Let $\mathbf{D}$ be an additive category, $\mathscr{F} : \mathrm{Ch}(\mathbf{A}) \to \mathbf{D}$ be an additive functor that takes quasi-isomorphisms to isomorphisms.

1. Let $C^\bullet$ be an exact complex in $\mathrm{Ch}(\mathbf{A})$. Then $\mathscr{F}(C^\bullet) = 0$, that is $\mathscr{F}(C^\bullet)$ is the zero object in $\mathbf{D}$.

2. Let $\alpha^\bullet : C^\bullet \to E^\bullet$ be a morphism of complexes that factors through an exact complex:

$$\begin{tikzcd} C^\bullet \arrow[rr, "\alpha^\bullet"] \arrow[dr] & & E^\bullet \\ & D^\bullet \arrow[ur] & \end{tikzcd}$$

   where $D^\bullet$ is exact. Then $\mathscr{F}(\alpha^\bullet)$ is the zero morphism.

***Proof*:**

1. Since $C^\bullet$ is exact, the zero morphism $0^\bullet : C^\bullet \to C^\bullet$ is a quasi-isomorphism, and so $\mathscr{F}(0^\bullet)$ maps to an isomorphism (i.e. an invertible morphism in $\mathbf{D}$). Thus:

$$\begin{tikzcd} \mathscr{F}(C^\bullet) \arrow[r, "\mathscr{F}(0^\bullet)"'] \arrow[rr, bend left, "\mathrm{id}_{\mathscr{F}(C^\bullet)}"] & \mathscr{F}(C^\bullet) \arrow[r, "\mathscr{F}(0^\bullet)^{-1}"'] & \mathscr{F}(C^\bullet) \end{tikzcd}$$

   Since $\mathscr{F}$ is additive functor, in particular a group homomorphism on hom-sets, $\mathscr{F}(0) = 0$. Thus:

$$\mathrm{id}_{\mathscr{F}(C^\bullet)} = 0$$

   It follows that $\mathscr{F}(C^\bullet)$ must be the zero object (as the reader should verify, see exercise ref:HERE)

2. Applying $\mathscr{F}$ to the diagram in the question, we get the diagram:

$$\begin{tikzcd} \mathscr{F}(C^\bullet) \arrow[rr] \arrow[dr] & & \mathscr{F}(E^\bullet) \\ & \mathscr{F}(D^\bullet) \arrow[ur] & \end{tikzcd}$$

   Since $\mathscr{F}(D^\bullet) = 0$ by the above result, we see that $\mathscr{F}(\alpha^\bullet)$ must be the zero morphisms, completing the proof.

Lemma 2.1.32: Derived Categories And Homotopies

Let $\mathscr{F} : \mathrm{Ch}(\mathbf{A}) \to \mathbf{D}$ be an additive functor that maps quasi-isomorphisms to isomorphisms. Let $f^\bullet, g^\bullet : C^\bullet \to D^\bullet$ be homotopic morphisms in $\mathrm{Ch}(\mathbf{A})$. Then

$$\mathscr{F}(f^\bullet) = \mathscr{F}(g^\bullet)$$

Note that this was shown explicitly when $\mathscr{F} = H^i(-)$.

***Proof*:**
Since $\mathscr{F}$ is additive, it suffices to show that if $f^\bullet \sim 0$, then $\mathscr{F}(f^\bullet) = 0$. By lemma 2.1.31, it suffices to show $f^\bullet$ factors through an exact complex. The exact complex that shall be used is the mapping cone $MC(\mathrm{id}_{C^\bullet})$ for the identity map $\mathrm{id}_{C^\bullet} : C^\bullet \to C^\bullet$, namely

$$MC(\mathrm{id}_{C^\bullet})^i = C^{i+1} \oplus C^i$$

$$d^i_{MC(\mathrm{id}_{C^\bullet})}(a,b) = (-d^{i+1}_{C^\bullet}(a), a + d^i_{C^\bullet}(b))$$

By proposition 2.1.21, since the identity map $\mathrm{id}_{C^\bullet}$ is (trivially) a quasi-isomorphism, $MC(\mathrm{id}_{C^\bullet})$ is an exact complex.

Now, the goal is to find morphism of complexes where:

$$C^\bullet \to MC(\mathrm{id}_{C^\bullet}) \to D^\bullet$$

the morphism of complexes $C^\bullet \to MC(\mathrm{id}_{C^\bullet})$ is readily available: for each i:

$$b \mapsto (0, b)$$

On the other hand, it is not immediately clear there is a morphism of complexes $MC(\mathrm{id}_{C^\bullet}) \to D^\bullet$. To find such a morphism, we take advantage of the homotopy $f^\bullet \sim 0$, that is there exists maps:

$$h^i : C^i \to D^{i-1}$$

such that $f^\bullet = fh + hd$. With these, define the maps:

$$\pi^i : MC(\mathrm{id}_{C^\bullet})i \to D^i \qquad (a,b) \mapsto h^{i+1}(a) + f^i(b)$$

If this is indeed a morphism of complexes, then the map $f : C^\bullet \to D^\bullet$ factored through an exact complex and hence must be trivial. What must be verified is that:

$$d^{i-1}_{D^\bullet} \circ \pi^{i+1} = \pi^i \circ d^{i-1}_{MC(\mathrm{id}_{C^\bullet})} \tag{2.4}$$

The left hand side of equation (2.4) gives:

$$(a,b) \mapsto h^i(a) + f^{i-1}(b) \mapsto d^{i-1}_{D^\bullet}(h^i(a) + f^{i-1}(b))$$

while the right hand side gives:

$$(a,b) \mapsto (-d^i_{C^\bullet}(a), a + d^{i-1}_{L^\bullet}(b)) \mapsto -h^{i+1}(d^i_{C^\bullet}(a) + f^i(a + d^{i-1}_{C^\bullet}(b))$$

These two sides must be equal. Simplifying the equality, we get:

$$(d^{i-1}_{D^\bullet} \circ h^i + h^{i+1} \circ d^i_{C^\bullet} - f^i)(a) = (f^i \circ d^{i-1}_{C^\bullet} - d^{i-1}_{D^\bullet} \circ f^{-1})(a)$$

for all $a \in C^{i+1}, b \in C^i$. But notice that both sides are zero: the right hand side b $f^\bullet$ is a morphism of complexes, while the left hand side because h is a homotopy and $f^\bullet \sim 0$. But then this gives us our desired result, completing the proof.

Thus, behave well for the desired universal property presented in equation 2.2. We formalize the category in question:

Definition 2.1.33: Homotopy Category

Let $\mathbf{A}$ be an abelian category. Then the *homotopy category* of $\mathbf{A}$, denoted $K(\mathbf{A})$ is the category whose objects ar ethe (co)chain complexes in $\mathbf{A}$ (i.e. $\mathrm{Ch}(\mathbf{A})$), and whose morphisms are:

$$\mathrm{Hom}_{K(\mathbf{A})}(C^\bullet, D^\bullet) = \mathrm{Hom}_{\mathrm{Ch}(\mathbf{A})}(C^\bullet, D^\bullet)/\sim$$

where $\sim$ is the homotopy relation, that is, the morphisms are morphisms of complexes up to homotopy.

These are indeed morphisms, namely $\sim$ is transitive, it respects composition (exercise). Naturally, the equivalent bounded variations $K^-(\mathbf{A})$ and $K^+(\mathbf{A})$ are obtained from $\mathrm{Ch}^+(\mathbf{A})$ and $\mathrm{Ch}^-(\mathbf{A})$.

Lemma 2.1.34: Homotopy Category Is Additive

Let $\mathbf{A}$ be an abelian category. Then $K(\mathbf{A})$ is an *additive category*

***Proof*:**
Recall from definition 2.1.33 that $K(\mathbf{A})$ has the same objects as $\mathrm{Ch}(\mathbf{A})$ and $\mathrm{Hom}_{K(\mathbf{A})}(C^\bullet, D^\bullet) = \mathrm{Hom}_{\mathrm{Ch}(\mathbf{A})}(C^\bullet, D^\bullet)/\sim$, the quotient by chain homotopy.

Step 1: The hom-sets are abelian groups. The null-homotopic morphisms form a subgroup of $\mathrm{Hom}_{\mathrm{Ch}(\mathbf{A})}(C^\bullet, D^\bullet)$: if $f = ds + sd$ and $f' = ds' + s'd$ then $f - f' = d(s - s') + (s - s')d$ is null-homotopic. Being a quotient of an abelian group by a subgroup, each hom-set of $K(\mathbf{A})$ is an abelian group, and $f \sim g$ exactly when $f - g$ is null-homotopic.

Step 2: Composition is well defined and bilinear. Suppose $f \sim f'$, say $f - f' = ds + sd$, and let g be a chain map composable on the left. Since g commutes with the differentials,

$$gf - gf' = g(ds + sd) = d(gs) + (gs)d$$

so $gf \sim gf'$; composing on the right with a chain map h gives $fh - f'h = d(sh) + (sh)d$ likewise. Hence composition descends to the quotient, and it is bilinear because it is bilinear in $\mathrm{Ch}(\mathbf{A})$ and the quotient maps are group homomorphisms.

Step 3: Zero object and biproducts. The zero complex is a zero object in $\mathrm{Ch}(\mathbf{A})$ and remains one after the quotient, since its hom-sets are already zero. For two complexes $C^\bullet$ and $D^\bullet$, the degreewise direct sum $C^\bullet \oplus D^\bullet$ carries the four structure morphisms i_C, i_D, p_C, p_D, which are chain maps and satisfy the biproduct identities $p_C i_C = \mathrm{id}$, $p_D i_D = \mathrm{id}$, $p_C i_D = 0$, $p_D i_C = 0$ and $i_C p_C + i_D p_D = \mathrm{id}$ in $\mathrm{Ch}(\mathbf{A})$. Identities between morphisms are preserved by the quotient functor, so they continue to hold in $K(\mathbf{A})$, and an object with such data is a biproduct there as well.

Together these are the axioms of an additive category, completing the proof.

Note that in general homotopic categories are *not* abelian. Indeed, homotopic maps $f^\bullet \sim g^\bullet$ need not have the same kernel/cokernel, giving no natural definition of these concepts. The homotopic category is a type of category that is stronger than additive but weaker than abelian, known as a *triangulated category* (recall box 2.1.20).

In $K(\mathbf{A})$, homotopy equivalences become isomorphisms. This comes straight from the definition, for if $f^\bullet \circ g^\bullet \sim \mathrm{id}$ in $\mathrm{Ch}(\mathbf{A})$, then $f^\bullet \circ g^\bullet = \mathrm{id}$ in $K(\mathbf{A})$, which can be thought of as formally inverting homotopy equivalences. Using this, we have the following almost universal property, or factorization

It looks like a universal property because it is one, though not the one for quasi-isomorphisms. Comparing with definition 1.2.1, the homotopy category is precisely the localization

$$K(\mathbf{A}) \cong \mathrm{Ch}(\mathbf{A})\big[\{\text{homotopy equivalences}\}^{-1}\big]$$

The proposition below is stated with the hypothesis that $\mathscr{F}$ inverts *quasi-isomorphisms*, which is strictly stronger than it needs to be, and that mismatch is what makes it read as an "almost" universal property. Its proof uses only that $\mathscr{F}$ inverts homotopy equivalences. Inverting all quasi-isomorphisms is the stronger requirement that produces $D(\mathbf{A})$ instead (corollary 1.2.6), and the gap between the two is exactly example 2.1.13: a quasi-isomorphism need not be a homotopy equivalence.

Proposition 2.1.35: Homotopy Category Factorization

Let $\mathscr{F} : \mathrm{Ch}(\mathbf{A}) \to D$ be an additive functor that maps quasi-isomorphisms to isomorphisms. Then $\mathscr{F}$ factors uniquely through $K(\mathbf{A})$:

$$\begin{tikzcd} \mathrm{Ch}(\mathbf{A}) \arrow[r, "\mathscr{F}"] \arrow[d] & D \\ K(\mathbf{A}) \arrow[ur, "\exists!"'] & \end{tikzcd}$$

***Proof*:**
Since every homotopy equivalence is a quasi-isomorphism (proposition 2.1.28), the hypothesis in particular sends homotopy equivalences to isomorphisms, and that is all the proof uses. Because $\mathscr{F}$ is defined on $\mathrm{Ch}(\mathbf{A})$ and $K(\mathbf{A})$ has the same objects, the only thing to establish is that $f \sim g$ implies $\mathscr{F}(f) = \mathscr{F}(g)$; the factorization and its uniqueness are then forced, since the quotient functor $\mathrm{Ch}(\mathbf{A}) \to K(\mathbf{A})$ is the identity on objects and surjective on morphisms.

Step 1: The cylinder of a complex. For a complex $C^\bullet$ define

$$\mathrm{Cyl}(C)^n = C^n \oplus C^{n+1} \oplus C^n, \qquad d(a,b,c) = (da - b,\ -db,\ dc + b)$$

This squares to zero, since $d^2(a,b,c) = (d(da-b) + db,\ d(db),\ d(dc+b) - db) = (0,0,0)$. Define chain maps

$$i_0(a) = (a,0,0), \qquad i_1(c) = (0,0,c), \qquad p(a,b,c) = a + c$$

Each is compatible with the differentials: $d\,i_0(a) = (da,0,0) = i_0(da)$, similarly for i_1, and $p(d(a,b,c)) = (da-b) + (dc+b) = d(a+c)$. Moreover $pi_0 = \mathrm{id}_C = pi_1$.

Step 2: p is a homotopy equivalence. The composite $i_0 p - \mathrm{id}$ sends (a,b,c) to $(c,-b,-c)$, and setting $s(a,b,c) = (0, -c, 0)$ gives

$$ds(a,b,c) = (c,\ dc,\ -c), \qquad sd(a,b,c) = (0,\ -(dc+b),\ 0)$$

whose sum is $(c, -b, -c)$. So $i_0 p \sim \mathrm{id}_{\mathrm{Cyl}(C)}$ while $pi_0 = \mathrm{id}_C$, making i_0 and p mutually inverse homotopy equivalences. The symmetric computation applies to i_1, with the contracting homotopy $s'(a,b,c) = (0, a, 0)$ in place of s, since $i_1 p - \mathrm{id}$ sends (a,b,c) to $(-a, -b, a)$.

Step 3: Homotopies are morphisms out of the cylinder. Let $f, g\colon C^\bullet \to D^\bullet$ with $g - f = ds + sd$. Define $H\colon \mathrm{Cyl}(C) \to D^\bullet$ by $H(a,b,c) = f(a) + s(b) + g(c)$. It is a chain map: expanding both sides, $H(d(a,b,c)) - dH(a,b,c) = (g - f - ds - sd)(b) = 0$. By construction $Hi_0 = f$ and $Hi_1 = g$.

Step 4: Conclusion. Apply $\mathscr{F}$. By Step 2 the morphisms i_0 and i_1 are homotopy equivalences, hence quasi-isomorphisms, so $\mathscr{F}(i_0)$ and $\mathscr{F}(i_1)$ are isomorphisms by hypothesis. From $pi_0 = \mathrm{id} = pi_1$ and functoriality, $\mathscr{F}(p)\mathscr{F}(i_0) = \mathrm{id} = \mathscr{F}(p)\mathscr{F}(i_1)$, and since $\mathscr{F}(i_0)$ is invertible so is $\mathscr{F}(p)$; cancelling it gives $\mathscr{F}(i_0) = \mathscr{F}(p)^{-1} = \mathscr{F}(i_1)$. Therefore

$$\mathscr{F}(f) = \mathscr{F}(Hi_0) = \mathscr{F}(H)\mathscr{F}(i_0) = \mathscr{F}(H)\mathscr{F}(i_1) = \mathscr{F}(Hi_1) = \mathscr{F}(g)$$

so $\mathscr{F}$ identifies homotopic morphisms and factors uniquely through $K(\mathbf{A})$, completing the proof.

Hence, in the case where $K(\mathbf{A})$ is *not* the universal object satisfying equation (2.2), there must be a unique functor from $K(\mathbf{A})$ to $D(\mathbf{A})$.

2.2 Projective and Injective Resolutions

It has been alluded to a few times that for special categories, homotopy equivalence and quasi-isomorphism coincide. This shall now be covered in this section. In particular, this shall linked to the idea of an abelian category $\mathbf{A}$ being bounded from above or bellow, and having *enough projective objects* or *enough injective objects*. These categories shall furthermore be shown to be universal in some appropriate way, making their choice not arbitrary. Recall that in $\mathbf{A} = R\text{-}\mathbf{Mod}$, an R-module M Is called projective if $\mathrm{Hom}_R(M, -)$ is exact, and injective if $\mathrm{Hom}_R(-, M)$ is exact. Recall that projective/injective modules nicely lend themselves to a splitting condition on short exact sequences for which they are one of the modules forming them (recall [7, Equivalencies for Projective Modules] and [7, Injective Modules Equivalences]). This will be important, due to the link of chain homotopies with split exact sequences.

Definition 2.2.1: Projective And Injective Objects

Let $\mathbf{A}$ be an abelian category. Then

1. An object $P \in \mathbf{A}$ is said to be *projective* if the functor $\mathrm{Hom}_A(P, -)$ is exact.
2. An object $Q \in \mathbf{A}$ is said to be *injective* if the functor $\mathrm{Hom}_A(-, Q)$ is exact

[7, Equivalencies for Projective Modules] and [7, Injective Modules Equivalences] hold for abelian categories more generally (in particular the splitting remarks), as the reader may venture to check.

Since the dual of an abelian category is abelian (see exercise ref:HERE), the theory on projective objects is dual to that of injective object, and the functor $(-)^{\mathrm{op}}$ maps projective/injectives to injectives/projectives respectively. Thus, many proofs shall simply proof the result for projectives, the result for injectives coming automatically. Note that as we've seen in [7, Extension Problem: Prelude to Homological Algebra] in a fixed [abelian] category $\mathbf{A}$, projective and injective may have very different features.

In what follows, let $\mathbf{P}$ denote the full subcategory of $\mathbf{A}$ determined by the projective objects, and let $\mathbf{I}$ denote the full subcategory of $\mathbf{A}$ determined by the injective objects. Let $K^-(\mathbf{P})$ denote the full subcategory of $K(\mathbf{A})$ consisting of bounded-above complexes of projective objects of $\mathbf{A}$, and $K^+(\mathbf{I})$ be the full subcategory of $K(\mathbf{A})$ consisting of bounded-below complexes of injective objects of $\mathbf{A}$. Note that these are generally not abelian categories, and furthermore projective and injective objects need not exist:

Example 2.2.2: Category With And Without Projective/Injective Resolutions

1. The category of finite abelian groups has no projective or injective, hence no projective/injective resolution.
2. The category of finitely generated abelian groups has projectives, but no injectives, namely since $\mathbb{Z}^n$ for every n are objects this category. Thus, it is possible to create a free, and hence projective, resolution.
3. the dual of the above category has enough injectives but not enough projectives.
4. The category of R-modules has both projective and injective resolutions (projective resolutions comes from free resolutions, while injective resolutions comes from repeated application of [7, Universal Property Of Injective Hulls])

It is thus helpful to add the following condition to an abelian category

Definition 2.2.3: Enough Projective And Enough Injectives

Let $\mathbf{A}$ be an abelian category. Then:

1. we say that $\mathbf{A}$ as *enough projectives* if for every object A of $\mathbf{A}$, there exists a projective object P of $\mathbf{A}$ and an epimorphism $P \twoheadrightarrow A$.
2. we say that $\mathbf{A}$ as *enough injectives* if for every object A of $\mathbf{A}$, there exists a injective object Q of $\mathbf{A}$ and a monomorphism $A \hookrightarrow Q$.

With the important definitions established, the main objective of this section may start going under way. The goal is to show that if we have a quasi-isomorphism between two bounded-above complexes of projectives $f^\bullet : P_0^\bullet \to P_1^\bullet$ (resp. quasi-isomorphism between two bounded-bellow complexes of injectives $f^\bullet : Q_0^\bullet \to Q_1^\bullet$), then $f^\bullet$ is in fact a homotopy-equivalence, from which it can be concluded that in $K^-(\mathbf{P})$ and $K^+(\mathbf{I})$ (homotopy-classes of) quasi-isomorphisms are isomorphisms, which shall give us the desired derived category $D^-(\mathbf{A})$ and $D^+(\mathbf{A})$ in the case where A has enough projectives (resp. enough injectives).

Let us now revisit this splitting condition. Recall that if $C^\bullet$ in $\mathrm{Ch}(\mathbf{A})$ is exact, then the identity map $\mathrm{id}_{C^\bullet}$ and the trivial map induce the same morphisms in (co)homology. However, there do exist exact complexes that are *not* homotopic to 0. As discussed earlier, this directly links to the splitting condition, namely the identity is homotopic to 0 if the sequence is split exact. Being projective is exactly the condition that causes splitting behavior. With this in mind, let us prove the first lemma:

Lemma 2.2.4: Zero-morphism In Cohomology And Homotopy

Let $\mathbf{A}$ be an abelian category, $P^\bullet \in \mathrm{Ch}^-(\mathbf{P})$ (i.e. $P^i = 0$ for $i > 0$), and let $C^\bullet$ be a complex in $\mathrm{Ch}(\mathbf{A})$ such that $H^i(C^\bullet) = 0$ for $i < 0$ (i.e. has trivial cohomology in negative coefficients). Let $f^\bullet : P^\bullet \to C^\bullet$ be a morphism of complexes that induces the zero-morphism in cohomology: $H^\bullet(f^\bullet) = 0$. Then

$$f^\bullet \sim 0$$

i.e., $f^\bullet$ is null-homotopic.

The injective version of this statement is comes from dualizing the appropriate terms.

***Proof*:**
To show $f^\bullet$ is null-homotopic, we must show there exists maps $h^i : P^i \to C^{i-1}$ such that

$$f^i = d^{i-1}_{C^\bullet} \circ h^i + h^{i+1} \circ d^i_{P^\bullet} \tag{2.5}$$

as a diagram:

$$\begin{tikzcd}
\cdots \arrow[r] & P^{-2} \arrow[r] \arrow[dl, "h^{-2}"'] \arrow[d, "f^{-2}"] & P^{-1} \arrow[r] \arrow[dl, "h^{-1}"'] \arrow[d, "f^{-1}"] & P^0 \arrow[r] \arrow[dl, "h^0"'] \arrow[d, "f^0"] & 0 \arrow[r] \arrow[dl, "h^1=0"'] \arrow[d] & \cdots \\
\cdots \arrow[r] & C^{-2} \arrow[r] & C^{-1} \arrow[r] & C^0 \arrow[r] & C^1 \arrow[r] & \cdots
\end{tikzcd}$$

Naturally, $h^i = 0$ for $i > 0$. For the rest, we shall use induction. The base case is $i = 0$. Since the cohomology induced by $f^\bullet$ is zero, f^0 factors through the image of $d^{-1}_{C^\bullet}$:

$$\begin{tikzcd}
 & P^0 \arrow[d, "f^0"] & \\
C^{-1} \arrow[r, "d^{-1}_{C^\bullet}"] & \mathrm{im} d^{-1}_{C^\bullet} \arrow[r] & 0
\end{tikzcd}$$

Now, since P^0 is projective, there exists a morphism $h^0 : P^0 \to C^{-1}$ such that $f^0 = d^{-1}_{C^\bullet} \circ h^0$, as needed.

Next, assume h^i and h^{i+1} satisfies equation (2.5). The goal is to show h^{i-1} satisfies it too. By equation (2.5) and the fact that $P^\bullet$ is a complex:

$$d^{i-2}_{C^\bullet} \circ h^i \circ d^{i-1}_{P^\bullet} = (f^i - h^{i+1} \circ d^{i-1}_{P^\bullet}) = f^i \circ d^{i-1}_{P^\bullet} - h^{i+1} \circ 0 = f^i \circ d^{i-1}_{P^\bullet}$$

which can be seen by staring at the following diagram:

$$\begin{tikzcd}
\cdots \arrow[r, dotted] & P^{i-1} \arrow[r, "d^{i-1}_{P^\bullet}"] \arrow[d, dotted] & P^i \arrow[r, "d^i_{P^\bullet}"] \arrow[dl, "h^i"'] \arrow[d, "f^i"] & P^{i+1} \arrow[r, dotted] \arrow[dl, "h^{i+1}"'] \arrow[d, dotted] & {} \\
\cdots \arrow[r, dotted] & C^{i-1} \arrow[r, "d^{i-1}_{C^\bullet}"'] & C^i \arrow[r, dotted] & C^{i+1} \arrow[r, dotted] & {}
\end{tikzcd}$$

Thus, since f^b is a morphism of cochain complexes:

$$d_{C^\bullet}^{i-1} \circ (f^{i-1} - h^i \circ d_{P^\bullet}^{i-1}) = d_{C^\bullet}^{i-1} \circ f^{i-1} - f^i \circ d_{P^\bullet}^{i-1} = 0$$

Importantly, $(f^{i-1} - h^i \circ d_{P^\bullet}^{i-1})$ factors through $\ker d_{C^\bullet}^{i-1}$. Since $C^\bullet$ is exact at C^{i-1} for $i \leq 0$, $\ker d_{C^\bullet} = \mathrm{im}_{C^\bullet}^{i-2}$. Thus, we get:

$$\begin{tikzcd} & P^{i-1} \arrow[d, "f^{i-1} - h^i \circ d_{P^\bullet}^{i-1}"] & \\ C^{i-2} \arrow[r, "d_{C^\bullet}^{i-2}"] & \mathrm{im}(d_{C^\bullet}^{i-2}) \arrow[r] & 0 \end{tikzcd}$$

Now, since P^{i-1} is projective, there exists a lift $h^{i-1} : P^{i-1} \to C^{i-2}$ such that

$$f^{i-1} - h^i \circ d_{P^\bullet}^{i-1} = d_{C^\bullet}^{i-2} \circ h^{i-1}$$

which, shuffling around, gives us

$$f^{i-1} = d_{C^\bullet}^{i-2} \circ h^{i-1} + h^i \circ d_{P^\bullet}^{i-1}$$

as we sought to show.

Corollary 2.2.5: Exactness Cohomology And Homotopy

Let $\mathbf{A}$ be an abelian category, $P^\bullet \in \mathrm{Ch}^-(\mathbf{A})$, and $C^\bullet \in \mathrm{Ch}(\mathbf{S})$ an exact complex. Then every morphism $P^\bullet \to L^\bullet$ is homotopic to 0

***Proof*:**
Since every morphism to an exact complex induces zero-morphism in cohomology, this is a direct application of lemma 2.2.4.

Note the importance of the bounding from above/bellow (depending on whether working with projectives or injective) in the inductive argument, without it there wouldn't be an appropriate base case. Using the above results, the first quasi-isomorphism to homotopy equivalence can be established:

Corollary 2.2.6: Projectives With Quasi-isomorphism And Zero Morphisms

Let $P^\bullet$ (resp. $Q^\bullet$) be a bounded-above exact complex of projectives (resp. a bounded bellow exact complex of injectives). Then $P^\bullet$ (resp. Q^bb) is homotopy-equivalent to the zero-complex.

***Proof*:**
Treat the projective case; the injective one is the same argument in the opposite category. A complex is homotopy equivalent to the zero complex exactly when its identity morphism is null-homotopic, since a homotopy inverse to the zero complex forces $\mathrm{id} \sim 0$ and conversely. By lemma 2.1.23 it therefore suffices to show that $P^\bullet$ is split exact.

Write $Z^n = \ker d^n$ and $B^n = \mathrm{im} d^{n-1}$; exactness gives $Z^n = B^n$, so for every n there is a short exact sequence

$$0 \to B^n \to P^n \to B^{n+1} \to 0$$

Since $P^\bullet$ is bounded above there is an N with $P^n = 0$ for $n > N$, whence $B^n = 0$ for $n > N + 1$.

Argue by descending induction that each B^n is projective. For $n > N + 1$ it is zero. Suppose B^{n+1} is projective. Then the displayed short exact sequence splits, because a surjection onto a projective object admits a section, so $P^n \cong B^n \oplus B^{n+1}$ and B^n is a direct summand of the projective object P^n, hence projective itself.

Every one of the short exact sequences therefore splits. Choosing a section $\sigma^{n+1} \colon B^{n+1} \to P^n$ of d^n for each n and letting $s^{n+1} \colon P^{n+1} \to P^n$ be σ^{n+1} composed with the projection $P^{n+1} \twoheadrightarrow B^{n+1}$ gives $dsd = d$, since $d^n\sigma^{n+1}$ is the identity on B^{n+1} and d factors through B^{n+1}. So $P^\bullet$ is split exact, and lemma 2.1.23 makes $\mathrm{id}_{P^\bullet}$ null-homotopic, completing the proof.

Another important result that will be used in the next section is that quasi-isomorphisms are "non-zero-divisors" up to homotopy wrt morphisms from commplexes of projectives:

Lemma 2.2.7: Quasi-isomorphisms Of Projectives And Zero-divisor

Let $\mathbf{A}$ be an abelian category, $f^\bullet : C^\bullet \to D^\bullet$ be a quasi-isomorphism. Let $P^\bullet$ be a bounded-above complex of projectives, and let $\alpha^\bullet : P^\bullet \to C^\bullet$ be a morphism of cochain complexes such that the composition:

$$P^\bullet \xrightarrow{\alpha^\bullet} C^\bullet \xrightarrow{f^\bullet} D^\bullet$$

is homotopic to the zero-morphism. Then $\alpha^\bullet \sim 0$

***Proof*:**
Let $h^i : P^i \to D^{i-1}$ define the homotopy between $f^\bullet \circ \alpha^\bullet$ and 0, namely:

$$-f^i \circ \alpha^i = d^{i-1}_{C^\bullet} \circ h^i + h^{i+1} \circ d^i_{P^\bullet}$$

Take the mapping cone $MC(f)^\bullet$ of $f^\bullet$ and define the morphisms:

$$\beta^i = (\alpha^, h^i) : P^i \to MC(f)^{i-1}$$

which may be visualized in the following diagram:

$$\begin{tikzcd}
\cdots \arrow[r] & P^{i-1} \arrow[r, "d^{i-1}_{P^\bullet}"] \arrow[d, dotted, "\beta^{i-1}"] & P^i \arrow[r, "d^i_{P^\bullet}"] \arrow[d, "\beta^i"'] & P^{i+1} \arrow[r] \arrow[d, "\beta^{i+1}"] & {} \\
\cdots \arrow[r] & C^{i-1} \oplus D^{i-2} \arrow[r, "d^{i-2}_{MC(f)^\bullet}"'] & C^i \oplus D^{i-1} \arrow[r, "d^{i-1}_{MC(f)^\bullet}"'] & C^{i+1} \oplus D^i \arrow[r] & {}
\end{tikzcd}$$

These give a morphism of cochain complexes $P^\bullet \to MC(f)[-1]^\bullet$. Indeed:

$$\beta^{i+1} \circ d^i_{P^\bullet} = (\alpha^{i+1} \circ d^i_{P^\bullet}, h^{i+1} \circ d^i_{P^\bullet})$$
$$-d^{i-1}_{MC(f)} \circ \beta^i = (d^i_{C^\bullet} \circ \alpha^i, -f^i \circ \alpha^i - d^{i-1}_{M^\bullet} \circ h^i)$$

Then by definition of h^i and by the morphism property of $\alpha^\bullet$, these are equal. Now, by hypothesis, $f^\bullet$ is a quasi-isomorphism. Then by proposition 2.1.21 $MC(f)^\bullet$ is an exact complex. Then by corollary 2.2.6 $\beta^\bullet$ is homotopic to 0. Since $\alpha^\bullet$ is the composition:

$$P^\bullet \xrightarrow{\beta^\bullet} MC(f)[-1]^\bullet = L^\bullet \oplus M[-1]^\bullet \to L^\bullet$$

it follows that $\beta^\bullet \sim 0$ implies $\alpha^\bullet \sim 0$, as we sought to show.

Though $\alpha^\bullet$ must be homotopic to 0, it need not be the case that $\alpha^\bullet = 0$. Take for example:

$$\begin{tikzcd}
\cdots \arrow[r] & 0 \arrow[r] \arrow[d] & \mathbb{Z} \arrow[r] \arrow[d, "\mathrm{id}"] & \mathbb{Z} \arrow[r] \arrow[d, "\cdot 2"] & 0 \arrow[r] \arrow[d] & \cdots \\
\cdots \arrow[r] & 0 \arrow[r] \arrow[d] & \mathbb{Z} \arrow[r, "\cdot 2"] \arrow[d] & \mathbb{Z} \arrow[r] \arrow[d, "\pi"] & 0 \arrow[r] \arrow[d] & \cdots \\
\cdots \arrow[r] & 0 \arrow[r] & 0 \arrow[r] & \mathbb{Z}/2\mathbb{Z} \arrow[r] & 0 \arrow[r] & \cdots
\end{tikzcd}$$

In this case, $f^\bullet$ is a quasi-isomorphism and $f^\bullet \circ \alpha^\bullet = 0$, and so $\alpha^\bullet \sim 0$, even though it is not equal to 0. With this, one more result is required before the desired goal:

Proposition 2.2.8: projectives, and one-sided inverses

Let $\mathbf{A}$ be an abelian category, $L^\bullet \in \mathrm{Ch}(\mathbf{A})$, $P^\bullet \in \mathrm{Ch}^-(\mathbf{P})$, and $f^\bullet : C^\bullet \to P^\bullet$ a quasi-isomorphism. Then there exists a morphism of complexes $g^\bullet : P^\bullet \to C^\bullet$ such that:

$$f^\bullet \circ g^\bullet \sim \mathrm{id}_{P^\bullet}$$

For injective, $Q^\bullet \in \mathrm{Ch}^+(\mathbf{I})$, $f^\bullet : Q^\bullet \to C^\bullet$ is a quasi-isomorphism, for which then there exists a morphism $g^\bullet : C^\bullet \to Q^\bullet$ such that $g^\bullet \circ f^\bullet \sim \mathrm{id}_{Q^\bullet}$. Notice that this is a one-sided inverse, hence is a slightly more general result than what is being built up to.

***Proof*:**
Since $f^\bullet : C^\bullet \to P^\bullet$ is a quasi-isomorphism, $MC(f)^\bullet = L[1]^\bullet \oplus P^\bullet$ is an exact complex. Define $\rho^\bullet = (0, \mathrm{id}_{P^\bullet})$ to be the morphism:

$$\rho^\bullet : P^\bullet \to MC(f)^\bullet$$

By corollary 2.2.5, since $MC(f)^\bullet$ is exact, $\rho^\bullet$ is homotopic to the zero morphism. Thus, there exists morphisms:

$$\widehat{h}^i : P^i \to MC(f)^{i-1} = L^i \oplus P^{i-1}$$

such that

$$\rho^i = d^{i-1}_{MC(f)^\bullet} \circ \widehat{h}^i + \widehat{h}^{i+1} \circ d^i_{P^\bullet}$$

Visually, we may see this as:

$$
\begin{tikzcd}
\cdots \arrow[r] & P^{i-1} \arrow[r, "d^{i-1}_{P^\bullet}"] \arrow[dl, "\widehat{h}^{i-1}"'] \arrow[d, "\rho^{i-1}"] & P^i \arrow[r, "d^i_{P^\bullet}"] \arrow[dl, "\widehat{h}^i"'] \arrow[d, "\rho^i"] & P^{i+1} \arrow[r] \arrow[dl, "\widehat{h}^{i+1}"'] \arrow[d, "\rho^{i+1}"] & \cdots \\
\cdots \arrow[r] & C^i \oplus P^{i-1} \arrow[r, "d^{i-2}_{MC(f)^\bullet}"'] & C^{i+1} \oplus P^i \arrow[r, "d^{i-1}_{MC(f)^\bullet}"'] & C^{i+2} \oplus P^{i+1} \arrow[r] & \cdots
\end{tikzcd}
$$

Now, write out $\widehat{h}^i$ out in components so that $\widehat{h}^i = (g^i, h^i)$ where $g^i : P^i \to C^i$ and $h^i : P^i \to P^{i-1}$. We'll show that the collection $\{g^i\}_{i\in\mathbb{Z}}$ is a morphism of cochain complexes (i.e. they form commutative squares $g^{i+1} \circ d^i_{P^\bullet} = d^i_{C^\bullet} \circ g^i$) such that $f^\bullet \circ g^\bullet \sim \mathrm{id}_{P^\bullet}$. Indeed, since

$$
\begin{aligned}
(0, \mathrm{id}_{P^i}) &= \rho^i \\
&= d^{i-1}_{MC(f)^\bullet} \circ \widehat{h}^i + \widehat{h}^{i+1} \circ d^i_{P^\bullet} \\
&= (d_{MC}(f)^\bullet)^{i-1} \circ (g^i, h^i) + (g^{i+1}, h^{i+1}) \circ d^i_{P^\bullet} \\
&= (-d^i_{C^\bullet} \circ g^i, f^i \circ g^i + d^{i-1}_{P^\bullet} \circ h^i) + (\beta^{i+1} \circ d^i_{P^\bullet}, h^{i+1} \circ d^i_{P^\bullet})
\end{aligned}
$$

It follows that:

$$
\begin{aligned}
g^{i+1} \circ d^i_{P^\bullet} - d^i_{C^\bullet} \circ g^i &= 0 \\
\mathrm{id}_{P^i} - f^i \circ g^i &= d^{i-1}_{P^\bullet} \circ h^i + h^{i+1} \circ d^i_{P^\bullet}
\end{aligned}
$$

which is exactly what we need, hence the morphism of complexes $g^\bullet : P^\bullet - > C^\bullet$ is the needed homotopy right-inverse of $f^\bullet$, as we sought to show.

With this, we can finally prove the desired result:

Theorem 2.2.9: Projective, Quasi-isomorphism Becomes Homotopy

Let $\mathbf{A}$ be an abelian category, and let $\alpha^\bullet : P_0^\bullet - > P_1^\bullet$ be a quasi-isomorphism between bounded-above complexes of projectives (resp. quasi-isomorphism between two bounded-bellow complexes of injectives $f^\bullet : Q_0^\bullet \to Q_1^\bullet$). Then $\alpha^b b$ is a homotopy-equivalence.

***Proof*:**
The argument for projectives shall be given, the injective argument following by appropriate dualizing. By proposition 2.2.8, since $P_1^\bullet$ is in $C^-(\mathbf{P})$ and $\alpha^\bullet : P_0^\bullet \to P_1^\bullet$ is a quasi-isomorphism, there exists a morphism of complexes $\beta^\bullet : P_1^\bullet \to p_0^\bullet$ such that

$$\alpha^\bullet \circ \beta^\bullet \sim \mathrm{id}^\bullet_{P_1}$$

Then by proposition 2.1.26, $H^\bullet(\alpha^\bullet) \circ H^\bullet(\beta^\bullet) = \mathrm{id}$. Since $H^\bullet(\alpha^\bullet)$ is invertible, so is $H^\bullet(\beta^\bullet)$, and so $\beta^\bullet$ must be a quasi-isomorphism. Since $P_0^\bullet$ is also in $\mathrm{Ch}^-(\mathbf{P})$, we have that there exists a morphism $(\alpha^\bullet)' : P_0^\bullet \to P_1^\bullet$ such that $\beta^\bullet \circ (\alpha^\bullet)' \sim \mathrm{id}_{P_0^\bullet}$. Since:

$$(\alpha^\bullet)' \sim (\alpha^\bullet \circ \beta^\bullet) \circ (\alpha^\bullet)' = \alpha^\bullet(\beta^\bullet \circ (\alpha^\bullet)') \sim \alpha^\bullet$$

it follows that $\beta^\bullet \sim \alpha^\bullet \sim_{P_0^\bullet}$ (exercise 4.6 in Aluffi). Thus, $\alpha^\bullet$ is indeed a homotopy equivalence, as we sought to show.

Corollary 2.2.10: Projective In Homotopy Category

In $K^{-}(\mathbf{P})$ and $K^{+}(\mathbf{I})$, (homotopy classes of) quasi-isomorphissm are isomorphisms.

***Proof*:**
follows immediately form theorem 2.2.9.

2.2.1 Resolutions

With the important equivalence between quasi-isomorphisms and homotopy-equivalence established in the case of complexes of projective (resp. projective) objects, we turn back our attention to the question of finding (co)homological invariants, focusing on categories with enough projective s(resp. Injectives). In particular, given an object A in an abelian category $\mathbf{A}$, it wouldb e desirable to associate to it a (co)chain of projective (resp. injective) objects in some universal way (up to homotopy) from which information can be extracted.

Let us first refine the definition of projective/injective resolutions of an object. Recall that there i a copy of $\mathbf{A}$ in $\mathrm{Ch}(\mathbf{A})$ given by mapping $A \mapsto \iota(A)$.

Definition 2.2.11: Projective And Injective Resolution

Let $A \in \mathrm{Obj}(\mathbf{A})$ be an object in an abelian category A. Then:

1. A *resolution* of A is a quasi-isomorphism $C^{\bullet} \to \iota(A)$ for some complex $C^{\bullet} \in \mathrm{Obj}(\mathrm{Ch}(\mathbf{A}))$.
2. A *projective resolution* of A is a quasi-isomorphism $P^{\bullet} \to \iota(A)$ where $P^{\bullet}$ is a complex in $\mathrm{Ch}^{\leq 0}(\mathbf{P})$
3. An *injective resolution* of A is a quasi-isomorphism $\iota(A) \to Q^{\bullet}$ where $Q^{\bullet}$ is a complex in $\mathrm{Ch}^{\geq 0}(\mathbf{I})$

Often, $P^{\bullet}$ and $Q^{\bullet}$ shall be referred to as the projective resolution and injective resolution respectively, omitting the quasi-isomorphism.

2.2.12 Remark There is a point of confusion that may arise. It may seem that the resolutions *themselves* are projective/injective objects in the abelian category $\mathrm{Ch}(\mathbf{A})$. However, this is not the case (exercise ref:HERE).

Recall further that any resolution $C^{\bullet}$ will be exact everywhere except possibly at C^{0} (exercise, or double-check example 2.1.13). Hence, the non-trivial cohomological information lies in $H^{0}(C^{\bullet})$. Let us now focus on resolutions for a particular object A: consider the category of homotopy classes of quasi-isomorphisms with *target* $\iota(A)$. Then we shall show that, if they exist, projective resolution are initial in this category.

$$\begin{tikzcd} P^{\bullet} \arrow[d, "\exists!"] \arrow[dr] & \\ C^{\bullet} \arrow[r, "\text{q-iso}"'] & \iota(A) \end{tikzcd}$$

Analogously, injective resolutions are final in homotopic category of quasi-isomorphisms with *source* $\iota(A)$. Given this universal property, this means that each projective resolution maps uniquely (in $K(\mathbf{A})$) to *every* resolution of A. The following lemma proves this:

Lemma 2.2.13: Projective Resolution As Initial Objects

Let $A \in \mathbf{A}$, let $C^\bullet$ be a resolution of A (so that $C^i = 0$ for $i > 0$, $H^i(C^\bullet) = 0$ for $i < 0$, and $H^0(C^\bullet) = A$), and let $P^\bullet \in \mathrm{Ch}^{\leq 0}(\mathbf{P})$ be a bounded-above complex of projectives. Then any morphism

$$\varphi : H^0(P^\bullet) \to H^0(C^\bullet) \; [= A]$$

is induced by a morphism of complexes $\varphi^\bullet : P^\bullet \to C^\bullet$, and any two such $\varphi^\bullet$ are homotopic.

***Proof*:**
Since both complexes vanish in positive degrees, $H^0(P^\bullet) = \mathrm{coker} d_P^{-1}$ and $H^0(C^\bullet) = \mathrm{coker} d_C^{-1}$; write $\pi_P \colon P^0 \twoheadrightarrow H^0(P^\bullet)$ and $\pi_C \colon C^0 \twoheadrightarrow A$ for the two quotient maps.

Step 1: Degree 0. The composite $\varphi \circ \pi_P \colon P^0 \to A$ is a morphism out of a projective object, and π_C is an epimorphism, so by definition 2.2.1 it lifts: there is $\varphi^0 \colon P^0 \to C^0$ with $\pi_C \circ \varphi^0 = \varphi \circ \pi_P$.

Step 2: Degree -1. Compute

$$\pi_C \circ \varphi^0 \circ d_P^{-1} = \varphi \circ \pi_P \circ d_P^{-1} = 0$$

since $\pi_P \circ d_P^{-1} = 0$. Hence $\varphi^0 \circ d_P^{-1}$ factors through $\ker \pi_C = \mathrm{im} d_C^{-1}$. As P^{-1} is projective and d_C^{-1} is an epimorphism onto its image, the map $P^{-1} \to \mathrm{im} d_C^{-1}$ lifts to $\varphi^{-1} \colon P^{-1} \to C^{-1}$ with $d_C^{-1} \circ \varphi^{-1} = \varphi^0 \circ d_P^{-1}$.

Step 3: The induction. Suppose φ^{-i} has been constructed for $i < n$ with every square commuting. Then

$$d_C^{-n+1} \circ \left(\varphi^{-n+1} \circ d_P^{-n}\right) = \varphi^{-n+2} \circ d_P^{-n+1} \circ d_P^{-n} = 0$$

so $\varphi^{-n+1} \circ d_P^{-n}$ lands in $\ker d_C^{-n+1}$, which for $n \geq 2$ equals $\mathrm{im} d_C^{-n}$ because $C^\bullet$ is exact in degree $-n+1 < 0$; the case $n = 1$ is Step 2, where $\ker \pi_C$ plays that role. Projectivity of P^{-n} again lifts it along the epimorphism $C^{-n} \twoheadrightarrow \mathrm{im} d_C^{-n}$, producing φ^{-n} with $d_C^{-n} \circ \varphi^{-n} = \varphi^{-n+1} \circ d_P^{-n}$. This constructs $\varphi^\bullet$ in every degree, and Step 1 arranges that it induces φ on H^0.

Step 4: Uniqueness up to homotopy. Let $\varphi^\bullet$ and $\psi^\bullet$ both induce φ, and set $f^\bullet = \varphi^\bullet - \psi^\bullet$. On H^0 it induces $\varphi - \varphi = 0$, and in every negative degree $H^i(C^\bullet) = 0$, so $H^i(f^\bullet) = 0$ there as well; in positive degrees both complexes vanish. Hence $H^\bullet(f^\bullet) = 0$. The hypotheses of lemma 2.2.4 are met, $P^\bullet$ being a bounded-above complex of projectives and $C^\bullet$ having vanishing cohomology in negative degrees, so $f^\bullet \sim 0$, that is $\varphi^\bullet \sim \psi^\bullet$, completing the proof.

Since these are initial (resp. final) objects in the category, we have a unique isomorphism beetween any two solutions to the universal problem. In this case, this may be stated as followed:

Proposition 2.2.14: Unique Projection Resolution

Let $\mathbf{A}$ be an abelian category. Then any two projective (resp. injective) resolutions of an object A in $\mathbf{A}$ are homotopy equivalent

***Proof*:**
Let $P^\bullet$ and $Q^\bullet$ be two projective resolutions of A, so that each is a bounded-above complex of projectives with $H^0 = A$. Apply lemma 2.2.13 twice to the morphism id_A: once with source $P^\bullet$ and target $Q^\bullet$, giving $f^\bullet\colon P^\bullet \to Q^\bullet$, and once the other way, giving $g^\bullet\colon Q^\bullet \to P^\bullet$. Each is unique up to homotopy.

The composite $g^\bullet \circ f^\bullet\colon P^\bullet \to P^\bullet$ induces id_A on H^0, and so does $\mathrm{id}_{P^\bullet}$. Both are therefore lifts of the same morphism id_A, and the uniqueness clause of lemma 2.2.13 gives $g^\bullet \circ f^\bullet \sim \mathrm{id}_{P^\bullet}$. Symmetrically $f^\bullet \circ g^\bullet \sim \mathrm{id}_{Q^\bullet}$. Hence $f^\bullet$ is a homotopy equivalence with homotopy inverse $g^\bullet$, as we sought to show. The injective case is dual.

A final important observation is that this lift is functorial!

Proposition 2.2.15: Projective Resolution Functoriality

Let $\mathbf{A}$ be an abelian category, and let A_0, A_1 be objects of $\mathbf{A}$. Let $P_0^\bullet$ and $P_1^\bullet$ be projective resolutions for A_0 and A_1 respectively. Then every morphism $\varphi : A_0 \to A_1$ in $\mathbf{A}$ induces a morphism $\alpha^\bullet : P_0^\bullet \to P_1^\bullet$ that is unique up to homotopy

***Proof*:**
Notice that φ can be interpreted as a morphism $\varphi : H^0(P_0^\bullet) \to H^0(P_1^\bullet)$. The complex $P_0^\bullet$ consists of projectives, and $P1^\bullet$ is a resolution of A_1. Thus, by lemma 2.2.13, there exists a list α^b that is unique up to homotopy, as we sought to show.

Corollary 2.2.16: Full Subcategory Of Homotopic Category

Let $\mathbf{A}$ be an abelian category with enough projectives (resp. enough injectives). Then $\mathbf{A}$ is equivalent to a full subcategory of $K^-(\mathbf{P})$ (resp. $K^+(\mathbf{I})$).

***Proof*:**
Treat the projective case. Choose for each object A of $\mathbf{A}$ a projective resolution $P_A^\bullet$, possible because $\mathbf{A}$ has enough projectives, and send $A \mapsto P_A^\bullet$. By proposition 2.2.15 every $\varphi\colon A_0 \to A_1$ lifts to a morphism of resolutions, well defined up to homotopy, so this is a functor $R\colon \mathbf{A} \to K^-(\mathbf{P})$: composites and identities are preserved because a lift of a composite and the composite of lifts both lift the same morphism and so agree up to homotopy, which is equality in $K^-(\mathbf{P})$.

R is faithful. If $R(\varphi) = 0$ then a lift $\varphi^\bullet$ of φ is null-homotopic, so it induces the zero morphism on cohomology by proposition 2.1.24. Since $H^0(\varphi^\bullet) = \varphi$ under the identifications $H^0(P_{A_i}^\bullet) = A_i$, this gives $\varphi = 0$.

R is full. Let $f^\bullet \colon P^\bullet_{A_0} \to P^\bullet_{A_1}$ be any morphism in $K^-(\mathbf{P})$, represented by a chain map. It induces $\varphi := H^0(f^\bullet) \colon A_0 \to A_1$, and $f^\bullet$ is then a lift of φ. By the uniqueness clause of lemma 2.2.13 any two lifts of φ are homotopic, so $R(\varphi) = [f^\bullet] = f^\bullet$ in $K^-(\mathbf{P})$.

A fully faithful functor is an equivalence onto its essential image, which is a full subcategory of $K^-(\mathbf{P})$. Hence $\mathbf{A}$ is equivalent to a full subcategory of $K^-(\mathbf{P})$, and dually to one of $K^+(\mathbf{I})$ when $\mathbf{A}$ has enough injectives, completing the proof.

2.2.2 Derived Category with Enough Projectives

Let us finally confirm that the solution to the universal problem in equation 2.2. Recall that the derived category is a category $D(\mathbf{A})$ where for any category D for which there exists an additive functor $\mathscr{F} : \mathrm{Ch}(\mathbf{A}) \to D$ where every quasi-isomorphism in $\mathrm{Ch}(\mathbf{A})$ becomes an isomorphism in D, there exists a unique functor such that the following diagram commutes:

$$\begin{tikzcd} \mathrm{Ch}(\mathbf{A}) \arrow[r, "\mathscr{F}"] \arrow[d] & D \\ D(\mathbf{A}) \arrow[ur, "\exists!"'] & \end{tikzcd}$$

In the case where $\mathbf{A}$ has enough projectives (resp. enough injectives) and is bounded from above (resp. from bellow), then $K^-(\mathbf{P})$ (resp. $K^+(\mathbf{I})$) is the solution to the above universal problem. In particular, there is an equivalence of categories between $K^-(\mathbf{P})$ and $D^-(\mathbf{A})$. The formal definition of the derived category shall be given in section ref:HERE, hence this equivalence shall not be proven, however the universal property is readily provable, and once the derived category is formally defined it will be the equivalence essentially come to proper manipulation of universal properties.

To start, a general procedure for finding projective/injective resolutions for any (co)chain must be explored. The discussion so far has been for projective/injective resolution, that is quasi-isomorphisms $P^\bullet \to \iota(A)$ or $\iota(A) \to Q^\bullet$. Is it possible to replace $\iota(A)$ with any bounded complex? The answer is yes, though rather technical. The importance of this will be the existence of a functor p that maps $\mathrm{Ch}^-(\mathbf{P})$ into $K^-(\mathbf{P})$, which shall be important for solving the universal problem above, as well as finding functors between derived categories. Such functors shall also be key in solving the "extension problem" that was mentioned all the way back in [7, Extension Problem: Prelude to Homological Algebra].

Theorem 2.2.17: Resolution And Complexes

Let $\mathbf{A}$ be an abelian category with enough projectives and $C^\bullet$ a complex in $C^-(\mathbf{A})$. Then there exists a founded-above complex of projective $P^\bullet$ and a quasi-isomorphism $P^\bullet \to C^\bullet$, where $P^\bullet$ is uniquely defined up to homotopy equivalence. Furthermore, every morphism $\alpha^\bullet$ of complexes in $C^-(\mathbf{A})$ lifts to a morphism of the corresponding projective resolutions in $K^-(\mathbf{P})$, which is uniquely defined up to homotopy.

***Proof Key ideas*:**
The uniqueness and lifting halves are already available; only the existence half needs a construction, and that construction is a descending induction whose bookkeeping is long without being conceptually informative, so it is given in outline here and in full in *An Introduction to Homological Algebra* [12].

Existence. Let $C^\bullet$ be bounded above, say $C^n = 0$ for $n > N$. Build P^n by descending induction on n, starting with $P^n = 0$ for $n > N$. At each stage one has already constructed the terms above degree n together with the maps to $C^\bullet$, and one chooses a projective object surjecting onto the pullback of C^n against the cycles constructed one degree up. That single choice simultaneously arranges surjectivity on cohomology in degree n and injectivity in degree $n+1$, which is what makes the resulting map a quasi-isomorphism; enough projectives is exactly what supplies the surjection at each stage.

Uniqueness up to homotopy equivalence. Two such projective complexes over $C^\bullet$ are homotopy equivalent by the argument of proposition 2.2.14, with its object-level input lemma 2.2.13 replaced by its complex-level counterpart proposition 2.2.8: each maps to the other over $C^\bullet$, and both composites lift the identity, hence are homotopic to it.

Lifting of morphisms. Given $\alpha^\bullet\colon C^\bullet \to D^\bullet$ and projective complexes $P^\bullet \to C^\bullet$, $Q^\bullet \to D^\bullet$, the composite $P^\bullet \to C^\bullet \to D^\bullet$ is a morphism from a bounded-above complex of projectives into a complex quasi-isomorphic to $Q^\bullet$. Existence of a factorization through $Q^\bullet$ comes from corollary 2.2.5, applied to the cone of $Q^\bullet \to D^\bullet$, and its uniqueness up to homotopy from the cancellation property lemma 2.2.7. Together they give a well-defined morphism in $K^-(\mathbf{P})$.

Thus, given an abelian category $\mathbf{A}$ with enough projective, there is a functor

$$\mathscr{P} : C^-(\mathbf{A}) \to K^-(\mathbf{P})$$

associating each (bounded-above) complex $C^\bullet$ to a projective resolution $\mathscr{P}(C^\bullet)$ in a functorial manner (so that morphisms $f^\bullet$, $g^\bullet$ are lifted and $\mathscr{P}(g^\bullet \circ f^\bullet) = \mathscr{P}(g^\bullet) \circ \mathscr{P}(f^\bullet)$ since both sides are classes of homotopy lifts of $g^\bullet \circ f^\bullet$). The uniqueness of $\mathscr{P}$ is up to natural transformation, that is if another functor $\mathscr{P}'$ is chosen that assigns a different choice of projective resolution, then by theorem 2.2.17 there is a homotopy equivalence between $\mathscr{P}(C^\bullet)$ and $\mathscr{P}'(C^\bullet)$, hence an isomorphism $\varphi_{C^\bullet} : \mathscr{P}(C^\bullet) \to \mathscr{P}'(C^\bullet)$ in $K^-(\mathbf{P})$. Since the identity on $C^\bullet$ lifts to a unique map up to homotopy, the isomorphism $\varphi_{C^\bullet}$ is unique. Furthermore, given a morphism $f^\bullet : C^\bullet \to D^\bullet$, the diagram:

$$\begin{tikzcd}
\mathscr{P}(C^\bullet) \arrow[r, "\mathscr{P}(f^\bullet)"] \arrow[d, "\varphi_{C^\bullet}"', "\sim"] & \mathscr{P}(D^\bullet) \arrow[d, "\sim"', "\varphi_{D^\bullet}"] \\
\mathscr{P}'(C^\bullet) \arrow[r, "\mathscr{P}'(f^\bullet)"'] & \mathscr{P}'(D^\bullet)
\end{tikzcd}$$

commutes (again by the uniqueness of lifts up to homotopy, can you see why)? Thus, we have the unique natural isomorphism for any two choices of functors giving a projective resolution.

Theorem 2.2.18: Universal Property Of Derived Category With Enough Projectives

Let $\mathbf{A}$ be an abelian category with enough projectives, and $\mathscr{P}$ a functor sending bounded-above cochains to projective resolutions. Then:

1. If $f^\bullet$ is a quasi-isomorphism in $C^-(\mathbf{A})$, then $\mathscr{P}(f^\bullet)$ is an isomorphism in $K^-(\mathbf{P})$
2. If $\mathscr{F} : C^-(\mathbf{A}) \to D$ is an additive functor that takes quasi-isomorphisms to isomorphisms, then there exists a unique functor $\widetilde{\mathscr{F}} : K^-(\mathbf{P}) \to D$ unique up to natural isomorphism such that the following diagram commutes up to natural isomorphism:

$$\begin{tikzcd} C^-(\mathbf{A}) \arrow[r, "\mathscr{F}"] \arrow[d, "\mathscr{P}"'] & D \\ K^-(\mathbf{P}) \arrow[ur, "\widetilde{\mathscr{F}}"'] & \end{tikzcd}$$

where commuting up to natural transformation means there exists a natural transformation $\nu : \widetilde{\mathscr{F}} \circ \mathscr{P} \rightsquigarrow \mathscr{F}$ such that:

$$\nu_{C^\bullet} : \widetilde{\mathscr{F}} \circ \mathscr{P}(C^\bullet) \xrightarrow{\sim} \mathscr{F}(C^\bullet)$$

is an isomorphism for every complex $C^\bullet$ in $C^-(\mathbf{A})$.

The fact that the commutative diagram commutes up to natural isomorphism reflects that the homotopic category is categorically equivalent to the derived category. If $\mathbf{A}$ has enough injective, then we may instead define the functor $\mathscr{I} : C^+(\mathbf{A}) \to K^+(\mathbf{I})$ that shall satisfy the dual universal property.

***Proof*:**

1. Let $f^\bullet : C^\bullet \to D^\bullet$ be any morphism in $C^-(\mathbf{A})$. By theorem 2.2.17, $f^\bullet$ lifts to a morphism of resolutions which commutes up to homotopy in the following diagram:

$$\begin{tikzcd} \mathscr{P}(C^\bullet) \arrow[r, "\mathscr{P}(f^\bullet)"] \arrow[d, "\mu_{C^\bullet}"'] & \mathscr{P}(C^\bullet) \arrow[d, "\mu_{D^\bullet}"] \\ C^\bullet \arrow[r, "f^\bullet"] & D^\bullet \end{tikzcd}$$

where $\mu_{C^\bullet}, \mu_{D^\bullet}$ are the quasi-isomorphisms of the projective resolutions. Since the diagram commute up to homootpy, it commutes after taking cohomology, thus since $f^\bullet$ is a quasi-isomorphism, so is $\mathscr{P}(f^\bullet)$. Then by corollary 2.2.10, $\mathscr{P}(f^\bullet)$ is an isomorphism.

2. Define $\widetilde{\mathscr{F}} : K^-(\mathbf{P}) \to D$ via:

$$\widetilde{\mathscr{F}}(P^\bullet) := \mathscr{F}(P^\bullet) \qquad \widetilde{\mathscr{F}}(f^\bullet) := \mathscr{F}(f^\bullet)$$

where $\widetilde{\mathscr{F}}(f^\bullet)$ can be any representative of $f^\bullet$. This map is well-defined, since homotopic morphisms have the same image in D (lemma 2.1.32). Naturally, $\widetilde{\mathscr{F}}$ is a functor. We need to check the commutativity up to morphism of the diagram and the uniqueness of $\widetilde{\mathscr{F}}$ up to natural isomorphism.

First, let $C^\bullet \in C^-(\mathbf{A})$. Then (by construction of $\mathscr{P}$, there exists a quasi-isomorphism

$$\mu_{C^\bullet} : \mathscr{P}(C^\bullet) \to C^\bullet$$

Since $\mathscr{F}$ maps quasi-isomorphisms to isomorphism, the induced morphism $\nu_{C^\bullet} := \mathscr{F}(\mu_{C^\bullet})$:

$$\nu_{C^\bullet} : \widetilde{\mathscr{F}}(\mathscr{P}(C^\bullet)) = \mathscr{F}(\mathscr{P}(C^\bullet)) \to \mathscr{F}(C^\bullet)$$

is an isomorphism. What's left to check is that $\nu_{C^\bullet}$ defines a natural transformation. Let $f^\bullet : C^\bullet \to D^\bullet$ in $C^-(\mathbf{A})$. We want to show the following diagram commutes:

$$\begin{tikzcd} \widetilde{\mathscr{F}}(\mathscr{P}(C^\bullet)) \arrow[r, "\widetilde{\mathscr{F}}(\mathscr{P}(f^\bullet))"] \arrow[d, "\nu_{C^\bullet}"'] & \widetilde{\mathscr{F}}(\mathscr{P}(D^\bullet)) \arrow[d, "\nu_{D^\bullet}"] \\ \mathscr{F}(C^\bullet) \arrow[r, "\mathscr{F}(f^\bullet)"] & \mathscr{F}(D^\bullet) \end{tikzcd}$$

This diagram is obtained by applying $\mathscr{F}$ to the diagram discussed above the theorem:

$$\begin{tikzcd} \mathscr{P}(C^\bullet) \arrow[r, "\mathscr{P}(f^\bullet)"] \arrow[d, "\varphi_{C^\bullet}"', "\sim"] & \mathscr{P}(D^\bullet) \arrow[d, "\varphi_{D^\bullet}", "\sim"'] \\ \mathscr{P}'(C^\bullet) \arrow[r, "\mathscr{P}'(f^\bullet)"'] & \mathscr{P}'(D^\bullet) \end{tikzcd}$$

This diagram commutes up to homotopy, hence the other diagram commutes directly (lemma 2.1.32).

For uniqueness up to isomorphism, say $\overline{\mathscr{F}} : K^-(\mathbf{P}) \to D$ is another functor satisfying the properties. Then there exists a natural isomorphism:

$$\overline{\nu} : \overline{\mathscr{F}} \circ \mathscr{P} \rightsquigarrow \mathscr{F}$$

Now, for any $P^\bullet \in K^-(\mathbf{P})$, take the composition:

$$\overline{ff}(P^\bullet) \underset{\sim}{\overset{\overline{ff}(\mu_{P^\bullet})^{-1}}{\to}} \overline{ff}(\mathscr{P}((P^\bullet)) \underset{\sim}{\overset{\overline{\nu}_{P^\bullet}}{\to}} = \widetilde{ff}(P^\bullet)$$

If φ is this composition, then it is easy to verify that $\varphi : \overline{\mathscr{F}} \to \widetilde{\mathscr{F}}$ is a natural isomorphism, completing the proof.

2.3 Derived Functors: Ext and Tor

Let $\mathscr{F} : \mathbf{A} \to \mathbf{B}$ be an additive functor. Often, we study information about $\mathbf{A}$ by passing it through $\mathbf{B}$ (think of $- \otimes N$ or $\mathrm{Hom}_R(N, -)$). As alluded to in [7, Ext and Tor], such information may be the number of extensions two objects may have, the amount of torsion an object may have. In [7, Ext and Tor], it was claimed that this comes from "fixing" the lack of exactness of a left- or right-exact functor using homological algebra. We are finally in a position to make this precise. In this section, we shall show how to take a (not-necessarily left/right exact) additive functor between two abelian categories with $\mathbf{A}$ having enough projectives (resp. injectives) $\mathscr{F} : \mathbf{A} \to \mathbf{B}$ and get a *derived functor*:

$$D^-(\mathscr{F}) : D^-(\mathbf{A}) \to D^-(\mathbf{B}) \qquad (\text{resp. } D^-(\mathscr{F}) : D^-(\mathbf{A}) \to D^-(\mathbf{B}))$$

This functor shall be used to develop a long-exact sequence in $\mathbf{B}$, given a short exact sequence of objects in $\mathbf{A}$.

2.3.1 Derived Functor

Let $\mathscr{F} : \mathbf{A} \to \mathbf{B}$ be an additive functor between two abelian categories. Then $\mathscr{F}$ extends to a functor:

$$\mathrm{Ch}(\mathscr{F}) : \mathrm{Ch}(\mathbf{A}) \to \mathrm{Ch}(\mathbf{B})$$

namely, if $C^\bullet$ is a complex:

$$\mathscr{F}(d^{i+1}_{C^\bullet}) \circ \mathscr{F}(d^i_{C^\bullet}) = \mathscr{F}(d^{i+1}_{C^\bullet} \circ d^i_{C^\bullet}) = \mathscr{F}(0) = 0$$

hence, $\mathrm{Ch}(\mathscr{F})(C^\bullet)$ is indeed a complex. Then since homotopic morphisms are sent to homotopic morphisms (lemma 2.1.29), $\mathrm{Ch}(\mathscr{F})$ induces a morphism $K(\mathscr{F})$:

$$K(\mathscr{F}) : K(\mathbf{A}) \to K(\mathbf{B})$$

From the results we have proved, we have the following commutative diagram:

$$\begin{tikzcd}
\mathbf{A} \arrow[r, "\mathscr{F}"] \arrow[d, hook, "\iota"'] & \mathbf{B} \arrow[d, hook, "\iota"] \\
K(\mathbf{A}) \arrow[r, "K(\mathscr{F})"] & K(\mathbf{B})
\end{tikzcd}$$

Furthermore, if the complexes are bounded, then certainly $\mathrm{Ch}(\mathscr{F})$ sends bounded complexes to bounded complexes, and hence the same holds for $K(\mathscr{F})$. Now, if $\mathbf{A}$ has enough projectives so that we may consider $K^-(\mathbf{P})$, can we restrict $K(\mathbf{A})$ to $K(P(\mathbf{A}))$, and map to $K(P(\mathbf{B}))$ (where $P(\mathbf{A})$ and $P(\mathbf{B})$ are the projectives of $\mathbf{A}$ and $\mathbf{B}$ respectively)? This is not the case in general; for example we saw that $- \otimes N$ in R-**Mod** does not preserve projectives. If $\mathbf{B}$ has enough projectives, then we may choose a projective resolution functor $\mathscr{P}_B : \mathrm{Ch}^-(B) \to K^-(\mathbf{P}^{-1})$ mapping each complex to a projective resolution in a functorial manner. Since $\mathscr{P}_B$ sends homotopic morphisms to homotopic morphisms (lemma 2.2.7), $\mathscr{P}_B$ descends to a functor $K(\mathscr{P}_B)$. Combining these facts, we may define the following functor:

Definition 2.3.1: Left-Derived (right-Derived) Functor

Let $\mathscr{F} : \mathbf{A} \to \mathbf{B}$ be an additive functor. Then the *left-derived functor* of $\mathscr{F}$ is the composition:

$$\begin{tikzcd}
K^-(P(\mathbf{A})) \arrow[r, "\iota_A"'] \arrow[rrr, bend left, "\mathbf{L}\mathscr{F}"] & K^-(\mathbf{A}) \arrow[r, "K(\mathscr{F})"'] & K^-(\mathbf{B}) \arrow[r, "K(\mathscr{P}_B))"'] & K^-(P(\mathbf{B}))
\end{tikzcd}$$

where ι_A is the inclusion of $K^-(P(\mathbf{A}))$ as a full subcategory of $K^-(\mathbf{A})$. The resulting functor is denoted $\mathbf{L}\mathscr{F}$.

It may seem that the natural notation would be $D^-(\mathscr{F})$, however the $\mathbf{L}\mathscr{F}$ notation was established before the abstraction given above, and has stuck. Similarly, if $\mathbf{A}$, $\mathbf{B}$ has enough injectives, we may define the right-derived functor $\mathbf{R}\mathscr{F}$.

It is natural to want to characterize $\mathbf{L}\mathscr{F}$ and $\mathbf{R}\mathscr{F}$ in terms of some universal property. Naturally, this universal property would be defined up to natural isomorphism, for $\mathscr{P}_A$ and $\mathscr{P}_B$ are chosen up to natural isomorphism. This turns to be a bit subtle. For example, there is no natural functor which shall always make the following diagram commute:

$$\begin{tikzcd}
\hat{\mathbf{A}} \arrow[r, dashed, "??"] \arrow[d, hook] & \hat{\mathbf{B}} \arrow[d, hook] \\
K^-(P(\mathbf{A})) \arrow[r, "\mathbf{L}\mathscr{F}"] & K^-(P(\mathbf{B}))
\end{tikzcd}$$

The reason is, it is not necessarily the case that a complex whose cohomology is concentrated at 0 would map via $\mathbf{L}\mathscr{F}$ to a complex with the same property. In fact, the reader may expect this, for the image of a functor $\mathscr{F}$ may present some non-trivial cohomological information, (think of $\mathscr{F} := \mathrm{Hom}_R(N, -)$ where N is non-projective). If the functor $\mathscr{F}$ is *exact*, then then there is essentially no "change in cohomology". This is captured in the following proposition:

Proposition 2.3.2: Derived Functors And Exactness

Let $\mathscr{F}$ be an additive *exact* functor. Then $\hat{\mathbf{A}}$ is in fact sent to $\hat{\mathbf{B}}$ via $\mathbf{L}\mathscr{F}$

***Proof*:**

Let $P^\bullet \in \hat{\mathbf{A}}$ so that $H^i(P^\bullet) = 0$ for all $i \neq 0$. Then $\mathbf{L}\mathscr{F}(P^\bullet)$ is given by choosing a projective resolution $P^\bullet_{\mathscr{F}(P^\bullet)}$ of $\mathscr{F}(P^\bullet)$. Since quasi-isomorphissm preserve cohomology and (key!) exact functors *commute* with cohomology (exercise ref:HERE):

$$H^i(\mathbf{L}\mathscr{F}(P^\bullet)) = H^i(P^\bullet_{\mathscr{F}(P^\bullet)}) \cong H^i(\mathscr{F}(P^\bullet)) \cong \mathscr{F}(H^i(P^\bullet)) = \mathscr{F}(0) \overset{i\neq 0}{=} 0$$

Thus, $\mathbf{L}\mathscr{F}(P^\bullet)$ does indeed map to an object of $\hat{\mathbf{B}}$.

Even more is true if $\mathscr{F}$ is exact: $H^i(\mathbf{L}\mathscr{F}(P^\bullet)) \cong \mathscr{F}(H^0(P^\bullet))$, saying that the restriction of $\mathbf{L}\mathscr{F}$ agrees with $\mathscr{F}$ (modulo equivalence of categories). Hence, in general we shall be interested in non-exact functors, usually functors that are only left or right exact.

Returning to finding an appropriate universal property for derived functors, another natural candidate would be given by taking a "step back" from the above diagram and consider instead if the following diagram commutes (up to natural isomorphism)

$$\begin{tikzcd}
K^-(\mathbf{A}) \arrow[r, "K(\mathscr{F})"] \arrow[d, "\mathscr{P}_A"] & K^-(\mathbf{B}) \arrow[d, "\mathscr{P}_B"] \\
K^-(P(\mathbf{A})) \arrow[r, "\mathbf{L}\mathscr{F}"'] & K^-(P(\mathbf{B}))
\end{tikzcd}$$

in particular, that there may be a natural isomorphism:

$$\mathbf{L}\mathscr{F} \circ \mathscr{P}_A \overset{\sim}{\rightsquigarrow} \mathscr{P}_B \circ K(\mathscr{F})$$

Unfortunately, this isn't in general the case

Example 2.3.3: Lacking Natural Isomorphism
(exercise 7.3 in Aluffi, should I put solution here?)

The following is the best we can hope for:

Proposition 2.3.4: Universal Property of (left-)Derived Functor

Let $\mathscr{F} : \mathbf{A} \to \mathbf{B}$ be an additive functor between abelian categories. Then $\mathbf{L}\mathscr{F}$ satisfies the following universal property:

1. There is a natural transformation:

$$\mathbf{L}\mathscr{F} \circ \mathscr{P}_A \overset{\sim}{\rightsquigarrow} \mathscr{P}_B \circ K(\mathscr{F})$$

2. for every functor $\mathscr{G} : K^-(P(\mathbf{A})) \to K^-(P(\mathbf{B}))$ and every natural tarnsformation $\gamma : \mathscr{G} \circ \mathscr{P}_A \rightsquigarrow \mathscr{P}_B \circ K(\mathscr{F})$, there is a unique (up to natural isomorphism) natural transformation $\mathscr{G} \rightsquigarrow \mathbf{L}\mathscr{F}$ inducign a factorization:

$$\gamma : \mathscr{G} \circ \mathscr{P}_A \rightsquigarrow \mathbf{L}\mathscr{F} \circ \mathscr{P}_A \rightsquigarrow \mathscr{P}_B \circ K(\mathscr{F})$$

A similar result holds for right-derived functors.

***Proof Key ideas*:**
The argument is the usual one for a Kan-extension-style universal property, and it is stated here rather than carried out because the ambient apparatus is not yet in place: the definition of $D(\mathbf{A})$ used below rests on the localization machinery that section 1.2 defers. The full treatment is in *An Introduction to Homological Algebra* [12].

For (1), the natural transformation is the comparison map induced by functoriality of projective resolutions: $\mathscr{P}_A$ picks a resolution, $K(\mathscr{F})$ applies $\mathscr{F}$ termwise, and proposition 2.2.15 makes the resulting square commute up to homotopy, which is commutativity in K^-.

For (2), given $\mathscr{G}$ and γ, the factorization is forced on objects because a projective resolution is initial among complexes of projectives mapping to the object (lemma 2.2.13), so any transformation out of $\mathscr{G} \circ \mathscr{P}_A$ is determined by its value on resolutions; uniqueness up to natural isomorphism is then the uniqueness up to homotopy of those comparison maps (proposition 2.2.14).

(a word on composition)

2.3.2 Long Exact Sequences of Derived Functors

With the derived functor established and being the best candidate for holding (co)homology information, we proceed with extracting this informaiton: we may take $\mathbf{L}_i\mathscr{F} := H^{-i} \circ \mathbf{L}\mathscr{F}$. The reason for taking H^{-i} instead of i is since the indexing works out better. This produced a functor

$\mathbf{L}_i\mathscr{F} : D^-(\mathbf{A}) \to B$. Notice that B doesn't need to have enough projectives for:

$$\mathbf{L}_i\mathscr{F}(P^\bullet) = H^{-i}(\mathscr{P}_B(\mathrm{Ch}(\mathscr{F})(P^\bullet))) \cong H^{-i}(\mathrm{Ch}(\mathscr{F})(P^\bullet))$$

where the last isomorphism comes from $\mathscr{P}_B(\mathrm{Ch}(\mathscr{F})(P^\bullet))$ being quasi-isomorphic to $\mathrm{Ch}(\mathscr{F})(P^\bullet)$. Though this abstract approach is useful, the following more concrete version of $\mathbf{L}_i\mathscr{F}$ shall prove more useful for computations It feels like I should just skip to this, idk what the above really gives me here

Definition 2.3.5: ith Left-Derived Functor

Let $\mathscr{F} : \mathbf{A} \to \mathbf{B}$ be an additive functor between two abelian categories where $\mathbf{A}$ has enough projectives. Then the *ith left-derived functor* $\mathbf{L}_i\mathscr{F} : A \to B$ is given by:

$$\mathbf{L}_i\mathscr{F} := H^{-1} \circ \mathbf{L}\mathscr{F} \circ \mathscr{P}_A \circ \iota_A$$

The complex in $\mathrm{Ch}(\mathbf{B})$ with $\mathbf{L}_i\mathscr{F}(M)$ in the $-i$th degree with vanishing differentials is denoted $\mathbf{L}_\bullet\mathscr{F}(M)$.

To be precise, if $M \in \mathbf{A}$, then $\mathbf{L}_i(M)$ is found by

1. finding any projective resolution $P_M^\bullet$ of M
2. Applying $\mathrm{Ch}(\mathscr{F})$ to $P_M^\bullet$, $\mathrm{Ch}(\mathscr{F})(P_M^\bullet)$
3. take the $(-i)$th cohomology of this complex, $H^{-i}(\mathrm{Ch}(\mathscr{F})(P_M^\bullet))$

With all the build-up done so far, the resulting cohomology is independent of choice, for:

1. Any two projective resolutions are homotopy-equivalent
2. the image of homotopy-equivalent complexes via additive functors have isomorphic cohomology

Naturally, the *ith right-derived functor* $\mathbf{R}^i\mathscr{F}$ of an additive functor $\mathscr{F}$ and the assocaited complex $\mathbf{R}^\bullet\mathscr{F}$ is defined simplify (given $\mathbf{A}$ has enough injectives), namely

$$\mathbf{R}^i\mathscr{F} := H^i \circ \mathbf{R}\mathscr{F} \circ \mathscr{Q}_A \circ \iota_A$$

A note should be said about *contravriant functors*: since they can always be viewed as *covariant* functor $\mathbf{A}^{\mathrm{op}} \to B$, $\mathbf{A}$ would instead have to have the appropriate injectives or projectives swapped. The right-derived functor of an additive *contravariant functor* would need $\mathbf{A}$ to have enough *projectives*, since $\mathbf{A}^{\mathrm{op}}$ would have to have enough injectives. The $\mathbf{L}_i$ and $\mathbf{R}_i$ notation can be thought of as "looking" at the cohomology provided by $\mathscr{F}$. As mentioned earlier, if $\mathscr{F}$ is exact, then the derived functors are *trivial* in terms of cohomology, hence the derived functors are only interesting if $\mathscr{F}$ is not exact, at which point these functors give you interesting information.

Example 2.3.6: ith Derived Functors

It is finally time to bring back [7, Ext and Tor]

1. Let $\mathbf{A} = R$-$\mathbf{Mod}$ for a commutative ring R, pick an R-module N, and define

$$- \otimes_R N$$

The left-derived functor of $-\otimes_R N$ is labeled

$$-\overset{\mathbf{L}}{\otimes}_R N : D^-(R\text{-Mod}) \to D^-(R\text{-Mod})$$

The ith derived left-functor of $-\otimes_R N$ is denoted:

$$\mathrm{Tor}_i^R(-\otimes_R N) = H^{-i}((-)_\bullet \otimes N)$$

where the right hand side of the inequality comes from the result elaborated in [7, Ext and Tor]. It was mentioned that a flat resolution may be used instead of a projective resolution, however this was does not currently fit within the theory. This shall be addressed in the next section.

2. Similarly, $\mathrm{Hom}_R(-,N)$ and $\mathrm{Hom}_R(N,-)$ admits a right derived functors

$$\mathrm{Ext}_R^i(M,N) = H^i(\mathrm{Hom}_R(M,N^\bullet))$$

For $\mathrm{Hom}_R(-,N)$, don't forget that one must take the opposite category and a projective resolution.

The most important part of left-/right-derived functors is their use in fixing exactness after applying a not necessarily exact functor $\mathscr{F}$. On a high level, since the derived category there is not an abelian category (hence not necessarily containing kernels and cokernels, nor is it really feasible to investigate such categories), the notion of *triangulated categories* is useful for categories where given a "short exact sequence", we can create a long exact sequence (that is, a category in which snakes lemma can be applied). We shall not investigate the idea of triangulated categories, but the following is a concrete results needed for any rigorous exploration of triangulated categories.

Assuming the whole that that our category $\mathbf{A}$ has enough projectives, take a short exact sequence:

$$0 \to L \to M \to N \to 0$$

Since there is enough projectives, can the projective resolutions be arranged to create a short exact sequence? The answer is yes, and given the following apt name:

Lemma 2.3.7: Horseshoe Lemma

Let
$$0 \to L \to M \to N \to 0$$
be an exact sequence in an abelian category $\mathbf{A}$ with enough projectives, and let $P_L^\bullet, P_N^\bullet$ be projective resolutions of L, N respectively. Then there exists an exact sequence:
$$0 \to P_L^\bullet \to P_M^\bullet \to P_N^\bullet \to 0$$
where $P_M^\bullet$ is a projective resolution of M, inducing the original sequence in cohomology (i.e. applying H^0). Moreover the sequence is *degreewise split*: the construction produces $P_M^i = P_L^i \oplus P_N^i$ in each degree, with the two displayed maps the inclusion and the projection.

The dual statement holds in an abelian category with enough injectives: given injective resolutions $L \to I_L^\bullet$ and $N \to I_N^\bullet$, there is a degreewise split exact sequence $0 \to I_L^\bullet \to I_M^\bullet \to I_N^\bullet \to 0$ with $I_M^\bullet$ an injective resolution of M. It is this form that theorem 4.4.2 uses, and it follows by applying the projective statement in the opposite category.

***Proof*:**
Visually, we want to find objects and maps to fill in this commutative diagram:

$$\begin{tikzcd}
 & \vdots \arrow[d] & \vdots \arrow[d, dotted] & \vdots \arrow[d] & \\
0 \arrow[r] & P_L^{-2} \arrow[r, dotted] \arrow[d] & P_M^{-2} \arrow[r, dotted] \arrow[d, dotted] & P_N^{-2} \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & P_L^{-1} \arrow[r, dotted] \arrow[d] & P_M^{-1} \arrow[r, dotted] \arrow[d, dotted] & P_N^{-1} \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & L \arrow[r] & M \arrow[r] & N \arrow[r] & 0
\end{tikzcd}$$

where the dotted lines and the P_M^i are the goal (the name of the lemma comes from this horseshoe shaped diagram). I claim that $P_M^i = P_L^i \oplus P_N^i$ (which the reader should recall or reprove is projective) with the natural inclusion and projection maps shall work. The argument is done via induction. Since P_N^0 is projective and $M \to N$ is an epimorphism, the morphism $P_N^0 \to N$ lifts to a isomorphism m $\beta : P_N^0 \to M$. On the other hand, P_L^0 maps to M via $\alpha : P_L^0 \to L \to M$. Thus, there is a morphism:
$$\pi_M := \alpha \oplus \beta : P_L^0 \oplus P_N^0 \to M$$

diving us the diagram

$$\begin{tikzcd}
 & \vdots \arrow[d] & \vdots \arrow[d, dotted] & \vdots \arrow[d] & \\
0 \arrow[r] & P_L^{-2} \arrow[r, dotted] \arrow[d] & P_M^{-2} \arrow[r, dotted] \arrow[d, dotted] & P_N^{-2} \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & P_L^{-1} \arrow[r, dotted] \arrow[d] & P_M^{-1} \arrow[r, dotted] \arrow[d, dotted] & P_N^{-1} \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & P_L^{0} \arrow[r] \arrow[d] & P_L^0 \oplus P_N^0 \arrow[r] \arrow[d] & P_N^{0} \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & L \arrow[r] & M \arrow[r] & N \arrow[r] & 0
\end{tikzcd}$$

The snake lemma and π_L being an epimorphism gives that the kernels of the vertical maps form an exact sequence, and hence we have an epimorphism from P_L^{-1} and P_N^{-1} to the respective kernels since the column of the original diagram are exact:

$$\begin{tikzcd}
 & P_L^{-1} \arrow[d, "\pi_L^{-1}"] & & P_N^{-1} \arrow[d, "\pi_N^{-1}"] & \\
0 \arrow[r] & \ker \pi_L \arrow[r] & \ker \pi_M \arrow[r] & \ker \pi_N \arrow[r] & 0
\end{tikzcd}$$

Now, repeat as before, defining $P_M^{-1} = P_L^{-1} \oplus P_N^{-1}$ and define $\pi_M^{-1} : P_L^{-1} \oplus P_N^{-1} \to \ker \pi_M \to P_L^0 \oplus P_N^0$ giving the diagram:

$$\begin{tikzcd}
0 \arrow[r] & P_L^{-1} \arrow[r] \arrow[d, "\pi_L^{-1}"] & P_L^{-1} \oplus P_N^{-1} \arrow[r] \arrow[d, "\pi_M^{-1}"] & P_N^{-1} \arrow[r] \arrow[d, "\pi_N^{-1}"] & 0 \\
0 \arrow[r] & P_L^{0} \arrow[r] \arrow[d, "\pi_L"] & P_L^0 \oplus P_N^0 \arrow[r] \arrow[d, "\pi_M"] & P_N^{0} \arrow[r] \arrow[d, "\pi_N"] & 0 \\
0 \arrow[r] & L \arrow[r] & M \arrow[r] & N \arrow[r] & 0
\end{tikzcd}$$

By then by the snake lemma, $P_L^{-1} \oplus P_N^{-1} \to \ker \pi_M$ is an epimorphism, and hence we get exactness at $P_L^0 \oplus P_N^0$. Continuing inductively constructs the rest of the diagram.

It is tempting to say from this result that the mapping $\mathbf{A} \to K(\mathbf{A})$ is exact, however $K(\mathbf{A})$ is not abelian and hence this is simply undefined. The above is should instead be interpreted as follows: $P_M^\bullet$ is the mapping cone of a morphism: $\rho^\bullet : P_N[-1]^\bullet \to P_L^\bullet$, giving the distinguished triangle:

$$\begin{tikzcd}
 & P_L^\bullet \arrow[dr] & \\
P_N^\bullet \arrow[ur, "+1"] & & MC(\rho)^\bullet [= P_M^\bullet] \arrow[ll]
\end{tikzcd}$$

This is as close as we can get to preserving exactness.

Corollary 2.3.8: Resolution Of Projective Resolutions

Let $M^\bullet$ be a complex bounded from above in a category $\mathbf{A}$ with enough projectives. Then there exists a complex $P^b_{M^\bullet}b$:

$$\cdots \to P^\bullet_{M^{-3}} \to P^\bullet_{M^{-2}} \to P^\bullet_{M^{-1}} \to\to P^\bullet_{M^0} \to 0$$

where each $P^\bullet_{M^i}$ is a projection resolution of M^i and inducing $M^\bullet$ in cohomology. If $M^\bullet$ is exact, then $P^\bullet_{M^\bullet}$ can be chosen to be exact.

***Proof*:**
Since $M^\bullet$ is a complex, it can be broken down into two short exact sequences:

$$0 \to K^i \to M^i \to I^{i+1} \to 0 \qquad 0 \to I^i \to K^i \to H^i \to 0$$

where K^i is the kernel of $M^i \to M^{i+1}$, I^{i+1} is its image, and H^i the cohomology at M^i. Chose arbitrary projective resolutions for each I^i and H^i for all i. Then by the horseshoe lemma, we get a projective resolution for K^i that fits within the short exact sequence:

$$0 \to P^\bullet_{I^i} \to P^\bullet_{K^i} \to P^\bullet_{H^i} \to 0$$

Applying the horseshoe lemma gain, we get a short exact sequence:

$$0 \to P^\bullet_{K^i} \to P^\bullet_{M^i} \to P^\bullet_{I^{i+1}} \to 0$$

Using these, the $P^\bullet_{M^i}$ can be used to construct a cochain complex, the differentials obtains by composition. If $M^\bullet$ is exact, then $I^i = K^i$, and so the construction gives $P^\bullet_{I^i} = P^\bullet_{K^i}$, so that $P^\bullet_{M^\bullet}$ is also exact.

Lemma 2.3.9: Projective Resolution and Functors

Let $\mathbf{A}$ be an abelian category and let

$$0 \to L^\bullet \to M^\bullet \to P^\bullet \to 0$$

be an exact sequence of complexes in $\mathbf{A}$, where P^i is projective for all i. Let $\mathscr{F} : \mathbf{A} \to \mathbf{B}$ be an additive functor between abelian categories. Then the sequence:

$$0 \to \mathscr{F}(L^b) \to \mathscr{F}(M^\bullet) \to \mathscr{F}(P^\bullet) \to 0$$

is exact

Note that there is nothing about $\mathscr{F}$ being an exact functor! This can be thought of as the generalization of the properties of projective modules which preserve exactness!!

***Proof*:**
Since P^i is projective, the sequence

$$0 \to L^i \to M^i \to P^i \to 0$$

splits. Thus

$$0 \to \mathscr{F}(L^i) \to \mathscr{F}(M^i) \to \mathscr{F}(P^i) \to 0$$

is split exact for all i (as the reader should check). Since exactness of a sequence of complexes is given by exactness at each degree, this gives the desired result.

Theorem 2.3.10: Long Exact Sequence From Derived Functor

Let $\mathscr{F} : \mathbf{A} \to \mathbf{B}$ be an additive functor of abelian categories, and assume $\mathbf{A}$ has enough projective. Then every short exact sequence in $\mathbf{A}$

$$0 \to L \to M \to N \to 0$$

[covariantly] functorially induces a long exact sequence in $\mathbf{B}$:

$$\begin{tikzcd}
0 \arrow[r] & L_2\mathscr{F}(L) \arrow[r] & L_2\mathscr{F}(M) \arrow[r] & L_2\mathscr{F}(N) \arrow[dll, "\delta_2"] & \\
& L_1\mathscr{F}(L) \arrow[r] & L_1\mathscr{F}(M) \arrow[r] & L_1\mathscr{F}(N) \arrow[dll, "\delta_1"] & \\
& L_0\mathscr{F}(L) \arrow[r] & L_0\mathscr{F}(M) \arrow[r] & L_0\mathscr{F}(N) \arrow[r] & 0
\end{tikzcd}$$

***Proof*:**
The functorial assignment will be left as an exercise. Given the short exact sequence, we get short exact sequence of projective resolutions, which by lemma 2.3.9 creates a short exact sequence:

$$0 \to \mathscr{F}(P_L^\bullet) \to \mathscr{F}(P_M^\bullet) \to \mathscr{F}(P_N^\bullet) \to 0$$

Since the cohomology of these complexes is precisely the left-derifevd functors of $\mathscr{F}$, the long exact cohomology sequence is given by Snakes lemma, completing the proof.

The change from short exact sequence to long exact sequence can be unified under the notion of triangles, namely this theorem can be seen as the triangle:

$$\begin{tikzcd}
& \iota(L) \arrow[dr] & \\
\iota(N) \arrow[ur, "+1"] & & \iota(M) \arrow[ll]
\end{tikzcd}$$

inducing the triangle

$$\begin{tikzcd} & \mathbf{L}_\bullet(L) \arrow[dr] & \\ \mathbf{L}_\bullet(N) \arrow[ur, "-1", "\delta"'] & & \mathbf{L}_\bullet(M) \arrow[ll] \end{tikzcd}$$

where the vertices of this triangle are complexes in $C^{\leq 0}(\mathbf{B})$. Naturally, the dual situation happens for right derived functors for categories that have enough injectives, iin which case the above triangle induces a triangle:

$$\begin{tikzcd} & \mathbf{R}^\bullet(R) \arrow[dr] & \\ \mathbf{R}^\bullet(N) \arrow[ur, "-1", "\delta"'] & & \mathbf{R}^\bullet(M) \arrow[ll] \end{tikzcd}$$

where the vertices of this triangle are complexes in $C^{\geq 0}(\mathbf{B})$.

These consideration were for general additive functors. In most cases we saw, there is usually a left-exact or right-exact functor (indeed, the nature of the left-derived and right-derived series almost force us to choose “a side”). If a functor is given to be either left- or right-exact, then it can be directly recovered from their derived functors, as we shall now show.

Proposition 2.3.11: Right-Exactness And Derived Functors

L $\mathbf{A}$ be an abelian category with enough projective and $\mathscr{F} : \mathbf{A} \to \mathbf{B}$ be a right-exact additive functor. Then

1. $\mathbf{L}_i\mathscr{F} = 0$ for $i < 0$
2. $L_0\mathscr{F}$ is naturally isomorphic to $\mathscr{F}$

The dual result holds for the category with enough injectives. Do note that the *right*-exact functor is associated to the *left* derived functors, and vice-versa for left-exact functors which would have the associated *right* derived functors. This is because the derived functors are “fixing” the exactness on the other side.

***Proof*:**
Since projective resolution $P^\bullet$ of an object M in $\mathbf{A}$ is in $C^{\leq 0}(\mathbf{A})$, $C(\mathscr{F})(P^b b)$ is 0 in positive degree, and hence so is its cohomology. Since $H_i = H^{-i}$, the first claim is immediate.

For the second, an exact sequence:

$$P^{-1} \to P^0 \to M \to 0$$

will map to a exact sequence by $\mathscr{F}$ (due to right-exactness)

$$\mathscr{F}(P^{-1}) - > \mathscr{F}(P^0) - > \mathscr{F}(M) - > 0$$

This induces an isomorphism: $\nu_L : \mathbf{L}_0\mathscr{F}(M) = H^0(C(\mathscr{F})(P^\bullet)) \xrightarrow{\sim} \mathscr{F}(M)$. If $\varphi : M_0 - > M_1$ is a morphism, then there exists a morphism of projective resolutions $\alpha^\bullet : P_0^\bullet - > P_1^b b$:

$$\begin{tikzcd} \cdots \arrow[r] & P_0^{-1} \arrow[r] \arrow[d, "\alpha^{-1}"] & P_1^0 \arrow[r] \arrow[d, "\alpha^0"] & M_0 \arrow[r] \arrow[d, "\varphi"] & 0 \\ \cdots \arrow[r] & P_1^{-1} \arrow[r] & P_1^0 \arrow[r] & M_1 \arrow[r] & 0 \end{tikzcd}$$

induced by φ where $H^0(\alpha^b b)$ agrees with φ. Applying $\mathscr{F}$ and truncating to $\alpha^\bullet$, we get the commutative diagram:

$$\begin{tikzcd} \mathscr{F}(P_0^{-1}) \arrow[r] \arrow[d, "\mathscr{F}(\alpha^{-1})"] & \mathscr{F}(P_1^0) \arrow[r] \arrow[d, "\mathscr{F}(\alpha^0)"] & \mathscr{F}(M_0) \arrow[r] \arrow[d, "\mathscr{F}(\varphi)"] & 0 \\ \mathscr{F}(P_1^{-1}) \arrow[r] & \mathscr{F}(P_1^0) \arrow[r] & \mathscr{F}(M_1) \arrow[r] & 0 \end{tikzcd}$$

with exact rows since $\mathscr{F}$ is right-exact. But then the following diagram commutes:

$$\begin{tikzcd} \mathbf{L}_0\mathscr{F}(M_0) \arrow[r, "\mathbf{L}_0\mathscr{F}(\varphi)"] \arrow[d, "\nu_{M_0}"] & \mathbf{L}_0\mathscr{F}(M_1) \arrow[d, "\nu_{M_1}"] \\ \mathscr{F}(M_0) \arrow[r, "\mathscr{F}(\varphi)"'] & \mathscr{F}(M_1) \end{tikzcd}$$

completing the proof.

This finally gives the desired recovery of exactness promised way back. Given a short exact sequence

$$0 \to L \to M \to N \to 0$$

then applying a right-exact functor

$$\mathscr{F}(L) \to \mathscr{F}(M)\mathscr{F}(N) \to 0$$

will give an exact sequence, which can be stretched out to form a long exact sequence using the derived functors:

$$\begin{tikzcd} 0 \arrow[r] & L_2\mathscr{F}(L) \arrow[r] & L_2\mathscr{F}(M) \arrow[r] & L_2\mathscr{F}(N) \arrow[dll, "\delta_2"] \\ & L_1\mathscr{F}(L) \arrow[r] & L_1\mathscr{F}(M) \arrow[r] & L_1\mathscr{F}(N) \arrow[dll, "\delta_1"] \\ & \mathscr{F}(L) \arrow[r] & \mathscr{F}(M) \arrow[r] & \mathscr{F}(N) \arrow[r] & 0 \end{tikzcd}$$

In this way, $\mathbf{L}_1\mathscr{F}$ measures the extent to which $\mathscr{F}$ fails to be left-exact, and each progressive $\mathbf{L}_i\mathscr{F}$ gives further measurement, further information, about this failure. When in a geometrical setting, this failure is often thought of as some form of obstruction (famously, in algebraic topology, this obstruction is interpreted as "'holes", in K-theory).

Similarly, $\mathbf{R}^-\mathscr{F}$ is naturally isomorphic to $\mathscr{F}$ if $\mathscr{F}$ is left-exact and the right-derived functors fit into the diagram:

$$\begin{tikzcd}
0 \arrow[r] & \mathscr{F}(L) \arrow[r] & \mathscr{F}(M) \arrow[r] & \mathscr{F}(N) \arrow[dll, "\delta_1"] \\
& R^1\mathscr{F}(R) \arrow[r] & R^1\mathscr{F}(M) \arrow[r] & R^1\mathscr{F}(N) \arrow[dll, "\delta_0"] \\
& R^2\mathscr{F}(R) \arrow[r] & R^2\mathscr{F}(M) \arrow[r] & R^2\mathscr{F}(N) \arrow[r] & \cdots
\end{tikzcd}$$

Thus, we have successfully captured the cohomological information that a functor may provide!

Acyclic Functors

There still remain some questions to be answered, one that was mentioned was that instead of using a projective resolution for Tor, a flat resolution may be used instead and give the same answer.

Definition 2.3.12: Acyclic Functor

Let $\mathscr{F}$ be an additive functor. An object M of $\mathbf{A}$ is $\mathscr{F}$*-acyclic* (with respect to left-derived functors) if $\mathbf{L}_i\mathscr{F}(M) = 0$ for all $i \neq 0$.

Similarly, an object M is $\mathscr{F}$-acyclic with respect to right-derived functors if $\mathbf{R}^i\mathscr{F}(M) = 0$ for $i \neq 0$. Naturally, projectives are acyclic for every right-exact functor, and injective acyclic for every left-exact functor, however such a functor may admit other $\mathscr{F}$-acyclic objects.

Example 2.3.13: Functor-acyclic Objects

Let R be a commutative ring. Recall that an R-module M is *flat* if $- \otimes_R M$ is exact (equivalently, by symmetry of $\otimes$ that $M \otimes_R -$ is exact). Then flat modules are *acyclic* with respect to tensor products, namely if N is flat:

$$\mathrm{Tor}^i_R(M, N) = 0 \qquad \forall i \neq 0,\ \forall M$$

and symmetrically if M is flat. Thus, flat modules are $(- \otimes_R N)$-acyclic for all modules N.

What shall be built up to now is that $\mathscr{F}$-acyclic resolutions can be used to compute $\mathbf{L}_i\mathscr{F}$.

Theorem 2.3.14: Derived Functors And Acylcic Functors

Let $\mathscr{F} : \mathbf{A} \to \mathbf{B}$ be a right-exact functor of abelian groups where $\mathbf{A}$ has enough projectives. Let $A^\bullet$ be a resolution of M where each A^i is $\mathscr{F}$-acyclic. Then for all i:

$$\mathbf{L}_i\mathscr{F}(M) \cong H^{-i}(\mathrm{Ch}(\mathscr{F})(A^\bullet))$$

Note that if each A^i is projective, the statement follows by definition.

***Proof*:**
The 0th and -1th position will be computed directly, and each successive position comes from induction. Take the exact sequence

$$A^{-1} \to A^0 \to M \to 0$$

which creates another exact sequence in $\mathbf{B}$ after applying the right-exact functor $\mathscr{F}$:

$$\mathscr{F}(A^{-1}) \to \mathscr{F}(A^0) \to \mathscr{F}(M) \to 0$$

Then:

$$H^0(\mathrm{Ch}(\mathscr{F})(A^\bullet)) \cong \mathscr{F}(M) \cong \mathbf{L}_0\mathscr{F}(M)$$

Now, if K is the kernel of $A^0 \to M$, then we have a short exact sequence:

$$0 \to K \to A^0 \to M \to 0$$

Thus applying the left-derived functors we form the long exact sequence:

$$\mathbf{L}_1\mathscr{F}(A^0) \to \mathbf{L}_1\mathscr{F}(M) \to \mathscr{F}(K) \to \mathscr{F}(A^0) \to \mathscr{F}(M) \to 0$$

Since A^0 is acyclic, $L_1\mathscr{F}(A^0) = 0$, hence by exactness

$$\mathbf{L}_1\mathscr{F}(M) \cong \ker(\mathscr{F}(K) \to \mathscr{F}(A^0)) \cong H^{-1}(\mathrm{Ch}(\mathscr{F})(A^\bullet))$$

where the last isomorphism is some manipulation of definition. For the rest, we do induction. Assume now that for all object of $\mathbf{A}$ and all indices $< i$. Truncating and shifting $\mathbf{A}^\bullet$ so that A^{-1} is in the 0 degree:

$$\cdots \to A^{-3} \to A^{-2} \to A^{-1} \to 0 \to \cdots$$

we get an $\mathscr{F}$-acyclic resolution of K, and so by the induction hypothesis $\mathbf{L}_{i-1}\mathscr{F}(K)$ can be computed by applying $\mathrm{Ch}(\mathscr{F})$ to the complex and taking cohomology, namely:

$$\mathbf{L}_{i-1}\mathscr{F}(K) \cong H^{-i}(A^\bullet)$$

On the other hand, the other terms in the long exact sequence obtained using the left derived functors gives:

$$\cdots \to \mathbf{L}_i\mathscr{F}(A^0) \to L_i\mathscr{F}(M) \to \mathbf{L}_{i-1}\mathscr{F}(K) \to \mathbf{L}_{i-1}\mathscr{F}(A^0) \to \cdots$$

since A^0 is $\mathscr{F}$-acyclic and $i > 1$, we get:

$$\mathbf{L}_i\mathscr{F}(M) \cong \mathbf{L}_{i-1}\mathscr{F}(K)$$

giving the desired result.

Thus, for the Tor functor, we may use a flat resolution to compute it the left-derived functors. Since $\mathscr{F}$-acyclic objects suffice in the computation, one can imagine that there is not need for $\mathbf{A}$ to have enough projectives to compute it. This is indeed the case, given that $\mathbf{A}$ has "enough $\mathscr{F}$-acyclic's". (this is an exercise in Aluffi, 8.14, with bifunctors).

2.4 Double Complexes

This final section shall be the basis for being able to compute many complexes. It was mentioned in remark ref:HERE that $\mathrm{Ext}^i_R(M, N)$ may be computed by using either a projective resolution of M or an injective resolution on N, both give the same answer. More generally, we may compute the resolution of either M or N and get the same value. This section will now prove this result. This will make many computations much simpler, for usually one object in is readily understandable while the other is being analyzed.

To motivate this, switching from projective to $\mathscr{F}$-acyclic resolutions can be viewed in the following important alternative way. Consider the following diagram where the top row and right column are cochain:

$$\begin{tikzcd}
\cdots \arrow[r] & \mathscr{F}(A^{-2}) \arrow[r] & \mathscr{F}(A^{-1}) \arrow[r] & \mathscr{F}(A^{-0}) \arrow[r] & \mathscr{F}(M) \\
\cdots \arrow[r, dotted] & {} \arrow[r, dotted] \arrow[u, dotted] & {} \arrow[r, dotted] \arrow[u, dotted] & {} \arrow[r, dotted] \arrow[u, dotted] & \mathscr{F}(P_M^0) \arrow[u] \\
\cdots \arrow[r, dotted] & {} \arrow[r, dotted] \arrow[u, dotted] & {} \arrow[r, dotted] \arrow[u, dotted] & {} \arrow[r, dotted] \arrow[u, dotted] & \mathscr{F}(P_M^{-1}) \arrow[u] \\
\cdots \arrow[r, dotted] & {} \arrow[r, dotted] \arrow[u, dotted] & {} \arrow[r, dotted] \arrow[u, dotted] & {} \arrow[r, dotted] \arrow[u, dotted] & \mathscr{F}(P_M^{-2}) \arrow[u] \\
& \vdots \arrow[u, dotted] & \vdots \arrow[u, dotted] & \vdots \arrow[u, dotted] & \vdots \arrow[u]
\end{tikzcd}$$

in some way, there was a reason to go from the top chain to the right chain, with the intermediate values "interpolating" the result. The natural candidate for this would be a complex of complex. Needing to bound it in one direction, say we are looking at $\mathrm{Ch}^{\leq 0}(\mathrm{Ch}^{\leq 0}(\mathbf{A}))$. Such an object would be of the form:

$$\cdots \to M^{-3,\bullet} \xrightarrow{d_h^{-3,\bullet}} M^{-2,\bullet} \xrightarrow{d_h^{-2,\bullet}} M^{-1,\bullet} \xrightarrow{d_h^{-1,\bullet}} M^{0,\bullet} \to 0 \to \cdots$$

The h in the subscript is to represent the fact we are going *horizontally*, and the respective morphism of each $d_h^{-i,\bullet}$ will be going vertically and are denoted $d_v^{i,j}$, or simply d_v to simply show the function's direction:

$$\begin{tikzcd}
\cdots \arrow[r] & M^{-3,0} \arrow[r] & M^{-2,0} \arrow[r] & M^{-1,0} \arrow[r] & M^{0,0} \\
\cdots \arrow[r] & M^{-3,-1} \arrow[r] \arrow[u] & M^{-2,-1} \arrow[r] \arrow[u] & M^{-1,-1} \arrow[r] \arrow[u] & M^{0,-1} \arrow[u] \\
\cdots \arrow[r] & M^{-3,-2} \arrow[r] \arrow[u] & M^{-2,-2} \arrow[r] \arrow[u] & M^{-1,-2} \arrow[r] \arrow[u] & M^{0,-2} \arrow[u] \\
& \vdots \arrow[u] & \vdots \arrow[u] & \vdots \arrow[u] & \vdots \arrow[u]
\end{tikzcd}$$

Given a complex of complex, it would be nice to reduce this to being a complex, a more familiar setting. One strategy the reader may come up with is to take:

$$\bigoplus_{i+j=n} M^{i,j}$$

with the natural morphisms between them. However these morphisms *almost* complexes: instead of commutativity for each square, they are *anti-commutative*. To fix this, create a new type of object that will satisfy this condition:

Definition 2.4.1: Double Complex

Let $A = \{A_{p,q}\}$ be a family of objects from $\mathbf{A}$. Then A together with maps

$$d^h : A_{p,q} \to A_{p-1,q} \qquad \delta^v : A_{p,q} \to A_{p,q-1}$$

satisfying $d^h \circ d^h = 0$, $\delta^v \circ \delta^v = 0$ and $d^h \circ \delta^v + \delta^v \circ d^h = 0$ is called a *double complex* or *bicomplex*.
Both differentials *lower* an index; this homological convention is the one used throughout, and in particular in chapter 4. A cochain double complex, whose differentials raise indices, is the same thing read through $A_{p,q} = A^{-p,-q}$.

We may visualize a double complex like so, with d^h running to the right and δ^v running downwards, each lowering its index:

$$\begin{tikzcd}
 & \vdots \arrow[d] & \vdots \arrow[d] & \vdots \arrow[d] & \\
\cdots \arrow[r] & A_{p+1,q+1} \arrow[r] \arrow[d] & A_{p,q+1} \arrow[r] \arrow[d] & A_{p-1,q+1} \arrow[r] \arrow[d] & \cdots \\
\cdots \arrow[r] & A_{p+1,q} \arrow[r] \arrow[d] & A_{p,q} \arrow[r] \arrow[d] & A_{p-1,q} \arrow[r] \arrow[d] & \cdots \\
\cdots \arrow[r] & A_{p+1,q-1} \arrow[r] \arrow[d] & A_{p,q-1} \arrow[r] \arrow[d] & A_{p-1,q-1} \arrow[r] \arrow[d] & \cdots \\
 & \vdots & \vdots & \vdots &
\end{tikzcd} \tag{2.6}$$

Note this is not a commutative diagram but an *anti-commutative* one. That is a convention, not a restriction, and it is worth pinning down once because both forms occur in practice. A grid whose squares *commute*, which is what a complex of complexes literally gives, is converted into a double complex in the sense above by replacing the vertical differential with

$$\delta^v_{p,q} := (-1)^p \, \tilde{\delta}^v_{p,q}$$

where $\tilde{\delta}^v$ is the commuting one. The sign makes the squares anticommute, which is exactly what makes $d^h + \delta^v$ square to zero on the total complex. Every later construction, here and in chapter 4, assumes the anticommuting form; when a source of double complexes produces commuting squares, this twist is understood to have been applied.

As mentioned, we can form a total complex from this:

Proposition 2.4.2: Total Complex

Let $A = \{A^{p,q}\}$ be a double complex. Then define the total complex $\mathrm{Tot}^{\Pi}(A)$ and $\mathrm{Tot}^{\oplus}(A)$ by:

$$\mathrm{Tot}^{\Pi}(A)_n = \prod_{p+q=n} A_{p,q} \qquad \mathrm{Tot}^{\oplus}(A)_n = \bigoplus_{p+q=n} A_{p,q}$$

with $d^{\mathrm{tot}} = d^h + \delta^v$, which squares to zero precisely because the two differentials anticommute. The two agree whenever only finitely many $A_{p,q}$ are nonzero on each diagonal $p + q = n$, which is the case for every double complex considered here.

***Proof*:**
Computing:

$$d_{tot} \circ d_{tot} = (d_h + \delta_v) \circ (d_h + \delta_v) = d_h \circ d_h + d_h \circ \delta_v + \delta_v \circ d_h + \delta_v \circ \delta_v = 0$$

Note that the total complex does not exist in all categories, namely it may not exist in categories which are not closed under infinite products and coproducts (ex. in the category of finite abelian groups). A category which is closed for infinite products is called *complete* and *cocomplete* if it is closed for infinite sums. We can visualize a total complex by taking the diagram in equation (2.6) and moving it like so:

$$\begin{tikzcd}
 & & & M^{0,0} \\
 & & M^{-1,0} \arrow[ur, "d_h"] & M^{0,-1} \arrow[u, "\delta_v"'] \\
 & M^{-2,0} & M^{-1,-1} \arrow[u, "\delta_v"] \arrow[ur, "d_h"] & M^{0,-2} \arrow[u, "\delta_v"'] \\
M^{-3,0} \arrow[ur, "d_h"] & M^{-2,-1} \arrow[u] \arrow[ur, "d_h"] & M^{-1,-2} \arrow[u, "\delta_v"] \arrow[ur, "d_h"] & M^{0,-2} \arrow[u, "\delta_v"'] \\
\vdots \arrow[u, "\delta_v"] \arrow[ur, "d_h"'] & \vdots \arrow[u, "\delta_v"] \arrow[ur, "d_h"'] & \vdots \arrow[u, "\delta_v"] \arrow[ur, "d_h"'] & \vdots \arrow[u, "\delta_v"']
\end{tikzcd}$$

It is then easy to visualize how we get a total complex form this.

Example 2.4.3: Total Complexs

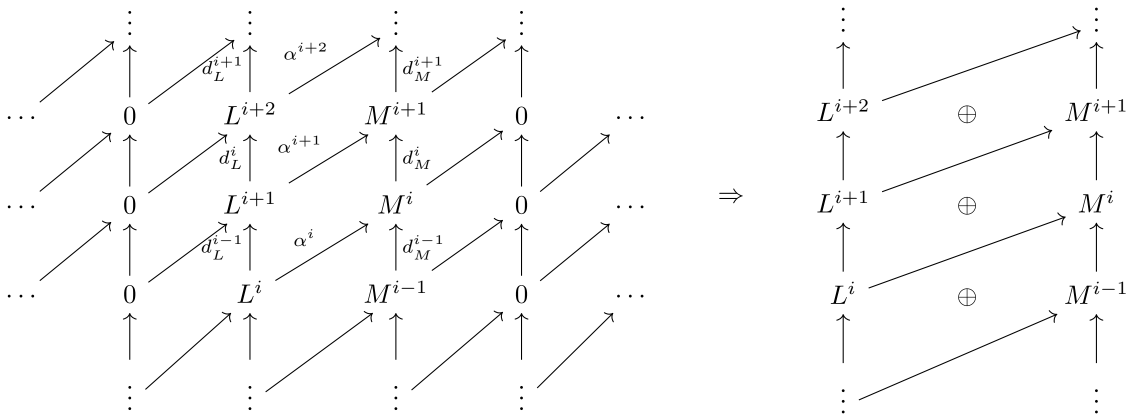
1. Consider a double complex:

$$\cdots \to 0 \to L^{\bullet} \xrightarrow{\alpha^{\bullet}} M^{\bullet} \to 0 \to \cdots$$

which we may represent as:

$$\begin{tikzcd}
 & \vdots & \vdots & \vdots & \vdots & \\
\cdots \arrow[r] & 0 \arrow[r] \arrow[u] & L^{i+1} \arrow[r, "\alpha^{i+1}"] \arrow[u, "d_L^{i+1}"] & M^{i+1} \arrow[r] \arrow[u, "d_M^{i+1}"'] & 0 \arrow[r] \arrow[u] & \cdots \\
\cdots \arrow[r] & 0 \arrow[r] \arrow[u] & L^{i} \arrow[r, "\alpha^{i}"] \arrow[u, "d_L^{i}"] & M^{i} \arrow[r] \arrow[u, "d_M^{i}"'] & 0 \arrow[r] \arrow[u] & \cdots \\
\cdots \arrow[r] & 0 \arrow[r] \arrow[u] & L^{i-1} \arrow[r, "\alpha^{i-1}"] \arrow[u, "d_L^{i-1}"] & M^{i-1} \arrow[r] \arrow[u, "d_M^{i-1}"'] & 0 \arrow[r] \arrow[u] & \cdots \\
 & \vdots \arrow[u] & \vdots \arrow[u] & \vdots \arrow[u] & \vdots \arrow[u] &
\end{tikzcd}$$

we shall get the following:



which is exactly the definition of a mapping cone!

2. The operators $\mathrm{Hom}_A(-,-)$ and $-\otimes-$ both give rise to double complexes. Focusing on the $\mathrm{Hom}_A(-,-)$ example, consider a chain in $L^\bullet \in \mathrm{Ch}^{\leq 0}(\mathbf{A})$ and $M^\bullet \in \mathrm{Ch}^{\geq 0}(\mathbf{A})$. These can be used to form a double complex, or isomorphically a complex in $\mathrm{Ch}^{\leq 0}(\mathrm{Ch}^{\leq 0}(\mathbf{A}))$:

$$\cdots \to \mathrm{Hom}_A(L^\bullet, M^0) \to \mathrm{Hom}_A(L^\bullet, M^1) \to \mathrm{Hom}_A(L^\bullet, M^2) \to \cdots$$

which creates the following upper-quadrant grid:

$$\begin{tikzcd}
\vdots & \vdots & \vdots & \\
\mathrm{Hom}_A(L^{-2}, M^0) \arrow[r] \arrow[u] & \mathrm{Hom}_A(L^{-2}, M^1) \arrow[r] \arrow[u] & \mathrm{Hom}_A(L^{-2}, M^2) \arrow[r] \arrow[u] & \cdots \\
\mathrm{Hom}_A(L^{-1}, M^0) \arrow[r] \arrow[u] & \mathrm{Hom}_A(L^{-1}, M^1) \arrow[r] \arrow[u] & \mathrm{Hom}_A(L^{-1}, M^2) \arrow[r] \arrow[u] & \cdots \\
\mathrm{Hom}_A(L^{0}, M^0) \arrow[r] \arrow[u] & \mathrm{Hom}_A(L^{0}, M^1) \arrow[r] \arrow[u] & \mathrm{Hom}_A(L^{0}, M^2) \arrow[r] \arrow[u] & \cdots
\end{tikzcd}$$

Flipping the above diagram across the main diagonal, we would get the double complex:

$$\cdots \to \mathrm{Hom}_A(L^0, M^\bullet) \to \mathrm{Hom}_A(L^{-1}, M^\bullet) \to \mathrm{Hom}_A(L^{-2}, M^\bullet) \to \cdots$$

These two double complexes are isomorphic (as the reader should check). Due to this, by slight abuse of notation the total complex of both can be denoted $TC(\mathrm{Hom}_A(L, M))^\bullet$, where the degree i term is:

$$\mathrm{Hom}_A(L^{-i}, M^0) \oplus \mathrm{Hom}_A(L^{-i+1}, M^1) \oplus \cdots \oplus 0\mathrm{Hom}_A(L^{-1}, M^{i-1}) \oplus 0\mathrm{Hom}_A(L^0, M^i)$$

The reader should verify that:

(a) $\ker d^i_{tot}$ is equal to morphisms of cochains $L^\bullet \to M[i]^\bullet$

(b) The image $\mathrm{im} d^{i-1}_{tit}$ consists of morphisms that are homotopy equivalent to 0

(c) Thus, the cohomology of $TC(\mathrm{Hom}_A(M, N))^\bullet$ "paramaterizes" the morphisms in the homotopic category

To ground this, consider $TC(\mathrm{Hom}_A(M, N))^0 = \mathrm{Hom}_A(L^0, M^0)$. Then $d_{tot}(\alpha) = 0$ means that both $\alpha \circ d^{-1}_{L^\bullet} \in \mathrm{Hom}_A(L^{-1}, M^0)$ and $d^0_{M^\bullet} \in \mathrm{Hom}_A(L^0, M^1)$ both vanish. The reader should draw out this diagram, and notice that this will make α have the condition to be a morphism of cochain complexes $L^\bullet \to M^\bullet$!

$$H^0(TC(\mathrm{Hom}_A(M, N))^\bullet) = \mathrm{Hom}_{\mathrm{Ch}(\mathbf{A})}(L^\bullet, M^\bullet) \overset{!}{=} \mathrm{Hom}_{K(\mathbf{A})}(L^\bullet, M^\bullet)$$

where the $\overset{!}{=}$ equality comes from the fact that there is no nontrivial homotopies in this case! For $H^i(TC(\mathrm{Hom}_A(M, N))^\bullet)$, with some care for signs, it can be identified with $\mathrm{Hom}_{K(\mathbf{A})}(L^\bullet, M[i]^\bullet)$

The cohomology of double complexes requires some book-keeping through the use of *spectral sequences*, which shall be explored in greater depth in chapter 4. For the current goal, it is enough to consider when a double-complex is exact:

Proposition 2.4.4: Exactness Of Complex Of Complex

Let $\mathbf{A}$ be an abelian category and let

$$\cdots \to M^{-3,\bullet} \to M^{-2,\bullet} \to M^{-1,\bullet} \to M^{0,\bullet} \to 0$$

be a complex in $\mathrm{Ch}^{\leq 0}(\mathrm{Ch}^{\leq 0}(\mathbf{A}))$, regarded as a double complex $M^{\bullet,\bullet}$ with $M^{p,q} = 0$ unless $p \leq 0$ and $q \leq 0$, and let $T^\bullet$ denote its total complex. Then $T^\bullet$ is exact if *either* of the following holds:

1. the displayed complex is exact in $\mathrm{Ch}^{\leq 0}(\mathbf{A})$, equivalently every row $M^{\bullet,q}$ is exact;
2. every column $M^{p,\bullet}$ is exact.

Neither hypothesis implies the other, and either one alone suffices; this is the form in which the result is used below and in chapter 4. The same statement holds in $\mathrm{Ch}^{\geq 0}(\mathrm{Ch}^{\geq 0}(\mathbf{A}))$, by reversing the direction of every index. Note further that only one of the two directions need be bounded: case (2) uses only $p \leq 0$ and case (1) only $q \leq 0$, the bound being what makes the following staircase terminate; with only one bound the diagonals are infinite, so there the total complex must be taken to be $\mathrm{Tot}^{\oplus}$, since the staircase needs a least index in the support of a given element. The result is known as the *acyclic assembly lemma*; see *An Introduction to Homological Algebra* [12] for the variants in the other quadrants and for the distinction between Tot^{Π} and $\mathrm{Tot}^{\oplus}$, which does not arise here because each diagonal of $M^{\bullet,\bullet}$ carries only finitely many nonzero terms.

***Proof*:**
Step 1: Reduction to case (2). Transposing the double complex, that is exchanging the two indices and the two differentials, converts rows into columns and leaves the total complex unchanged up to the sign of the differential, which does not affect exactness. So case (1) follows from case (2) applied to the transpose, and only the latter needs proof.

Step 2: The staircase. Assume every column is exact, write d_h and d_v for the two differentials and $d = d_h + d_v$ for the differential of $T^\bullet$. Let $x \in T^n$ satisfy $dx = 0$; the claim is that $x = dy$ for some $y \in T^{n-1}$. Write $x = \sum_{p+q=n} x^{p,q}$, a finite sum.

Let p_1 be least with $x^{p_1,q_1} \neq 0$, where $q_1 = n - p_1$; if there is no such p_1 then $x = 0$ and there is nothing to prove. Consider the component of the equation $dx = 0$ in bidegree $(p_1, q_1 + 1)$. Only two terms can contribute to it, namely $d_v x^{p_1,q_1}$ and $d_h x^{p_1-1,q_1+1}$, and the second vanishes because $p_1 - 1 < p_1$ and p_1 was least. Hence

$$d_v x^{p_1,q_1} = 0$$

so x^{p_1,q_1} is a cocycle in the column $M^{p_1,\bullet}$. That column is exact, so there is $y^{p_1,q_1-1} \in M^{p_1,q_1-1}$ with $d_v y^{p_1,q_1-1} = x^{p_1,q_1}$.

Step 3: Termination. Replace x by

$$x' = x - d\big(y^{p_1,q_1-1}\big) = x - d_v y^{p_1,q_1-1} - d_h y^{p_1,q_1-1}$$

This is again a cocycle, since it differs from x by a coboundary. Its component in bidegree (p_1, q_1) is $x^{p_1,q_1} - d_v y^{p_1,q_1-1} = 0$, and $d_h y^{p_1,q_1-1}$ lives in bidegree $(p_1 + 1, q_1 - 1)$, so no component with $p < p_1$ is disturbed and all of those were already zero. The least index with a nonzero component has therefore increased by at least one.

Because $M^{p,q} = 0$ for $p > 0$, this least index cannot exceed 0, so after at most $|p_1| + 1$ repetitions every component vanishes. Summing the finitely many y's produced along the way gives $y \in T^{n-1}$ with $dy = x$, so $T^\bullet$ is exact at degree n. As n was arbitrary, this completes the proof.

Example 2.4.5: Exactness Of Double Complex

Let $P^\bullet \in C^{\leq 0}(\mathbf{A})$ be a complex and $L^\bullet \in C^{\geq 0}(\mathbf{A})$ an exact complex. Since each P^i is projective, $\mathrm{Hom}_{\mathbf{A}}(P^i, L^\bullet)$ for each i. Hence, the complex:

$$0 \to \mathrm{Hom}_{\mathbf{A}}(P^0, L^\bullet) \to \mathrm{Hom}_{\mathbf{A}}(P^{-1}, L^\bullet) \to \mathrm{Hom}_{\mathbf{A}}(P^{-2}, L^\bullet) \to \cdots$$

is exact. Using example 2.4.3, we see that:

$$\mathrm{Hom}_{K(\mathbf{A})}(P^\bullet, L[i]^\bullet) = 0 \qquad \forall i \in \mathbb{Z}$$

This implies that every cochain morphism from a bounded-above projective complex to a bounded-bellow exact complex is homotopic to 0. This just proved corollary 2.2.5 (with an extra boundedness condition)! In some textbooks, due to the speed at which some results can be proved using double-complexes, they are introduced earlier. However, the more "down-to-earth" introduction has is favored in this chapter since it demonstrated key homological concepts.

Next, we look at resolutions of complexes.. As we saw in definition 2.2.11, a (projective) resolution is a quasi-isomorphism $M^\bullet \to \iota(A)$ for some complex $M^\bullet \in \mathrm{Obj}(\mathrm{Ch}(\mathbf{A}))$, and more generally a projective resolution for a general complex is a quasi-isomorphism $P^\bullet \to C^\bullet$ for some projective complex $P^\bullet$. Generalizing the definition of resolution of the first kind, $\iota(A)$, and say that resolution of a complex $N^\bullet \in \mathrm{Ch}^\bullet(A) = A'$ is a complex of complex $M^{\bullet,\bullet} \in \mathrm{Ch}^{\leq 0}(A')$ and a quasi-isomorphism

$$\begin{tikzcd} \cdots \arrow[r] & M^{-2,\bullet} \arrow[r, "d_h"] \arrow[d] & M^{-2,\bullet} \arrow[r, "d_h"] \arrow[d] & M^{-2,\bullet} \arrow[r] \arrow[d] & 0 \arrow[r] \arrow[d] & \cdots \\ \cdots \arrow[r] & 0 \arrow[r] & 0 \arrow[r] & N^\bullet \arrow[r] & 0 \arrow[r] & \cdots \end{tikzcd} \tag{2.7}$$

where the diagram commutes. We shall show that this definition of a resolution induces a resolution as given in theorem 2.2.17. To see this, notice that in equation (2.7), $\alpha^\bullet \circ d_h = 0$, hence we can collapse the diagram like so:

$$\cdots \to M^{-2,\bullet} \xrightarrow{-d_h} M^{-1,\bullet} \xrightarrow{d_h} M^{0,\bullet} \xrightarrow{\alpha^\bullet} N^\bullet \to 0 \cdots$$

Call this complex $M_N^{\bullet,\bullet}$. Two important diagrams can be derived here. The first is by taking the total complex of $M^{\bullet,\bullet}$ and taking a morphism from this to $N^\bullet$:

$$\begin{tikzcd}
 & & & & 0 \arrow[r] & 0 \\
 & & & & M^{0,0} \arrow[u] \arrow[r] & N^0 \arrow[u] \\
 & & M^{-1,0} \arrow[urr] & \oplus & M^{0,-1} \arrow[u] \arrow[r] & N^{-1} \arrow[u] \\
M^{-2,0} \arrow[urr] & \oplus & M^{-1,-1} \arrow[u] \arrow[urr] & \oplus & M^{0,-2} \arrow[u] \arrow[r] & N^{-2} \arrow[u] \\
{} \arrow[u] \arrow[urr] & & {} \arrow[u] \arrow[urr] & & {} \arrow[u] & {} \arrow[u]
\end{tikzcd}$$

This gives a morphism:

$$TC(\alpha^\bullet) : TC(M^{\bullet,\bullet}) \to N^\bullet$$

where the morphism is defined since $TC(-)$ is a functorial. Another diagram we can create is by considering the total complex of $M_N^{\bullet,\bullet}$:

$$\begin{tikzcd}
 & & & & & & N^0 \\
 & & & & M^{0,0} \arrow[urr] & \oplus & N^{-1} \arrow[u] \\
 & & M^{-1,0} \arrow[urr] & \oplus & M^{0,-1} \arrow[u] \arrow[urr] & \oplus & N^{-2} \arrow[u] \\
M^{-2,0} \arrow[urr] & \oplus & M^{-1,-1} \arrow[u] \arrow[urr] & \oplus & M^{0,-2} \arrow[u] \arrow[urr] & \oplus & N^{-3} \arrow[u] \\
{} \arrow[u] \arrow[urr] & & {} \arrow[u] \arrow[urr] & & {} \arrow[u] \arrow[urr] & & {} \arrow[u]
\end{tikzcd}$$

These two are related in the following way:

Proposition 2.4.6: Total Complex Of Resolution

Let $M^{\bullet,\bullet}$ be a resolution of $N^\bullet$ and let $M_N^{\bullet,\bullet}$ be defined as above. Then

$$TC(M_N)^\bullet \cong MC\big(TC(\alpha)\big)[-1]^\bullet$$

canonically, equivalently $TC(M_N)[1]^\bullet \cong MC(TC(\alpha))^\bullet$.

***Proof*:**
Both sides are computed degreewise, so it suffices to match the terms and the differentials.

Step 1: The terms. The complex $M_N^{\bullet,\bullet}$ is $M^{\bullet,\bullet}$ with $N^\bullet$ appended in the column $p = 1$, to the right of $M^{0,\bullet}$ and joined to it by the augmentation $\alpha^\bullet$. A term N^j therefore sits in total degree

$1+j$, so its contribution to total degree n is N^{n-1} and

$$TC(M_N)^n = \Big(\bigoplus_{i+j=n} M^{i,j}\Big) \oplus N^{n-1} = TC(M)^n \oplus N^{n-1}$$

which is what the diagram above the proposition displays, $M^{0,0}$ being grouped with N^{-1}. On the other side $TC(\alpha)\colon TC(M)^\bullet \to N^\bullet$ is the totalized augmentation, and since definition 2.1.18 puts the *source* of a morphism in the shifted slot,

$$MC\big(TC(\alpha)\big)^n = TC(M)^{n+1} \oplus N^n$$

Comparing the two, $TC(M_N)^n = MC(TC(\alpha))^{n-1}$, which is the shift asserted in the statement.

Step 2: The differentials. On $TC(M_N)^\bullet$ the differential is the total differential of the augmented double complex: on the M part it is $d_h + d_v$, the component carrying $M^{0,\bullet}$ into the appended column is $\alpha^\bullet$, and on the N part it is $-d_N$, the sign being the $(-1)^p$ twist at $p=1$ that makes the augmented grid anticommute. So $d_{TC(M_N)}(m,n) = (d_{TC(M)}m,\ \alpha(m) - d_N n)$. By definition 2.1.18 the cone differential is $d_{MC}(l,n) = (-d_{TC(M)}l,\ \alpha(l) + d_N n)$, so under the identification of Step 1 the two differ by an overall sign.

The shift convention $d_{[r]} = (-1)^r d$ gives $d_{MC(TC(\alpha))[-1]}(m,n) = -\big(-d_{TC(M)}m,\ \alpha(m)+d_N n\big) = (d_{TC(M)}m,\ -\alpha(m) - d_N n)$, so the two differentials agree in the $TC(M)$ component and differ exactly in the sign of the α-component. They are therefore not equal on the nose, but the map

$$\varphi(m,n) = (m,-n)$$

is an isomorphism of complexes intertwining them: $\varphi\big(d_{TC(M_N)}(m,n)\big) = (d_{TC(M)}m,\ -\alpha(m) + d_N n) = d_{MC(TC(\alpha))[-1]}\big(\varphi(m,n)\big)$. Hence $TC(M_N)^\bullet \cong MC(TC(\alpha))[-1]^\bullet$, canonically, completing the proof.

Using this, we can see how the resolution in $\mathrm{Ch}^{\leq 0}(A')$ leads to a resolution as defined in theorem 2.2.17:

Theorem 2.4.7: Total Complexes And Resolution Of Complexes

Let $\mathbf{A}$ be an abelian category, $\mathbf{A}' = \mathrm{Ch}^{\leq 0}(\mathbf{A})$, and $N^\bullet \in A'$. Let $M^{\bullet,\bullet} \in \mathrm{Ch}^{\leq 0}(\mathbf{A}')$ with total complex $T^\bullet$.

1. Assume that $M^{\bullet,\bullet}$ is a resolution of $N^\bullet$ in $\mathrm{Ch}^{\leq 0}(\mathbf{A}')$. Then $T^\bullet$ is quasi-isomorphic to $N^\bullet$ in $\mathbf{A}'$.

2. Assuming the cohomology of $M^{i,\bullet}$ is concentrated at 0. Then $T^\bullet$ is quasi-isomorphic to the complex of cohomology objects induced by the complex $M^{\bullet,\bullet}$:

$$\cdots \to H^0(M^{-3,\bullet}) \to H^0(M^{-2,\bullet}) \to H^0(M^{-1,\bullet}) \to H^0(M^{0,\bullet}) \to 0 \cdots$$

Notice that this theorem is a direct extensions of proposition 2.4.4. This is also a special case of results we shall see in chapter 4

***Proof*:**
Notice that (2) follows from (1) by flipping the corresponding double-complex across the diagonal, hence we prove the first. For the first, if $M_N^{\bullet,\bullet}$ is quasi-isomorphic, then $TC(M_N)^\bullet$ is exact by proposition 2.4.4, which by proposition 2.4.6 is the mapping cone of the morphism $T^\bullet \to N^\bullet$, which by proposition 2.1.21 shows that $\alpha^\bullet : T^\bullet \to N^\bullet$ is a quasi-isomorphism, as we sought to show.

Using this theorem, a simple alternative to resolving complexes (theorem 2.2.17). By the horseshoe lemma, every bounded-above complex $N^\bullet$ in an abelian category with enough projectives may be viewed as a complex induced by H^0 from a complex:

$$\cdots \to P^{-3,\bullet} \to P^{-2,\bullet} \to P^{-1,\bullet} \to P^{0,\bullet} \to 0$$

where $P^{i,\bullet}$ are projective resolutions of N^i. Then by theorem 2.4.7(2) the total complex of $P^{\bullet,\bullet}$ is a projective resolution of $N^\bullet$. This then can give theorem 2.2.17. It is still worth presenting theorem 2.2.17 to firmly grasp the fundamentals of homological algebra.

2.4.1 Applications to Computations of Ext and Tor

Let us now show that we may compute $\mathrm{Ext}(M,N)$ or $\mathrm{Tor}(M,N)$ using a resolution of either term. Going back to the $\mathscr{F}$-acyclic resolution of an object M:

$$\cdots \to A^{-2} \to A^{-1} \to A^0 \to M \to 0 \to \cdots$$

We may create a complex of projective resolution by corollary 2.3.8:

$$\cdots \to P^{-2,\bullet} \to P^{-1,\bullet} \to P^{0,\bullet} \to P_M^\bullet \to 0 \to \cdots$$

where $P^{i,\bullet}$ is a projective resolution of A^{-i}. Since the resolution by projective resolution's is exact, the rows of the double-complex corresponding to $P^{\bullet,\bullet}$ are projective resolutions of P_M^i. Now, applying the functor $\mathscr{F}$, we get the diagram:

$$\begin{tikzcd}
\cdots \arrow[r] & \mathscr{F}(A^{-2}) \arrow[r] & \mathscr{F}(A^{-1}) \arrow[r] & \mathscr{F}(A^0) \arrow[r] & \mathscr{F}(M) \\
\cdots \arrow[r] & \mathscr{F}(P_M^{-2,0}) \arrow[r] \arrow[u] & \mathscr{F}(P_M^{-1,0}) \arrow[r] \arrow[u] & \mathscr{F}(P_M^{0,0}) \arrow[r] \arrow[u] & \mathscr{F}(P_M^0) \arrow[u] \\
\cdots \arrow[r] & \mathscr{F}(P_M^{-2,-1}) \arrow[r] \arrow[u] & \mathscr{F}(P_M^{-1,-1}) \arrow[r] \arrow[u] & \mathscr{F}(P_M^{0,-1}) \arrow[r] \arrow[u] & \mathscr{F}(P_M^{-1}) \arrow[u] \\
\cdots \arrow[r] & \mathscr{F}(P_M^{-2,-2}) \arrow[r] \arrow[u] & \mathscr{F}(P_M^{-1,-2}) \arrow[r] \arrow[u] & \mathscr{F}(P_M^{0,-2}) \arrow[r] \arrow[u] & \mathscr{F}(P_M^{-2}) \arrow[u] \\
& \vdots \arrow[u] & \vdots \arrow[u] & \vdots \arrow[u] & \vdots \arrow[u]
\end{tikzcd}$$

Since each row and column of this diagram (excluding the top row and right column) is exact, by theorem 2.4.7 the total complex of:

$$\cdots \mathscr{F}(P^{-2,\bullet}) \to \mathscr{F}(P^{-1,\bullet}) \to \mathscr{F}(P^{0,\bullet}) \to 0 \to \cdots$$

is quasi-isomorphic to both $\mathrm{Ch}(\mathscr{F})(A^\bullet)$ and $\mathrm{Ch}(\mathscr{F})(P^\bullet)$, that is:

$$TC(\mathscr{F}(P))^\bullet \sim \mathrm{Ch}(\mathscr{F})(A^\bullet) \sim \mathrm{Ch}(\mathscr{F})(P^\bullet)$$

Thus, by definition of quasi-isomorphism

$$\mathbf{L}_i\mathscr{F}(M) \cong H^i(\mathrm{Ch}(\mathscr{F})(A^\bullet)) \cong H^i(\mathrm{Ch}(\mathscr{F})(P^\bullet))$$

Applying these results to Ext and Tor will finally give us the following results:

Theorem 2.4.8: Computation Of Tor and Ext

Let M, N be R-modules for a commutative ring R, and let $P_M^\bullet$ and $P_N^\bullet$ be projections resolutions of M and N respectively, and $Q_N^\bullet$ an injective resolution of N. Then:

$$\mathrm{Tor}_i^R(M,N) \cong H^i(P_M^\bullet \otimes_R N) \cong H^i(M \otimes_R P_N^\bullet)$$

and

$$\mathrm{Ext}_R^i(M,N) \cong H^i(\mathrm{Hom}_R(P_M^\bullet, N)) \cong H^i(\mathrm{Hom}_R(M, Q_N^\bullet))$$

***Proof*:**
We shall show the result for Tor, the result following very similarly for Ext. Take the complex:

$$\cdots \to P_M^{-2} \otimes_R P_N^\bullet \to P_M^{-1} \otimes_R P_N^\bullet \to P_M^0 \otimes_R P_N^\bullet \to 0 \cdots$$

Since each P_N^j is projective, the cohomology of the complex is concentrated at 0 and equals $M \otimes_R P_N^\bullet$. By theorem 2.4.7 we have that :

$$TC(P_M^\bullet \otimes_R P_N^\bullet)^\bullet \qquad \text{is quasi-isomorphic to} \qquad M \otimes_R P_N^\bullet$$

Similarly, since each P_M^i is projective, the complex $P_M^i \otimes_R P_N^\bullet$ is a resolution of $P_M^i \otimes N$, i.e. the cohomology of each $P_M^i \otimes_R P_N^\bullet$ is concentrated at degree 0 and equals $P_M^i \otimes_R N$. By theorem 2.4.7 we have that :

$$TC(P_M^\bullet \otimes_R P_N^\bullet)^\bullet \qquad \text{is quasi-isomorphic to} \qquad P_M^\bullet \otimes_R N$$

but then, the cohomologies are identical, as we sought to show.

2.5 *Further Topic in Homological Algebra

look at commented line here:

In this bonus section, we shall introduce formally give the definition of a derived category and introduce the notion of triangulated categories.

2.5.1 Derived Categories, Formally

As defined in this chapter, derived categories required enough projectives/injectives. However, it was mentioned that derived categories may be defined without this restriction, allowing for a category that satisfies the universal property for abelian categories more generally. The theoretical build-up needed for this was carried out in section 1.2: the general derived category is the localization $D(\mathbf{A}) = \mathrm{Ch}(\mathbf{A})[\mathrm{qis}^{-1}]$, which exists by theorem 1.2.2 and is recorded as corollary 1.2.6. What follows revisits that construction concretely, in the roof-diagram form that theorem 1.2.5 justifies, so that the reader sees the mechanism rather than only the universal property. This section is heavily inspired by *Aluffi Chapter 0* which presented these idea's very concisely.

We shall have the objects still be $\mathrm{C}(\mathbf{A})$, however we need the quasi-isomorphisms to be invertible. Recall lemma 2.2.7 where it was shown that quasi-isomorphisms of projective resolutions have the property of being acting like non-zero divisors. We shall take full advantage of this idea, and essentially "formally invert" quasi-isomorphisms in the same vein as the construction of localization's for rings and modules. In fact, from an abstract perspective, the study of homological algebra can be thought of as the study of the localization of the category of (co)chain complexes (localized at the class of quasi-isomorphisms). When there is enough projectives, the localization of $\mathrm{C}(\mathbf{A})$ becomes $\mathrm{K}(\mathbf{a})$.

For the more general case, consider the following *roof diagram*:

$$\begin{tikzcd} & Z \arrow[dl, "\alpha"'] \arrow[dr, "\beta"] & \\ A & & B \end{tikzcd}$$

Then we may formally add a morphism $\beta \circ \alpha^{-1} : A \to B$:

$$\begin{tikzcd} & Z \arrow[dl, "\alpha"'] \arrow[dr, "\beta"] & \\ A \arrow[rr, dashed] & & B \end{tikzcd}$$

This can be thought of as the fraction. There are many technicalities being swept under the rug, like how do we insure these are well-defined morphisms (ex. how would we compose roofs?), while more fundamental problems like the amount of morphisms shouldn't exceed a *set*-amount (note that $\mathbf{A}$ and $\mathbf{B}$ are classes). Furthermore, we may ask if this is possible on any possible category, or should we restrict to abelian categories? Can any morphism be invertible, or do we need to restrict to a special subset (similar to a multiplicative set)? Giving this technical difficulties, we may imagine that from this the diagram

$$\begin{tikzcd} & Z \arrow[dl, "\alpha"'] \arrow[dr, "\mathrm{id}_Z"] & \\ A \arrow[rr, dashed] & & Z \end{tikzcd}$$

would formally invert a morphism. For $\mathrm{D}(\mathbf{A})$, we would localize at the class of quasi-isomorphism of $\mathrm{K}(\mathbf{A})$. The category $\mathrm{D}(\mathbf{A})$ shall be the homotopic category of $\mathrm{C}(\mathbf{A})$ where a morphism $L^\bullet \to M^\bullet$ can be represented as a roof:

$$\begin{tikzcd} & N^\bullet \arrow[dl, "\text{quasi-iso}"'] \arrow[dr] & \\ L^\bullet & & M^\bullet \end{tikzcd}$$

(A word on how many abelian categories may induce the same derived categories. However, there is a special "triangulatedd structure", or t-structure which allows you to recover A)

2.5.2 Triangulated Categories

Homotopic categories are certainly almost never not abelian, for the kernels and cokernels of homotopic maps need not match (hence not being well-defined in the equivalence class). However, a derived functors $\mathrm{D}(\mathscr{F})$ always has a long exact sequence associated to it, which shows some weaker property these categories possess. These considerations are explored in this chapter. Take a short exact sequence of complexes:

$$0 \to L^\bullet \to M^\bullet \to N^\bullet \to 0$$

Then snakes lemma says that after taking homology, there is an interesting connection between $N^\bullet$ and $L^\bullet$. When working with the complex, there is no interesting connection between $N^\bullet$ and $L^\bullet$. We may connect these two complexes via the zero morphism: $N^\bullet \xrightarrow{0} L^\bullet$. From this, the short exact sequence of complexes may be interpreted as these three exact sequences:

$$N^\bullet \xrightarrow{0} L^\bullet \to M^\bullet$$
$$L^\bullet \to M^\bullet \to N^\bullet$$
$$M^\bullet \to N^\bullet \xrightarrow{0} L^\bullet$$

These three exact sequences are essentially can be combined if we fold any of them into the following triangle:

Definition 2.5.1: Exact Triangle

Let $L^\bullet, M^\bullet, N^\bullet \in \mathrm{Ch}(\mathbf{A})$ be three complexes. Then if there exists morphisms st the following diagram is exact (at each vertex):

$$\begin{tikzcd} & L^\bullet \arrow[dr] & \\ N^\bullet \arrow[ur] & & M^\bullet \arrow[ll] \end{tikzcd}$$

Then the diagram is called an *exact triangle* of $\mathrm{Ch}(\mathbf{A})$. This diagram is also sometimes denoted

$$L^\bullet \to M^\bullet \to N^\bullet \to$$

if it is easier to write it out flat.

Hence, the particular exact triangle given by folding any of the above exact sequences is:

$$\begin{tikzcd} & L^\bullet \arrow[dr] & \\ N^\bullet \arrow[ur, "+1"] & & M^\bullet \arrow[ll] \end{tikzcd}$$

where the $+1$ morphism represents the zero map. The $+1$ notation is to hint that this map will be a sort of shift from $N^\bullet$ to $L^\bullet[1]$. This can be stretched out into a single long sequence:

$$\cdots \to N^{i-1} \xrightarrow{0} L^i \to M^i \to N^i \xrightarrow{0} L^{i+1} \to \cdots$$

More explicitly, the triangle can be "unwrapped" to create the following nice diagram:

here, v good visual intuition

Applying the cohomology functor to the above long exact sequence and replacing the zero maps with the connecting morphism, we get:

$$\cdots \to H^{i-1}(N^\bullet) \xrightarrow{\delta^{i-1}} H^i(L^\bullet) \xrightarrow{0} H^i(M^\bullet) \xrightarrow{0} H^i(N^\bullet) \xrightarrow{\delta^i} H^{i+1}(L^\bullet) \to \cdots$$

Note that the maps δ^i did not come from the cohomology functor. In a sense, this is because $\mathrm{Ch}(\mathbf{A})$ is the wrong category to apply the cohomology functor, rather it is better to apply it in $\mathrm{D}(\mathbf{A})$. To make this connection, the notion of *distinguished triangles* would have to be developed in some level of detail. I do wish to do so since I'm interested in some of the connections they have, but for now I shall follow Aluffi's example and simply state the ingredients necessary on a shallow enough level to explain how the morphism δ would arise without needing any special considerations.

Let $\mathbf{A}$ be any Abelian category. We shall define the notion of a *distinguished triangle*. We first define a functor which is called the *translation functor*. Though it is called a translation functor, it shall be characterized by the axioms of a distinguished triangle. For the categories $\mathrm{K}(\mathbf{A})$ and $\mathrm{D}(\mathbf{A})$ this is the shift functor $L^\bullet \xrightarrow{+1} L^\bullet[1]$. Distinguished triangle are triangles that must satisfy 4 axioms:

1. The triangle:

$$\begin{tikzcd} & 0 \arrow[dr] & \\ A \arrow[ur, "+1"] & & A \arrow[ll, "\mathrm{id}_A"] \end{tikzcd}$$

   has to be part of the collection of distinguished triangles

2. every morphism $\alpha : A \to B$ must be the side of a distinguished triangle.

3. Triangles that are isomorphic (as triangle diagrams) to distinguished triangles must also be distinguished

4. Distinguished triangles must be able to rotate, namely if:

$$\begin{tikzcd} & A \arrow[dr, "\alpha"] & \\ C \arrow[ur, "\gamma^{+1}"] & & B \arrow[ll, "\beta"] \end{tikzcd}$$

   is a distinguished triangle if and only if

$$\begin{tikzcd} & A \arrow[dr, "\beta"] & \\ A[1] \arrow[ur, "-\alpha^{+1}"] & & B \arrow[ll, "\gamma"] \end{tikzcd}$$

   is a distinguished triangle (think of visual ref:HERE, the 3D view, to intuit this axiom)

5. Finally, distinguished triangles must satisfy an axiom called the *octahedral axiom* which will not be necessary to cover for the remarks this section endeavors to explicate.

A category that contains a collection of distinguished triangles is called a *triangulated category.* The category $\mathrm{K}(\mathbf{A})$ is an example of such a category where the translation functor is the shift functor and all distinguished triangles are isomorphic to:

$$\begin{tikzcd}
& L^\bullet \arrow[dr, "\alpha^\bullet"] & \\
MC(\alpha^\bullet) \arrow[ur, "+1"] & & M^\bullet \arrow[ll]
\end{tikzcd}$$

Note how this diagram also fulfills the 3rd axiom: any map $\alpha^\bullet : L^\bullet \to M^\bullet$ is the edge of the above distinguished triangle where the other morphisms are the natural ones. Note also how the axioms prevent $\mathrm{Ch}(\mathbf{A})$ from being a triangulated category. Since

$$\begin{tikzcd}
& 0 \arrow[dr] & \\
A \arrow[ur, "+1"] & & A \arrow[ll, "\mathrm{id}_A"]
\end{tikzcd}$$

needs to be a distinguished triangle, then since $L^\bullet$ is the mapping cone of the zero-morphism $0 \to L^\bullet$ the rotation would axiom would require

$$\begin{tikzcd}
& L^\bullet \arrow[dr, "\mathrm{id}_{L^\bullet}"] & \\
0 \arrow[ur, "+1"] & & L^\bullet \arrow[ll]
\end{tikzcd}$$

The mapping cone of the identity is not 0, and hence it cannot be a distinguished category. However, the mapping cone of the identity is *homotopically-equivalent* to 0, and hence the above triangle is indeed distinguished in $\mathrm{K}(\mathbf{A})$. Furthermore, a triangulated category, the cohomology functor will map distinguished triangles to exact triangles (in the case of $\mathrm{K}(\mathbf{A})$, this is because all distinguished triangles are isomorphic to the triangles with mapping cones), and hence give long exact sequences. Two notable examples of cohomological functors that map quasi-isomorphisms to isomorphisms for any triangulated category $\mathbf{A}$ are $\mathrm{Hom}(A, -)$ and $\mathrm{Hom}(=, A)$ for any object $A \in \mathbf{A}$.

The derived category $\mathrm{D}(\mathbf{A})$ is naturally a triangulated category, and the natural functor $\mathrm{K}(\mathbf{A}) \to \mathrm{D}(\mathbf{A})$ maps distinguished triangles to distinguished triangles and maps the translation functor to the translation functor. Derived categories are less restrictive than homotopic categories in the following important way: if

$$0 \to L^\bullet \xrightarrow{\alpha^\bullet} M^\bullet \xrightarrow{\beta^\bullet} N^\bullet \to 0$$

is a short exact sequence, then in general there is no distinguished triangle

$$\begin{tikzcd}
& L^\bullet \arrow[dr, "\alpha^\bullet"] & \\
N^\bullet \arrow[ur, "\gamma^\bullet\ \ +1"] & & M^\bullet \arrow[ll, "\beta^\bullet"]
\end{tikzcd}$$

In $\mathrm{K}(\mathbf{A})$ for any choice $\gamma^\bullet$ (even more is the case, $N^\bullet$ need not be homotopically equivalent to $MC(\alpha^\bullet)$). However, such a triangle *does* always exist in $\mathrm{D}(\mathbf{A})$), namely because there does exist a quasi-isomorphism $MC(\alpha)^\bullet \to N^\bullet$. Then applying the cohomology functor to this triangle shall give

$$\begin{tikzcd} & H^\bullet(L^\bullet) \arrow[dr] & \\ H^\bullet((N^\bullet) \arrow[ur, "+1"] & & H^\bullet((M^\bullet) \arrow[ll] \end{tikzcd}$$

where $+1$ is the connection homomorphism δ up to isomorphism. Furthermore, the 0 morphisms that connect the above in when applying the $H^\bullet(-)$ turned out to be particular choices in the more general case.

Finally, it should be noted that any functor $\mathscr{F} : \mathbf{A} \to \mathbf{B}$ between Abelian categories shall induce a functor $\mathrm{K}(\mathscr{F}) : \mathrm{K}(\mathbf{A}) \to \mathrm{K}(\mathbf{B})$, and this functor is in fact a functor of triangulated categories. Hence, the bounded equivalents, as well as $\mathrm{L}(\mathscr{F}), \mathrm{R}(\mathscr{F}) : \mathrm{D}^\pm(\mathbf{A}) \to \mathrm{D}^\pm(\mathbf{B})$ are functor of triangulated categories. From this perspective, the results proven about derived functors is a consequence of this fact.

2.5.3 Stable Module Theory

I really want to talk about $f, g : M \to N$ being "homotopic" if $h = f - g$ can factor through a projective module $h : M \to P \to N$; and how this is equivalent to the homotopy equivalence we talked about earlier. This brings back the intuition from the geometric setting were we didn't need a resolution of modules to have two modules be homotopy equivalent! This also naturally leads to naturally define stably equivalent, namely M, N are stably equivalent if there exists a P such that:

$$M \oplus P \cong N \oplus P$$

This is further developed in [11], but it is definitely worth mentioning it here since it is an excellent prelude!

3

Applications of (co)Homology

With cohomology theory more securely under our belt, it is time to use it! All the work in this chapter can be learnt without the need for any algebraic topology background, as it is the goal of the book to minimize the need for external material. However, the proceeding ideas are abstractions of ideas in algebraic topology, and ignoring the source of inspiration for these ideas is itself a disservice to the pedagogy of the material. Therefore, the section will be written in such a way that a background in homology/cohomology will be advantageous. Namely, it is best if the reader is familiar with simplicial and singular homology, how cohomology is derived from $\mathrm{Hom}(C_\bullet(X), G) = C^\bullet(X; G)$ and the relation of homology to cohomology (ex. the universal coefficient theorem for cohomology). The reader may choose to consult [9] or [8] for the details.

This chapter does *not* present an in-depth coverage of any of the material, in particular these subjects are a large source of modern development in mathematics (as of the early 21st century). The material presented is to demonstrate to the reader the power of cohomological theories so that the reader has an easier time when reading the more dense books containing this information. For a book in the EYNTKA series covering group and galois cohomology, see [6], and for K-theory see [11].

In this chapter we:

1. Shall use the above theory to find cohomological invariants of groups
2. Shall show some ways of interpreting Ext and Tor
3. Shall show that the notion of dimension from the geometric setting can be brought over to the algebraic!
4. Some K-theory will certainly interest the reader

We shall not do sheaf cohomology, as (naturally) it required sheaf theory that was not covered, and it is an exploration for EYNTKA Algebraic Geometry [1].

3.1 Group Cohomology

We start with no link to algebraic topology, and shall include the relation to singular cohomology before the examples.

Given a (not necessarily abelian) group G, what would it mean to extract cohomological information from it? A (co)chain would have to naturally be formed, and a functor must be found to derive. Groups themselves don't form an abelian category, but it was shown that representations, that is $k[G]$-modules, do form them. This is perhaps a first place in which we may think we can study groups, however this category is too nice (think of (co)homology with $\mathbb{Z}$ vs. $\mathbb{R}/\mathbb{C}$ coefficients). What is more useful is it's generalization:

Definition 3.1.1: G-Modules

Let G be an arbitrary group, and M an abelian group. Then M is a *G-module* if there exists a [left] group-action on M. Equivalently, if there exists a $\mathbb{Z}[G]$-module structure on M.

The reader is free to parse through the technicalities of the group action. All $k[G]$-modules presented in [4, Representation of Finite Groups] are $\mathbb{Z}[G]$-modules with the action restricted. An example of a G-module that is not a representation is the action of the group $\mathrm{Gal}_K(L)$ on the module L/K (a galois extension of L over K). All finite groups G have a non-trivial G-module given by the action of G on the group-ring $k[G]$ (note the addition is abelian). From a geometric perspective, modules represent sheaves (or in special cases vector bundles) over some schemes,

As we have seen many times at this point, the object acts on an abelian group can give lots of information on the object acting on it and the object being acted on. Given a G-module M, we can define the following:

Definition 3.1.2: G-Invariant subgroup

Let M be a G-module. Then the *G-invariant* set of M is the set:

$$M^G := \{m \in M \mid g \cdot m = m, \ \forall g \in G\}$$

which is the largest trivial G-submodule of M.

It is easy to see that $(-)^G$ is a functor. The following will help in computing cohomology

Corollary 3.1.3: Hom Representation of G-invariant

$$M^G \cong \mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}, M^G) \cong \mathrm{Hom}_G(\mathbb{Z}, M)$$

***Proof*:**
Give $\mathbb{Z}$ the trivial G-action, which is the action the statement requires.

For any abelian group N, evaluation at 1 is an isomorphism $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}, N) \to N$, with inverse sending x to the map $m \mapsto mx$. Taking $N = M^G$ gives the first isomorphism.

For the second, let $\varphi\colon \mathbb{Z} \to M$ be $\mathbb{Z}[G]$-linear and put $m = \varphi(1)$. Equivariance gives $\varphi(g \cdot 1) = g\varphi(1)$, and $g \cdot 1 = 1$ because the action on $\mathbb{Z}$ is trivial, so $m = gm$ for every $g \in G$, that is $m \in M^G$. Conversely, any $m \in M^G$ defines a $\mathbb{Z}[G]$-linear map $\mathbb{Z} \to M$, $k \mapsto km$, since $g(km) = k(gm) = km$. These assignments are mutually inverse and additive, so $\mathrm{Hom}_G(\mathbb{Z}, M) \cong M^G$, completing the proof.

Similarly, we can define:

Definition 3.1.4: G-Coinvariant subgroup

Let M be a G-module. Then the *G-coinvariant* of M is the set:

$$M_G := \frac{M}{\langle (gm - m) \mid g \in G, m \in M \rangle}$$

and is the maximal invariant quotient module with the trivial action.

Just like $(-)^G$, $(-)_G$ is also a functor. The reader should further verify that

Proposition 3.1.5: Tensor Representation of Coinvariant Module

$$M_G \cong_G \mathbb{Z} \otimes_{\mathbb{Z}[G]} M$$

Proposition 3.1.6: Invariants Is Left-exact

The functor $(-)^G$ is left-exact, the functor $(-)_G$ is right exact.

***Proof*:**
First, prove that $(-)^G$ is right-adjoint to the trivial-action functor.

$$\mathrm{Hom}(T(M), N) \cong \mathrm{Hom}(M, N^G)$$

And similarly for $(-)_G$; it is left-adjoint to the trivial-action functor. As $T(-)$ is an exact functor, $(-)^G$ and $(-)_G$ are left-exact and right-exact respectively, completing the proof.

Both of these functors are not exact:

Example 3.1.7: Lack Of Exactness's
Let $G = C_2$ be the cyclic group of order 2 and:

$$0 \to I_G \to \mathbb{Z}[G] \xrightarrow{\epsilon} \mathbb{Z} \to 0$$

where $\epsilon(1 \cdot a + \sigma \cdot b) = a + b$ and I_G is the kernel. This sequence is exact. Take $(-)^G$:

$$0 \to (I_G)^G \to (\mathbb{Z}[G])^G \xrightarrow{\epsilon^G} (\mathbb{Z})^G \to 0$$

i.e.:

$$0 \to 0 \to \Delta \xrightarrow{\epsilon^G} \mathbb{Z} \to 0$$

where $\Delta = \{(a,a) \mid a \in \mathbb{Z}\}$. Then ϵ^G is no longer surjective as $\epsilon^G(a,a) = 2a$. The above example can also be used to show $(-)_G$ is not left-exact. More generally, if $0 \to M_2^G \to M_2 \to M_3 \to 0$ is exact, taking $(-)^G$ doesn't change $M - 2^G$, but now we get an isomorphism between the 1st and second terms, so it cannot be surjective.

More generally, let G act trivially on a coset $[m]$ of a quotient M/L. Can this action always be lifted so that G acts trivially on a representative m?

Thus, it is fruitful to take the ith right-derived functor of $(-)^G$ (similarly the left-derived functors of $(-)_G$):

$$\mathbf{R}^i(-)^G := H^i(G,-)$$

which would give us the long exact sequence of abelian groups:

$$0 \to L^G \to M^G \to N^G \to H^1(G,L) \to H^1(G,M) \to \cdots$$

Since $M^G = \mathrm{Hom}_G(\mathbb{Z}, M)$ by taking an injective resolution of M, or a projective resolution of $\mathbb{Z}$, we can write:

$$H^i(G,M) = \mathrm{Ext}^i_{\mathbb{Z}[G]}(\mathbb{Z}, M)$$

Note that $H^0(G,M) = M^G$, hence the higher order cohomology groups represent some failure for points to be fixed. Let us compute some examples to show the power of the machinery developed in chapter 2 before concretely demonstrating how the cohomology groups represent the obstruction of points being fixed.

Example 3.1.8: Group Cohomologies – Homological Algebra Computations

1. The trivial group has trivial cohomology
2. (Integers) Let $G = \mathbb{Z}$. Then $\mathbb{Z}[G] \cong \mathbb{Z}[x, x^{-1}]$. Then since $\mathbb{Z}[x,x^{-1}]/(1-x) \cong_G \mathbb{Z}$ with the trivial action, define the following free (hence projective) resolution:

   $$\cdots \to 0 \to \mathbb{Z}[x,x^{-1}] \xrightarrow{\cdot(1-x)} \mathbb{Z}[x,x^{-1}] \to 0 \to \cdots$$

   Applying the (contravariant) $\mathrm{Hom}_{\mathbb{Z}[x,x^{-1}]}(-,M)$, we can compute the cohomology of

   $$\cdots \to 0 \to M \xrightarrow{\cdot(1-x)} M \to 0 \to \cdots$$

   Then we can conclude by direct computation:

   $$H^0(\mathbb{Z}, M) \cong \ker(M \xrightarrow{\cdot(1-x)} M) = M^{\mathbb{Z}}$$

   and:

   $$H^1(\mathbb{Z}, M) = \frac{M}{(1-x)M}$$

   and furthermore $H^i(\mathbb{Z}, M) = 0$ for $i \notin \{0,1\}$
3. Let $G = C_n$ be a finite cyclic group of over n with generator σ. Let us form a free resolution of $\mathbb{Z}[G]$ by taking advantage of the fact that $\sigma - 1$ and $N = \sum_{i=0}^{n-1} \sigma^i$ are zero divisors:

   $$N \cdot (\sigma - 1) = (1 + \sigma + \cdots + \sigma^{n-1})(\sigma - 1) = \sigma^n - 1 = 1 - 1 = 0$$

This gives a resolution:

$$\cdots \xrightarrow{\sigma-1} \mathbb{Z}[G] \xrightarrow{N} \mathbb{Z}[G] \xrightarrow{\sigma-1} \mathbb{Z}[G] \xrightarrow{\epsilon}\!\!\to \mathbb{Z} \to 0$$

where

$$\epsilon\left(\sum_g a_g g\right) = \sum_g a_g$$

is the usual augmentation map. Then:

$$H^k(C_n, M) = \begin{cases} \frac{M^G}{NM} & k \text{ even, } k \geq 2 \\ \frac{{}_N M}{(\sigma-1)M} & k \text{ odd, } k \geq 1 \end{cases} \qquad \text{where} \qquad {}_N M = \{m \in M \mid N \cdot m = 0\}$$

For example, if $M = \mathbb{Z}$, then:

$$H^k(C_n, \mathbb{Z}) = \begin{cases} \mathbb{Z}/m\mathbb{Z} & k \text{ even, } k \geq 2 \\ 0 & \text{otherwise} \end{cases}$$

4. Let $G = *_{s\in S}\mathbb{Z}$ be the free product of S copies of $\mathbb{Z}$. This has a simple free resolution, namely take:

$$\epsilon : \mathbb{Z}[G] \to \mathbb{Z}$$

where the kernel is:

$$I_S = \sum_{s\in S} \mathbb{Z}[G] \cdot (s-1)$$

It should be verified that I_S is a free G-module, hence we have a free-resoluion:

$$0 \to I_S \to \mathbb{Z}[G] \xrightarrow{\epsilon}\!\!\to \mathbb{Z} \to 0$$

and cohomology:

$$H^k(*_{s\in S}\mathbb{Z}, \mathbb{Z}^G) = \begin{cases} \mathbb{Z} & k = 0 \\ \bigoplus_{s\in S} \mathbb{Z} & k = 1 \\ 0 & k \geq 2 \end{cases}$$

5. (If you know some algebraic geometry) *Galois Cohomology*: Take the category of varieties W over k such that when extending scalars to $\overline{k}$ we have $V/\overline{k} \cong W/\overline{k}$. Let's label it $\mathbf{Var}_{k\to\overline{k}}$, and say that the objects are:

$$\mathrm{Obj}(\mathbf{Var}_{k\to\overline{k}}) = \left\{W/k \;\middle|\; V/\overline{k} \cong W/\overline{k}\right\}$$

and the morphism are regular morphisms. Then I claim that:

$$\mathbf{Var}_{k\to\overline{k}} \cong H^1(\mathrm{Gal}_k(\overline{k}), \mathrm{Aut}(V/\overline{k}))$$

To see this, first note that an element $\sigma \in \mathrm{Gal}_k(\overline{k})$ acts on a $\overline{k}$-variety $W/\overline{k} = W_k \times_{\mathrm{Spec}\,(k)} \mathrm{Spec}\,(\overline{k})$ via:

$$\mathrm{id}_{W_k} \times \mathrm{Spec}\,(\sigma^{-1}) : W/\overline{k} \to W/\overline{k}$$

Or, for those less comfortable with fibered produced from algebraic geometry, they can think of this map as the "equivalent":

$$\mathrm{id}_{W_k} \otimes_k \sigma^{-1} : k[W] \otimes_k \overline{k}$$

With the action defined, let us look at the correspondence. Take $W_k \in \mathbf{Var}_{k\to\overline{k}}$ so that $\varphi : W_{\overline{k}} \xrightarrow{\sim} V_{\overline{k}}$ is an isomorphism. Then notice that this isomorphism induces a map $\varphi \circ \sigma^{-1} \circ \varphi^{-1}$, which is an automorphism on $W_{\overline{k}}$. The collection of all these maps as σ varies can uniquely define W_k (k is the fixed subfield over all σ). This is true up to a choice of $\mathrm{Aut}_k(V)$, and so we quotient out by it. Then we can see this creates the correspondence between these two categories, and shows how far away the category $\mathbf{Var}_{k\to\overline{k}}$ is from being invariant to the group-action of the galois group.

6. (if you know elliptic curves) An excellent example of where galois cohomology is immediately useful is in the theory of elliptic curves. On a high level, it is relatively easy to find the solution of an elliptic curve over $\mathbb{C}$, but it becomes much more difficult to solve it over finite extensions of $\mathbb{Q}$, or $\mathbb{Q}$ itself. If the reader has read (or is reading) [5], then the have encountered the short exact sequence of $\mathrm{Gal}_K(\overline{K})$-modules:

$$0 \to E[m] \to E(\overline{K}) \xrightarrow{[m]} E(\overline{K}) \to 0$$

where $E[m]$ are the m-torsion points of an elliptic curve ($m \geq 2$), K is a number field, $E(\overline{K})$ are the points of the elliptic curve over $\overline{K}$. Let $G = \mathrm{Gal}_K(\overline{K})$. Then when considering solutions over K, the short exact sequence sequence becomes a long exact sequences:

$$0 \to E(K)[m] \to E(K) \xrightarrow{[m]} E(K) \xrightarrow{\delta} H^1(G, E[m])$$
$$\to H^1(G, E(\overline{K})) \xrightarrow{[m]} H^1(G, E(\overline{K}) \to \cdots$$

This is the foundation on which the torsion-free points of E/K will be computed!

7. If the reader knows about the classifying space BG, then:

$$H^n(BG, \mathbb{Z}) \cong H^n(G, \mathbb{Z})$$

where he left hand side is singular cohomology and the right hand side is group cohomology.

What obstruction information does this capture? Since there is a failure in surjectivity $\varphi : M \twoheadrightarrow N$ when applying $(-)^G$, this means that a fixed element $n \in N^G$ does *not* necessarily come form a fixed element $m \in M^G$. Thus, even though $\varphi(m) = n$ exists, it need not be the case that there exists an $m' \in M^G$ such that $\varphi^G(m') = n$. To measure this failure, take the short exact sequence

$$0 \to L \xrightarrow{\iota} M \xrightarrow{\varphi} N \to 0$$

and let $n \in N^G$. As $N^G \subseteq N$, there exists an $m \in M$ such that $\varphi(m) = n$. Consider any $g \in G$ and take $g \cdot m - m$. Then applying the G-homomorphism:

$$\varphi(g \cdot m - m) = g \cdot \varphi(m) - \varphi(m) = g \cdot n - n \overset{!}{=} n - n = 0$$

where $\overset{!}{=}$ comes from $n \in N^G$. By exactness $\ker \varphi = \mathrm{im} \iota$, so there must exist an element $\lambda_m(g) \in L$ such that

$$\iota(\lambda_m(g)) = g \cdot m - m$$

Now, $n \in N^G$ can be lifted if and only if $g \cdot m - m = 0$ for all $g \in G$, since then $m \in M^G$. In terms of elements of L:

$$\forall g \in G,\ \iota(\lambda_m(g)) = 0$$

by injectivity, it must be that $\lambda_m(g) = 0$ for all $g \in G$. Thus, the function $\lambda_m : G \to L$ is non-trivial if the lift does not exist. Let us explore the properties of λ_m. First, if $g_1, g_2 \in G$ then:

$$\begin{aligned}
\iota(\lambda_m(g_1 g_2)) &= g_1 g_2 \cdot m - m \\
&= g_1 g_2 \cdot m - g_1 \cdot m + g_1 \cdot m - m \\
&= g_1 \cdot (g_2 \cdot m - m) + (g_1 \cdot m - m) \\
&= g_1 \iota(\lambda_m(g_2)) + \iota(\lambda_m(g_1)) \\
&= \iota(g_1 \lambda_m(g_2)) + \iota(\lambda_m(g_1)) \\
&= \iota(g_1 \lambda_m(g_2) + \lambda_m(g_1))
\end{aligned}$$

Thus, as f is injective (more specifically a monomorphism) we get:

$$\lambda_m(g_1 g_2) = \lambda_m(g_1) + g_1 \lambda_m(g_2)$$

A function with this property is given a name:

Definition 3.1.9: 1-Cocycle

A function $f : G \to M$ is a called a *1-cocycle*, *1-derivation*, or *crossed homomorphism* if:

$$f(gh) = f(g) + gf(h)$$

The set of all derivations is denoted $Z^1(G, M)$. If it is clear from context, the "1-" is dropped.

Now, the above λ_m was dependent on a choice of m. But $M \hookrightarrow N$ is surjective, and hence there may be another m' such that $\varphi(m') = n$. As φ is a G-homomorphism

$$\varphi(m - m') = c - c = 0$$

and so $m - m' \in \ker \varphi = \operatorname{im} \iota$. Thus, there is an $\ell \in L$ such that

$$m - m' = \iota(\ell) \qquad \text{or} \qquad m' = m + \iota(\ell)$$

Thus, let us relate $\lambda_{m'}$ to λ_m:

$$\begin{aligned}
\iota(\lambda_{m'}(g)) &= g \cdot m' - m' \\
&= g \cdot (m + \iota(\ell)) - m - \iota(\ell) \\
&= g \cdot m + g \cdot \iota(\ell)) - m - \iota(\ell) \\
&= (g \cdot m - m) + g \cdot \iota(\ell) - \iota(\ell) \\
&= \iota(\lambda_m(g)) + \iota(g \cdot \ell) - \iota(\ell) \\
&= \iota(\lambda_m(g) + g \cdot \ell - \iota(\ell))
\end{aligned}$$

Thus, as ι is a monomorphism:

$$\lambda_{m'}(g) = \lambda_m(g) + g \cdot \ell - \ell$$

Thus, the term $g \cdot \ell - \ell$ can be seen as the "trivial" part of the obstruction, as it does not change the difficulty in finding an obstruction. This is given a name:

Definition 3.1.10: 1-Coboundary

A function $f_m : G \to M$ is a called a *1-coboundary* or *principal derivation* if:

$$f_m(g) = g \cdot m - m$$

The set of all boundary maps is denoted $B^1(G, M)$.If it is clear from context, the "1-" is dropped.

Note that a coboundary is a cocycle:

$$f_m(gh) = gh \cdot m - m = gh \cdot m - h \cdot m + h \cdot m - m = f_m(g) + g f_m(h)$$

Hence $B^1(G, M) \subseteq Z^1(G, M)$. It shall be shown in a moment that:

$$H^1(G, M) \cong \frac{Z^1(G, M)}{B^1(G, M)}$$

First, let us show there is a map $L^G \to H^1(G, M)$. Indeed there is a well-defined G-module homomorphism:

$$\ell \mapsto [g \mapsto g \cdot m - m]$$

where m is in the fiber of ℓ.

Let us next motivate the 2nd order cohomology group. Let us assume the cocycle-representation and the derived functor representation are isomorphic for the next part. Let us start with the long exact sequence we know is given:

$$0 \to L^G \xrightarrow{\alpha} M^G \xrightarrow{\beta} N^G \xrightarrow{\delta^0} H^1(G, L) \xrightarrow{\alpha_*} H^1(G, M) \xrightarrow{\beta_*} H^1(G, N) \xrightarrow{\delta^1} H^2(G, L) \cdots$$

Consider $[\nu] \in H^1(G, N)$; let us call it a 1-obstruction. Then can it be lifted some $[\mu] \in H^1(G, M)$, namely does there exist $[\mu]$ such that $\beta_*([\mu]) = [\nu]$. To find out, first take some representative cocycle $\nu : G \to N$ (notice we are now working with a map, not an element). As $M \hookrightarrow N$ is surjective, we have a diagram:

$$\begin{tikzcd} & M \arrow[d, two heads, "\beta"] \\ G \arrow[ur, dashed, "\mu?"] \arrow[r, "\nu"'] & N \end{tikzcd}$$

We can certainly lift each point *locally*, namely given a point $\nu(g) = n \in N$ there is a point (or points) m such that $\beta(m) = n$[1]. But this lifting can fail *globally*, namely such a lift is not guaranteed to be a 1-cocycle. This mirrors the general pattern in cohomology where having local data of a function does not imply the existence of a global function that restricts to all the local data. Let us measure this failure of lifting. If the lifted μ is a cocycle, it would satisfy:

$$\mu(g_1 g_2) - \mu(g_1) - g_1 \mu(g_2) = 0$$

Thus, define the function:

$$\psi : G \times G \to M, \qquad \psi(g_1, g_2) = \mu(g_1 g_2) - \mu(g_1) - g_1 \mu(g_2)$$

[1] Technically, the axiom of choice is invoked here for making a potentially uncountable number of choices

Let us show that for any $(g_1, g_2) \in G \times G$, $\psi(g_1, g_2) \in \ker \beta$. Indeed:

$$\begin{aligned}
\beta(\psi(g_1, g_2)) &= \beta(\mu(g_1 g_2) - \mu(g_1) - g_1\mu(g_2)) \\
&= \beta(\mu(g_1 g_2)) - \beta(\mu(g_1)) - \beta(g_1\mu(g_2)) \\
&\overset{!}{=} \nu(g_1 g_2)) - \nu(g_1) - g_1\nu(g_2)) \\
&= 0 && \nu \text{ is a cocycle}
\end{aligned}$$

where $\overset{!}{=}$ comes from μ being defined as a lift of ν and hence $\beta \circ \mu = \nu$. Thus, by exactness $\ker \beta = \mathrm{im}\alpha$ and so for each pair $(g_1, g_2) \in G \times G$, there exists a unique element in L that maps to $\psi(g_1, g_2)$, namely there exists a function $\Omega : G \times G \to L$ such that

$$\alpha(\Omega(g_1, g_2)) = \psi(g_1, g_2)$$

This Ω would now be our new 2-cocycle and are denoted $Z^2(G, M)$. Doing a similar but longer computations such as the one for the 1-cycle, we can find that:

$$g_1\Omega(g_2, g_3) - \Omega(g_1 g_2, g_3) + \Omega(g_1, g_2 g_3) - \Omega(g_1, g_2) = 0$$

Similarly, an explicit computation would fine that 2-boundaries (given a 1-chain ψ) are of the form

$$\Omega(g_1, g_2) = g_1 \cdot \psi(g_2) - \psi(g_1 g_2) + \psi(g_1)$$

These are label $B^2(G, M)$ and:

$$H^2(G, M) \cong \frac{Z^2(G, M)}{B^2(G, M)}$$

The same process continues for higher order cohomology. For the reader who knows some algebraic topology, the following informal discussion grounds the higher order group-cohomology as the singular (or simplicial) cohomology of BG, the classifying space of G with the property that $\pi_1(BG) = G$ and $\pi_n(BG) = \{0\}$ for all $n > 1$ (see [9]). In particular:

$$H^n_{\mathrm{Sing}}(BG, M) \cong H^n(G, M)$$

Note that if M has a non-trivial G-module action, it corresponds to cohomology with local coefficients, but the geometric picture remains the same (the details are left for [6]). Using the nerve or bar construction on BG and considering the usual covering $EG \to BG$ we get the following correspondence:

Geometric Object in EG	Correspondence in BG	Algebraic equivalent
0-Simplices (Vertices)	A single 0-simplex, base point $*$.	elements of G
1-Simplices (Edges)	Directed edge for each element $g \in G$. An arrow from $*$ to $*$ labeled by g.	Functions on G
2-Simplices (Triangles)	Oriented triangle for each pair $(g_1, g_2) \in G \times G$. Edges are the 1-simplices for g_1, g_2, and the product $g_1 g_2$.	Functions on $G \times G$
3-Simplices (Tetrahedral)	Oriented tetrahedron for each triple $(g_1, g_2, g_3) \in G \times G \times G$. Faces are the triangles for (g_2, g_3), $(g_1, g_2 g_3)$, etc.	Functions on $G \times G \times G$

n-Simplices	n-tuple $(g_1, \ldots, g_n) \in G^n$.	Functions on G^n

Where "functions" here means set-function (to take into account the "failed" lifting), and the transition maps d_i captures the G-module homomorphism information. Geometrically, the G-module homomorphisms correspond to G-equivariant maps $f : G^n \to M$.

Using this intuition, we can construct a standard way of defining a projective resolution of $\mathbb{Z}$ by G-modules. First, let $\mathbb{Z}[G^n]$ be a G-module with the diagonal action given by multiplying on the left, or more specifically:

$$g \cdot (g_1, ..., g_n) = (gg_1, ..., gg_n)$$

Note that this diagonal map corresponds to the natural action of left multiplication by g on EG. These modules are free G-modules; clearly $\mathbb{Z}[G]$ is a $\mathbb{Z}$-module with basis given by the elements G. Then for $n \geq 0$:

$$\mathbb{Z}[G^{n+1}] \cong \bigoplus_{(g_1,...,g_n)\in G^n} \mathbb{Z}[G] \cdot (1, g_1, ..., g_n)$$

which shows that these are free. From this, we can consider the following projective (even free) resolution:

Definition 3.1.11: Standard Resolution

Let G be a group. Then the *standard resolution* of $\mathbb{Z}$ by G-modules is the exact sequence of G-module homomorphism:

$$\cdots \to \mathbb{Z}[G^{n+1}] \xrightarrow{d_n} \mathbb{Z}[G^n] \xrightarrow{d_{n-1}} \cdots \xrightarrow{d_1} \mathbb{Z}[G] \xrightarrow{\epsilon} \mathbb{Z} \to 0$$

where $\mathbb{Z}[G^{n+1}]$ carries the diagonal G-action and, on the basis G^{n+1},

$$d_n(g_0, ..., g_n) = \sum_{i=0}^{n} (-1)^i (g_0, ..., \hat{g}_i, ..., g_n)$$

the term $\hat{g}_i$ being omitted from the tuple, while $\epsilon(g) = 1$ for every $g \in G$. The map ϵ is also called the *augmentation map*.

Note that each $\mathbb{Z}[G^{n+1}]$ is a free $\mathbb{Z}[G]$-module, the diagonal action permuting a basis freely. We claimed this is an exact sequence, however this ought to be verified.

Lemma 3.1.12: Standard Resolution is Exact

The standard resolution of $\mathbb{Z}$ by G-modules is exact

This shows that the nomenclature of cocycles and coboundaries above is used correctly.

***Proof*:**
Exactness is a statement about the underlying abelian groups, so it is enough to produce a contracting homotopy by $\mathbb{Z}$-linear maps; these need not be G-equivariant, and indeed the ones below are not.

Step 1: The maps. Write $e \in G$ for the identity element and define

$$s_{-1}\colon \mathbb{Z} \to \mathbb{Z}[G], \quad 1 \mapsto (e), \qquad s_n\colon \mathbb{Z}[G^{n+1}] \to \mathbb{Z}[G^{n+2}], \quad (g_0,\dots,g_n) \mapsto (e, g_0, \dots, g_n)$$

extended $\mathbb{Z}$-linearly.

Step 2: The contracting identity in positive degrees. Fix $n \geq 1$ and compute $d_{n+1}s_n(g_0,\dots,g_n) = d_{n+1}(e, g_0, \dots, g_n)$. Omitting the entry in position 0 gives $(g_0,\dots,g_n)$ with sign $+1$; omitting the entry in position $j \geq 1$, which is g_{j-1}, gives $(e, g_0, \dots, \hat{g}_{j-1}, \dots, g_n)$ with sign $(-1)^j$. Reindexing by $i = j-1$ and using $(-1)^{i+1} = -(-1)^i$,

$$d_{n+1}s_n(g_0,\dots,g_n) = (g_0,\dots,g_n) - \sum_{i=0}^{n} (-1)^i (e, g_0, \dots, \hat{g}_i, \dots, g_n)$$

The subtracted sum is exactly $s_{n-1}d_n(g_0,\dots,g_n)$, since s_{n-1} prepends e to each term of $d_n(g_0,\dots,g_n) = \sum_i (-1)^i (g_0,\dots,\hat{g}_i,\dots,g_n)$. Hence

$$d_{n+1}s_n + s_{n-1}d_n = \mathrm{id}$$

Step 3: The bottom of the complex. At the augmentation, $\epsilon s_{-1}(1) = \epsilon(e) = 1$, so $\epsilon s_{-1} = \mathrm{id}_{\mathbb{Z}}$. In degree zero,

$$d_1 s_0(g_0) = d_1(e, g_0) = (g_0) - (e), \qquad s_{-1}\epsilon(g_0) = s_{-1}(1) = (e)$$

so $d_1 s_0 + s_{-1}\epsilon = \mathrm{id}$ as well.

Step 4: Conclusion. The augmented complex therefore has its identity morphism null-homotopic as a complex of abelian groups. By Step 1 of the proof of lemma 2.1.23, a complex whose identity is null-homotopic is exact. Hence the standard resolution is exact, completing the proof.

Hence, after taking the contravariant functor $\mathrm{Hom}_{\mathbb{Z}[G]}(-, M)$, we get the cochain complex:

$$0 \to \mathrm{Hom}_{\mathbb{Z}[G]}(\mathbb{Z}[G], M) \to \cdots \to \mathrm{Hom}_{\mathbb{Z}[G^n]}(-, M) \xrightarrow{d_n^*} \mathrm{Hom}_{\mathbb{Z}[G^{n+1}]}(-, M) \to \cdots$$

where d_n^* maps $\varphi \mapsto \varphi \circ d_n$. Similarly, we can take the covariant functor

$$\cdots \to \mathbb{Z}[G^3] \otimes_{\mathbb{Z}[G]} A \xrightarrow{d_{2*}} \mathbb{Z}[G^2] \otimes_{\mathbb{Z}[G]} A \xrightarrow{d_{1*}} \mathbb{Z}[G] \otimes_{\mathbb{Z}[G]} A \to 0$$

where d_{n*} is given by $(g_0, ..., g_n) \otimes a \mapsto d_n(g_0, ..., g_n) \otimes a$, the map ϵ is the augmentation map given by $\sum a_g g \mapsto \sum a_g$, and we view the modules as *right* $\mathbb{Z}[G]$-modules with the action

$$(g_1, ..., g_n) \cdot g = (g_1 g, ..., g_n g)$$

Let us now better understand these hom-sets and relate to the above discussion. Let

Definition 3.1.13: coChain

Let
$$C^n(G,M) = (\mathrm{Hom}_{\mathbf{Set}}(G^n, M), +)$$
with group operation defined on the maps given by point-wise addition. This group is called the *n-cochain group*.

From this, we can define a boundary map:

$$\begin{aligned} d^n : C^n(G,M) &\to C^{n+1}(G,M) \\ d^n(f)(g_0,...,g_n) &= g_0 f(g_1,...,g_n) - f(g_0g_1, g_2, ..., g_n) \\ &\quad + f(g_0, g_1g_2, ..., g_n) - \cdots + (-1)^n f(g_0, ..., g_{n-2}, g_{n-1}g_n) + (-1)^{n+1} f(g_0, ..., g_{n-1}) \end{aligned}$$

Then $d^{n+1} \circ d^n = 0$, giving us a cochain complex. We can then have:

Definition 3.1.14: Cohomology Group – using Cocycles and Coboundaries

$$H^n(G,M) = \frac{\ker d^n}{\mathrm{im} d^{n-1}} = \frac{Z^n(G,M)}{B^n(G,M)}$$

We sum up many of the results that we've shown before using the new notation:

1. $C^0(G,M) = M$, and $H^0(G,M) \cong M^G$
2. $d^0 : C^0(G,M) \to C^1(G,M)$ is given by $d^0(a)(g) = ga - a$
3. $H^0(G,M) \cong \ker d^0 = M^G$
4. $\mathrm{im} d^0 = \{f : G \to M \mid \exists a \in M \text{ such that } f(g) = ga - a, \ \forall g \in G\}$ (principal crossed homomorphism)
5. $d^1 : C^1(G,M) \to C^2(G,M)$ given by $d^1(f)(g,h) = gf(h) - f(gh) + f(g)$
6. $\ker d^1 = \{f : G \to M \mid f(gh) = f(g) + gf(h)\}$ (crossed homomorphisms)
7. If $0 \to A \to B \to C \to 0$ is exact, we get an exact sequence of cochains for every n:
$$0 \to C^n(G,A) \to C^n(G,B) \to C^n(G,C) \to 0$$

We shall show that the collection of complexes given by the above is isomorphic to those given by the standard resolution

Proposition 3.1.15: Characterizing Hom of Resolution

Let M be a G-module. Then for every $n \geq 0$ there is an isomorphism:

$$\Phi^n : \mathrm{Hom}_{\mathbb{Z}[G]}(\mathbb{Z}[G^{n+1}], M) \xrightarrow{\cong} C^n(G, M)$$

where the map $\varphi : \mathbb{Z}[G^{n+1}] \to M$ goes to $f : G^n \to M$ given by:

$$f(g_1, ..., g_n) = \varphi(1, g_1, g_1g_2, g_1g_2g_3, ..., g_1g_2g_3 \cdots g_n)$$

The isomorphism Φ^n commutes with boundary maps, that is

$$\Phi^{n+1} \circ d^*_{n+1} = d^n \circ \Phi^n$$

Hence the two cochain complexes agree: the one built from the standard resolution and the one built from cocycles and coboundaries are isomorphic, so the two descriptions of $H^n(G, M)$ coincide.

***Proof*:**
Step 1: Φ^n is a bijection. The group G acts diagonally on the basis G^{n+1} of $\mathbb{Z}[G^{n+1}]$, and this action is free, since $g \cdot (h_0, \ldots, h_n) = (h_0, \ldots, h_n)$ forces $gh_0 = h_0$ and hence $g = e$. Each orbit contains exactly one tuple whose first entry is e, namely $h_0^{-1} \cdot (h_0, \ldots, h_n)$. A $\mathbb{Z}[G]$-linear map out of $\mathbb{Z}[G^{n+1}]$ is therefore determined freely and without constraint by its values on the tuples $(e, k_1, \ldots, k_n)$, so restriction to those tuples is a bijection onto $\mathrm{Hom}_{\mathbf{Set}}(G^n, M) = C^n(G, M)$.

The substitution in the statement is the reparametrization $k_i = g_1g_2 \cdots g_i$ of that indexing. It is a bijection $G^n \to G^n$, with inverse $g_1 = k_1$ and $g_i = k_{i-1}^{-1}k_i$ for $i \geq 2$. Composing the two bijections gives Φ^n, and it is additive because both steps are, so Φ^n is an isomorphism of abelian groups.

Step 2: Compatibility with the differentials. Let $\varphi \in \mathrm{Hom}_{\mathbb{Z}[G]}(\mathbb{Z}[G^{n+1}], M)$ and set $f = \Phi^n(\varphi)$. Write $h_i = g_1 \cdots g_i$, so that $f(g_1, \ldots, g_n) = \varphi(e, h_1, \ldots, h_n)$. Applying Φ^{n+1} to $d^*_{n+1}\varphi = \varphi \circ d_{n+1}$ and evaluating at $(g_0, \ldots, g_n)$ gives $\varphi\big(d_{n+1}(e, k_1, \ldots, k_{n+1})\big)$ where $k_i = g_0g_1 \cdots g_{i-1}$.

Expand d_{n+1} by definition 3.1.11. The term omitting position 0 is $(k_1, \ldots, k_{n+1})$; pulling out the common left factor $k_1 = g_0$ by equivariance of φ turns it into $g_0 \cdot \varphi(e, g_0^{-1}k_2, \ldots) = g_0f(g_1, \ldots, g_n)$. The term omitting position j for $1 \leq j \leq n$ merges g_{j-1} and g_j, contributing $(-1)^j f(g_0, \ldots, g_{j-1}g_j, \ldots, g_n)$, and the term omitting the last position drops g_n, contributing $(-1)^{n+1}f(g_0, \ldots, g_{n-1})$. Summing,

$$\Phi^{n+1}(d^*_{n+1}\varphi)(g_0, \ldots, g_n) = g_0f(g_1, \ldots, g_n) + \sum_{j=1}^{n}(-1)^j f(g_0, \ldots, g_{j-1}g_j, \ldots, g_n) + (-1)^{n+1}f(g_0, \ldots, g_{n-1})$$

which is precisely $d^n(f)(g_0, \ldots, g_n)$ as defined above. Hence $\Phi^{n+1} \circ d^*_{n+1} = d^n \circ \Phi^n$.

The maps $\Phi^\bullet$ are therefore isomorphisms commuting with the differentials, that is an isomorphism of cochain complexes, completing the proof.

Due to the relevance of the standard representation, we have the following definition:

Definition 3.1.16: Induced and Coinduced G-module

Let G be a group and A an abelian group. Then:

$$\mathrm{CoInd}^G(A) = \mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}[G], A)$$

with the G-action defined as $(g\varphi)(z) = \varphi(zg)$ is called the *coinduced G-module* associated to A. Similarly

$$\mathrm{Ind}^G(A) = \mathbb{Z}[G] \otimes_{\mathbb{Z}} A$$

with G-action defined by $g(z \otimes a) = (gz) \otimes a$ is the is the induced G-module associated to A.

A result that is common in many Cohomologies theories is some version of the following:

Lemma 3.1.17: Induced and Coinduced Equivalence

Let G be a finite group and A an abelian group. Then

$$\mathrm{Ind}^G(A) \cong \mathrm{CoInd}^G(A)$$

As a consequence, if $B = \mathrm{Ind}^G(A)$ or $B = \mathrm{CoInd}^G(A)$, then

$$H_0(G, B) \cong H^0(G, B) \cong A$$

and they are both zero for $n \geq 1$.

***Proof*:**
Define the map: $\alpha : \mathrm{Coind}^G(A) \to \mathrm{Ind}(G)(A)$ given by:

$$\varphi \mapsto \sum_{g \in G} g^{-1} \otimes \varphi(g)$$
$$(g^{-1} \mapsto a) \leftarrow\!\shortmid g \otimes a$$

with $(g^{-1} \mapsto a)(h) = 0$ for $h \in G \setminus \{g^{-1}\}$. These are certainly inverses, and easily verifiable to be G-module homomorphisms (this is a similar technique used in Maschkle's Theorem (see [4, Maschke's Theorem]).

The last part is left for exercise ref:HERE.

Proposition 3.1.18: Torsion of H^1

if G has order n and is finite, then every element of $H^1(G, M)$ has order divisible by n.

***Proof*:**
Let $\delta \in Z^1$ so that $\delta(gh) = \delta(g) + g\delta(h)$ for all $g \in G$. Sum both sides over $h \in G$ keeping g

fixed. Then as multiplying by g is translation:

$$\sum_{h\in G}\delta(gh)=\sum_{h\in G}\delta(h)=S$$

As $\delta(g)$ is a constant, we can re-arrange and get:

$$(1-g)S=|G|\delta(g)$$

Now, re-write it suggestively as:

$$g(-S)-(-S)=|G|\delta(g)$$

But now the left hand side is a coboundary, and hence the equation is 0 in $H^1(G,M)$ showing that the order of the group zero's any cocycle, completing the proof.

Proposition 3.1.19: M finitely generated, then H1 finite

Let M be a finitely generated G-module and G finite. Then $H^1(G,M)$ is finite.

***Proof*:**
Let $\langle m_1,...,m_r\rangle = M$. Then C^1 is finitely generated as a $\mathbb{Z}$-module (note that G had to be finite). Then $Z^1(G,M)$ (as $\mathbb{Z}$-submodule) is finitely generated. But now by proposition 3.1.19, every element has order divisible by $n=|G|$. Thus, take the lcm of the order of the generators, giving a bound of the order of H^1, completing the proof.

Finally, let us prove the foreshadowing in [7, Foreshadowing]. Let L/K be a cyclic extension so that $G=\mathrm{Gal}_K(L)\cong\mathbb{Z}/m\mathbb{Z}$. The G acts on $L^\times$ in the natural way, and hence we find it's group cohomology. First:

Lemma 3.1.20: Cocycles using Norm and Trace

Let L/K be a cyclic extension $G=\mathrm{Gal}_K(L)\cong\mathbb{Z}/m\mathbb{Z}$ with generator σ, and $Z^1(G,L^\times)$ the 1-cocycles. Let $U_{L/K}=\left\{l\in L \;\middle|\; \mathrm{Nm}_{L/K}(l)=1\right\}$ (sometimes this is denoted L^1). Then as groups:

$$\begin{gathered}Z^1(G,L^\times)\cong U_{L/K}\\ c\mapsto c(\sigma)\end{gathered}$$

Similarly:

$$Z^1(G,L)\cong T_{L/K}$$

where $T_{L/K}$ is the traceless elements[a]

[a] There is less common notation, sometimes it is denoted L^0

***Proof*:**
Let us first show the map is well-defined. Note first that a cocycle is determined by where it maps σ, namely for an arbitrary $c \in Z^1(G, L^\times)$:

$$\begin{aligned} c(\sigma^2) &= c(\sigma) \cdot \sigma(c(\sigma)) \\ c(\sigma^3) &= c(\sigma) \cdot \sigma(c(\sigma^2)) \\ &\vdots \end{aligned}$$

Next, note that $c(\sigma^m) = c(e) = 1$ since:

$$c(e) = c(e \cdot e) = c(e) \cdot e(\sigma(e)) = c(e) \cdot c(e)$$

as these map to $L^\times$, all elements are invertible, hence:

$$c(e) = 1 \in L^\times$$

Then:

$$1 = c(\sigma^n) = c(\sigma) \cdot \sigma(c(\sigma)) \cdot \sigma^2(c(\sigma)) \cdots \sigma^{n-1}(c(\sigma))$$

Letting $\beta = c(\sigma)$:

$$1 = c(\sigma^n) = \beta \cdot \sigma(\beta) \cdot \sigma^2(\beta) \cdots \sigma^{n-1}(\beta) = \prod_{g \in G} g(\beta) = \mathrm{Nm}_{L/K}(\beta)$$

showing the map is well-defined. As it is determined by where σ maps to, if $c_1, c_2 \in Z^1(G, L^\times)$ and $c_1(\sigma) = c_2(\sigma)$ then they agree on each σ^k and hence $c_1 = c_2$, showing injectivity. Surjectivity can be shown by defining $c(\sigma) = \beta$ for $\beta \in U_{L/K}$ and showing it satisfies the cocycle condition, which will be left as an exercise.

A similar argument can be done using the trace.

Theorem 3.1.21: Hilbert's 90th Theorem – Cohomology Upgrade

Using the above lemma's notation:

$$H^1(G, L^\times) \cong \{1\}$$

***Proof*:**
By Hilbert's 90th theorem ([7, Hilbert's 90th Theorem]), every norm 1 element is of the form:

$$\frac{\alpha}{\sigma^k(\alpha)}$$

But this is a coboundary element, hence all cocycles are coboundaries, completing the proof.

Note this theorem generalizes to finite galois extensions in general, see ref:HERE. This result should now make intuitive sense: the elements of K are *exactly* the $\mathrm{Gal}_K(L)$-invariant element. Note that if there were more complicated groups instead of $L^\times$, there can be more interesting cohomology groups, see [6] for more details.

Exercise 3.1.1

1. Let A, B be any G-modules. Show that $H^n(G, A \oplus B) \cong H^n(G, A) \oplus H^n(G, B)$
2. Show that $H^0(G, \mathrm{CoInd}^G(A)) \cong A$ and $H^n(G, \mathrm{CoInd}^G(A)) \cong 0$ for all $n \geq 1$ (hint: consider $\zeta : \mathrm{Hom}_{\mathbb{Z}[G]}(\mathbb{Z}[G^n], \mathrm{CoInd}^G(A)) \to \mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}[G^n], A)$ given by $\varphi \mapsto (z \mapsto \varphi(z)(1))$. What is the inverse of this map?)
3. Show that $H_0(G, \mathrm{Ind}^G(A)) \cong A$ and $H_n(G, \mathrm{Ind}^G(A)) \cong 0$ for all $n \geq 1$.

3.2 Homological Dimension

We now turn to measuring the "complexity" of modules and rings using the length of their resolutions. Before defining these measures rigorously, it is worth stepping back to 1890 to understand why mathematicians began counting the length of resolutions in the first place.

Historical Motivation: Invariant Theory

In the late 19th century, algebra was dominated by *Invariant Theory*. Imagine a group G acting on a polynomial ring $S = \mathbb{C}[x_1, \ldots, x_n]$ (for example, by linear substitutions of variables). Mathematicians were looking for *invariant ring* S^G:

$$S^G = \{f \in S \big| g \cdot f = f \text{ for all } g \in G\}$$

The central question was if S^G finitely generated? Can we find a finite set of basic invariant polynomials $f_1, \ldots, f_m$ such that every other invariant is just a polynomial combination of these?

Hilbert showed we can map a polynomial ring onto the invariants:

$$\varphi : \mathbb{C}[y_1, \ldots, y_m] \twoheadrightarrow S^G \qquad y_i \mapsto f_i$$

and then the kernel $I = \ker(\varphi)$ represents the *relations* between the invariants. Hence, the study of invariants reduced to the study of the kernel. To find these invariants, 19th-century computationalists (like Paul Gordan) asked tried computing them and found the following problem:

1. **First Syzygies:** The ideal I describes the relations. By the Basis Theorem, I is finitely generated.
2. **Second Syzygies:** Are these relations independent? Or are there "relations between the relations"? If we have a relation $Ar_1 + Br_2 = 0$, that is a second syzygy.
3. **Higher Syzygies:** We can keep going. We can form a module of second syzygies and ask for its generators, then ask for the relations between *those* generators.

Mathematicians feared this chain of relations might continue forever, becoming infinitely complex. Hilbert's *Syzygy Theorem* proved that if you are working with a polynomial ring in n variables, this process *must stop*. Specifically, after at most n steps, the module of relations becomes free—there are no more "hidden" relations.

This was part of David Hilbert's seminal paper laying the foundation of commutative geometry. In particular, he proved three theorems: The *Basis Theorem*, the *Nullstellensatz*, and the *Syzygy Theorem*.

3.2.1 Etymology The word *Syzygy* comes from the Greek *syzygia*, meaning "yoked together". In astronomy, it refers to the alignment of three celestial bodies (like the Sun, Earth, and Moon). In algebra, it refers to a relation that "yokes" generators together to zero.

3.2.1 Projective and Global Dimension

We can now formalize Hilbert's discovery using the language of resolutions.

Definition 3.2.2: Projective Dimension

Let M be an R-module. The *projective dimension* of M, denoted $\mathrm{pd}_R(M)$, is the minimal length n of a projective resolution of M:

$$0 \to P_n \to \cdots \to P_1 \to P_0 \to M \to 0$$

If no finite resolution exists, we say $\mathrm{pd}_R(M) = \infty$.

Definition 3.2.3: Global Dimension

The *global dimension* of a ring R, denoted $\mathrm{gldim}(R)$, is the supremum of the projective dimensions of all R-modules:

$$\mathrm{gldim}(R) = \sup\{\mathrm{pd}_R(M) \big| M \in R\text{-}\mathbf{Mod}\}$$

Theorem 3.2.4: Global Dimension of Polynomial Rings

Let R be a commutative ring. Then

$$\mathrm{gldim}(R[x]) = \mathrm{gldim}(R) + 1$$

***Proof*:**
We will prove this by establishing both the upper bound $\mathrm{gldim}(R[x]) \leq \mathrm{gldim}(R) + 1$ and the lower bound $\mathrm{gldim}(R[x]) \geq \mathrm{gldim}(R) + 1$. If $\mathrm{gldim}(R) = \infty$, the equality trivially holds, so we may assume $\mathrm{gldim}(R) = n < \infty$.

Step 1: The Characteristic Exact Sequence Let M be any $R[x]$-module. We can view M as an R-module by restricting the scalars. We can then construct the extended $R[x]$-module $R[x] \otimes_R M$.

Define a sequence of $R[x]$-modules:

$$0 \to R[x] \otimes_R M \xrightarrow{\alpha} R[x] \otimes_R M \xrightarrow{\beta} M \to 0$$

where the maps are defined by:

- $\alpha(f(x) \otimes m) = xf(x) \otimes m - f(x) \otimes xm$
- $\beta(f(x) \otimes m) = f(x)m$

We must prove this sequence is exact.

- *Surjectivity of β:* For any $m \in M$, $\beta(1 \otimes m) = m$, so β is surjective.
- *Exactness at the middle:* Clearly $\beta(\alpha(1 \otimes m)) = \beta(x \otimes m - 1 \otimes xm) = xm - xm = 0$, so $\mathrm{im}(\alpha) \subseteq \ker(\beta)$. Conversely, any element in $R[x] \otimes_R M$ can be written as $\sum_{i=0}^{d} x^i \otimes m_i$. If it lies in $\ker(\beta)$, then $\sum_{i=0}^{d} x^i m_i = 0$. We can rewrite our original element as:

$$\sum_{i=0}^{d} x^i \otimes m_i = \sum_{i=0}^{d} (x^i \otimes m_i - 1 \otimes x^i m_i)$$

  Notice that $x^i \otimes m - 1 \otimes x^i m = \alpha\left(\sum_{j=0}^{i-1} x^{i-1-j} \otimes x^j m\right)$. Thus, the element is in the image of α.

- *Injectivity of α:* Suppose $\sum_{i=0}^{d} x^i \otimes m_i \in \ker(\alpha)$. Then:

$$\sum_{i=0}^{d} x^{i+1} \otimes m_i - \sum_{i=0}^{d} x^i \otimes xm_i = 0$$

  Because $R[x]$ is free over R, we can compare the coefficients of $x^k \otimes -$ degree by degree. For degree 0, we have $-1 \otimes xm_0 = 0 \implies xm_0 = 0$. For degree $k > 0$, we have $m_{k-1} - xm_k = 0 \implies m_{k-1} = xm_k$. For degree $d+1$, we have $m_d = 0$. Working backward from $m_d = 0$, we get $m_{d-1} = xm_d = 0$, and by induction, $m_i = 0$ for all i. Hence, α is strictly injective.

Step 2: The Upper Bound
Let M be an $R[x]$-module. View M as an R-module, and take a projective resolution $P_\bullet \to M \to 0$ over R of length at most n (since $\mathrm{gldim}(R) = n$).

Because $R[x]$ is free (and thus flat) over R, tensoring the resolution with $R[x]$ preserves exactness. Therefore, $R[x] \otimes_R P_\bullet$ is a projective resolution of $R[x] \otimes_R M$ over $R[x]$. This proves that:

$$\mathrm{pd}_{R[x]}(R[x] \otimes_R M) \leq \mathrm{pd}_R(M) \leq n$$

Now, apply the functor $\mathrm{Ext}^*_{R[x]}(-, N)$ for an arbitrary $R[x]$-module N to our characteristic exact sequence. We obtain the long exact sequence:

$$\cdots \to \mathrm{Ext}^i_{R[x]}(R[x] \otimes_R M, N) \to \mathrm{Ext}^{i+1}_{R[x]}(M, N) \to \mathrm{Ext}^{i+1}_{R[x]}(R[x] \otimes_R M, N) \to \cdots$$

Since $\mathrm{pd}_{R[x]}(R[x] \otimes_R M) \leq n$, the Ext groups vanish for all indices $\geq n+1$. Therefore, setting $i = n+1$, we get:

$$0 \to \mathrm{Ext}^{n+2}_{R[x]}(M, N) \to 0$$

Since $\mathrm{Ext}^{n+2}_{R[x]}(M, N) = 0$ for all modules N, we conclude $\mathrm{pd}_{R[x]}(M) \leq n+1$. Because this holds for all $R[x]$-modules, $\mathrm{gldim}(R[x]) \leq n+1$.

Step 3: The Lower Bound
Since $\mathrm{gldim}(R) = n$, there exists an R-module M and an R-module K such that $\mathrm{Ext}^n_R(M, K) \neq 0$. We can elevate both M and K to the status of $R[x]$-modules by declaring that x acts trivially on them (i.e., $xM = 0$ and $xK = 0$).

Apply the characteristic exact sequence to M (viewed as this specific $R[x]$-module) and take the long exact sequence for $\mathrm{Hom}_{R[x]}(-, K)$. By the Tensor-Hom Adjunction, there is a natural isomorphism:

$$\mathrm{Hom}_{R[x]}(R[x] \otimes_R P_\bullet, K) \cong \mathrm{Hom}_R(P_\bullet, K)$$

Because $R[x] \otimes_R P_\bullet$ resolves $R[x] \otimes_R M$ over $R[x]$, taking the cohomology of both sides gives an isomorphism of the derived functors:

$$\mathrm{Ext}^i_{R[x]}(R[x] \otimes_R M, K) \cong \mathrm{Ext}^i_R(M, K) \quad \text{for all } i \geq 0$$

Let us examine the connecting map $\alpha^* : \mathrm{Ext}^i_{R[x]}(R[x] \otimes_R M, K) \to \mathrm{Ext}^i_{R[x]}(R[x] \otimes_R M, K)$ induced by α. On the level of the 0-th Hom group, for any $R[x]$-linear map $f : R[x] \otimes_R M \to K$:

$$(\alpha^* f)(1 \otimes m) = f(\alpha(1 \otimes m)) = f(x \otimes m - 1 \otimes xm)$$

Since x acts trivially on M, $1 \otimes xm = 0$. By $R[x]$-linearity, $f(x \otimes m) = x \cdot f(1 \otimes m)$. Because x also acts trivially on the codomain K, we get $x \cdot f(1 \otimes m) = 0$. Thus, the map α^* is identically the zero map on the entire cochain complex, and consequently, it is the zero map on the Ext groups.

Because the maps α^* are zero, our long exact sequence splits into short exact sequences for each i:

$$0 \to \mathrm{Ext}^i_R(M, K) \to \mathrm{Ext}^{i+1}_{R[x]}(M, K) \to \mathrm{Ext}^{i+1}_R(M, K) \to 0$$

Setting $i = n$, we get:

$$0 \to \mathrm{Ext}^n_R(M, K) \to \mathrm{Ext}^{n+1}_{R[x]}(M, K) \to \mathrm{Ext}^{n+1}_R(M, K) \to 0$$

Since $\mathrm{Ext}^n_R(M, K) \neq 0$, it directly follows that $\mathrm{Ext}^{n+1}_{R[x]}(M, K) \neq 0$. This proves that $\mathrm{pd}_{R[x]}(M) \geq n + 1$.

Combining the upper and lower bounds gives $\mathrm{gldim}(R[x]) = \mathrm{gldim}(R) + 1$, completing the proof.

Hilbert's theorem is effectively the calculation of the global dimension of polynomial rings.

Theorem 3.2.5: Hilbert's Syzygy Theorem

Let k be a field and $R = k[x_1, \ldots, x_n]$ be the polynomial ring in n variables. Then every finitely generated R-module has finite projective dimension, and:

$$\mathrm{gldim}(R) = n$$

***Proof*:**
While Hilbert's original proof was highly computational, the modern approach is much cleaner. We must show two things: that the global dimension is at least n, and that it is at most n.

First, we show $\mathrm{gldim}(R) \geq n$ by finding a module with projective dimension exactly n. This module is the residue field $k \cong R/(x_1, \ldots, x_n)$. We can explicitly construct a free resolution for k called the *Koszul Complex*. Let V be a k-vector space of dimension n with basis $e_1, \ldots, e_n$. The resolution is given by the exterior algebra $\bigwedge^\bullet V \otimes_k R$:

$$0 \to \bigwedge^n V \otimes R \xrightarrow{d_n} \ldots \xrightarrow{d_2} \bigwedge^1 V \otimes R \xrightarrow{d_1} R \xrightarrow{\epsilon} k \to 0$$

where the differential is given by $d_1(e_i \otimes 1) = x_i$. Because the variables $x_1, \ldots, x_n$ form a regular sequence, this complex is exact. It has length exactly n, so $\mathrm{pd}_R(k) = n$.

Second, to show that *no* module has a resolution longer than n, we proceed by induction on n. For $n = 0$, $R = k$ is a field; every module is free, so the dimension is 0. For $n = 1$, $R = k[x]$ is a PID. Submodules of free modules are free, so any resolution stops at step 1. For the inductive step, we write $R = S[x_n]$ where $S = k[x_1, \ldots, x_{n-1}]$.

By theorem 3.2.4 ($\mathrm{gldim}(S[x]) = \mathrm{gldim}(S) + 1$) and our inductive hypothesis, $\mathrm{gldim}(S) = n - 1$, which forces $\mathrm{gldim}(R) \leq n$, completing the proof.

This theorem has implications beyond just counting relations:

1. *Degree of the Hilbert Polynomial:* Hilbert originally used this theorem to prove that the Hilbert Function $H_M(t) = \dim_k(M_t)$ of a graded module eventually becomes a polynomial. The finite resolution allows us to calculate $H_M(t)$ as an alternating sum of the Hilbert functions of free modules, which are essentially binomial coefficients $\binom{t+k}{k}$.

2. *Structure of Projective Modules:* A celebrated problem posed by Serre (related to this theorem) asked if every projective module over $k[x_1, \ldots, x_n]$ is free. This is the algebraic analogue of "vector bundles over affine space are trivial". This was solved affirmatively in 1976 (the *Quillen-Suslin Theorem*). This implies that over polynomial rings, "projective resolution" and "free resolution" are synonymous.

3.2.2 Dimension of Local Rings

Computing the global dimension for general rings seems daunting, as one must check *every* module. However, we can simplify this task using derived functors.

Proposition 3.2.6: Detecting Projective Dimension with Tor

Let R be a ring and M an R-module. The following are equivalent:

1. $\mathrm{pd}_R(M) \leq n$
2. $\mathrm{Tor}_i^R(M, N) = 0$ for all R-modules N and all $i > n$.
3. $\mathrm{Tor}_{n+1}^R(M, N) = 0$ for all R-modules N.

***Proof*:**
(1) $\Rightarrow$ (2): If $\mathrm{pd}_R(M) \leq n$, there is a resolution $P_\bullet$ of length n. Computing Tor using this resolution involves terms P_i which are zero for $i > n$, hence the homology vanishes. (2) $\Rightarrow$ (3): Trivial. (3) $\Rightarrow$ (1): Let $\cdots \to P_1 \to P_0 \to M \to 0$ be a projective resolution.

Let $K_n = \ker(P_{n-1} \to P_{n-2})$. A standard dimension shifting argument (using the long exact sequence of Tor) shows that for any N:

$$\mathrm{Tor}_1^R(K_n, N) \cong \mathrm{Tor}_{n+1}^R(M, N)$$

By assumption (3), $\mathrm{Tor}_1^R(K_n, N) = 0$ for all N. This implies K_n is flat (and if R is Noetherian and M finitely generated, K_n is projective). Thus the resolution can stop at K_n, giving a length n projective resolution $0 \to K_n \to P_{n-1} \cdots \to M \to 0$.

It is more common that we consider rings that already have certain properties and verify how these conditions affect the global dimension. For example, given a local ring, the work is significantly reduced. In the local case, we can verify condition (3) by checking a *single* module: the residue field.

Lemma 3.2.7: Projective Dimension over Local Rings

Let $(R, \mathfrak{m}, k)$ be a local Noetherian ring and M a finitely generated R-module. Then:

$$\mathrm{pd}_R(M) \leq n \iff \mathrm{Tor}_{n+1}^R(M, k) = 0$$

Consequently, $\mathrm{pd}_R(M)$ is the smallest n such that $\mathrm{Tor}_i^R(M, k) = 0$ for all $i > n$.

***Proof*:**
The forward direction is immediate from proposition 3.2.6. For the converse, let $\cdots \to P_1 \to P_0 \to M \to 0$ be a *minimal* free resolution (meaning the matrices of the maps have entries in $\mathfrak{m}$). Then the differentials in the complex $P_\bullet \otimes_R k$ are all zero maps. Thus:

$$\mathrm{Tor}_i^R(M, k) \cong H_i(P_\bullet \otimes k) \cong P_i \otimes k \cong k^{\beta_i}$$

where β_i is the rank of the free module P_i. If $\mathrm{Tor}_{n+1}^R(M, k) = 0$, then $P_{n+1} \otimes k = 0$. By Nakayama's Lemma, this implies $P_{n+1} = 0$, so the resolution stops at n.

This leads to the fundamental equality attributed to Auslander, Buchsbaum, and Serre, which connects the homological algebra back to the geometry of the ring.

Theorem 3.2.8: Auslander-Buchsbaum-Serre Theorem

Let $(R, \mathfrak{m}, k)$ be a local Noetherian ring. Then the global dimension is determined by the residue field:

$$\mathrm{gldim}(R) = \mathrm{pd}_R(k)$$

Furthermore, $\mathrm{gldim}(R)$ is finite if and only if R is a *Regular Local Ring*.

This theorem is the bridge between algebra and geometry: "Regular" is a geometric condition (non-singular point), while Global Dimension is a homological condition (finite resolutions).

***Proof*:**
By definition, $\mathrm{gldim}(R) \geq \mathrm{pd}_R(k)$. We must show the reverse. Let $n = \mathrm{pd}_R(k)$. Let M be any finitely generated R-module. We check its projective dimension using lemma 3.2.7.

$$\mathrm{Tor}_{n+1}^R(M, k)$$

Since Tor is symmetric, $\mathrm{Tor}_{n+1}^R(M,k) \cong \mathrm{Tor}_{n+1}^R(k,M)$. Since $\mathrm{pd}_R(k) = n$, by proposition 3.2.6, $\mathrm{Tor}_{n+1}^R(k,-)$ is the zero functor. Thus $\mathrm{Tor}_{n+1}^R(M,k) = 0$. By lemma 3.2.7, this implies $\mathrm{pd}_R(M) \leq n$. Since this holds for all finitely generated M, $\mathrm{gldim}(R) = n$ (a standard limit argument extends this to non-finitely generated modules).

3.2.3 Beyond Regularity: The Hierarchy of Singularities

The Auslander-Buchsbaum-Serre theorem (theorem 3.2.8) presents a stark dichotomy: a local ring either has finite global dimension (and is geometrically smooth), or its global dimension is completely infinite.

For decades, algebraists sought to understand the "infinite" case. If a ring is singular, just *how* singular is it? How chaotic do the infinite free resolutions become? This inquiry led to a classification of local Noetherian rings, a hierarchy that measures the severity of singularities based on homological behavior.

To properly measure the complexity of rings that fail to be regular, we need to formalize the notion of a regular sequence:

Definition 3.2.9: Regular Sequence

Let R be a ring and M an R-module. A sequence of elements $x_1, \dots, x_n \in R$ is called an *M-regular sequence* if:

1. $(x_1, \dots, x_n)M \neq M$
2. For $i = 1, \dots, n$, the element x_i is not a zero-divisor on the quotient module $M/(x_1, \dots, x_{i-1})M$.

Intuitively, a regular sequence is a sequence of elements that "cut down" the dimension of the module as cleanly and efficiently as possible, without creating any hidden nilpotent. This allows us to define a new homological invariant.

Definition 3.2.10: Depth

Let $(R, \mathfrak{m})$ be a local Noetherian ring and M a finitely generated R-module. The *depth* of M, denoted $\mathrm{depth}(M)$, is the maximum length of an M-regular sequence contained in the maximal ideal $\mathfrak{m}$.

Depth is always bounded above by the Krull dimension ($\mathrm{depth}(R) \leq \mathrm{kdim}(R)$). The gap between a ring's depth and its dimension is one of the primary ways we measure how "bad" a singularity is.

Using this, we can make the Auslander-Buchsbaum-Serre theorem more refined by looking at modules over R:

Theorem 3.2.11: Auslander-Buchsbaum Formula

Let $(R, \mathfrak{m}, k)$ be a local Noetherian ring, and let M be a non-zero finitely generated R-module with finite projective dimension. Then:

$$\mathrm{pd}_R(M) + \mathrm{depth}(M) = \mathrm{depth}(R)$$

***Proof Key ideas*:**
The proof is an induction on $\mathrm{depth}(M)$ whose steps are standard commutative algebra rather than homological algebra; it is given in full in *An Introduction to Homological Algebra* [12].

The base case is $\mathrm{depth}(M) = 0$, where $\mathfrak{m}$ is an associated prime of M and one shows $\mathrm{pd}_R(M) = \mathrm{depth}(R)$ directly from a minimal free resolution: the length of that resolution is detected by $\mathrm{Tor}_i^R(M, k)$ against the residue field (lemma 3.2.7), and the depth of R bounds where those groups can survive.

The inductive step reduces the depth of M by one. If $\mathrm{depth}(M) > 0$ and $\mathrm{depth}(R) > 0$, prime avoidance applied to $\mathrm{Ass}(R) \cup \mathrm{Ass}(M)$ produces an $x \in \mathfrak{m}$ regular on both M and R; passing to M/xM over $R/(x)$ drops both $\mathrm{depth}(M)$ and $\mathrm{depth}(R)$ by one while leaving pd unchanged, so the formula for the quotient gives the formula for M. The remaining case is $\mathrm{depth}(R) = 0$, handled separately: there finite projective dimension forces M to be free, so both sides of the formula vanish.

Using depth and other structural properties, local rings can be classified into a strict chain of implications:

$$\text{Regular} \implies \text{Complete Intersection} \implies \text{Gorenstein} \implies \text{Cohen-Macaulay}$$

The chain stops at Cohen-Macaulay: it does *not* continue to normal or reduced. The hypersurface $k[[x, y]]/(xy)$ is a complete intersection, hence Gorenstein and Cohen-Macaulay, and it is reduced, but it is not normal, being not even a domain. By Serre's criterion normality is the conjunction of R_1 and S_2; Cohen-Macaulay supplies S_2 and says nothing about R_1, which is where the implication fails. The correct companion chain runs

$$\text{Regular} \implies \text{Normal} \implies \text{Reduced}$$

While these definitions are strictly for *local* rings, we say a general commutative ring has one of these properties if its localization at every prime ideal has it.

Complete Intersections: These are the "nicest" singularities. Geometrically, they are spaces cut out by exactly the correct number of equations (e.g., a 1D curve in $\mathbb{C}^3$ defined by exactly two equations). Algebraically, R is the quotient of a regular local ring by a regular sequence. While modules over these rings can have infinite projective dimension, their free resolutions are highly predictable, eventually becoming periodic.

Gorenstein Rings: These rings represent spaces with "symmetric" singularities. They are exactly the spaces where a generalized form of Poincaré Duality still holds. Algebraically, a local ring is Gorenstein if it has finite injective dimension over itself. The free resolutions over Gorenstein rings exhibit a mirror symmetry when the functor $\mathrm{Hom}_R(-, R)$ is applied.

Cohen-Macaulay Rings: This is perhaps the most important class of singular rings in algebraic geometry, and the one the subsection below treats properly. A local ring is Cohen-Macaulay if its depth equals its Krull dimension (definition 3.2.12). Geometrically, these spaces are "equidimensional" and lack hidden, embedded fuzzy components. Here, the Auslander-Buchsbaum formula perfectly aligns with the geometric dimension.

Normal Rings: A ring is normal if it is integrally closed in its total ring of fractions. Geometrically, this means the space is "smooth in codimension 1". For example, a 2D normal surface can have an isolated singular point (like the tip of a cone), but it cannot have a singular crease or folded line running through it.

Reduced Rings: A ring is reduced if it contains no non-zero nilpotents ($x^n = 0 \implies x = 0$). Geometrically, this simply means the space has no "thickness" or "fuzz"; it is a classical geometric space uniquely determined by the values of functions at its points.

3.2.4 Cohen-Macaulay Rings

Of the classes just listed, Cohen-Macaulay is the one downstream geometry actually consumes, and it deserves more than a description. Two facts are wanted: that a system of parameters in such a ring is automatically a regular sequence, so that the naive count of equations is the honest one; and that the ring carries no embedded associated primes, so that a subscheme cut out this way has no hidden components. Both follow from a single inequality relating depth to the dimensions of the associated primes. Associated primes, prime avoidance, and the identification of the zero-divisors with the union of the associated primes are those of [3], a co-requisite of this book.

Definition 3.2.12: Cohen-Macaulay Ring

A Noetherian local ring $(R, \mathfrak{m})$ is *Cohen-Macaulay* if $\mathrm{depth}(R) = \mathrm{kdim}(R)$. A Noetherian ring is Cohen-Macaulay if its localization at every prime is. More generally a finitely generated module $M \neq 0$ over a Noetherian local ring is *Cohen-Macaulay* if $\mathrm{depth}(M) = \mathrm{kdim}(M)$, the dimension of $R/\mathrm{Ann}(M)$.

Lemma 3.2.13: Depth Drops by One Modulo a Non-Zero Divisor

Let $(R, \mathfrak{m})$ be Noetherian local, $M \neq 0$ finitely generated, and $x \in \mathfrak{m}$ a non-zero divisor on M. Then $\mathrm{depth}(M/xM) = \mathrm{depth}(M) - 1$.

***Proof*:**
If $x, y_1, \ldots, y_{n-1}$ is an M-regular sequence in $\mathfrak{m}$ of length n then $y_1, \ldots, y_{n-1}$ is an M/xM-regular sequence, so $\mathrm{depth}(M/xM) \geq \mathrm{depth}(M) - 1$ once one knows some maximal regular sequence may be taken to start with x; and it may, because prepending x to any M/xM-regular sequence yields an M-regular sequence, which gives the reverse inequality $\mathrm{depth}(M) \geq \mathrm{depth}(M/xM) + 1$ directly from definition 3.2.10. Combining the two, both are equalities.

Proposition 3.2.14: Depth Is Bounded by the Dimension of Every Associated Prime

Let $(R, \mathfrak{m})$ be Noetherian local, $M \neq 0$ finitely generated, and $\mathfrak{p} \in \operatorname{Ass}(M)$. Then

$$\operatorname{depth}(M) \;\leq\; \operatorname{kdim}(R/\mathfrak{p})$$

***Proof*:**
Induct on $n = \operatorname{depth}(M)$. For $n = 0$ there is nothing to prove. Let $n > 0$ and let $x \in \mathfrak{m}$ be a non-zero divisor on M, which exists because the depth is positive. Since $\mathfrak{p}$ consists of zero-divisors on M, [3, The Zero-Divisors Are the Union of the Associated Primes] gives $x \notin \mathfrak{p}$.

Write $\mathfrak{p} = \operatorname{Ann}(m)$ with $m \neq 0$. We first arrange $m \notin xM$. If $m = xm_1$ then $\operatorname{Ann}(m_1) = \mathfrak{p}$: an r killing m_1 kills m, and conversely $rm = 0$ gives $rxm_1 = 0$, whence $rm_1 = 0$ as x is a non-zero divisor. Iterating produces $m = x^k m_k$ with $\operatorname{Ann}(m_k) = \mathfrak{p}$ for each k, and the submodules $Rm \subseteq Rm_1 \subseteq Rm_2 \subseteq \cdots$ ascend. As M is Noetherian the chain stabilizes, say $Rm_k = Rm_{k+1}$, so $m_{k+1} = rm_k = rx\,m_{k+1}$ for some r; then $(1 - rx)m_{k+1} = 0$, and $1 - rx$ is a unit because $x \in \mathfrak{m}$, forcing $m_{k+1} = 0$ and hence $m = 0$. That contradiction means the iteration halts, so we may take $m \notin xM$.

Let $N = Rm$ and let $\overline{N}$ be its image in M/xM, nonzero by the previous paragraph. Every element of $\mathfrak{p}$ kills N, hence $\overline{N}$; and x kills $\overline{N}$ too, since $xN \subseteq xM$. So $\operatorname{Ann}(\overline{N}) \supseteq \mathfrak{p} + (x)$. Choose $\mathfrak{q} \in \operatorname{Ass}(\overline{N}) \subseteq \operatorname{Ass}(M/xM)$, nonempty by [3, Zero-Divisors and Annihilators]. Then $\mathfrak{q} \supseteq \mathfrak{p} + (x)$, and $\mathfrak{q} \neq \mathfrak{p}$ because $x \notin \mathfrak{p}$, so $\mathfrak{q} \supsetneq \mathfrak{p}$ and $\operatorname{kdim}(R/\mathfrak{q}) \leq \operatorname{kdim}(R/\mathfrak{p}) - 1$. By lemma 3.2.13 and the inductive hypothesis applied to M/xM and $\mathfrak{q}$,

$$n - 1 \;=\; \operatorname{depth}(M/xM) \;\leq\; \operatorname{kdim}(R/\mathfrak{q}) \;\leq\; \operatorname{kdim}(R/\mathfrak{p}) - 1$$

which is the assertion.

Corollary 3.2.15: Depth Is at Most the Dimension

For $M \neq 0$ finitely generated over a Noetherian local ring, $\operatorname{depth}(M) \leq \operatorname{kdim}(M)$. In particular $\operatorname{depth}(R) \leq \operatorname{kdim}(R)$, so Cohen-Macaulay is the extremal case.

***Proof*:**
$\operatorname{Ass}(M)$ is nonempty ([3, Zero-Divisors and Annihilators]); pick $\mathfrak{p}$ in it. Then $\mathfrak{p} \supseteq \operatorname{Ann}(M)$, so $\operatorname{kdim}(R/\mathfrak{p}) \leq \operatorname{kdim}(R/\operatorname{Ann}(M)) = \operatorname{kdim}(M)$, and proposition 3.2.14 bounds the depth by the left-hand side.

This is the inequality asserted informally above, and it is what makes the definition of Cohen-Macaulay a demand rather than a coincidence.

Corollary 3.2.16: A Cohen-Macaulay Ring Has No Embedded Associated Primes

Let $(R, \mathfrak{m})$ be a Cohen-Macaulay local ring of dimension d. Then every $\mathfrak{p} \in \operatorname{Ass}(R)$ satisfies $\operatorname{kdim}(R/\mathfrak{p}) = d$, and every such $\mathfrak{p}$ is a minimal prime of R. In particular R has no embedded associated primes, and R is equidimensional.

***Proof*:**
For $\mathfrak{p} \in \mathrm{Ass}(R)$, proposition 3.2.14 gives $d = \mathrm{depth}(R) \leq \mathrm{kdim}(R/\mathfrak{p}) \leq d$, so $\mathrm{kdim}(R/\mathfrak{p}) = d$. If $\mathfrak{p}$ were not minimal it would strictly contain a prime $\mathfrak{p}_0$, and then $\mathrm{kdim}(R/\mathfrak{p}) < \mathrm{kdim}(R/\mathfrak{p}_0) \leq d$, a contradiction. An associated prime that is minimal is by definition not embedded.

Theorem 3.2.17: A System of Parameters in a Cohen-Macaulay Local Ring Is a Regular Sequence

Let $(R, \mathfrak{m})$ be a Cohen-Macaulay local ring of dimension d and let $x_1, \ldots, x_d \in \mathfrak{m}$ be a system of parameters, that is d elements generating an $\mathfrak{m}$-primary ideal. Then $x_1, \ldots, x_d$ is a regular sequence, and $R/(x_1, \ldots, x_d)$ is Cohen-Macaulay of dimension zero.

***Proof*:**
Induct on d, the case $d = 0$ being the empty sequence. Let $d > 0$. Since $x_1, \ldots, x_d$ generate an $\mathfrak{m}$-primary ideal, the images of $x_2, \ldots, x_d$ generate an $\mathfrak{m}$-primary ideal of $R/(x_1)$, so $\mathrm{kdim} R/(x_1) \leq d - 1$ by the dimension theorem of [3]; and $\mathrm{kdim} R/(x_1) \geq d - 1$ by Krull's height theorem there. So $\mathrm{kdim} R/(x_1) = d - 1$.

Now x_1 is a non-zero divisor. Suppose $x_1 \in \mathfrak{p}$ for some $\mathfrak{p} \in \mathrm{Ass}(R)$. By corollary 3.2.16, $\mathrm{kdim}(R/\mathfrak{p}) = d$; but $\mathfrak{p} \supseteq (x_1)$ gives a surjection $R/(x_1) \twoheadrightarrow R/\mathfrak{p}$ and hence $\mathrm{kdim} R/(x_1) \geq d$, contradicting the previous paragraph. So x_1 lies outside every associated prime and is a non-zero divisor by [3, The Zero-Divisors Are the Union of the Associated Primes].

By lemma 3.2.13, $\mathrm{depth}\, R/(x_1) = d - 1 = \mathrm{kdim} R/(x_1)$, so $R/(x_1)$ is Cohen-Macaulay of dimension $d - 1$, and the images of $x_2, \ldots, x_d$ are a system of parameters for it. The inductive hypothesis makes them a regular sequence there, which is exactly the statement that $x_1, \ldots, x_d$ is a regular sequence on R and that the final quotient is Cohen-Macaulay of dimension zero.

3.3 The Local Criterion for Flatness

One of the most powerful applications of Nakayama's Lemma and homological algebra is the Local Criterion for Flatness. Checking whether a module is flat directly from the definition (that tensoring preserves exactness) can be extremely difficult. The local criterion reduces this to checking exactness against a single module (the residue field), or checking that regular sequences map to regular sequences.

Theorem 3.3.1: Local Criterion for Flatness

Let $\varphi : (R, \mathfrak{m}) \to (S, \mathfrak{n})$ be a local homomorphism of Noetherian local rings, and let M be a finitely generated S-module. Then the following are equivalent:

1. M is flat over R.
2. $\mathrm{Tor}_1^R(M, R/\mathfrak{m}) = 0$.
3. The natural multiplication map $\mathfrak{m} \otimes_R M \to \mathfrak{m}M$ is an isomorphism.

***Proof*:**
The implication (1) $\Rightarrow$ (2) is trivial since M is flat implies $\mathrm{Tor}_i^R(M, -) = 0$ for all $i > 0$. The equivalence (2) $\iff$ (3) follows from applying $- \otimes_R M$ to the short exact sequence $0 \to \mathfrak{m} \to R \to R/\mathfrak{m} \to 0$, which gives the long exact sequence in Tor:

$$0 = \mathrm{Tor}_1^R(R, M) \to \mathrm{Tor}_1^R(R/\mathfrak{m}, M) \to \mathfrak{m} \otimes_R M \to R \otimes_R M \to R/\mathfrak{m} \otimes_R M \to 0$$

Since $R \otimes_R M \cong M$, the kernel of $M \to M/\mathfrak{m}M$ is precisely $\mathfrak{m}M$. Thus $\mathrm{Tor}_1^R(R/\mathfrak{m}, M) = 0$ if and only if $\mathfrak{m} \otimes_R M \to \mathfrak{m}M$ is injective (and it is always surjective).

For (2) $\Rightarrow$ (1), we must show that $\mathrm{Tor}_1^R(M, N) = 0$ for all finitely generated R-modules N. We proceed by induction on the length of N. By Nakayama's Lemma ([3, Nakayama's Lemma I]), if $N \neq 0$, we can find a non-zero quotient $N \to R/\mathfrak{m}$. Let K be the kernel, so we have $0 \to K \to N \to R/\mathfrak{m} \to 0$. The long exact sequence in Tor gives:

$$\mathrm{Tor}_1^R(K, M) \to \mathrm{Tor}_1^R(N, M) \to \mathrm{Tor}_1^R(R/\mathfrak{m}, M) = 0$$

Thus, $\mathrm{Tor}_1^R(K, M)$ surjects onto $\mathrm{Tor}_1^R(N, M)$. By iterating this process (using the Artin-Rees lemma and the $\mathfrak{m}$-adic filtration on N; [3, Artin-Rees Lemma]), one can show that $\mathrm{Tor}_1^R(N, M) \subseteq \mathfrak{m}\mathrm{Tor}_1^R(N, M)$. Since M is a finitely generated S-module and S is Noetherian, $\mathrm{Tor}_1^R(N, M)$ is a finitely generated S-module. By Nakayama's Lemma (applied over S, since $\mathfrak{m}S \subseteq \mathfrak{n}$), we conclude $\mathrm{Tor}_1^R(N, M) = 0$.

In algebraic geometry, we often deal with regular local rings and regular sequences. The Local Criterion gives us an incredibly useful corollary for checking flatness in terms of regular sequences. This is the algebraic engine behind an infamous result known as the "Miracle Flatness" theorem, an important "shortcut" for proving flatness in [1].

Corollary 3.3.2: Regular Sequence Criterion for Flatness

Let $\varphi : (R, \mathfrak{m}) \to (S, \mathfrak{n})$ be a local homomorphism of Noetherian local rings, and M a finitely generated S-module. Let $x \in \mathfrak{m}$ be a non-zero divisor on R. Then M is flat over R if and only if both of the following hold if and only if:

1. x is a non-zero divisor on M, and
2. M/xM is flat over $R/(x)$.

By induction, if $x_1, \ldots, x_d \in \mathfrak{m}$ is a regular sequence in R, then M is flat over R if and only if $x_1, \ldots, x_d$ forms a regular sequence on M and $M/(x_1, \ldots, x_d)M$ is flat over $R/(x_1, \ldots, x_d)$.

***Proof*:**
($\Rightarrow$): If M is flat over R, applying $- \otimes_R M$ to the exact sequence $0 \to R \xrightarrow{\cdot x} R \to R/(x) \to 0$ yields the exact sequence $0 \to M \xrightarrow{\cdot x} M \to M/xM \to 0$. Thus x is a non-zero divisor on M. Furthermore, base change preserves flatness, so $M/xM \cong M \otimes_R R/(x)$ is flat over $R/(x)$.

($\Leftarrow$): Assume x is regular on M and M/xM is flat over $R/(x)$. We want to show M is flat. Since M/xM is flat over $R/(x)$, $\mathrm{Tor}_1^{R/(x)}(R/\mathfrak{m}, M/xM) = 0$. Combined with the fact that x is regular on M, this gives $\mathrm{Tor}_1^R(R/\mathfrak{m}, M) = 0$: taking a free resolution $F_\bullet \to M$ over R, regularity of x on both R and M makes $F_\bullet \otimes_R R/(x)$ a free resolution of M/xM over $R/(x)$, so $\mathrm{Tor}_i^{R/(x)}(N, M/xM) \cong \mathrm{Tor}_i^R(N, M)$ for every $R/(x)$-module N, and $N = R/\mathfrak{m}$ is one. By the Local Criterion for Flatness (theorem 3.3.1), M is flat over R.

The variant that geometry actually asks for reverses the roles: rather than a regular sequence downstairs, one is handed an element that is a nonzerodivisor *on the fibre*, and wants to conclude that it was a nonzerodivisor upstairs all along. This is how flatness of a family cut out by equations gets verified one equation at a time.

Corollary 3.3.3: Fibrewise Nonzerodivisor Criterion for Flatness

Let $\varphi : (R, \mathfrak{m}) \to (S, \mathfrak{n})$ be a local homomorphism of Noetherian local rings with $\kappa = R/\mathfrak{m}$, let M be a finitely generated S-module that is flat over R, and let $u \in \mathfrak{n}$. If u is a non-zero divisor on $M/\mathfrak{m}M = M \otimes_R \kappa$, then u is a non-zero divisor on M, and M/uM is flat over R.

***Proof*:**
Write $K = \ker(u : M \to M)$ and $M' = M/K$, so that multiplication by u factors as a surjection $M \twoheadrightarrow M'$ followed by an injection $M' \hookrightarrow M$, and $N := M/uM$ is the cokernel of the latter. Both sequences

$$0 \to K \to M \to M' \to 0, \qquad 0 \to M' \xrightarrow{u} M \to N \to 0$$

are exact by construction.

Step 1: N is flat. Apply $- \otimes_R \kappa$ to the second sequence. Flatness of M gives $\mathrm{Tor}_1^R(\kappa, M) = 0$, so the long exact sequence begins

$$0 \to \mathrm{Tor}_1^R(\kappa, N) \to M' \otimes_R \kappa \to M \otimes_R \kappa \to N \otimes_R \kappa \to 0$$

Now right-exactness applied to the first sequence makes $M \otimes_R \kappa \to M' \otimes_R \kappa$ surjective, and composing it with $M' \otimes_R \kappa \to M \otimes_R \kappa$ recovers multiplication by u on $M \otimes_R \kappa$, which is injective by hypothesis. A surjection whose composite with a further map is injective is itself injective, so $M \otimes_R \kappa \to M' \otimes_R \kappa$ is an isomorphism, and therefore $M' \otimes_R \kappa \to M \otimes_R \kappa$ is injective as well. Hence $\mathrm{Tor}_1^R(\kappa, N) = 0$, and N is flat over R by theorem 3.3.1.

Step 2: $K = 0$. Since M and N are both flat over R, the second sequence forces M' to be flat too: for any R-module T the long exact sequence reads $0 = \mathrm{Tor}_2^R(T, N) \to \mathrm{Tor}_1^R(T, M') \to \mathrm{Tor}_1^R(T, M) = 0$. In particular $\mathrm{Tor}_1^R(\kappa, M') = 0$, so applying $- \otimes_R \kappa$ to the first sequence leaves $K \otimes_R \kappa \to M \otimes_R \kappa$ injective. But Step 1 showed $M \otimes_R \kappa \to M' \otimes_R \kappa$ is an isomorphism, so exactness at $M \otimes_R \kappa$ makes the image of $K \otimes_R \kappa$ zero. An injective map with zero image is zero, so $K/\mathfrak{m}K = 0$. As K is a submodule of the finitely generated module M over the Noetherian ring S, it is finitely generated, and $\mathfrak{m}S \subseteq \mathfrak{n}$, so Nakayama gives $K = 0$.

Exercise 3.3.1

1. Let $R = k[x, y]$. Find a free resolution for the module $M = R/(x^2, xy, y^2)$. What is the length of this resolution?
2. Show that over $R = \mathbb{Z}$, the module $M = \mathbb{Q}$ has projective dimension 1, but does not have a

finite *free* resolution if we require finitely generated free modules (which is impossible since $\mathbb{Q}$ isn't finitely generated).

3. Why does the Syzygy theorem fail for $R = k[x_1, x_2, \dots]$ (infinite variables)? Find a module with infinite projective dimension.

4

Spectral Sequences

The derived functors of chapter 2 measure the failure of a single functor to be exact, and the long exact sequence of theorem 2.3.10 turns that measurement into something computable. Both tools carry the same limitation: each relates the cohomology of one object to the cohomology of two others, and no more. An object assembled from many pieces produces not one long exact sequence but a whole family of them at once, one for each stage of the assembly, and no finite amount of splicing turns that family into an answer. A module carrying a filtration $M = M_0 \supseteq M_1 \supseteq \cdots$ ([3, Filtrations]) is the basic case; a functor that factors through a third abelian category is another, and the double complexes of section 2.4 are a third.

A spectral sequence is the bookkeeping that assembles such a family. It produces a sequence of *pages* $E_1, E_2, E_3, \ldots$, each the cohomology of the one before it, beginning with a crude first approximation to the answer and correcting it repeatedly until the corrections run out. This chapter builds the machine and then supplies the three standard sources of one. Section 4.1 isolates the algebraic mechanism, Massey's exact couple, from which every page and every differential is generated by a single operation. Section 4.2 constructs the spectral sequence of a filtered complex and proves the convergence and comparison theorems that make it usable. Section 4.3 treats the two spectral sequences of a double complex, completing the computations of section 2.4.1 that were deferred to this chapter. Section 4.4 constructs the spectral sequence of a composite of derived functors, the form in which the machine is most often applied. Section 4.5 closes with hypercohomology, the derived functors of a functor applied to a complex rather than an object.

4.1 Exact Couples and Derived Couples

The obstacle a spectral sequence exists to overcome is visible already in the simplest case. A short exact sequence of complexes $0 \to A_\bullet \to C_\bullet \to C_\bullet/A_\bullet \to 0$ yields a long exact sequence relating the three cohomologies, but that sequence does not determine $H_\bullet(C)$ from $H_\bullet(A)$ and $H_\bullet(C/A)$: it determines only the two ends of a short exact sequence

$$0 \to \mathrm{coker}\partial \to H_n(C) \to \ker \partial \to 0$$

and the extension itself is extra information the sequence does not carry. With one layer the ambiguity is a single extension problem. With a filtration of many layers there is one such problem per layer, and there are also correction terms of a different kind: a cohomology class of one layer may fail to extend over the next layer but succeed over the one after, and nothing in any individual long exact sequence detects that.

The device that organizes all of these at once packages every long exact sequence of the filtration into a single self-regenerating triangle. Its two vertices carry different roles: one is the answer being approximated, the other the auxiliary object that remembers how the layers are glued.

Definition 4.1.1: Exact Couple

Let $\mathcal{A}$ be an abelian category. An *exact couple* in $\mathcal{A}$ is a pair of objects D and E together with three morphisms

$$\begin{tikzcd} D \arrow[rr, "i"] & & D \arrow[dl, "j"] \\ & E \arrow[ul, "k"] & \end{tikzcd}$$

such that the triangle is exact at each of its three corners:

$$\mathrm{im} i = \ker j, \qquad \mathrm{im} j = \ker k, \qquad \mathrm{im} k = \ker i$$

The composite $d := j \circ k \colon E \to E$ is the *differential* of the couple, and the couple is written (D, E, i, j, k).

The data should be read as a compression of a long exact sequence, or rather of all of them simultaneously. In the filtered case treated in section 4.2, the object D collects the cohomologies of the stages of the filtration, the object E collects the cohomologies of the successive layers, the map i is induced by the inclusion of one stage in the next, the map j is the passage to the layer, and k is the connecting homomorphism. Exactness at the three corners is precisely the exactness of every one of those long exact sequences. Of the two objects only E is meant to be looked at, since it is the approximation to the answer; D is the auxiliary object, and it is the auxiliary object that makes the approximation improvable.

Two of the three exactness conditions are spent constructing the next stage, and one is spent immediately, on the claim that d deserves its name.

Lemma 4.1.2: The Differential of an Exact Couple Squares to Zero

Let (D, E, i, j, k) be an exact couple. Then $d \circ d = 0$, so (E, d) is a differential object and its cohomology

$$H(E, d) = \frac{\ker d}{\mathrm{im} d}$$

is defined.

***Proof*:**
Expand the composite and regroup:

$$d \circ d = (j \circ k) \circ (j \circ k) = j \circ (k \circ j) \circ k$$

Exactness at the corner E gives $\mathrm{im} j = \ker k$, so every morphism factoring through j is annihilated by k, that is $k \circ j = 0$. Hence $d \circ d = j \circ 0 \circ k = 0$, as we sought to show.

Only exactness at E was used. The remaining two conditions are exactly what is needed to promote $H(E, d)$ from a bare cohomology object to the E-term of a new exact couple, and it is that promotion which makes the construction repeat. The new auxiliary object is the image of i, and the new maps are induced by the old ones.

4.1.3 Convention: Elements in an Abelian Category The constructions below are phrased with elements, in the style of a diagram chase. This is legitimate in any abelian category: by the Freyd-Mitchell theorem (theorem 1.1.15), a small abelian category admits a fully faithful exact embedding into R-Mod for some ring R, and a diagram whose exactness is verified in R-Mod is exact in $\mathcal{A}$ because the embedding reflects exactness. Every statement in this section may therefore be proved for modules and read off for a general $\mathcal{A}$. The standard sources of exact couples, filtered complexes and double complexes of modules or of sheaves, are in any case module-like.

Definition 4.1.4: Derived Couple

Let (D, E, i, j, k) be an exact couple with differential $d = j \circ k$. Its *derived couple* (D', E', i', j', k') consists of the objects

$$D' := \mathrm{im} i, \qquad E' := H(E, d) = \frac{\ker d}{\mathrm{im} d}$$

together with the morphisms

1. $i' := i|_{D'} : D' \to D'$, the restriction of i to its image;
2. $j' : D' \to E'$ defined by $j'(i(x)) := [\, j(x)\,]$ for $x \in D$;
3. $k' : E' \to D'$ defined by $k'([e]) := k(e)$ for $e \in \ker d$.

The name is older than, and unrelated to, the derived categories of section 2.5; it originates in Massey's work in algebraic topology. Three of the four items above require an argument before the definition is even meaningful: that i carries D' into itself, that j' is independent of the choice of preimage, and that k' both lands in D' and is independent of the representative cycle. Those checks and the exactness of the result are one theorem.

Theorem 4.1.5: The Derived Couple is an Exact Couple

Let (D, E, i, j, k) be an exact couple. Then the morphisms i', j' and k' of definition 4.1.4 are well defined, and the triangle

$$\begin{tikzcd} D' \arrow[rr, "i'"] & & D' \arrow[dl, "j'"] \\ & E' \arrow[ul, "k'"] & \end{tikzcd}$$

is an exact couple.

***Proof*:**
Step 1: The three maps are well defined. For i', note that $i(D') = i(i(D)) \subseteq i(D) = D'$, so restricting i to D' does indeed produce an endomorphism of D'.

For j', two things must be checked. First, $j(x)$ is a cycle for every $x \in D$: applying the differential gives $d(j(x)) = j(k(j(x))) = 0$, since $k \circ j = 0$ by exactness at E. So the class $[j(x)] \in E'$ makes sense. Second, the class does not depend on the choice of x. Suppose $i(x) = i(x')$. Then $x - x' \in \ker i = \mathrm{im} k$ by exactness at the target corner, say $x - x' = k(e)$ for some $e \in E$. Applying j gives

$$j(x) - j(x') = j(k(e)) = d(e) \in \mathrm{im} d$$

so $[j(x)] = [j(x')]$ in $E' = \ker d / \mathrm{im} d$.

For k', again two checks. First, k carries cycles into D': if $d(e) = 0$ then $j(k(e)) = 0$, so $k(e) \in \ker j = \mathrm{im} i = D'$ by exactness at the source corner. Second, the value is independent of the representative. If $e - e' = d(f) = j(k(f))$ for some $f \in E$, then

$$k(e) - k(e') = k(j(k(f))) = (k \circ j)(k(f)) = 0$$

again because $k \circ j = 0$. So k' is well defined.

Step 2: Exactness at the corner receiving i', that is $\mathrm{im} i' = \ker j'$. For the inclusion $\mathrm{im} i' \subseteq \ker j'$, take $y = i(x) \in D'$ and compute

$$j'(i'(y)) = j'(i(i(x))) = [\,j(i(x))\,] = [0] = 0$$

the last equality because $\mathrm{im} i = \ker j$. For the reverse inclusion, let $y = i(x) \in D'$ satisfy $j'(y) = 0$, that is $[j(x)] = 0$, so $j(x) = d(e) = j(k(e))$ for some $e \in E$. Then $j(x - k(e)) = 0$, so $x - k(e) \in \ker j = \mathrm{im} i$, say $x - k(e) = i(z)$. Applying i and using $i \circ k = 0$ (exactness at the corner where k arrives) gives

$$y = i(x) = i(k(e)) + i(i(z)) = i(i(z)) = i'(i(z))$$

and $i(z) \in D'$, so $y \in \mathrm{im} i'$.

Step 3: Exactness at E', that is $\mathrm{im} j' = \ker k'$. For $\mathrm{im} j' \subseteq \ker k'$, take $y = i(x) \in D'$ and compute

$$k'(j'(y)) = k'([\,j(x)\,]) = k(j(x)) = 0$$

using $k \circ j = 0$. For the reverse inclusion, let $[e] \in E'$ satisfy $k'([e]) = k(e) = 0$. Then $e \in \ker k = \operatorname{im} j$, say $e = j(x)$ for some $x \in D$. The element $i(x)$ lies in D' and $j'(i(x)) = [j(x)] = [e]$, so $[e] \in \operatorname{im} j'$.

Step 4: Exactness at the corner receiving k', that is $\operatorname{im} k' = \ker i'$. For $\operatorname{im} k' \subseteq \ker i'$, take $[e] \in E'$; then $i'(k'([e])) = i(k(e)) = 0$ since $\operatorname{im} k = \ker i$. For the reverse inclusion, let $y = i(x) \in D'$ satisfy $i'(y) = 0$, that is $i(i(x)) = 0$. Then $i(x) \in \ker i = \operatorname{im} k$, say $i(x) = k(e)$ for some $e \in E$. This e is a cycle:

$$d(e) = j(k(e)) = j(i(x)) = 0$$

because $\operatorname{im} i = \ker j$. Hence $[e] \in E'$ and $k'([e]) = k(e) = i(x) = y$, so $y \in \operatorname{im} k'$.

All six inclusions hold, so the derived triangle is exact at each corner, completing the proof.

The theorem is the basis of the entire chapter. Because the derived couple is again an exact couple, the construction applies to it in turn, and iterating produces an infinite sequence of couples whose E-terms form a sequence of objects each of which is the cohomology of its predecessor. That sequence is the object the chapter is named for.

Definition 4.1.6: Spectral Sequence

Let $\mathcal{A}$ be an abelian category. A *spectral sequence* in $\mathcal{A}$ is a family $\{(E_r, d_r)\}_{r \geq r_0}$ of objects E_r of $\mathcal{A}$, each equipped with an endomorphism $d_r \colon E_r \to E_r$ satisfying $d_r \circ d_r = 0$, together with isomorphisms

$$E_{r+1} \cong H(E_r, d_r) = \frac{\ker d_r}{\operatorname{im} d_r}$$

for every $r \geq r_0$. The object E_r is the *r-th page*, d_r is the *r-th differential*, and r_0 is the *initial page*.
The spectral sequence is *bigraded* if each E_r carries a decomposition $E_r = \bigoplus_{p,q} E_r^{p,q}$ and each d_r is homogeneous of a fixed bidegree. Two conventions are in use. In the *homological* convention the pages are written $E^r_{p,q}$ and

$$d_r \colon E^r_{p,q} \to E^r_{p-r,\, q+r-1}$$

while in the *cohomological* convention they are written $E_r^{p,q}$ and

$$d_r \colon E_r^{p,q} \to E_r^{p+r,\, q-r+1}$$

Both differentials change the total degree $p + q$ by exactly one, downward in the homological convention and upward in the cohomological one. The two are exchanged by $E_r^{p,q} = E^r_{-p,-q}$.

The bidegrees are not a separate convention imposed on the definition; they are forced by the exact couple that generates the sequence, as the next result records. Every spectral sequence appearing in this book, and every one cited from it elsewhere in the series, arises from an exact couple in the manner described.

Theorem 4.1.7: The Spectral Sequence of an Exact Couple

Let $(D, E, i, j, k) = (D^1, E^1, i^{(1)}, j^{(1)}, k^{(1)})$ be an exact couple, and for $r \geq 1$ let $(D^{r+1}, E^{r+1}, i^{(r+1)}, j^{(r+1)}, k^{(r+1)})$ be the derived couple of the r-th one. Then $\{(E^r, d_r)\}_{r\geq 1}$, with $d_r = j^{(r)} \circ k^{(r)}$, is a spectral sequence, and $D^{r+1} = i^r(D)$ for every r.
Suppose in addition that D and E are bigraded and that i, j, k are homogeneous of bidegrees

$$\deg i = (1, -1), \qquad \deg j = (0, 0), \qquad \deg k = (-1, 0)$$

with the pages of the derived couples indexed by $D^{r+1}_{p,q} = i\big(D^r_{p-1,q+1}\big)$. Then

$$\deg i^{(r)} = (1, -1), \qquad \deg j^{(r)} = \big(-(r-1),\, r-1\big), \qquad \deg k^{(r)} = (-1, 0)$$

and consequently d_r is homogeneous of bidegree $(-r,\, r-1)$, so the sequence is bigraded in the homological convention of definition 4.1.6.

***Proof*:**
Step 1: The pages form a spectral sequence. By theorem 4.1.5 each derived couple is again an exact couple, so the construction may be iterated and every d_r satisfies $d_r \circ d_r = 0$ by lemma 4.1.2. By definition 4.1.4 the E-term of the derived couple is $E^{r+1} = H(E^r, d_r)$, which is the required isomorphism. The identity $D^{r+1} = i^r(D)$ follows by induction: it holds for $r = 0$ by convention, and if $D^r = i^{r-1}(D)$ then $D^{r+1} = i^{(r)}(D^r) = i(i^{r-1}(D)) = i^r(D)$, since $i^{(r)}$ is the restriction of i.

Step 2: The bidegree of $i^{(r)}$ and $k^{(r)}$. Each $i^{(r)}$ is a restriction of i to a subobject, so it is homogeneous of the same bidegree $(1, -1)$ for every r. Similarly $E^{r+1}_{p,q}$ is a subquotient of $E^r_{p,q}$ in the same bidegree, and $k^{(r+1)}$ is induced by $k^{(r)}$ on representatives, so $\deg k^{(r)} = (-1, 0)$ for every r by induction.

Step 3: The bidegree of $j^{(r)}$ shifts by $(-1, 1)$ at each stage. Fix r and suppose $\deg j^{(r)} = (a, b)$. By definition 4.1.4 the map $j^{(r+1)}$ is defined on an element of D^{r+1} written as $i(x)$ with $x \in D^r$, by $j^{(r+1)}(i(x)) = [\,j^{(r)}(x)\,]$. With the stated indexing, an element of $D^{r+1}_{p,q}$ is of the form $i(x)$ for $x \in D^r_{p-1,q+1}$, and $j^{(r)}(x)$ then lies in $E^r_{p-1+a,\, q+1+b}$. Hence $j^{(r+1)}$ carries $D^{r+1}_{p,q}$ into $E^{r+1}_{p+a-1,\, q+b+1}$, that is $\deg j^{(r+1)} = (a-1, b+1)$. Since $\deg j^{(1)} = (0, 0)$, induction gives $\deg j^{(r)} = (-(r-1), r-1)$.

Step 4: The bidegree of d_r. Combining Steps 2 and 3,

$$\deg d_r = \deg j^{(r)} + \deg k^{(r)} = \big(-(r-1),\, r-1\big) + (-1, 0) = (-r,\, r-1)$$

so $d_r \colon E^r_{p,q} \to E^r_{p-r,\, q+r-1}$, which is the homological convention of definition 4.1.6, completing the proof.

The bidegree computation is worth pausing over, because it explains what the higher pages are for. The differential d_1 moves one step in the filtration direction, d_2 moves two steps, and d_r moves r steps. Each derivation of the couple replaces D by $i(D)$, which is to say it discards the classes that survive only one stage further and keeps those that survive r stages; the reach of the differential grows in step with that. A class killed by d_r and by no earlier differential encodes a gluing that spans exactly r layers of the filtration, information no single long exact sequence of the family can see. This is the precise sense in which the spectral sequence contains more than the collection of long exact sequences it was built from.

The simplest instance is a filtration with only two stages, where there is a single differential and the machine reduces to something already familiar.

Proposition 4.1.8: The Two-Step Filtration Recovers the Long Exact Sequence

Let $0 \to A_\bullet \xrightarrow{\iota} C_\bullet \xrightarrow{\pi} C_\bullet/A_\bullet \to 0$ be a short exact sequence of complexes in $\mathcal{A}$, and let $D = H_\bullet(A) \oplus H_\bullet(C)$ and $E = H_\bullet(A) \oplus H_\bullet(C/A)$ be the terms of the exact couple attached to the two-step filtration $0 \subseteq A_\bullet \subseteq C_\bullet$. Then:

1. the first page is $E^1 = H_\bullet(A) \oplus H_\bullet(C/A)$, with d_1 the connecting homomorphism $\partial\colon H_n(C/A) \to H_{n-1}(A)$ of theorem 2.1.15;
2. $E^2 = E^\infty$, that is $d_r = 0$ for all $r \geq 2$;
3. E^2 is the associated graded of $H_\bullet(C)$ for the induced two-step filtration, namely

$$E^2 \cong \operatorname{coker}\partial \oplus \ker \partial$$

with $\operatorname{coker}\partial$ the image of $H_n(A) \to H_n(C)$ and $\ker\partial$ its quotient.

***Proof*:**
Step 1: Identification of the first page. The filtration has stages $F_0 = A_\bullet$ and $F_1 = C_\bullet$, with $F_{-1} = 0$ and $F_p = C_\bullet$ for $p \geq 1$. The layers are $F_0/F_{-1} = A_\bullet$ and $F_1/F_0 = C_\bullet/A_\bullet$, and all other layers vanish. Since E^1 is by construction the sum of the cohomologies of the layers, $E^1 = H_\bullet(A) \oplus H_\bullet(C/A)$. The map k is the connecting homomorphism of the long exact sequence attached to the layer and j is the projection onto the layer, so on the summand $H_n(C/A)$ the composite $d_1 = j \circ k$ is the connecting homomorphism ∂ followed by the identification of $H_{n-1}(A)$ with the layer at filtration level zero, which is the identity. Hence $d_1 = \partial$ on that summand, and $d_1 = 0$ on $H_\bullet(A)$ because k vanishes there (its target is the cohomology of a layer at filtration level -1, which is zero).

Step 2: Degeneration after the second page. By theorem 4.1.7 the differential d_r has bidegree $(-r, r-1)$, so it decreases the filtration index by r. Only the filtration indices 0 and 1 carry nonzero terms, so for $r \geq 2$ the source and target of d_r cannot both be nonzero: a nonzero source at index 1 would have target at index $1 - r \leq -1$, and a source at index 0 would have target at index $-r < 0$. Hence $d_r = 0$ for $r \geq 2$ and the pages stabilize at E^2.

Step 3: Identification of the limit. By Step 1 the second page is the cohomology of $d_1 = \partial$, which is $\ker\partial$ in filtration index 1 and $\operatorname{coker}\partial$ in filtration index 0. The long exact sequence of theorem 2.1.15 reads

$$\cdots \to H_n(A) \xrightarrow{\iota_*} H_n(C) \xrightarrow{\pi_*} H_n(C/A) \xrightarrow{\partial} H_{n-1}(A) \to \cdots$$

Exactness at $H_n(C)$ gives $\mathrm{im}\iota_* = \ker \pi_*$, and exactness at $H_n(A)$ identifies $\mathrm{im}\iota_* \cong H_n(A)/\mathrm{im}\partial = \mathrm{coker}\partial$ in degree n; exactness at $H_n(C/A)$ identifies the quotient $H_n(C)/\mathrm{im}\iota_* \cong \mathrm{im}\pi_* = \ker \partial$. These two are precisely the graded pieces of $H_n(C)$ for the filtration induced by $A_\bullet \subseteq C_\bullet$, and they agree with the two summands of E^2 computed above, as we sought to show.

The proposition is the sanity check for the whole apparatus: in the one case where the answer is already known, the spectral sequence returns exactly the long exact sequence and nothing more. It also isolates what the machine does *not* do. The second page records $\ker \partial$ and $\mathrm{coker}\partial$ as separate graded pieces, but the extension

$$0 \to \mathrm{coker}\partial \to H_n(C) \to \ker \partial \to 0$$

is not part of the output. A spectral sequence delivers the associated graded of the answer, never the answer itself; recovering the latter requires solving the extension problems left behind. This is a permanent feature, not a defect of the two-step case, and it is stated in general in theorem 4.2.6.

A worked instance makes the bookkeeping concrete.

Example 4.1.9: The Spectral Sequence of a Filtered Complex of Abelian Groups
Let $C_\bullet$ be the complex of abelian groups concentrated in degrees 1 and 0,

$$C_1 = \mathbb{Z} \xrightarrow{\cdot 2} C_0 = \mathbb{Z}$$

and let $A_\bullet \subseteq C_\bullet$ be the subcomplex with $A_1 = 0$ and $A_0 = 2\mathbb{Z}$. This is a subcomplex because $\partial(A_1) = 0 \subseteq A_0$. Filter by $F_0 = A_\bullet$ and $F_1 = C_\bullet$.

The cohomology of $C_\bullet$ itself is immediate: $H_1(C) = \ker(\cdot 2) = 0$ and $H_0(C) = \mathbb{Z}/2\mathbb{Z}$. The spectral sequence recomputes this from the two layers. The layer at level 0 is $A_\bullet$, with $H_0(A) = 2\mathbb{Z} \cong \mathbb{Z}$ and $H_1(A) = 0$. The layer at level 1 is $C_\bullet/A_\bullet$, which is $\mathbb{Z}$ in degree 1 and $\mathbb{Z}/2\mathbb{Z}$ in degree 0, with induced differential $1 \mapsto 2 \bmod 2 = 0$; so $H_1(C/A) = \mathbb{Z}$ and $H_0(C/A) = \mathbb{Z}/2\mathbb{Z}$. Indexing by $E^1_{p,q} = H_{p+q}(F_p/F_{p-1})$, the nonzero entries of the first page are

$$E^1_{0,0} = H_0(A) \cong \mathbb{Z}, \qquad E^1_{1,0} = H_1(C/A) \cong \mathbb{Z}, \qquad E^1_{1,-1} = H_0(C/A) \cong \mathbb{Z}/2\mathbb{Z}$$

The only possibly nonzero differential on this page is $d_1 \colon E^1_{1,0} \to E^1_{0,0}$, the connecting homomorphism $H_1(C/A) \to H_0(A)$. Compute it on the generator: the class of $1 \in (C/A)_1$ lifts to $1 \in C_1$, whose boundary is $2 \in C_0$, and 2 is the generator of $A_0 = 2\mathbb{Z}$. So d_1 sends a generator to a generator and is an isomorphism $\mathbb{Z} \to \mathbb{Z}$.

The second page therefore has $E^2_{1,0} = E^2_{0,0} = 0$, while $E^2_{1,-1} = \mathbb{Z}/2\mathbb{Z}$ survives untouched (the differential out of the spot $(1,-1)$ lands in $E^1_{0,-1} = H_{-1}(A) = 0$, and the one into it comes from $E^1_{2,-1} = 0$). By proposition 4.1.8 the pages stabilize here. Reading off total degrees: the diagonal $p+q = 1$ is entirely zero, giving $H_1(C) = 0$, and the diagonal $p+q = 0$ contains the single group $E^\infty_{1,-1} = \mathbb{Z}/2\mathbb{Z}$, giving $H_0(C) = \mathbb{Z}/2\mathbb{Z}$. Both agree with the direct computation, and because each diagonal contains at most one nonzero group there is no extension problem to solve.

Not every exact couple comes from a filtration. A single short exact sequence of complexes, read the other way round, produces one whose iteration does not terminate and whose limit is a statement about torsion.

Example 4.1.10: The Bockstein Exact Couple

Let p be a prime and $C_\bullet$ a complex of free abelian groups. Multiplication by p is injective on each C_n, so there is a short exact sequence of complexes

$$0 \to C_\bullet \xrightarrow{\cdot p} C_\bullet \to C_\bullet/pC_\bullet \to 0$$

Its long exact sequence in cohomology is exactly an exact couple, with

$$D = H_\bullet(C), \qquad E = H_\bullet(C/pC), \qquad i = (\cdot p), \qquad j = \text{reduction}, \qquad k = \partial$$

The differential $d_1 = j \circ k$ is the *Bockstein homomorphism*. By theorem 4.1.7 the r-th auxiliary object is $D^{r+1} = i^r(D) = p^r H_\bullet(C)$, so passing to the derived couple repeatedly divides out by higher and higher powers of p. When $H_\bullet(C)$ is finitely generated, an element of p-power torsion is annihilated by p^r for some r and therefore disappears from D^{r+1}, while a free generator survives every stage; the limit page is $\big(H_\bullet(C)/\text{torsion}\big) \otimes \mathbb{F}_p$.

The smallest case exhibits the mechanism. Take $C_1 = \mathbb{Z} \xrightarrow{\cdot p} C_0 = \mathbb{Z}$, so that $H_1(C) = 0$ and $H_0(C) = \mathbb{Z}/p\mathbb{Z}$ is pure torsion. Then $C_\bullet/pC_\bullet$ has zero differential, so $E^1 = H_\bullet(C/pC)$ is $\mathbb{F}_p$ in degrees 1 and 0. The connecting map k sends the degree-one generator to the class of 1 in $H_0(C) = \mathbb{Z}/p\mathbb{Z}$ (lift $\bar{1}$ to $1 \in C_1$, take the boundary p, and divide by p), and j sends that class to a generator of $H_0(C/pC) = \mathbb{F}_p$. So d_1 is an isomorphism $E^1_1 \to E^1_0$, giving $E^2 = 0 = E^\infty$. This matches the general description: $H_\bullet(C)$ is entirely torsion, so $(H_\bullet(C)/\text{torsion}) \otimes \mathbb{F}_p = 0$.

Exercise 4.1.1

1. Let (D, E, i, j, k) be an exact couple in which i is an isomorphism. Show that $E = 0$. Deduce that every page of the associated spectral sequence vanishes.
2. Let (D, E, i, j, k) be an exact couple. Show that $d = j \circ k$ vanishes if and only if $\ker i \subseteq \operatorname{im} i$. Then exhibit an exact couple with $d = 0$ but $k \neq 0$, so that a vanishing differential does not force the connecting map to vanish. (Take $D = \mathbb{Z}/4\mathbb{Z}$ and i multiplication by 2.)
3. Verify directly from definition 4.1.4 that the derived couple of the derived couple has $D'' = i^2(D)$, and that i'' is the restriction of i to $i^2(D)$, without appealing to theorem 4.1.7.
4. Let $C_\bullet$ be the complex $C_1 = \mathbb{Z} \xrightarrow{\cdot 6} C_0 = \mathbb{Z}$ filtered by $F_0 = A_\bullet$ with $A_1 = 0$, $A_0 = 3\mathbb{Z}$, and $F_1 = C_\bullet$. Compute the first page, the differential d_1, and the limit page, and check the answer against a direct computation of $H_\bullet(C)$.
5. In the Bockstein couple of example 4.1.10, take $C_\bullet$ to be $C_1 = \mathbb{Z} \xrightarrow{\cdot p^2} C_0 = \mathbb{Z}$. Show that $d_1 = 0$ but $d_2 \neq 0$, and that $E^3 = E^\infty = 0$. Explain how this illustrates the statement that d_r detects gluing spanning r stages.
6. Let $\mathcal{A}$ be an abelian category and (D, E, i, j, k) an exact couple with $E = 0$. Show that i is an isomorphism, so that the converse of exercise 4.1.1 (1) holds.

4.2 The Spectral Sequence of a Filtered Complex

The exact couples of section 4.1 were treated as abstract data. Their principal source is a complex equipped with a filtration, and this section carries out that construction, describes the pages explicitly enough to compute with, and proves the two theorems that make the output usable: a convergence theorem saying what the limit page computes, and a comparison theorem saying when two spectral sequences agree. These are the results cited from this chapter across the series, so both are proved in the generality in which they are used, for increasing filtrations of chain complexes and, in the restatement at the end of the section, for decreasing filtrations of cochain complexes.

Definition 4.2.1: Filtered Complex

Let $\mathcal{A}$ be an abelian category. A *filtered complex* is a chain complex $C_\bullet$ in $\mathcal{A}$ together with a family of subcomplexes $\{F_pC_\bullet\}_{p\in\mathbb{Z}}$ satisfying $F_pC_\bullet \subseteq F_{p+1}C_\bullet$ for every p. The filtration is called

1. *exhaustive* if $\bigcup_p F_pC_n = C_n$ for every n;
2. *bounded below* if for each n there is an $s(n)$ with $F_{s(n)}C_n = 0$;
3. *bounded* if it is bounded below and for each n there is a $t(n)$ with $F_{t(n)}C_n = C_n$.

The *associated graded* of the filtration is the complex $\mathrm{gr}_pC_\bullet = F_pC_\bullet/F_{p-1}C_\bullet$, and the filtration induced on homology is

$$F_pH_n(C) = \mathrm{im}\big(H_n(F_pC) \to H_n(C)\big)$$

Each inclusion $F_{p-1}C_\bullet \hookrightarrow F_pC_\bullet$ is a subcomplex inclusion in the sense of definition 2.1.7, so $\mathrm{gr}_pC_\bullet$ is again a complex.

Boundedness is the hypothesis that does the work later, and it is worth seeing why before it is used. In each fixed degree n a bounded filtration passes from 0 to all of C_n in finitely many steps, so any bookkeeping indexed by the filtration terminates in that degree after finitely many stages. Without some such condition the pages of the spectral sequence need not stabilize and the limit need not compute anything.

Every filtered complex produces an exact couple in one step: the long exact sequence of each pair $(F_pC_\bullet, F_{p-1}C_\bullet)$ supplies the three maps, and the filtration index supplies the bigrading.

Theorem 4.2.2: The Exact Couple of a Filtered Complex

Let $(C_\bullet, F_\bullet)$ be a filtered complex. Set

$$D^1_{p,q} = H_{p+q}(F_pC), \qquad E^1_{p,q} = H_{p+q}(\mathrm{gr}_pC)$$

and let i be induced by the inclusion $F_{p-1}C \hookrightarrow F_pC$, let j be induced by the projection $F_pC \to \mathrm{gr}_pC$, and let k be the connecting homomorphism of theorem 2.1.15 applied to

$$0 \to F_{p-1}C_\bullet \to F_pC_\bullet \to \mathrm{gr}_pC_\bullet \to 0$$

Then (D^1, E^1, i, j, k) is an exact couple with

$$\deg i = (1, -1), \qquad \deg j = (0, 0), \qquad \deg k = (-1, 0)$$

Consequently theorem 4.1.7 yields a spectral sequence with first page $E^1_{p,q} = H_{p+q}(\mathrm{gr}_pC)$ and differentials $d_r\colon E^r_{p,q} \to E^r_{p-r,\,q+r-1}$.
It is convenient to record one page earlier. Set

$$E^0_{p,q} := \mathrm{gr}_pC_{p+q}$$

with d_0 the differential induced by ∂ on the associated graded; then $E^1 = H(E^0, d_0)$, which is the identification just made, and the sequence is indexed from $r = 0$.

***Proof*:**
Step 1: Exactness. The displayed short exact sequence of complexes gives, by theorem 2.1.15, the long exact sequence

$$\cdots \to H_n(F_{p-1}C) \xrightarrow{i} H_n(F_pC) \xrightarrow{j} H_n(\mathrm{gr}_pC) \xrightarrow{k} H_{n-1}(F_{p-1}C) \to \cdots$$

Assembling these over all p and all n, the three maps i, j, k are defined on all of D^1 and E^1, and exactness of the triangle at each corner is exactness of the long exact sequences at the three types of term. This is exactly definition 4.1.1.

Step 2: Bidegrees. Write $n = p + q$. The map i carries $H_n(F_{p-1}C) = D^1_{p-1,q+1}$ to $H_n(F_pC) = D^1_{p,q}$, since $(p-1) + (q+1) = n$; hence $\deg i = (1, -1)$. The map j carries $H_n(F_pC) = D^1_{p,q}$ to $H_n(\mathrm{gr}_pC) = E^1_{p,q}$, so $\deg j = (0, 0)$. The connecting map k carries $H_n(\mathrm{gr}_pC) = E^1_{p,q}$ to $H_{n-1}(F_{p-1}C) = D^1_{p-1,q}$, since $(p-1) + q = n - 1$; hence $\deg k = (-1, 0)$. These are the hypotheses of theorem 4.1.7, which then gives the spectral sequence and the stated bidegree of d_r, completing the proof.

The construction is quick, but the pages it produces are described only implicitly, as iterated subquotients of the first one. Computing with them, and proving anything about the limit, requires a description of E^r in terms of the original complex. The right notion is that of a chain which is a cycle only up to a given depth in the filtration.

Definition 4.2.3: Approximate Cycles

Let $(C_\bullet, F_\bullet)$ be a filtered complex and write $n = p + q$. For $r \geq 0$ set

$$Z^r_{p,q} = \{\, x \in F_p C_n \;:\; \partial x \in F_{p-r} C_{n-1} \,\}$$

the object of chains at filtration level p whose boundary drops r levels. By convention $Z^{-1}_{p,q} = F_p C_n$.

The objects $Z^r_{p,q}$ decrease in r, and their intersection over all r is, when the filtration is bounded below, the cycles of $C_\bullet$ lying in $F_p C_n$. An element of $Z^r_{p,q}$ is a class that survives to the r-th page: it is a cycle in the associated graded to accuracy r, and its failure to be a cycle of $C_\bullet$ is what d_r records.

Proposition 4.2.4: Explicit Description of the Pages

Let $(C_\bullet, F_\bullet)$ be a filtered complex and let $\{(E^r, d_r)\}$ be the spectral sequence of theorem 4.2.2. For every $r \geq 0$ there are isomorphisms

$$E^r_{p,q} \cong \frac{Z^r_{p,q}}{Z^{r-1}_{p-1,q+1} + \partial Z^{r-1}_{p+r-1,\,q-r+2}}$$

under which the differential d_r is induced by the boundary map ∂ of $C_\bullet$.

***Proof*:**
Write $\mathcal{E}^r_{p,q}$ for the right-hand side and $n = p + q$. The proof is an induction on r; the inductive statement is that $E^r \cong \mathcal{E}^r$ compatibly with the differentials, the differential on $\mathcal{E}^r$ being the one induced by ∂.

Step 1: The map induced by ∂ is a differential on $\mathcal{E}^r$. If $x \in Z^r_{p,q}$ then $\partial x \in F_{p-r} C_{n-1}$ and $\partial(\partial x) = 0 \in F_{p-2r} C_{n-2}$, so $\partial x \in Z^r_{p-r,\,q+r-1}$. Hence ∂ carries $Z^r_{p,q}$ into $Z^r_{p-r,q+r-1}$, and it carries the denominator at (p, q) into the denominator at $(p-r, q+r-1)$: the summand $Z^{r-1}_{p-1,q+1}$ is sent to $\partial Z^{r-1}_{p-1,q+1}$, which is literally the second summand of the target denominator, and the summand $\partial Z^{r-1}_{p+r-1,q-r+2}$ is sent to zero because $\partial\partial = 0$. So ∂ induces a map $\mathcal{E}^r_{p,q} \to \mathcal{E}^r_{p-r,q+r-1}$ squaring to zero.

Step 2: The base case $r = 0$. With the convention $Z^{-1}_{p,q} = F_p C_n$, the numerator is $Z^0_{p,q} = F_p C_n$ and the denominator is $F_{p-1} C_n + \partial F_{p-1} C_{n+1} = F_{p-1} C_n$, since $\partial F_{p-1} C_{n+1} \subseteq F_{p-1} C_n$. So $\mathcal{E}^0_{p,q} = F_p C_n / F_{p-1} C_n = \mathrm{gr}_p C_n$, which is $E^0_{p,q}$, and the induced differential is the one ∂ induces on the associated graded. Taking homology recovers $E^1_{p,q} = H_n(\mathrm{gr}_p C)$, matching theorem 4.2.2; the differential $d_1 = j \circ k$ is by construction the connecting homomorphism, which on representatives is ∂.

Step 3: The inductive step. Assume $E^r \cong \mathcal{E}^r$ compatibly with differentials. Then $E^{r+1} = H(E^r, d_r) \cong H(\mathcal{E}^r, \partial)$, so it suffices to identify the latter with $\mathcal{E}^{r+1}$. Abbreviate the denominator $D = Z^{r-1}_{p-1,q+1} + \partial Z^{r-1}_{p+r-1,q-r+2}$.

First the kernel. Let $x \in Z^r_{p,q}$ represent a class killed by d_r, so that ∂x lies in the denominator at $(p-r, q+r-1)$, namely $\partial x = z + \partial w$ with $z \in Z^{r-1}_{p-r-1,q+r}$ and $w \in Z^{r-1}_{p-1,q+1}$. Then $x - w$ still lies in F_pC_n, because $w \in F_{p-1}C_n$, and

$$\partial(x-w) = \partial x - \partial w = z \in F_{p-r-1}C_{n-1}$$

so $x - w \in Z^{r+1}_{p,q}$. Since $w \in D$, the classes of x and $x-w$ agree in $\mathcal{E}^r_{p,q}$. Hence $\ker d_r = (Z^{r+1}_{p,q} + D)/D$.

Next the image. A class in $\mathcal{E}^r_{p+r,q-r+1}$ is represented by some $y \in Z^r_{p+r,q-r+1}$, and d_r sends it to the class of ∂y. Hence $\mathrm{im} d_r = (\partial Z^r_{p+r,q-r+1} + D)/D$, and therefore

$$H(\mathcal{E}^r, \partial)_{p,q} = \frac{Z^{r+1}_{p,q} + D}{\partial Z^r_{p+r,q-r+1} + D} \cong \frac{Z^{r+1}_{p,q}}{Z^{r+1}_{p,q} \cap \left(\partial Z^r_{p+r,q-r+1} + D\right)}$$

the isomorphism being the second isomorphism theorem.

It remains to show that the intersection appearing there equals $Z^r_{p-1,q+1} + \partial Z^r_{p+r,q-r+1}$, which is exactly the denominator of $\mathcal{E}^{r+1}_{p,q}$. For the inclusion $\supseteq$, note $Z^r_{p-1,q+1} \subseteq Z^{r+1}_{p,q}$ (an element lies in $F_{p-1} \subseteq F_p$ and its boundary lies in F_{p-1-r}) and $Z^r_{p-1,q+1} \subseteq Z^{r-1}_{p-1,q+1} \subseteq D$; while $\partial Z^r_{p+r,q-r+1} \subseteq Z^{r+1}_{p,q}$ because such a boundary lies in F_pC_n and is itself a cycle. For the inclusion $\subseteq$, let $x \in Z^{r+1}_{p,q}$ be written as $x = \partial y + z + \partial w$ with $y \in Z^r_{p+r,q-r+1}$, $z \in Z^{r-1}_{p-1,q+1}$ and $w \in Z^{r-1}_{p+r-1,q-r+2}$. Then $z = x - \partial y - \partial w$ lies in $F_{p-1}C_n$ and $\partial z = \partial x \in F_{p-r-1}C_{n-1}$, so $z \in Z^r_{p-1,q+1}$. Moreover $w \in Z^{r-1}_{p+r-1,q-r+2}$ satisfies $w \in F_{p+r-1} \subseteq F_{p+r}$ and $\partial w \in F_{(p+r-1)-(r-1)} = F_p$, so $w \in Z^r_{p+r,q-r+1}$ and hence $\partial w \in \partial Z^r_{p+r,q-r+1}$. Therefore $x = z + \partial(y+w) \in Z^r_{p-1,q+1} + \partial Z^r_{p+r,q-r+1}$, as required.

Substituting gives $H(\mathcal{E}^r, \partial)_{p,q} \cong \mathcal{E}^{r+1}_{p,q}$, which closes the induction and completes the proof.

The formula repays reading slowly, because it makes the interpretive claim of section 4.1 precise. The numerator at page r consists of the chains whose boundary drops r filtration levels, and the denominator discards those already accounted for one level down together with the boundaries of chains that drop $r-1$ levels. Passing from page r to page $r+1$ tightens the requirement on the boundary by one level. A class that survives to page r and dies there is one that could be corrected to a cycle modulo F_{p-r} but not modulo F_{p-r-1}, which is the precise sense in which d_r detects gluing across r layers.

With the pages described, the limit can be identified. This is the theorem the rest of the series cites from this section.

Definition 4.2.5: Convergence of a Spectral Sequence

A bigraded spectral sequence $\{(E^r, d_r)\}$ *degenerates* at the spot (p, q) if the differentials into and out of that spot vanish for all sufficiently large r; the common value of $E^r_{p,q}$ for large r is then written $E^\infty_{p,q}$. The spectral sequence *converges* to a graded object $H_\bullet$ equipped with a filtration $F_\bullet H_\bullet$, written

$$E^1_{p,q} \implies H_{p+q}$$

if it degenerates at every spot, the filtration $F_\bullet H_n$ is bounded for each n (that is $F_p H_n = 0$ for p small and $F_p H_n = H_n$ for p large), and $E^\infty_{p,q} \cong F_p H_{p+q}/F_{p-1} H_{p+q}$ for all p, q. The boundedness clause is what lets a vanishing family of graded pieces be read back as a vanishing abutment; without it the graded pieces constrain only the successive quotients.

Theorem 4.2.6: Convergence for a Bounded Filtration

Let $(C_\bullet, F_\bullet)$ be a filtered complex whose filtration is bounded in the sense of definition 4.2.1. Then the spectral sequence of theorem 4.2.2 converges to $H_\bullet(C)$: it degenerates at every spot, and

$$E^\infty_{p,q} \cong \frac{F_p H_{p+q}(C)}{F_{p-1} H_{p+q}(C)}$$

for the filtration $F_p H_n(C) = \operatorname{im}(H_n(F_p C) \to H_n(C))$ of definition 4.2.1. Moreover that filtration of $H_n(C)$ is itself bounded, so $H_n(C)$ has a finite filtration whose graded pieces are the terms of the limit page on the diagonal $p + q = n$.

***Proof*:**
Fix n and let $s(m), t(m)$ be the bounds of definition 4.2.1 in degree m.

Step 1: Degeneration. Fix a spot (p, q) with $p + q = n$. The differential out of it lands in $E^r_{p-r,q+r-1}$, a subquotient of $E^1_{p-r,q+r-1} = H_{n-1}(\mathrm{gr}_{p-r} C)$. Once r is large enough that $p-r \le s(n-1)$, the complex $\mathrm{gr}_{p-r} C$ vanishes in degree $n-1$, so the target is zero. The differential into (p, q) comes from $E^r_{p+r,q-r+1}$, a subquotient of $H_{n+1}(\mathrm{gr}_{p+r} C)$; once $p + r - 1 \ge t(n + 1)$ one has $F_{p+r-1} C_{n+1} = F_{p+r} C_{n+1} = C_{n+1}$, so $\mathrm{gr}_{p+r} C$ vanishes in degree $n + 1$ and the source is zero. Both hold for all large r, so $E^{r+1}_{p,q} = E^r_{p,q}$ eventually and $E^\infty_{p,q}$ is defined.

Step 2: The large-r value of the explicit description. Take r large enough that $F_{p-r} C_{n-1} = 0$ and $F_{p+r-1} C_{n+1} = C_{n+1}$, which is possible by boundedness. Write $Z_n = \ker(\partial \colon C_n \to C_{n-1})$ and $B_n = \operatorname{im}(\partial \colon C_{n+1} \to C_n)$. Then:

1. $Z^r_{p,q} = \{x \in F_p C_n : \partial x \in F_{p-r} C_{n-1} = 0\} = Z_n \cap F_p C_n$;
2. $Z^{r-1}_{p-1,q+1} = \{x \in F_{p-1} C_n : \partial x \in F_{p-r} C_{n-1} = 0\} = Z_n \cap F_{p-1} C_n$;
3. $\partial Z^{r-1}_{p+r-1,q-r+2} = \partial\{y \in F_{p+r-1} C_{n+1} : \partial y \in F_p C_n\} = \partial(C_{n+1}) \cap F_p C_n = B_n \cap F_p C_n$, using $F_{p+r-1} C_{n+1} = C_{n+1}$.

Substituting into proposition 4.2.4,

$$E^\infty_{p,q} \cong \frac{Z_n \cap F_p C_n}{(Z_n \cap F_{p-1} C_n) + (B_n \cap F_p C_n)}$$

Step 3: The same object is the graded piece of homology. By definition $F_p H_n(C)$ is the image of $H_n(F_p C)$ in $H_n(C)$, that is

$$F_p H_n(C) = \frac{(Z_n \cap F_p C_n) + B_n}{B_n} \cong \frac{Z_n \cap F_p C_n}{B_n \cap F_p C_n}$$

the isomorphism being the second isomorphism theorem. Taking the quotient by $F_{p-1} H_n(C)$, which under this identification is the image of $Z_n \cap F_{p-1} C_n$, gives

$$\frac{F_p H_n(C)}{F_{p-1} H_n(C)} \cong \frac{Z_n \cap F_p C_n}{(Z_n \cap F_{p-1} C_n) + (B_n \cap F_p C_n)}$$

which is the object computed in Step 2. Hence $E^\infty_{p,q} \cong \mathrm{gr}_p H_n(C)$.

Both sides were computed as subquotients of $F_p C_n$ by the same formula, and the isomorphism between them is induced by the identity on representatives. A morphism of filtered complexes carries representatives to representatives and preserves every object appearing in that formula, so the identification is natural in $(C_\bullet, F_\bullet)$; this is what theorem 4.2.9 uses when it compares the two filtrations of homology.

Step 4: The induced filtration is bounded. If $p \leq s(n)$ then $F_p C_n = 0$, so $H_n(F_p C) = 0$ and $F_p H_n(C) = 0$. If $p \geq t(n)$ then $F_p C_n = C_n$; taking p also at least $t(n+1)$ makes $F_p C_{n+1} = C_{n+1}$ as well, so every cycle and every boundary of degree n already lies at filtration level p and $F_p H_n(C) = H_n(C)$. Hence the filtration of $H_n(C)$ passes from 0 to everything in finitely many steps, completing the proof.

Two features of the statement deserve emphasis, both of them permanent. The limit page computes the *associated graded* of $H_n(C)$, not $H_n(C)$ itself; recovering the latter means solving the extension problems between successive graded pieces, exactly as in the two-step case of proposition 4.1.8. And convergence is a theorem about the filtration, not about the complex: boundedness is what forces the pages to stabilize, and without it the limit page can fail to compute anything.

The applications elsewhere in the series are stated for cochain complexes with decreasing filtrations, so the translation is recorded explicitly rather than left to the reader.

Corollary 4.2.7: Convergence for a Decreasing Filtration of a Cochain Complex

Let $(C^\bullet, d)$ be a cochain complex with a decreasing filtration $C^\bullet = F^0C^\bullet \supseteq F^1C^\bullet \supseteq \cdots$ compatible with d, that is $d(F^pC^n) \subseteq F^pC^{n+1}$, and suppose the filtration is exhaustive and bounded, so that for each n there is a P with $F^PC^n = 0$. Then there is a spectral sequence with

$$E_0^{p,q} = \frac{F^pC^{p+q}}{F^{p+1}C^{p+q}}, \qquad d_r\colon E_r^{p,q} \to E_r^{p+r,\,q-r+1}$$

each page the cohomology of the one before it, converging to $H^{p+q}(C)$:

$$E_1^{p,q} \implies H^{p+q}(C), \qquad E_\infty^{p,q} \cong \operatorname{gr}_F^p H^{p+q}(C) = \frac{F^pH^{p+q}(C)}{F^{p+1}H^{p+q}(C)}$$

where $F^pH^n(C) = \operatorname{im}\big(H^n(F^pC^\bullet) \to H^n(C^\bullet)\big)$.

***Proof*:**
Set $C_n = C^{-n}$ and $F_pC_\bullet = F^{-p}C^\bullet$. The differential d raises cochain degree by one and therefore lowers chain degree by one, so $(C_\bullet, F_\bullet)$ is a filtered chain complex in the sense of definition 4.2.1: the filtration is increasing because $F^\bullet$ is decreasing, it is exhaustive because $F^\bullet$ is, and it is bounded because for each n there is a P with $F^PC^n = 0$, giving $F_{-P}C_{-n} = 0$, while $F^0C^n = C^n$ gives $F_0C_{-n} = C_{-n}$.

Theorem 4.2.6 applies and gives a convergent spectral sequence with $E^0_{p,q} = \operatorname{gr}_p C_{p+q}$ and d_r of bidegree $(-r, r-1)$. Re-indexing by $E_r^{p,q} = E^r_{-p,-q}$, as in definition 4.1.6, turns the bidegree of d_r into $(r, -r+1)$ and turns the graded piece $\operatorname{gr}_{-p}H_{-p-q}(C_\bullet)$ into $\operatorname{gr}_F^p H^{p+q}(C^\bullet)$, which is the statement, as we sought to show.

The same re-indexing carries every other result of this section across, and it is worth saying so once rather than hedging at each use, since section 4.3 onward works exclusively with cochain complexes.

Corollary 4.2.8: The Cochain Form of the Section

Let $\mathcal{T}$ denote the translation $C_n = C^{-n}$, $F_p = F^{-p}$, $E_r^{p,q} = E^r_{-p,-q}$ of corollary 4.2.7. Under $\mathcal{T}$:

1. Theorem 4.2.9 holds for morphisms of cochain complexes with bounded decreasing filtrations: a morphism inducing an isomorphism on E_{r_0} for some r_0 induces one on E_∞ and on $H^\bullet$.

2. Definition 4.2.3, proposition 4.2.4 and definition 4.2.5 hold with the *bigrading* negated, the page index r unchanged. Explicitly $Z_r^{p,q} = \{x \in F^pC^{p+q} : dx \in F^{p+r}C^{p+q+1}\}$ and

$$E_r^{p,q} \cong \frac{Z_r^{p,q}}{Z_{r-1}^{p+1,q-1} + d\,Z_{r-1}^{p-r+1,\,q+r-2}}$$

***Proof*:**
$\mathcal{T}$ is an isomorphism between the category of cochain complexes with bounded decreasing filtrations and the category of chain complexes with bounded increasing filtrations, carrying morphisms to morphisms and the spectral sequence of one to the spectral sequence of the other, by the computation in the proof of corollary 4.2.7. A statement proved for one is therefore proved for the other; no hypothesis in theorem 4.2.9 or proposition 4.2.4 refers to the direction of the grading beyond boundedness, which $\mathcal{T}$ preserves, completing the proof.

The remaining tool is the one that lets a spectral sequence be recognized rather than computed: if two filtered complexes are compared by a map, and the comparison is an isomorphism on any single page, then it is an isomorphism on the abutment. This is what makes a spectral sequence an identification technique and not only a calculation.

Theorem 4.2.9: Comparison Theorem

Let $f\colon (C_\bullet, F_\bullet) \to (C'_\bullet, F'_\bullet)$ be a morphism of filtered complexes, that is a chain map with $f(F_pC_\bullet) \subseteq F'_pC'_\bullet$, and suppose both filtrations are bounded. Then f induces a morphism of spectral sequences $f^r\colon E^r_{p,q} \to E'^r_{p,q}$ commuting with the differentials, and if f^{r_0} is an isomorphism for some $r_0 \geq 0$, then:

1. f^r is an isomorphism for every $r \geq r_0$, and hence so is f^∞;
2. $f_*\colon H_n(C) \to H_n(C')$ is an isomorphism for every n.

***Proof*:**
Step 1: Functoriality. On the zeroth page f induces f^0 on $\mathrm{gr}_p C_{p+q}$ directly, and the identification $E^1 = H(E^0, d_0)$ of theorem 4.2.2 is natural in the filtered complex, so the case $r_0 = 0$ reduces to $r_0 = 1$. For the remaining pages: because f preserves the filtration, it carries the short exact sequences of theorem 4.2.2 for C to those for C', and the connecting homomorphism of theorem 2.1.15 is natural. So f induces a morphism of exact couples, and the construction of the derived couple in definition 4.1.4 is built from i, j, k alone and is therefore natural in the couple. Hence f induces f^r on every page, commuting with d_r.

Step 2: Isomorphism on all later pages. Suppose f^r is an isomorphism. Since f^r commutes with the differentials, it carries $\ker d_r$ isomorphically onto $\ker d'_r$ and $\mathrm{im} d_r$ isomorphically onto $\mathrm{im} d'_r$, so the induced map on the quotients is an isomorphism. That induced map is f^{r+1}, so the claim follows by induction from r_0.

Step 3: Isomorphism on the limit page. Both filtrations are bounded, so by Step 1 of the proof of theorem 4.2.6 each spot degenerates: there is an R depending on (p, q) with $E^r_{p,q} = E^\infty_{p,q}$ and $E'^r_{p,q} = E'^\infty_{p,q}$ for $r \geq R$. Taking $r \geq \max(R, r_0)$ and applying Step 2 gives $f^\infty_{p,q}$ an isomorphism.

Step 4: Isomorphism on homology. Fix n. By theorem 4.2.6 the filtrations $F_\bullet H_n(C)$ and $F'_\bullet H_n(C')$ are finite, and f_* respects them. Proceed by induction up the filtration. The bottom stage is zero on both sides. If f_* is an isomorphism $F_{p-1}H_n(C) \to F'_{p-1}H_n(C')$, then in the commutative diagram of short exact sequences

$$\begin{tikzcd}
0 \arrow[r] & F_{p-1}H_n(C) \arrow[r] \arrow[d] & F_pH_n(C) \arrow[r] \arrow[d] & E^{\infty}_{p,n-p} \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & F'_{p-1}H_n(C') \arrow[r] & F'_pH_n(C') \arrow[r] & E'^{\infty}_{p,n-p} \arrow[r] & 0
\end{tikzcd}$$

the outer vertical maps are isomorphisms, the left one by the inductive hypothesis and the right one by Step 3. Each row is already a five-term exact sequence, so lemma 2.1.17 applies directly, with the outer maps $0 \to 0$ trivially epi and mono, and forces the middle map to be an isomorphism. Since the filtration is finite and exhausts $H_n(C)$, finitely many steps give that $f_*: H_n(C) \to H_n(C')$ is an isomorphism, completing the proof.

A first application of the machinery, and the one that recurs in section 4.5, is the crudest filtration a complex carries: truncate it.

Example 4.2.10: The Stupid Filtration of a Cochain Complex

Let $(C^\bullet, d)$ be a cochain complex concentrated in degrees $0 \le n \le N$, and define the *stupid filtration* (the *filtration bête*) by

$$F^pC^n = \begin{cases} C^n & n \ge p \\ 0 & n < p \end{cases}$$

This is a decreasing filtration compatible with d, since d raises degree, and it is exhaustive and bounded because $F^0C^\bullet = C^\bullet$ and $F^{N+1}C^\bullet = 0$. The graded pieces are $\mathrm{gr}^p_F C^n = C^n$ if $n = p$ and 0 otherwise, so the zeroth page has a single nonzero entry in each column,

$$E_0^{p,q} = \begin{cases} C^p & q = 0 \\ 0 & q \neq 0 \end{cases}$$

The differential d_0 has bidegree $(0, 1)$ and so runs between entries with q differing by one; every such pair has a zero member, so $d_0 = 0$ and $E_1 = E_0$. The differential d_1 has bidegree $(1, 0)$ and is the map induced by d, namely $d: C^p \to C^{p+1}$. Hence

$$E_2^{p,q} = \begin{cases} H^p(C^\bullet) & q = 0 \\ 0 & q \neq 0 \end{cases}$$

and every later differential has target or source zero, so $E_2 = E_\infty$. By corollary 4.2.7 the sequence converges to $H^{p+q}(C^\bullet)$, and indeed the single surviving entry on the diagonal $p+q = n$ is $H^n(C^\bullet)$, with no extension problem. The stupid filtration therefore returns the cohomology of the complex and nothing more, which is the correct answer and a useful check on the indexing conventions. It becomes informative only when the entries C^p are themselves replaced by objects with cohomology of their own, which is what section 4.5 does.

Exercise 4.2.1

1. Let $(C_\bullet, F_\bullet)$ be a filtered complex whose filtration has exactly three nonzero stages, $0 = F_{-1} \subseteq F_0 \subseteq F_1 \subseteq F_2 = C_\bullet$. Show that $d_r = 0$ for $r \ge 3$, so that $E^3 = E^\infty$. Generalize to a filtration

with N stages.

2. Show that the filtration induced on $H_n(C)$ in definition 4.2.1 is increasing, and that a morphism of filtered complexes $f\colon (C_\bullet, F_\bullet) \to (C'_\bullet, F'_\bullet)$ satisfies $f_*\big(F_p H_n(C)\big) \subseteq F'_p H_n(C')$. (This is the compatibility used in Step 4 of the proof of theorem 4.2.9.)

3. A spectral sequence *degenerates at* E^r if $d_s = 0$ for all $s \geq r$. Show that if the spectral sequence of a bounded filtered complex degenerates at E^1, then for every n the object $H_n(C)$ has a finite filtration with graded pieces $H_n(\mathrm{gr}_p C)$. Show by example that this does not force $H_n(C) \cong \bigoplus_p H_n(\mathrm{gr}_p C)$.

4. Verify proposition 4.2.4 directly at $r = 1$ by computing both sides for the filtered complex of example 4.1.9, and confirm that the differential induced by ∂ is the map computed there.

5. Let $f\colon (C_\bullet, F_\bullet) \to (C'_\bullet, F'_\bullet)$ be a morphism of bounded filtered complexes which is an isomorphism on E^2 but not on E^1. Show that f_* is still an isomorphism on homology, and construct such an f with $E^1 \neq E'^1$.

4.3 The Two Spectral Sequences of a Double Complex

A double complex carries two filtrations of its total complex, one by columns and one by rows, and section 4.2 converts each into a spectral sequence. Both converge to the homology of the same total complex, and that coincidence is the source of most of the identifications homological algebra makes: whenever one filtration degenerates for an easy reason and the other computes something of interest, the two are forced to agree. Section 2.4 promised this machinery twice, once for the cohomology of double complexes and once for the balancing of Tor and Ext, and this section delivers both.

Throughout, $A = \{A_{p,q}\}$ is a double complex in the sense of definition 2.4.1, with horizontal differential $d^h\colon A_{p,q} \to A_{p-1,q}$ and vertical differential $\delta^v\colon A_{p,q} \to A_{p,q-1}$ satisfying $d^h d^h = 0$, $\delta^v\delta^v = 0$ and $d^h\delta^v + \delta^v d^h = 0$. The total complex is $\mathrm{Tot}(A)_n = \bigoplus_{p+q=n} A_{p,q}$ with differential $d = d^h + \delta^v$, as in proposition 2.4.2; anticommutativity is exactly what makes $d \circ d = 0$. Write

$$H^h_p(A)_\bullet, \qquad H^v_q(A)_\bullet$$

for the homology of the rows and of the columns respectively, each of which is again a complex under the differential the other direction induces.

Definition 4.3.1: The Column and Row Filtrations

Let A be a double complex. The *column filtration* and the *row filtration* of $\mathrm{Tot}(A)$ are

$$^{\mathrm{I}}F_p\mathrm{Tot}(A)_n = \bigoplus_{\substack{p'+q'=n \\ p' \leq p}} A_{p',q'}, \qquad ^{\mathrm{II}}F_p\mathrm{Tot}(A)_n = \bigoplus_{\substack{p'+q'=n \\ q' \leq p}} A_{p',q'}$$

The double complex is *first-quadrant* if $A_{p,q} = 0$ whenever $p < 0$ or $q < 0$.

Both are filtrations by subcomplexes, and for the reason that each differential moves in only one of the two directions: d^h lowers p' and leaves q' alone, while δ^v lowers q' and leaves p' alone. So d carries $^{\mathrm{I}}F_p$ into itself, and likewise for $^{\mathrm{II}}F_p$. The two differ only in which coordinate is being watched, which is why the same total complex acquires two different, and generally inequivalent, approximations to its homology.

Theorem 4.3.2: The Two Spectral Sequences of a Double Complex

Let A be a first-quadrant double complex. Then the column and row filtrations of $\mathrm{Tot}(A)$ give two spectral sequences, both converging to $H_\bullet(\mathrm{Tot}(A))$, with second pages

$$^{\mathrm{I}}E^2_{p,q} = H^h_p\big(H^v_q(A)\big) \implies H_{p+q}(\mathrm{Tot}(A))$$

$$^{\mathrm{II}}E^2_{p,q} = H^v_p\big(H^h_q(A)\big) \implies H_{p+q}(\mathrm{Tot}(A))$$

***Proof*:**
Step 1: Boundedness. Fix n. Since A is first-quadrant, $A_{p',q'} = 0$ unless $0 \le p' \le n$ on the diagonal $p' + q' = n$. Hence $^{\mathrm{I}}F_{-1}\mathrm{Tot}(A)_n = 0$ and $^{\mathrm{I}}F_n\mathrm{Tot}(A)_n = \mathrm{Tot}(A)_n$, so the column filtration is bounded in the sense of definition 4.2.1; the same argument applies to the row filtration. Theorem 4.2.6 therefore applies to each and gives convergence to $H_\bullet(\mathrm{Tot}(A))$ once the pages are identified.

Step 2: The zeroth and first pages of the column filtration. By definition 4.2.1 the graded piece is

$$\mathrm{gr}_p\mathrm{Tot}(A)_n = \frac{^{\mathrm{I}}F_p\mathrm{Tot}(A)_n}{^{\mathrm{I}}F_{p-1}\mathrm{Tot}(A)_n} = A_{p,\,n-p}$$

so $^{\mathrm{I}}E^0_{p,q} = A_{p,q}$. The induced differential is the part of $d = d^h + \delta^v$ that preserves the filtration index p. The summand d^h lowers p, so it vanishes on the graded piece; the summand δ^v preserves p. Hence $d^0 = \delta^v$ and

$$^{\mathrm{I}}E^1_{p,q} = H_q\big(A_{p,\bullet},\, \delta^v\big) = H^v_q(A)_p$$

the homology of the p-th column.

Step 3: The second page of the column filtration. By theorem 4.2.2 the differential d^1 has bidegree $(-1, 0)$, so it runs $^{\mathrm{I}}E^1_{p,q} \to {}^{\mathrm{I}}E^1_{p-1,q}$; by proposition 4.2.4 it is induced by the differential of $\mathrm{Tot}(A)$ on representatives. A representative of a class in $^{\mathrm{I}}E^1_{p,q}$ is an element of $A_{p,q}$ killed by δ^v, and applying $d = d^h + \delta^v$ to it gives d^h of it, an element of $A_{p-1,q}$. So d^1 is the map induced by d^h on vertical homology, and

$$^{\mathrm{I}}E^2_{p,q} = H_p\big(H^v_q(A)_\bullet,\, d^h\big) = H^h_p\big(H^v_q(A)\big)$$

Step 4: The row filtration. The argument is identical with the two coordinates exchanged. The graded piece of the row filtration at index p in total degree n is $A_{n-p,\,p}$, so $^{\mathrm{II}}E^0_{p,q} = A_{q,p}$, and the differential preserving the index p is now d^h. Hence $^{\mathrm{II}}E^1_{p,q} = H_q(A_{\bullet,p}, d^h) = H^h_q(A)_p$, and the next differential is induced by δ^v, giving $^{\mathrm{II}}E^2_{p,q} = H^v_p(H^h_q(A))$. Both converge to the homology of the same total complex by Step 1, completing the proof.

The two sequences are not related by any map; they are two different routes to one answer, and the content of the theorem is that the answer is the same. The standard way to exploit this is to arrange for one of them to collapse. A double complex whose columns are exact except in one spot, for instance, has a first page concentrated in a single row, and then every differential from the second page onward has a zero source or a zero target.

Lemma 4.3.3: Collapse Along a Single Row

Let $\{(E^r, d_r)\}$ be a spectral sequence with $E^2_{p,q} = 0$ for all $q \neq 0$. Then $E^2 = E^\infty$, and if the sequence converges to $H_\bullet$ then $H_n \cong E^2_{n,0}$ for every n.

***Proof*:**
The differential d_r has bidegree $(-r, r-1)$ by theorem 4.1.7. For $r \geq 2$ the differential out of the spot $(p, 0)$ lands in $(p-r,\ r-1)$, which has second index $r - 1 \geq 1$ and so vanishes; the differential into $(p, 0)$ comes from $(p+r,\ 1-r)$, which has negative second index and so vanishes. Every other spot is already zero. Hence $d_r = 0$ for all $r \geq 2$ and $E^2 = E^\infty$.

For the abutment, the diagonal $p + q = n$ of the limit page contains the single nonzero group $E^\infty_{n,0}$. By definition 4.2.5 the filtration of H_n is bounded and has these as graded pieces; a bounded filtration whose graded pieces vanish outside one step has exactly one nonzero step, so $H_n \cong E^\infty_{n,0} = E^2_{n,0}$, as we sought to show.

Both results are used in section 4.4 and section 4.5 for cochain double complexes, whose differentials raise indices. As with corollary 4.2.8, the translation is recorded once.

Corollary 4.3.4: The Cochain Form of the Double-Complex Sequences

Let $A^{\bullet,\bullet}$ be a cochain double complex concentrated in non-negative bidegrees, with differentials raising the two indices. Then the column and row filtrations of $\mathrm{Tot}(A)$ give two spectral sequences converging to $H^{p+q}(\mathrm{Tot}(A))$, with second pages

$$^{\mathrm{I}}E_2^{p,q} = H_h^p\big(H_v^q(A)\big), \qquad ^{\mathrm{II}}E_2^{p,q} = H_v^p\big(H_h^q(A)\big)$$

and lemma 4.3.3 holds in the form: if $E_2^{p,q} = 0$ for all $q \neq 0$ then $E_2 = E_\infty$ and $H^n \cong E_2^{n,0}$; symmetrically if $E_2^{p,q} = 0$ for all $p \neq 0$ then $H^n \cong E_2^{0,n}$.

***Proof*:**
Apply the translation $\mathcal{T}$ of corollary 4.2.8 in both indices, setting $A_{p,q} = A^{-p,-q}$. This turns a cochain double complex concentrated in non-negative bidegrees into a chain double complex concentrated in non-positive ones, which is bounded on diagonals in the sense of exercise 4.3.1 (5); theorem 4.3.2 and lemma 4.3.3 apply to it, and negating every index returns the displayed statements. The column form of the collapse is exercise 4.3.1 (4), translated the same way, completing the proof.

The lemma turns theorem 4.3.2 into an identification technique, and the standard application is the one section 2.4 deferred: that Tor may be computed by resolving either variable.

Corollary 4.3.5: Balancing Tor

Let R be a ring, M a right R-module and N a left R-module, and let $P_\bullet \to M$ and $Q_\bullet \to N$ be projective resolutions. Then

$$H_n\big(P_\bullet \otimes_R N\big) \cong H_n\big(\mathrm{Tot}(P_\bullet \otimes_R Q_\bullet)\big) \cong H_n\big(M \otimes_R Q_\bullet\big)$$

for every n. In particular $\mathrm{Tor}_n^R(M, N)$ may be computed by resolving either variable, recovering the commutative case of theorem 2.4.8 as the degenerate case of theorem 4.3.2.

***Proof*:**
Let $A_{p,q} = P_p \otimes_R Q_q$, with horizontal differential induced by that of $P_\bullet$ and vertical differential induced by that of $Q_\bullet$, the latter twisted by the sign $(-1)^p$ so that the two anticommute as definition 2.4.1 requires. Both resolutions are supported in non-negative degrees, so A is first-quadrant and theorem 4.3.2 applies.

Step 1: The column filtration. Each P_p is projective, hence a direct summand of a free module, hence flat, so the functor $P_p \otimes_R -$ is exact. Therefore it commutes with homology and

$${}^{\mathrm{I}}E^1_{p,q} = H_q\big(P_p \otimes_R Q_\bullet\big) \cong P_p \otimes_R H_q(Q_\bullet)$$

Since $Q_\bullet \to N$ is a resolution, $H_q(Q_\bullet)$ is N for $q = 0$ and zero otherwise. So the first page is concentrated in the row $q = 0$, where it is $P_p \otimes_R N$, and the second page is

$${}^{\mathrm{I}}E^2_{p,0} = H_p\big(P_\bullet \otimes_R N\big)$$

with ${}^{\mathrm{I}}E^2_{p,q} = 0$ for $q \neq 0$. By lemma 4.3.3, $H_n(\mathrm{Tot}(A)) \cong H_n(P_\bullet \otimes_R N)$.

Step 2: The row filtration. Symmetrically, each Q_q is flat, so ${}^{\mathrm{II}}E^1_{p,q} = H_q(P_\bullet \otimes_R Q_p) \cong H_q(P_\bullet) \otimes_R Q_p$, which is $M \otimes_R Q_p$ for $q = 0$ and zero otherwise. Hence ${}^{\mathrm{II}}E^2_{p,0} = H_p(M \otimes_R Q_\bullet)$ with the other rows zero, and lemma 4.3.3 gives $H_n(\mathrm{Tot}(A)) \cong H_n(M \otimes_R Q_\bullet)$.

Both computations describe the homology of the same total complex, so the two are isomorphic, completing the proof.

The argument is worth contrasting with the one given in section 2.4. There the comparison was made by hand, through an explicit quasi-isomorphism between the total complex and each of its edges. Here it is a two-line consequence of a general theorem, and the same two lines apply verbatim to Ext, to group cohomology, and to any other derived functor of a bifunctor. That is the characteristic economy of the method: the work moves once into theorem 4.2.6 and is then reused without repetition.

Exercise 4.3.1

1. Let A be a first-quadrant double complex all of whose columns are exact. Show that $\mathrm{Tot}(A)$ is acyclic. Deduce proposition 2.4.4 for first-quadrant double complexes.
2. State and prove the analogue of corollary 4.3.5 for Ext, using a projective resolution of M

and an injective resolution of N. Identify which functor plays the role of $P_p \otimes_R -$ in Step 1, and which exactness property replaces flatness.

3. Let A be the first-quadrant double complex with $A_{0,0} = A_{1,0} = \mathbb{Z}$, $A_{0,1} = A_{1,1} = \mathbb{Z}$, horizontal differentials the identity, and vertical differentials the identity, with the sign twist making it anticommute. Compute both spectral sequences and $H_\bullet(\mathrm{Tot}(A))$, and check that the two agree.

4. Show that a spectral sequence with $E^2_{p,q} = 0$ for all $p \neq 0$ also collapses, and that $H_n \cong E^2_{0,n}$. (No quadrant hypothesis is needed.) (This is the column analogue of lemma 4.3.3.)

5. The first-quadrant hypothesis in theorem 4.3.2 is used in exactly one place. Identify it, and show that the theorem still holds for any double complex that is *bounded on diagonals*, meaning that for each n only finitely many $A_{p,q}$ with $p + q = n$ are nonzero. Then show that a double complex with $A_{p,q} \neq 0$ for every $p, q \in \mathbb{Z}$ fails this condition, so that theorem 4.2.6 does not apply to either filtration.

4.4 The Grothendieck Spectral Sequence

Derived functors were built in section 2.3 one functor at a time. In practice a functor of interest is often a composite: global sections of a sheaf factor through pushforward along a map, invariants under a group factor through invariants under a normal subgroup, and global Hom factors through sheaf Hom. In each case the derived functors of the two factors are understood and the derived functors of the composite are what is wanted. The relation between them is not an equality, and it is not captured by any exact sequence; it is a spectral sequence, and it is the form in which the machinery of this chapter is most often used elsewhere in the series.

The construction needs one tool beyond section 4.3: a way to resolve a whole complex, rather than a single object, by injectives, in a manner compatible with the complex's own cohomology. That is the Cartan-Eilenberg resolution, and it is built by applying the horseshoe lemma twice.

Definition 4.4.1: Cartan-Eilenberg Resolution

Let $\mathcal{A}$ be an abelian category with enough injectives and $C^\bullet$ a cochain complex in $\mathcal{A}$ bounded below. A *Cartan-Eilenberg resolution* of $C^\bullet$ is a double complex $I^{\bullet,\bullet}$ of injective objects, bounded below, together with an augmentation $C^p \to I^{p,0}$ for each p, such that for every p the induced augmentations make each of

$$I^{p,\bullet}, \qquad B^p(I^{\bullet,\bullet}), \qquad Z^p(I^{\bullet,\bullet}), \qquad H^p(I^{\bullet,\bullet})$$

an injective resolution of C^p, of $B^p(C^\bullet)$, of $Z^p(C^\bullet)$ and of $H^p(C^\bullet)$ respectively, where B^p, Z^p and H^p are taken in the first (horizontal) direction.

The three extra conditions are what make the resolution usable: without them a levelwise injective resolution of each C^p carries no information about the cohomology of $C^\bullet$, and the whole point is that the resolution should compute that cohomology as well as the individual terms.

Theorem 4.4.2: Existence of Cartan-Eilenberg Resolutions

Every bounded-below cochain complex in an abelian category with enough injectives admits a Cartan-Eilenberg resolution.

***Proof*:**
Step 1: Choices. For each p, use that $\mathcal{A}$ has enough injectives (definition 2.2.3) to choose injective resolutions

$$B^p(C^\bullet) \to \mathcal{B}^{p,\bullet}, \qquad H^p(C^\bullet) \to \mathcal{H}^{p,\bullet}$$

Since $C^\bullet$ is bounded below, so are all of these.

Step 2: Resolving the cycles. The definition of cohomology gives a short exact sequence

$$0 \to B^p(C^\bullet) \to Z^p(C^\bullet) \to H^p(C^\bullet) \to 0$$

Its two ends are already resolved by Step 1, so the horseshoe lemma (lemma 2.3.7), in its injective form, produces an injective resolution $Z^p(C^\bullet) \to \mathcal{Z}^{p,\bullet}$ fitting into a degreewise split short exact sequence of complexes $0 \to \mathcal{B}^{p,\bullet} \to \mathcal{Z}^{p,\bullet} \to \mathcal{H}^{p,\bullet} \to 0$.

Step 3: Resolving the terms. The complex $C^\bullet$ also gives, for each p, a short exact sequence

$$0 \to Z^p(C^\bullet) \to C^p \to B^{p+1}(C^\bullet) \to 0$$

whose outer terms are resolved by $\mathcal{Z}^{p,\bullet}$ from Step 2 and $\mathcal{B}^{p+1,\bullet}$ from Step 1. Applying lemma 2.3.7 again yields an injective resolution $C^p \to I^{p,\bullet}$ with $I^{p,q} \cong \mathcal{Z}^{p,q} \oplus \mathcal{B}^{p+1,q}$ in each degree, compatibly with the two inclusions.

Step 4: The horizontal differential. Define $I^{p,q} \to I^{p+1,q}$ as the composite

$$I^{p,q} \twoheadrightarrow \mathcal{B}^{p+1,q} \hookrightarrow \mathcal{Z}^{p+1,q} \hookrightarrow I^{p+1,q}$$

using the splitting of Step 3 for the first map and the inclusion of Step 2 for the second. The composite of two consecutive such maps is zero, because the image of the first lands in $\mathcal{Z}^{p+1,q}$, which the projection to $\mathcal{B}^{p+2,q}$ annihilates. So $I^{\bullet,\bullet}$ is a double complex of injectives, after the usual sign twist making the two differentials anticommute as definition 2.4.1 requires.

Step 5: The conditions hold. By construction $I^{p,\bullet}$ is an injective resolution of C^p, which is the first condition. By Step 4 the horizontal cycles are $Z^p(I^{\bullet,q}) = \mathcal{Z}^{p,q}$ and the horizontal boundaries are $B^p(I^{\bullet,q}) = \mathcal{B}^{p,q}$, which are injective resolutions of $Z^p(C^\bullet)$ and $B^p(C^\bullet)$ by Steps 2 and 1. The quotient $\mathcal{Z}^{p,q}/\mathcal{B}^{p,q} \cong \mathcal{H}^{p,q}$ is an injective resolution of $H^p(C^\bullet)$ by Step 1 and the splitting of Step 2. The augmentations are compatible with the horizontal differential by construction, since d^h was defined through the factorization $C^p \twoheadrightarrow B^{p+1} \hookrightarrow Z^{p+1} \hookrightarrow C^{p+1}$ that the differential of $C^\bullet$ itself factors through. All four conditions of definition 4.4.1 hold, completing the proof.

Existence is only half of what the resolution is used for. The spectral sequences built from it in section 4.4 and section 4.5 are asserted to be natural, and proposition 4.5.2 compares the resolutions of two different complexes; both need to know that a morphism of complexes lifts to the resolutions. The object-level comparison theorem (lemma 2.2.13 and its dual) does not supply this, since a Cartan-Eilenberg resolution is constrained not only degreewise but also on cycles, boundaries and cohomology. The lift has to be built through the same two horseshoes that produced the resolution, and the first step is to record that the horseshoe construction is itself functorial.

Lemma 4.4.3: Functoriality of the Horseshoe Construction

Consider a morphism of short exact sequences in an abelian category with enough injectives,

$$\begin{tikzcd} 0 \arrow[r] & L \arrow[r] \arrow[d, "\lambda"] & M \arrow[r] \arrow[d, "\mu"] & N \arrow[r] \arrow[d, "\nu"] & 0 \\ 0 \arrow[r] & L' \arrow[r] & M' \arrow[r] & N' \arrow[r] & 0 \end{tikzcd}$$

and injective resolutions $I_L^\bullet, I_N^\bullet, I_{L'}^\bullet, I_{N'}^\bullet$ of the four outer objects, and injective resolutions $I_M^\bullet$, $I_{M'}^\bullet$ of M and M' sitting in *degreewise split* short exact sequences $0 \to I_L^\bullet \to I_M^\bullet \to I_N^\bullet \to 0$ and $0 \to I_{L'}^\bullet \to I_{M'}^\bullet \to I_{N'}^\bullet \to 0$ lifting the two given rows. (This is what lemma 2.3.7 produces, but the argument uses only the splitting, which is automatic whenever the outer terms are injective.) Given lifts $\lambda^\bullet\colon I_L^\bullet \to I_{L'}^\bullet$ and $\nu^\bullet\colon I_N^\bullet \to I_{N'}^\bullet$ of λ and ν, there is a lift $\mu^\bullet\colon I_M^\bullet \to I_{M'}^\bullet$ of μ compatible with both, that is making the induced diagram of resolutions commute.

***Proof*:**
By lemma 2.3.7 the middle resolutions are $I_M^q = I_L^q \oplus I_N^q$ and $I_{M'}^q = I_{L'}^q \oplus I_{N'}^q$ in each degree, with the displayed maps the inclusion and the projection. Compatibility with those two maps forces $\mu^\bullet$ to have the shape

$$\mu^q = \begin{pmatrix} \lambda^q & \sigma^q \\ 0 & \nu^q \end{pmatrix}$$

so the only freedom is the family $\sigma^q\colon I_N^q \to I_{L'}^q$, and the content of the lemma is that it can be chosen so that $\mu^\bullet$ is a chain map lifting μ.

Write the differential of $I_M^\bullet$ as $\begin{pmatrix} d_L & \tau \\ 0 & d_N \end{pmatrix}$ and that of $I_{M'}^\bullet$ as $\begin{pmatrix} d_{L'} & \tau' \\ 0 & d_{N'} \end{pmatrix}$, where τ and τ' are the off-diagonal terms the horseshoe construction produces. The chain-map condition $\mu^{q+1}d = d\mu^q$ reads, in the only entry that is not automatic,

$$\lambda^{q+1}\tau^q + \sigma^{q+1}d_N^q = d_{L'}^q\sigma^q + \tau'^q\nu^q$$

so, writing $\theta^q := d_{L'}^q\sigma^q + \tau'^q\nu^q - \lambda^{q+1}\tau^q$, the family σ must satisfy $\sigma^{q+1} \circ d_N^q = \theta^q$ for every q.

The base case. Write the augmentation of $I_M^\bullet$ through the splitting as $\varepsilon_M = (\varphi,\ \varepsilon_N\pi)^\mathsf{T}$, where $\pi\colon M \to N$ is the given epimorphism and $\varphi\colon M \to I_L^0$ restricts to ε_L on L; similarly for M'. The morphism σ^0 is not given by the hypothesis, it has to be built. Set $\theta^{-1} := \varphi'\mu - \lambda^0\varphi\colon M \to I_{L'}^0$. On the subobject L this is $\varepsilon_{L'}\lambda - \varepsilon_{L'}\lambda = 0$, so θ^{-1} kills $L = \ker\pi$ and factors as $\bar\theta \circ \pi$ for a unique $\bar\theta\colon N \to I_{L'}^0$. Since $\varepsilon_N\colon N \to I_N^0$ is a monomorphism and $I_{L'}^0$ is injective, $\bar\theta$ extends along it to $\sigma^0\colon I_N^0 \to I_{L'}^0$ with $\sigma^0\varepsilon_N\pi = \theta^{-1}$, which is the degree-zero instance of the required identity and is the only place the compatibility of μ with λ and ν is used.

The inductive step. Suppose $\sigma^{\leq q}$ has been chosen. To extend θ^q along d_N^q it suffices that θ^q vanish on $\ker d_N^q = \operatorname{im} d_N^{q-1}$, and this is a direct computation. Squaring the differential of $I_M^\bullet$ gives $d_L\tau + \tau d_N = 0$, and likewise $d_{L'}\tau' + \tau' d_{N'} = 0$; using these, the inductive hypothesis $\sigma^q d_N^{q-1} = \theta^{q-1}$, the chain-map identities for λ and ν, and $d \circ d = 0$,

$$\theta^q \circ d_N^{q-1} = \left(\lambda^{q+1} d_L^q - d_{L'}^q \lambda^q\right) \circ \tau^{q-1} = 0$$

the last equality because $\lambda^\bullet$ is a chain map. For $q = 0$ the same conclusion holds with $\ker d_N^0 = \operatorname{im} \varepsilon_N$ in place of $\operatorname{im} d_N^{-1}$: applying θ^0 to $\varepsilon_N \pi$ and using $d_L^0 \varphi + \tau^0 \varepsilon_N \pi = 0$, which is the first component of $d_M^0 \varepsilon_M = 0$, together with the base case, gives $\theta^0 \varepsilon_N \pi = (\lambda^1 d_L^0 - d_{L'}^0 \lambda^0)\varphi = 0$, and π is an epimorphism. Hence in every degree θ^q factors through $\operatorname{im} d_N^q$, and since $\operatorname{im} d_N^q \hookrightarrow I_N^{q+1}$ is a monomorphism and $I_{L'}^{q+1}$ is injective, definition 2.2.1 extends the factored morphism to $\sigma^{q+1} \colon I_N^{q+1} \to I_{L'}^{q+1}$ with $\sigma^{q+1} d_N^q = \theta^q$, as required.

The resulting $\mu^\bullet$ is a chain map by construction, is compatible with the inclusion and projection by its shape, and lifts μ because it does so in degree -1, that is on the augmentations, completing the proof.

Theorem 4.4.4: Comparison Theorem for Cartan-Eilenberg Resolutions

Let $f^\bullet \colon C^\bullet \to \widetilde{C}^\bullet$ be a morphism of bounded-below cochain complexes in an abelian category with enough injectives, and let $J^{\bullet,\bullet}$ and $\widetilde{J}^{\bullet,\bullet}$ be Cartan-Eilenberg resolutions of $C^\bullet$ and $\widetilde{C}^\bullet$. Then $f^\bullet$ lifts to a morphism of double complexes

$$\tilde{f}^{\bullet,\bullet} \colon J^{\bullet,\bullet} \to \widetilde{J}^{\bullet,\bullet}$$

compatible with the augmentations, and any two such lifts are homotopic. Consequently the induced maps on Z^p, B^p and H^p in the horizontal direction are the resolutions of $Z^p(f)$, $B^p(f)$ and $H^p(f)$.

***Proof*:**
Step 1: Lifts on the building blocks. The morphism $f^\bullet$ induces $B^p(f) \colon B^p(C^\bullet) \to B^p(\widetilde{C}^\bullet)$ and $H^p(f) \colon H^p(C^\bullet) \to H^p(\widetilde{C}^\bullet)$ for every p. By definition 4.4.1 the objects $\mathcal{B}^{p,\bullet}$ and $\mathcal{H}^{p,\bullet}$ occurring in J are injective resolutions of $B^p(C^\bullet)$ and $H^p(C^\bullet)$, and likewise for $\widetilde{J}$. The dual of lemma 2.2.13 therefore lifts $B^p(f)$ and $H^p(f)$ to morphisms of those resolutions, each unique up to homotopy.

Step 2: Lifts on cycles. Apply lemma 4.4.3 to the morphism of short exact sequences

$$\left(0 \to B^p(C) \to Z^p(C) \to H^p(C) \to 0\right) \longrightarrow \left(0 \to B^p(\widetilde{C}) \to Z^p(\widetilde{C}) \to H^p(\widetilde{C}) \to 0\right)$$

induced by $f^\bullet$, with the lifts of Step 1 on the two ends. This produces a lift $\mathcal{Z}^{p,\bullet} \to \widetilde{\mathcal{Z}}^{p,\bullet}$ of $Z^p(f)$.

Step 3: Lifts on the terms. Apply lemma 4.4.3 again, now to

$$\left(0 \to Z^p(C) \to C^p \to B^{p+1}(C) \to 0\right) \longrightarrow \left(0 \to Z^p(\widetilde{C}) \to \widetilde{C}^p \to B^{p+1}(\widetilde{C}) \to 0\right)$$

with the lift of Step 2 on the left end and that of Step 1 on the right. This produces $\tilde{f}^{p,\bullet}\colon J^{p,\bullet} \to \widetilde{J}^{p,\bullet}$ lifting f^p.

Step 4: Compatibility with the horizontal differential. For any Cartan-Eilenberg resolution the horizontal differential factors as $J^{p,q} \twoheadrightarrow \mathcal{B}^{p+1,q} \hookrightarrow \mathcal{Z}^{p+1,q} \hookrightarrow J^{p+1,q}$: it factors through its image $\mathcal{B}^{p+1,q}$ by definition, and that image lies in $\mathcal{Z}^{p+1,q}$ because $d^h \circ d^h = 0$. Each of the three maps is respected by the lifts of Steps 1 to 3, the projection and the two inclusions being the structure maps of the splittings. Hence $\tilde{f}^{\bullet,\bullet}$ commutes with d^h, and it commutes with d^v because each $\tilde{f}^{p,\bullet}$ is a morphism of resolutions. So it is a morphism of double complexes.

Step 5: Uniqueness up to homotopy. Let $\tilde{f}$ and $\tilde{g}$ both lift $f^\bullet$. In each column their difference lifts the zero morphism $C^p \to \widetilde{C}^p$, so by the dual of lemma 2.2.4 it is null-homotopic: there are $s^{p,\bullet}$ with $d^v s + s d^v = \tilde{f} - \tilde{g}$ in each column.

Producing a homotopy of *double* complexes, that is one which additionally satisfies $sd^h = d^h s$, requires choosing those column homotopies compatibly. This is a third induction of the same shape as lemma 4.4.3: choose null-homotopies on $\mathcal{B}^{p,\bullet}$ and $\mathcal{H}^{p,\bullet}$ by the object-level uniqueness of Step 1, extend over the first horseshoe to $\mathcal{Z}^{p,\bullet}$ by an obstruction computation of the same shape as the one for θ^q, and extend again over the second to $J^{p,\bullet}$. The bookkeeping is routine and carries no idea not already in lemma 4.4.3, so it is not repeated; it is carried out in *An Introduction to Homological Algebra* [12].

For the final clause, let $\tilde{f}$ be any lift. Its restriction to horizontal cocycles, coboundaries and cohomology is determined by its compatibility with the augmentations, since those subquotients are computed from d^h and $\tilde{f}$ commutes with d^h by Step 4; so on H^p it is a lift of $H^p(f)$, whether or not it is the one built in Steps 1 to 3. This completes the proof.

With the tool in hand, the theorem itself is a two-filtration argument of exactly the kind section 4.3 was written for. The hypothesis to watch is the one relating the two functors: the first must carry injectives to objects on which the second's higher derived functors vanish.

Definition 4.4.5: Acyclic for a Left-Exact Functor

Let $G\colon \mathcal{B} \to \mathcal{C}$ be a left-exact additive functor between abelian categories. An object B of $\mathcal{B}$ is *G-acyclic* if $R^i G(B) = 0$ for all $i \geq 1$. This is the right-derived counterpart of definition 2.3.12. Injective objects are G-acyclic for every left-exact G, since an injective I has $0 \to I \to I \to 0$ as an injective resolution; and a resolution by G-acyclic objects computes $R^\bullet G$. The latter is the dual of theorem 2.3.14, which is stated there for right-exact functors and left-derived functors; the statement needed here is its image under passage to the opposite categories, where projectives become injectives and $\mathbf{L}_i$ becomes R^i.

Theorem 4.4.6: The Grothendieck Spectral Sequence

Let $\mathcal{A}$, $\mathcal{B}$, $\mathcal{C}$ be abelian categories with $\mathcal{A}$ and $\mathcal{B}$ having enough injectives, and let

$$F\colon \mathcal{A} \to \mathcal{B}, \qquad G\colon \mathcal{B} \to \mathcal{C}$$

be left-exact additive functors. Suppose F carries injective objects of $\mathcal{A}$ to G-acyclic objects of $\mathcal{B}$. Then for every object A of $\mathcal{A}$ there is a first-quadrant cohomological spectral sequence, natural in A,

$$E_2^{p,q} = R^p G\big(R^q F(A)\big) \implies R^{p+q}(G \circ F)(A)$$

***Proof*:**
Step 1: Setup. Choose an injective resolution $A \to I^\bullet$ in $\mathcal{A}$. Then $R^q F(A) = H^q(F(I^\bullet))$ by definition of the right-derived functors (definition 2.3.1 in its dual form). The complex $F(I^\bullet)$ is bounded below, so by theorem 4.4.2 it admits a Cartan-Eilenberg resolution $J^{\bullet,\bullet}$. Apply G to obtain a first-quadrant double complex $G(J^{\bullet,\bullet})$ in $\mathcal{C}$, and consider the two spectral sequences of corollary 4.3.4 converging to $H^\bullet(\mathrm{Tot}(G(J^{\bullet,\bullet})))$.

Step 2: The filtration that collapses. Take first the filtration whose first page is obtained by computing cohomology in the resolution direction. For each fixed p, the complex $J^{p,\bullet}$ is an injective resolution of $F(I^p)$, so

$$H^q\big(G(J^{p,\bullet})\big) = R^q G\big(F(I^p)\big)$$

By hypothesis I^p is injective and F carries injectives to G-acyclics, so $F(I^p)$ is G-acyclic and this vanishes for $q \geq 1$, leaving $G(F(I^p))$ when $q = 0$. Hence this first page is concentrated in the single row $q = 0$, where it is the complex $G(F(I^\bullet))$, and its second page is

$$H^p\big(GF(I^\bullet)\big) = R^p(G \circ F)(A)$$

the last equality because $I^\bullet$ is an injective resolution of A. By corollary 4.3.4,

$$H^n\big(\mathrm{Tot}(G(J^{\bullet,\bullet}))\big) \cong R^n(G \circ F)(A)$$

Step 3: The filtration that computes. Take now the other filtration, whose first page is the cohomology in the p direction. Because $J^{\bullet,\bullet}$ is a Cartan-Eilenberg resolution, the horizontal cocycles, coboundaries and cohomology are injective resolutions of those of $F(I^\bullet)$. The short exact sequences relating them therefore have injective outer terms in each degree, hence are degreewise split, exactly as spelled out in Step 2 of theorem 4.5.3. A functor preserves split exactness, so applying G commutes with taking horizontal cohomology:

$$H^p\big(G(J^{\bullet,q})\big) \cong G\big(H^p(J^{\bullet,q})\big)$$

By definition 4.4.1 the complex $H^a(J^{\bullet,\bullet})$ is an injective resolution of $H^a(F(I^\bullet)) = R^a F(A)$, where a denotes the degree in $F(I^\bullet)$ and the resolution degree is the remaining direction. This is the row filtration of corollary 4.3.4, whose index swap makes the filtration index, here the resolution degree, the *first* spectral index. Writing p for that resolution degree and q for a, taking cohomology in the resolution direction therefore gives

$$E_2^{p,q} = R^p G\big(R^q F(A)\big)$$

Step 4: Conclusion. Both spectral sequences converge to the cohomology of the same total complex by corollary 4.3.4. Step 2 identifies that cohomology as $R^n(G \circ F)(A)$ and Step 3 identifies the second page of the other as $R^pG(R^qF(A))$, giving

$$E_2^{p,q} = R^pG\big(R^qF(A)\big) \implies R^{p+q}(G \circ F)(A)$$

For naturality in A, let $a\colon A \to A'$ be a morphism. The dual of lemma 2.2.13 lifts it to a morphism of injective resolutions $I^\bullet \to I'^\bullet$, unique up to homotopy; applying F gives a morphism $F(I^\bullet) \to F(I'^\bullet)$, and theorem 4.4.4 lifts that to a morphism of the chosen Cartan-Eilenberg resolutions, again unique up to homotopy. Applying G and totalizing yields a morphism of filtered complexes, hence a morphism of spectral sequences by Step 1 of theorem 4.2.9. Homotopic lifts induce the same map on E_r for every $r \geq 2$. This needs an argument, because the homotopy has bidegree $(0, -1)$ and the filtration in play is by resolution degree, so it does *not* preserve the filtration: it lowers it by one. The standard filtration-shift estimate covers the gap. For a class represented by $x \in Z_r^p$, the two totalized lifts differ by $Ds(x) + s(Dx)$, where s is the totalized homotopy and D the total differential; here $s(x) \in F^{p-1}$ so $Ds(x) \in dF^{p-1}$, which dies in E_r for $r \geq 2$, and $Dx \in F^{p+r}$ so $s(Dx) \in F^{p+r-1}$, which also dies in E_r for $r \geq 2$. Hence the morphism of spectral sequences from page two onward depends only on a, which is all the naturality statement asserts. This completes the proof.

The shape of the conclusion deserves comment, because it is easy to misread. The spectral sequence does not say that $R^n(GF)$ is built from the groups $R^pG(R^qF)$ by any simple rule; it says there is a filtration of $R^n(GF)(A)$ whose graded pieces are subquotients of those groups, obtained after running the differentials. Only in the corners is the information immediate, and the standard exact sequence extracted from those corners is the following, which is how the theorem is most often applied in low degrees.

Nothing in that extraction uses where the spectral sequence came from. It uses only that the sequence is first-quadrant and converges, so it is worth recording at that generality: the same five terms appear for the spectral sequence of a double complex, or of any filtered complex meeting the hypotheses of corollary 4.2.7, and consumers outside this book need it in that form.

Both maps at the ends of that sequence are instances of a general construction worth naming, because downstream books identify them with concrete maps and need a statement to point at. In a first-quadrant spectral sequence the corner spots $(p, 0)$ and $(0, q)$ each meet the abutment in only one way.

Proposition 4.4.7: Edge Morphisms

Let $E_2^{p,q} \Rightarrow H^{p+q}$ be a first-quadrant cohomological spectral sequence converging in the sense of definition 4.2.5. Then for every $p, q \geq 0$ there are canonical morphisms

$$E_2^{p,0} \longrightarrow H^p, \qquad H^q \longrightarrow E_2^{0,q}$$

called the *edge morphisms*, given respectively by the composites

$$E_2^{p,0} \twoheadrightarrow E_\infty^{p,0} = F^p H^p \hookrightarrow H^p, \qquad H^q \twoheadrightarrow H^q/F^1 H^q = E_\infty^{0,q} \hookrightarrow E_2^{0,q}$$

They are natural in the spectral sequence.
For the Grothendieck spectral sequence of theorem 4.4.6 the two edge morphisms are the natural maps

$$R^p G\big(F(A)\big) \longrightarrow R^p(G \circ F)(A), \qquad R^q(G \circ F)(A) \longrightarrow G\big(R^q F(A)\big)$$

***Proof*:**
Step 1: The two composites are defined. Fix p. Every differential out of $(p, 0)$ lands in $(p+r, 1-r)$, which has negative second index for $r \geq 2$, and every differential into it comes from $(p-r, r-1)$, which has negative first index once $r > p$. So $E_\infty^{p,0}$ is a quotient of $E_2^{p,0}$, the successive pages being obtained by killing images. The filtration of H^p is finite with $F^{p+1}H^p = 0$, because $E_\infty^{a,b} = 0$ for $a > p$ on the diagonal $a + b = p$ by first-quadrantness, so $E_\infty^{p,0} = F^p H^p / F^{p+1} H^p = F^p H^p$, a subobject of H^p. Composing gives the first morphism.

Dually, every differential into $(0, q)$ comes from $(-r, q + r - 1)$, which has negative first index, so $E_\infty^{0,q}$ is a subobject of $E_2^{0,q}$, obtained by taking successive kernels. And $E_\infty^{0,q} = F^0 H^q / F^1 H^q = H^q / F^1 H^q$, since $F^0 H^q = H^q$ by exhaustiveness. Composing gives the second morphism.

Step 2: Naturality. A morphism of spectral sequences compatible with the abutments respects the pages, hence the successive kernels and images defining E_∞, and respects the filtration of the abutment. Each edge morphism is built only from those data, so the two squares commute.

Step 3: The Grothendieck case. Substituting $E_2^{p,q} = R^p G(R^q F(A))$ and $H^n = R^n(G \circ F)(A)$ gives $E_2^{p,0} = R^p G(R^0 F(A)) = R^p G(F(A))$, since $R^0 F = F$ by left exactness of F, and $E_2^{0,q} = R^0 G(R^q F(A)) = G(R^q F(A))$ likewise. So the two edge morphisms take the displayed shape. That they are the *natural* maps, that is the ones induced by the comparison of an injective resolution of A with a Cartan-Eilenberg resolution of F applied to it, is Step 2 together with the naturality of theorem 4.4.6, completing the proof.

Proposition 4.4.8: The Five-Term Exact Sequence of a First-Quadrant Spectral Sequence

Let $E_2^{p,q} \Rightarrow H^{p+q}$ be a first-quadrant cohomological spectral sequence, so that $E_r^{p,q} = 0$ whenever $p < 0$ or $q < 0$, converging to a filtered graded object $H^\bullet$ in the sense of definition 4.2.5. Then there is an exact sequence

$$0 \to E_2^{1,0} \to H^1 \to E_2^{0,1} \xrightarrow{d_2} E_2^{2,0} \to H^2$$

natural in the spectral sequence: a morphism of first-quadrant spectral sequences compatible with the abutment induces a morphism of these five-term sequences.

***Proof*:**
The spectral sequence is first-quadrant, so the filtration of H^n has graded pieces $E_\infty^{p,q}$ with $p+q=n$ and $0 \leq p \leq n$.

Step 1: Degree one. The diagonal $p+q=1$ contains the two spots $(1,0)$ and $(0,1)$. On the second page, the differentials touching $(1,0)$ run to $(3,-1)$ and from $(-1,1)$, both outside the first quadrant, so $E_\infty^{1,0} = E_2^{1,0}$. The differentials touching $(0,1)$ are $d_2\colon E_2^{0,1} \to E_2^{2,0}$ and one arriving from $(-2,2)$, which is zero; so $E_\infty^{0,1} = \ker\bigl(d_2\colon E_2^{0,1} \to E_2^{2,0}\bigr)$. The filtration of H^1 therefore gives a short exact sequence

$$0 \to E_\infty^{1,0} \to H^1 \to E_\infty^{0,1} \to 0$$

that is $0 \to E_2^{1,0} \to H^1 \to \ker d_2 \to 0$. The first map here is the edge morphism $E_2^{1,0} \to H^1$ of proposition 4.4.7 and the second is the edge morphism $H^1 \to E_2^{0,1}$ followed by nothing, both by construction; this is what lets consumers identify the two ends of the five-term sequence with concrete maps.

Step 2: Splicing. Substituting $E_\infty^{0,1} = \ker d_2$ turns the short exact sequence into exactness of

$$0 \to E_2^{1,0} \to H^1 \to E_2^{0,1} \xrightarrow{d_2} E_2^{2,0}$$

the last map being d_2 and exactness at $E_2^{0,1}$ being the statement that the image of H^1 is exactly $\ker d_2$.

Step 3: The final term. It remains to show exactness at $E_2^{2,0}$, that is that the kernel of $E_2^{2,0} \to H^2$ is the image of d_2. The filtration of H^2 has $E_\infty^{2,0}$ as its bottom graded piece, and the map $E_2^{2,0} \to H^2$ is the composite $E_2^{2,0} \twoheadrightarrow E_\infty^{2,0} \hookrightarrow H^2$. The second map is a monomorphism, so the kernel of the composite is the kernel of the first, which is the image of the only differential arriving at $(2,0)$, namely $d_2\colon E_2^{0,1} \to E_2^{2,0}$. Hence the sequence is exact there.

Step 4: Naturality. Let $\{E_r\} \to \{E'_r\}$ be a morphism of first-quadrant spectral sequences compatible with the abutments. Every object in the five-term sequence was constructed from data the morphism respects: the terms $E_2^{1,0}$, $E_2^{0,1}$, $E_2^{2,0}$ receive the induced map on the second page; the map d_2 is respected because a morphism of spectral sequences commutes with the differentials; $E_\infty^{p,q}$ receives the induced map because it is a subquotient of $E_2^{p,q}$ cut out by those same differentials; and H^1, H^2 receive the given map on abutments, which by hypothesis is compatible with the filtrations and hence with the identifications $E_\infty^{p,q} \cong F^pH^{p+q}/F^{p+1}H^{p+q}$. Each of the four maps in the sequence was defined from exactly these, so all four squares commute, completing the proof.

Specializing to theorem 4.4.6 recovers the form in which the sequence is usually quoted, and it is the one example 4.4.10 and the group-cohomology literature use.

Corollary 4.4.9: The Five-Term Exact Sequence of a Composite

Under the hypotheses of theorem 4.4.6, there is an exact sequence

$$0 \to R^1G\big(F(A)\big) \to R^1(GF)(A) \to G\big(R^1F(A)\big) \to R^2G\big(F(A)\big) \to R^2(GF)(A)$$

natural in A.

***Proof*:**
The Grothendieck spectral sequence is first-quadrant and converges to $R^\bullet(GF)(A)$, so proposition 4.4.8 applies. Substituting its second page, $E_2^{p,q} = R^pG(R^qF(A))$, gives $E_2^{1,0} = R^1G(F(A))$ since $R^0F = F$, then $E_2^{0,1} = G(R^1F(A))$ since $R^0G = G$, and $E_2^{2,0} = R^2G(F(A))$; and H^1, H^2 are $R^1(GF)(A)$, $R^2(GF)(A)$. Naturality in A follows from naturality of the spectral sequence in A, as we sought to show.

The machine can now be put to work. The instance closest to hand uses the group cohomology of section 3.1: invariants under a group factor through invariants under a normal subgroup.

Example 4.4.10: The Lyndon-Hochschild-Serre Spectral Sequence
Let G be a group, $N \trianglelefteq G$ a normal subgroup, and M a G-module. Taking N-invariants produces a G/N-module, because N acts trivially on M^N and the residual action of G factors through G/N, and taking G/N-invariants of that returns M^G. So the functor $(-)^G$ factors as

$$\mathrm{Mod}_G \xrightarrow{(-)^N} \mathrm{Mod}_{G/N} \xrightarrow{(-)^{G/N}} \mathbf{Ab}$$

Both factors are left exact: for $(-)^{G/N}$ this is proposition 3.1.6, and for $(-)^N : \mathrm{Mod}_G \to \mathrm{Mod}_{G/N}$ the same argument applies, exactness being tested on the underlying abelian groups. That $R^p(-)^{G/N} = H^p(G/N, -)$ is section 3.1 by definition.

The identification $R^q(-)^N = H^q(N, -)$ needs one step more than it looks. The left-hand side is computed from an injective resolution in Mod_G, while the right-hand side is computed from one in Mod_N, so the two agree only because *restriction to a subgroup preserves injectives*. That holds here because $\mathbb{Z}[G]$ is free as a $\mathbb{Z}[N]$-module, on any set of coset representatives, so the relative induction functor $\mathrm{Ind}_N^G = \mathbb{Z}[G] \otimes_{\mathbb{Z}[N]} -$, which is the evident generalization of the absolute one in definition 3.1.16, is exact and is left adjoint to restriction; a right adjoint to an exact functor preserves injectives, by the adjunction $\mathrm{Hom}_N(-, \mathrm{Res}I) \cong \mathrm{Hom}_G(\mathrm{Ind}(-), I)$, which is then a composite of two exact functors. Hence restricting an injective G-module gives an injective N-module, and an injective resolution in Mod_G restricts to one in Mod_N, computing $H^q(N, -)$.

The hypothesis of theorem 4.4.6 is that $(-)^N$ carries injective G-modules to $(-)^{G/N}$-acyclic G/N-modules, and in fact it carries them to injectives. Let I be an injective G-module, let $0 \to A \to B$

be a monomorphism of G/N-modules, and let $\varphi\colon A \to I^N$ be G/N-equivariant. Regard A and B as G-modules through $G \twoheadrightarrow G/N$; then $A \to I^N \hookrightarrow I$ is G-equivariant, so injectivity of I extends it to $\psi\colon B \to I$. For $b \in B$ and $n \in N$ one has $n \cdot \psi(b) = \psi(n \cdot b) = \psi(b)$, since N acts trivially on B; so ψ lands in I^N and is the required extension. Hence I^N is injective, and injectives are acyclic for any left-exact functor by definition 4.4.5.

Theorem 4.4.6 therefore gives a first-quadrant spectral sequence

$$E_2^{p,q} = H^p\big(G/N,\, H^q(N,M)\big) \implies H^{p+q}(G,M)$$

the *Lyndon-Hochschild-Serre spectral sequence*, and corollary 4.4.9 specializes to the inflation-restriction sequence

$$0 \to H^1(G/N, M^N) \to H^1(G,M) \to H^1(N,M)^{G/N} \to H^2(G/N, M^N) \to H^2(G,M)$$

using $H^0(G/N, H^1(N,M)) = H^1(N,M)^{G/N}$. The same argument runs verbatim for a profinite group acting on discrete modules, once the category of discrete modules is known to have enough injectives; that is the setting in which the sequence is used in Galois cohomology.

Exercise 4.4.1

1. Show that if $R^qF(A) = 0$ for all $q \geq 1$, that is if A is F-acyclic, then the Grothendieck spectral sequence collapses and $R^n(GF)(A) \cong R^nG(F(A))$ for every n.
2. Show that if G is exact, then $R^n(GF)(A) \cong G(R^nF(A))$ for every n. Identify which page of the spectral sequence collapses and why.
3. In the situation of example 4.4.10 take $N = G$, so that G/N is trivial. Verify directly that the spectral sequence degenerates and returns $H^n(G,M) = H^n(G,M)$, and explain which hypothesis makes the E_2 page concentrated in one column.
4. Prove the dual statement of theorem 4.4.6 for right-exact functors and left-derived functors: if $F\colon \mathcal{A} \to \mathcal{B}$ and $G\colon \mathcal{B} \to \mathcal{C}$ are right exact, $\mathcal{A}$ and $\mathcal{B}$ have enough projectives, and F carries projectives to G-acyclics, then $E^2_{p,q} = L_pG(L_qF(A)) \implies L_{p+q}(G \circ F)(A)$.
5. Show that the hypothesis that F carries injectives to G-acyclics cannot simply be dropped, by explaining exactly which step of the proof of theorem 4.4.6 fails without it and what the first filtration computes in its absence.

4.5 Hypercohomology

The derived functors of section 2.3 take an object as input. Many constructions instead present a whole complex and ask for the derived functor of that: a complex of sheaves rather than a single sheaf, a complex of modules rather than a module. Replacing the complex by its cohomology objects loses information, as the last exercise of section 4.4 showed, and the correct construction resolves the complex as a complex. The tool is again the Cartan-Eilenberg resolution, and the output carries two spectral sequences relating it to the two things one might naively have computed instead.

Definition 4.5.1: Hypercohomology

Let $F\colon \mathcal{A} \to \mathcal{B}$ be a left-exact additive functor between abelian categories with $\mathcal{A}$ having enough injectives, and let $C^\bullet$ be a bounded-below cochain complex in $\mathcal{A}$. Choose a Cartan-Eilenberg resolution $J^{\bullet,\bullet}$ of $C^\bullet$ (theorem 4.4.2), where the first index is the degree in $C^\bullet$ and the second the resolution degree. The *hypercohomology* of $C^\bullet$ with respect to F is

$$\mathbb{R}^n F(C^\bullet) := H^n\Big(\mathrm{Tot}\big(F(J^{\bullet,\bullet})\big)\Big)$$

The definition is only worth making if it does not depend on the resolution, and the comparison theorem of section 4.2 is exactly the instrument for showing that it does not.

Proposition 4.5.2: Hypercohomology is Well Defined

The objects $\mathbb{R}^n F(C^\bullet)$ of definition 4.5.1 are independent of the choice of Cartan-Eilenberg resolution, up to canonical isomorphism, and a quasi-isomorphism $C^\bullet \to \widetilde{C}^\bullet$ of bounded-below complexes induces an isomorphism $\mathbb{R}^n F(C^\bullet) \cong \mathbb{R}^n F(\widetilde{C}^\bullet)$ for every n.

***Proof*:**
Let $J^{\bullet,\bullet}$ and $\widetilde{J}^{\bullet,\bullet}$ be Cartan-Eilenberg resolutions of $C^\bullet$ and $\widetilde{C}^\bullet$ respectively, and let $f\colon C^\bullet \to \widetilde{C}^\bullet$ be a quasi-isomorphism; taking f to be the identity handles the first claim.

Step 1: A comparison map exists. By theorem 4.4.4, f lifts to a morphism of double complexes $J^{\bullet,\bullet} \to \widetilde{J}^{\bullet,\bullet}$, uniquely up to homotopy, and the lift induces $H^q(f)$ on the horizontal cohomology resolutions. Applying F and totalizing gives a map of filtered complexes $\mathrm{Tot}(F(J)) \to \mathrm{Tot}(F(\widetilde{J}))$ compatible with the filtration by resolution degree.

Step 2: The comparison is an isomorphism on a page. By theorem 4.5.3 below, whose proof uses only definition 4.4.1 and not the present proposition, the filtration by resolution degree has second page $R^pF(H^q(C^\bullet))$. By the final clause of theorem 4.4.4 the lift of Step 1 restricts on horizontal cohomology to a resolution of $H^q(f)$, so the map it induces on that second page is exactly $R^pF(H^q(f))$. Since f is a quasi-isomorphism, $H^q(f)$ is an isomorphism for every q, and hence so is $R^pF(H^q(f))$ by functoriality of the derived functors. So the induced map is an isomorphism on the second page.

Step 3: Conclusion. Both complexes are bounded below and both Cartan-Eilenberg resolutions are first-quadrant, so both filtrations are bounded and theorem 4.2.9 applies with $r_0 = 2$, in the cochain form of corollary 4.2.8. It gives that the map is an isomorphism on the abutment, that is $\mathbb{R}^n F(C^\bullet) \cong \mathbb{R}^n F(\widetilde{C}^\bullet)$ for every n. Independence of the resolution is the case $f = \mathrm{id}$, and canonicity follows because the lift of Step 1 is unique up to homotopy and homotopic maps agree on cohomology, completing the proof.

The two filtrations of the defining double complex give the two spectral sequences that make hypercohomology computable. One approximates it by the derived functors of the cohomology of the complex, the other by the cohomology of the complex of derived functors; neither is generally the answer, and the difference between them is the reason hypercohomology is not redundant.

Theorem 4.5.3: The Two Hypercohomology Spectral Sequences

With the hypotheses of definition 4.5.1, there are two spectral sequences with bounded filtrations, first-quadrant when $C^\bullet$ is concentrated in non-negative degrees, converging to $\mathbb{R}^{p+q}F(C^\bullet)$:

$$ {}^{\mathrm{I}}E_2^{p,q} = R^pF\big(H^q(C^\bullet)\big) \implies \mathbb{R}^{p+q}F(C^\bullet) $$

$$ {}^{\mathrm{II}}E_1^{p,q} = R^qF\big(C^p\big) \implies \mathbb{R}^{p+q}F(C^\bullet) $$

the first filtered by resolution degree, the second by the degree in $C^\bullet$.

Proof:
Both are instances of corollary 4.3.4 applied to the double complex $F(J^{\bullet,\bullet})$, which is a cochain double complex, and the work is in identifying the pages.

Step 1: Filtration by the degree in $C^\bullet$. The graded piece at level p is the column $F(J^{p,\bullet})$, and its cohomology in the resolution direction is

$$ H^q\big(F(J^{p,\bullet})\big) = R^qF(C^p) $$

because $J^{p,\bullet}$ is an injective resolution of C^p by definition 4.4.1. This is the first page of that filtration, so ${}^{\mathrm{II}}E_1^{p,q} = R^qF(C^p)$, with d_1 induced by the differential of $C^\bullet$.

Step 2: Filtration by resolution degree. The graded piece at level q is the row $F(J^{\bullet,q})$, whose cohomology is taken in the direction of the degree in $C^\bullet$. By definition 4.4.1 the short exact sequences

$$ 0 \to Z^p(J^{\bullet,q}) \to J^{p,q} \to B^{p+1}(J^{\bullet,q}) \to 0 $$

$$ 0 \to B^p(J^{\bullet,q}) \to Z^p(J^{\bullet,q}) \to H^p(J^{\bullet,q}) \to 0 $$

have injective outer terms and are therefore split. An additive functor preserves split exact sequences, so F commutes with forming Z^p, B^p and H^p in this direction:

$$ H^p\big(F(J^{\bullet,q})\big) \cong F\big(H^p(J^{\bullet,q})\big) $$

By definition 4.4.1 the complex $H^p(J^{\bullet,\bullet})$ is an injective resolution of $H^p(C^\bullet)$ as the resolution degree varies. Taking cohomology in that remaining direction therefore gives $R^\bullet F(H^p(C^\bullet))$, and indexing this sequence with the resolution degree as the filtration index yields ${}^{\mathrm{I}}E_2^{p,q} = R^pF(H^q(C^\bullet))$.

Step 3: Convergence. The complex $C^\bullet$ is bounded below and the resolution is bounded below, so $F(J^{\bullet,\bullet})$ is concentrated in non-negative bidegrees after a shift, and corollary 4.3.4 gives that both converge to the cohomology of $\mathrm{Tot}(F(J^{\bullet,\bullet}))$, which is $\mathbb{R}^{p+q}F(C^\bullet)$ by definition 4.5.1, completing the proof.

Two special cases recover the ordinary theory and explain the name. Both follow by collapsing one of the two sequences, and together they say that hypercohomology interpolates between the two computations one might have attempted directly.

Corollary 4.5.4: Hypercohomology Recovers the Ordinary Derived Functors

With the hypotheses of definition 4.5.1:

1. If $C^\bullet$ is concentrated in degree 0, with $C^0 = A$, then $\mathbb{R}^n F(C^\bullet) \cong R^n F(A)$ for every n.
2. If every C^p is F-acyclic, then $\mathbb{R}^n F(C^\bullet) \cong H^n\big(F(C^\bullet)\big)$ for every n.

***Proof*:**

1. If $C^\bullet$ is concentrated in degree 0 then $H^q(C^\bullet) = 0$ for $q \neq 0$ and $H^0(C^\bullet) = A$. So ${}^{\mathrm{I}}E_2^{p,q} = R^p F(H^q(C^\bullet))$ vanishes unless $q = 0$, where it is $R^p F(A)$. The second page is concentrated in the row $q = 0$, so by corollary 4.3.4 the sequence collapses and the diagonal $p + q = n$ carries the single group $R^n F(A)$. Hence $\mathbb{R}^n F(C^\bullet) \cong R^n F(A)$.
2. If every C^p is F-acyclic then $R^q F(C^p) = 0$ for $q \geq 1$ by definition 4.4.5, so ${}^{\mathrm{II}}E_1^{p,q} = R^q F(C^p)$ vanishes unless $q = 0$, where it is $F(C^p)$. The first page is concentrated in the row $q = 0$, and its differential d_1 is induced by the differential of $C^\bullet$, so the second page is $H^p(F(C^\bullet))$ in that row and zero elsewhere. Collapsing by corollary 4.3.4 gives $\mathbb{R}^n F(C^\bullet) \cong H^n(F(C^\bullet))$.

This completes the proof.

The second case is what makes hypercohomology a practical tool: a complex of acyclic objects may be substituted for the original complex, and the functor is then applied termwise. It is also the precise sense in which the definition generalizes the ordinary theory, since a resolution of a single object is a complex of acyclics quasi-isomorphic to that object, and proposition 4.5.2 says the answer depends only on the quasi-isomorphism class.

Example 4.5.5: Hypercohomology of a Complex with Zero Differential

Let $C^\bullet$ be a bounded-below complex with zero differential. Then $H^q(C^\bullet) = C^q$, and the first spectral sequence reads

$${}^{\mathrm{I}}E_2^{p,q} = R^p F(C^q) \implies \mathbb{R}^{p+q} F(C^\bullet)$$

The second has ${}^{\mathrm{II}}E_1^{p,q} = R^q F(C^p)$ with d_1 induced by the zero differential, hence zero, so ${}^{\mathrm{II}}E_2 = {}^{\mathrm{II}}E_1$. A Cartan-Eilenberg resolution may be taken to be the direct sum of injective resolutions of the individual C^p, since with zero differential the conditions of definition 4.4.1 on Z^p, B^p and H^p reduce to the condition on C^p itself. The resulting double complex splits as a direct sum, so every differential from the second page onward vanishes and there are no extension problems:

$$\mathbb{R}^n F(C^\bullet) \cong \bigoplus_{p+q=n} R^p F(C^q)$$

The example is the sanity check for the definition: when the complex carries no gluing, hypercohomology is the direct sum of the ordinary derived functors of its terms, and the content of the general case is exactly the failure of that splitting.

Exercise 4.5.1

1. Show that if F is exact then $\mathbb{R}^n F(C^\bullet) \cong F(H^n(C^\bullet))$ for every n, using whichever of the two spectral sequences of theorem 4.5.3 collapses.
2. Let $C^\bullet$ be a bounded-below complex with $H^q(C^\bullet) = 0$ for all q. Show that $\mathbb{R}^n F(C^\bullet) = 0$ for every n, and explain why this is the expected answer given proposition 4.5.2.
3. Write down the five-term exact sequence attached to the first hypercohomology spectral sequence of theorem 4.5.3, by the argument of corollary 4.4.9.
4. Show that theorem 4.4.6 may be restated as the assertion that $\mathbb{R}^n G(F(I^\bullet)) \cong R^n(GF)(A)$ when F carries injectives to G-acyclics, where $I^\bullet$ is an injective resolution of A. Which part of corollary 4.5.4 is being used?
5. Let $C^\bullet$ be the two-term complex $A \xrightarrow{f} B$ in degrees 0 and 1. Show that $\mathbb{R}^0 F(C^\bullet) \cong F(\ker f)$, by both spectral sequences of theorem 4.5.3.

4.6 Filtered Complexes and the Filtered Derived Category

Section 4.2 extracted a spectral sequence from a filtered complex and stopped there. That is enough when one filtration is present. It is not enough when two are, and two is the case that arises: a complex carrying both a weight filtration and a Hodge filtration is the object mixed Hodge theory is built from, and the question there is whether the second filtration passes to the pages of the spectral sequence of the first without ambiguity. Answering it needs the filtered complexes themselves organised into a category, with the right notion of equivalence.

Definition 4.6.1: Filtered Object and Strict Morphism

Let $\mathcal{A}$ be an abelian category. A *filtered object* is a pair (A, F) with F a decreasing chain of subobjects $\cdots \supseteq F^p A \supseteq F^{p+1} A \supseteq \cdots$. A morphism $f\colon (A, F) \to (B, F)$ is a morphism $f\colon A \to B$ with $f(F^p A) \subseteq F^p B$ for all p; it is *strict* if

$$f(A) \cap F^p B = f(F^p A)$$

for all p, that is, if every element of the image that lies in $F^p B$ is the image of something already in $F^p A$.

Strictness is the condition that makes filtered algebra behave. Without it the category of filtered objects is not abelian: a morphism can be a monomorphism and an epimorphism without being an isomorphism, because the filtration on the image can be finer than the filtration inherited from the target.

Proposition 4.6.2: Strictness and Exactness

A morphism $f\colon (A,F) \to (B,F)$ of filtered objects is strict if and only if the induced sequence

$$0 \to F^p(\ker f) \to F^pA \to F^pB$$

has gr_F exact, equivalently if and only if gr_F applied to $0 \to \ker f \to A \to B$ stays exact. Strict morphisms are closed under composition only when the intermediate filtrations agree; strictness is not a two-out-of-three property.

***Proof*:**
If f is strict and $b = f(a) \in F^pB$, then $b = f(a')$ with $a' \in F^pA$, so $\mathrm{gr}_F^p f$ is surjective onto the image, and its kernel is $\mathrm{gr}_F^p(\ker f)$ because $a - a' \in \ker f$ lies in F^pA when both do. Conversely if gr_F preserves exactness, an element of $f(A) \cap F^pB$ lifts step by step through the graded pieces from a large p downwards, the filtration being exhaustive and separated in each degree.

For the last claim, $g \circ f$ can fail to be strict when f and g are: the image of f may meet F^pB in more than $f(F^pA)$ once g has collapsed part of it. The standard example is a two-step filtration on a three-dimensional space, and the failure is exactly what theorem 4.6.5 is designed to rule out.

Definition 4.6.3: Filtered Quasi-Isomorphism

A morphism $f\colon (K^\bullet, F) \to (L^\bullet, F)$ of filtered complexes is a *filtered quasi-isomorphism* if $\mathrm{gr}_F^p f$ is a quasi-isomorphism of complexes for every p.

This is strictly stronger than being a quasi-isomorphism: a map can induce an isomorphism on cohomology while distorting the filtration, and such maps must not become invertible, or the spectral sequence would not be an invariant of the object.

Theorem 4.6.4: The Filtered Derived Category

Let $\mathcal{A}$ be an abelian category with enough injectives. The localization of the homotopy category of bounded-below filtered complexes over $\mathcal{A}$ at the filtered quasi-isomorphisms exists; call it the *filtered derived category* $D^+F(\mathcal{A})$. The forgetful functor $D^+F(\mathcal{A}) \to D^+(\mathcal{A})$ is well defined, and for each r the assignment $(K,F) \mapsto E_r(K,F)$ of theorem 4.2.2 factors through $D^+F(\mathcal{A})$: a filtered quasi-isomorphism induces an isomorphism of spectral sequences from the E_1-page onwards.

***Proof*:**
Existence of the localization is corollary 1.2.6 applied to the filtered setting: filtered quasi-isomorphisms form a multiplicative system, the verification being the same calculus-of-fractions argument, with the octahedral completion carried out one graded piece at a time. The only point needing care is that a filtered mapping cone is filtered by the cones of the graded pieces, which holds because the cone construction is degreewise a direct sum and gr_F is additive.

For the second claim, the E_1-page of the spectral sequence of (K,F) is $E_1^{p,q} = H^{p+q}(\mathrm{gr}_F^p K)$ by theorem 4.2.2, and a filtered quasi-isomorphism induces isomorphisms on exactly these groups by definition 4.6.3. Since every later page is computed from E_1 by taking subquotients along differentials that are natural in the filtered complex, the isomorphism propagates. It does not extend to E_0, which sees $\mathrm{gr}_F K$ itself rather than its cohomology.

The lemma below is the reason this section exists. It is the tool that lets a complex carrying two filtrations be handled at all, and it is what mixed Hodge theory consumes: there W is a weight filtration, F is a Hodge filtration, and the conclusion is that F passes unambiguously to the pages of the spectral sequence of W.

Theorem 4.6.5: The Lemma on Two Filtrations

Let $K^\bullet$ be a bounded-below complex in an abelian category $\mathcal{A}$, carrying an increasing filtration W by subcomplexes that is biregular (in each degree, finite) and a decreasing filtration F by subcomplexes that is biregular as well. Give the subcomplex $F^pK^\bullet$ the filtration $W_n(F^pK) = W_nK \cap F^pK$ and the quotient $K^\bullet/F^pK^\bullet$ the filtration $W_n(K/F^pK) = \mathrm{im}(W_nK)$, and write ${}_WE_r(-)$ for the spectral sequences of theorem 4.2.2. The three filtrations F induces on ${}_WE_r(K)$ are

$$F^p_d = \mathrm{im}\big({}_WE_r(F^pK) \to {}_WE_r(K)\big), \qquad F^p_{d^*} = \ker\big({}_WE_r(K) \to {}_WE_r(K/F^pK)\big)$$

on ${}_WE_r(K)$ for r finite, and on ${}_WE_\infty(K)$ by the same two formulas with $r = \infty$, the maps being the stabilized ones; and the *recursive* filtration F_{rec}: on ${}_WE_0$ it is the filtration F induces on $\mathrm{gr}^W_n K^\bullet$, and on ${}_WE_{r+1} = H({}_WE_r, d_r)$ it is the filtration a subquotient inherits from F_{rec} on ${}_WE_r$. Then:

1. $F^p_d \subseteq F^p_{\mathrm{rec}} \subseteq F^p_{d^*}$ on ${}_WE_r$ for every $r \geq 0$ and on ${}_WE_\infty$, and the three coincide on ${}_WE_0$ and on ${}_WE_1$, with no hypothesis beyond the above;
2. let $r_0 \geq 0$ and suppose that d_s is strictly compatible with F_{rec} on ${}_WE_s$ for every $s \leq r_0$. Then for every p the sequence
$$0 \to {}_WE_r(F^pK) \to {}_WE_r(K) \to {}_WE_r(K/F^pK) \to 0$$
is exact for every $r \leq r_0 + 1$, and the three filtrations coincide on ${}_WE_r$ for every $r \leq r_0 + 2$;
3. if d_s is strictly compatible with F_{rec} for every $s \geq 0$, then the three filtrations coincide on every page and on ${}_WE_\infty$, the differential of $K^\bullet$ is strictly compatible with F in the sense of definition 4.6.1, and the spectral sequence of $(K^\bullet, F)$ degenerates at E_1.

The hypothesis of (2) at $s = 0$ says exactly that the differential of $\mathrm{gr}^W_n K^\bullet$ is strictly compatible with the filtration induced by F, for every n; on its own it therefore gives coincidence on ${}_WE_1$ and ${}_WE_2$, and example 4.6.6 shows that it gives nothing more.

***Proof*:**
Conventions. By theorem 1.1.15 the arguments below may be run with elements. Index the spectral sequence of W by the weight n and the cohomological degree m: setting $C_m = K^{-m}$ and $F_pC_\bullet = W_pK^{-\bullet}$ presents $(K^\bullet, W)$ as a filtered chain complex in the sense of definition 4.2.1,

and theorem 4.2.2 produces

$$ {}_W E_0^{n,m} = \mathrm{gr}_n^W K^m, \qquad {}_W E_1^{n,m} = H^m(\mathrm{gr}_n^W K^\bullet), \qquad d_r\colon {}_W E_r^{n,m} \to {}_W E_r^{n-r,\,m+1} $$

Biregularity of W and boundedness below of $K^\bullet$ make that filtration bounded in each degree, so theorem 4.2.6 applies: the pages stabilize at every spot and ${}_W E_\infty^{n,m} \cong \mathrm{gr}_n^W H^m(K^\bullet)$ for the filtration $W_n H^m(K) = \mathrm{im}\big(H^m(W_n K) \to H^m(K)\big)$, itself finite in each degree. The same holds for $F^p K^\bullet$ and for $K^\bullet / F^p K^\bullet$: a biregular filtration restricts and corestricts to biregular filtrations, and both complexes are bounded below.

Fix p for the whole argument except where stated, and abbreviate

$$ A_r = {}_W E_r(F^p K^\bullet), \qquad B_r = {}_W E_r(K^\bullet), \qquad C_r = {}_W E_r(K^\bullet / F^p K^\bullet) $$

writing $\alpha_r\colon A_r \to B_r$ and $\beta_r\colon B_r \to C_r$ for the maps induced by the inclusion and the projection. The construction of theorem 4.2.2 is functorial in the filtered complex, so α_r and β_r commute with d_r, and α_{r+1}, β_{r+1} are the maps they induce on d_r-cohomology. By definition $F_d^p B_r = \mathrm{im}\alpha_r$ and $F_{d^*}^p B_r = \ker \beta_r$.

Step 1: the sequence at $r = 0$, and the three filtrations there. For every n and m the sequence

$$ 0 \to W_n K^m \cap F^p K^m \to W_n K^m \to W_n(K/F^p K)^m \to 0 $$

is exact, by the definition of the two induced filtrations. These sequences are compatible with the inclusions $W_{n-1} \subseteq W_n$, and the vertical maps from the sequence at $n-1$ to the sequence at n are monomorphisms. Applying theorem 2.1.15 to that map of short exact sequences, all three kernels vanish and the six-term sequence collapses to

$$ 0 \to \mathrm{gr}_n^W (F^p K)^m \to \mathrm{gr}_n^W K^m \to \mathrm{gr}_n^W (K/F^p K)^m \to 0 $$

which is therefore exact. This is $0 \to A_0 \to B_0 \to C_0 \to 0$ at the spot (n, m), and the maps commute with d_0, so it is a short exact sequence of complexes in the cohomological degree m, one for each n.

For the filtrations, $F_d^p B_0 = \mathrm{im}\alpha_0$ is $\big(W_n K^m \cap F^p K^m + W_{n-1} K^m\big)/W_{n-1} K^m$, which is the filtration F induces on $\mathrm{gr}_n^W K^m$, that is $F_{\mathrm{rec}}^p B_0$ by definition; and $F_{d^*}^p B_0 = \ker \beta_0$ is the same subobject by the exactness just proved. So all three agree on ${}_W E_0$, and in particular the phrase "d_0 is strict" is unambiguous and says exactly that the differential of each $\mathrm{gr}_n^W K^\bullet$ is strict for the induced F.

Step 2: the two inclusions, and equality on ${}_W E_1$. We prove $F_d^p \subseteq F_{\mathrm{rec}}^p \subseteq F_{d^*}^p$ on B_r by induction on r, the case $r = 0$ being the equality of Step 1.

For the first inclusion let $\xi \in F_d^p B_{r+1}$, say $\xi = \alpha_{r+1}(\eta)$ with $\eta \in A_{r+1} = H(A_r, d_r)$. Write $\eta = [a]$ with $a \in A_r$, $d_r a = 0$. Then $\alpha_r(a)$ is killed by d_r and lies in $\mathrm{im}\alpha_r = F_d^p B_r$, which by induction lies in $F_{\mathrm{rec}}^p B_r$; and $\xi = [\alpha_r(a)]$. Thus ξ is the class of an element of $\ker d_r \cap F_{\mathrm{rec}}^p B_r$, which is precisely what $F_{\mathrm{rec}}^p B_{r+1}$ consists of.

For the second let $\xi \in F_{\mathrm{rec}}^p B_{r+1}$, say $\xi = [b]$ with $b \in \ker d_r \cap F_{\mathrm{rec}}^p B_r$. By induction $b \in F_{d^*}^p B_r =$

$\ker\beta_r$, so $\beta_{r+1}(\xi) = [\beta_r(b)] = 0$ and $\xi \in \ker\beta_{r+1} = F^p_{d^*}B_{r+1}$. Since each spot of each page equals its limit value for large r, and the three filtrations are defined at $r = \infty$ as those common values, the inclusions persist on ${}_WE_\infty$.

Equality on ${}_WE_1$ needs the short exact sequence of Step 1. For fixed n the objects $B_0^{n,\bullet}$ with d_0 form a complex with $H^m = B_1^{n,m}$, and likewise for A and C; theorem 2.1.15 gives the long exact sequence

$$\cdots \to A_1^{n,m} \xrightarrow{\alpha_1} B_1^{n,m} \xrightarrow{\beta_1} C_1^{n,m} \to A_1^{n,m+1} \to \cdots$$

Exactness at $B_1^{n,m}$ reads $\mathrm{im}\alpha_1 = \ker\beta_1$, that is $F^p_d = F^p_{d^*}$ on ${}_WE_1$, and F_{rec} is squeezed between them. This proves (1).

Step 3: two equivalences. (Throughout, theorem 2.1.15 is applied to the E_r-pages with differential d_r, which for each fixed residue class of the weight index form ordinary complexes, so the lemma applies as stated.) For $r \geq 0$ write

$$(\star_r)\colon \qquad 0 \to A_r \to B_r \to C_r \to 0 \quad \text{is exact}$$

a statement about complexes, since α_r and β_r commute with d_r. Step 1 is $(\star_0)$.

(a) $(\star_r)$ implies $F^p_d = F^p_{d^*}$ on ${}_WE_{r+1}$. By theorem 2.1.15 the sequence

$$\cdots \to A_{r+1} \xrightarrow{\alpha_{r+1}} B_{r+1} \xrightarrow{\beta_{r+1}} C_{r+1} \xrightarrow{\delta_r} A_{r+1} \to \cdots$$

is exact; exactness at B_{r+1} is $\mathrm{im}\alpha_{r+1} = \ker\beta_{r+1}$. Combined with Step 2 this forces all three filtrations to agree on ${}_WE_{r+1}$.

(b) Assume $(\star_r)$. Then $(\star_{r+1})$ holds if and only if d_r is strictly compatible with F^p_d on ${}_WE_r$, that is, if and only if $d_r(B_r) \cap F^p_dB_r = d_r(F^p_dB_r)$ at every spot. By the long exact sequence of (a), α_{r+1} is a monomorphism exactly when the δ_r landing in A_{r+1} vanishes and β_{r+1} is an epimorphism exactly when the δ_r leaving C_{r+1} vanishes; so $(\star_{r+1})$ is equivalent to $\delta_r = 0$. By $(\star_r)$ the map α_r is a monomorphism, so δ_r is described as follows: for $c \in C_r$ with $d_rc = 0$, choose $b \in B_r$ with $\beta_r(b) = c$; then $\beta_r(d_rb) = d_rc = 0$, so $d_rb \in \ker\beta_r = \mathrm{im}\alpha_r$ and there is a unique $a \in A_r$ with $\alpha_r(a) = d_rb$; it satisfies $d_ra = 0$, and $\delta_r[c] = [a]$.

Suppose d_r is strict for F^p_d, and let c, b, a be as above. Then $d_rb \in \mathrm{im}\alpha_r = F^p_dB_r$, so strictness gives $d_rb = d_rb'$ with $b' \in F^p_dB_r$, say $b' = \alpha_r(a')$. Then $\alpha_r(a) = d_r\alpha_r(a') = \alpha_r(d_ra')$, so $a = d_ra'$ and $\delta_r[c] = 0$. Conversely suppose $\delta_r = 0$ and let $b \in B_r$ with $d_rb \in F^p_dB_r = \ker\beta_r$. Then $d_r\beta_r(b) = \beta_r(d_rb) = 0$, so $c := \beta_r(b)$ is a d_r-cocycle and $\delta_r[c] = [a]$ with $\alpha_r(a) = d_rb$. Since $\delta_r = 0$ we get $a = d_ra'$ for some $a' \in A_r$, whence $d_rb = \alpha_r(d_ra') = d_r\big(\alpha_r(a')\big)$ with $\alpha_r(a') \in F^p_dB_r$. Thus $d_r(B_r) \cap F^p_dB_r \subseteq d_r(F^p_dB_r)$, and the reverse inclusion is automatic.

Step 4: the induction, giving (2). Assume d_s is strictly compatible with F_{rec} on ${}_WE_s$ for every $s \leq r_0$. We show by induction on $s \leq r_0 + 1$, simultaneously for every p, that

$$(\mathrm{I}_s)\colon (\star_s) \text{ holds}, \qquad (\mathrm{II}_s)\colon F_d = F_{\mathrm{rec}} = F_{d^*} \text{ on } {}_WE_s$$

Step 1 supplies (I_0) and (II_0). Let $s \le r_0$ and assume both at stage s. By (II_s) the hypothesis at stage s says that d_s is strict for F^p_d, so Step 3(b) upgrades ($\star_s$) to ($\star_{s+1}$), which is (I_{s+1}); and (I_s) with Step 3(a) gives (II_{s+1}). This delivers ($\star_r$) for every $r \le r_0+1$ and coincidence on ${}_WE_r$ for every $r \le r_0+1$. Applying Step 3(a) once more to ($\star_{r_0+1}$) gives coincidence on ${}_WE_{r_0+2}$ as well. Each stage consumes the strictness produced by the previous one and produces the exactness that makes the next stage's strictness hypothesis unambiguous, which is why the two conclusions cannot be separated.

Step 5: part (3), first the limit page. Assume now that d_s is strict for every $s \ge 0$. Then ($\star_r$) holds for every r and every p, and the three filtrations agree on every page. Fix p and a spot (n,m). By convergence there is an R with $A_r^{n,m} = A_\infty^{n,m}$, $B_r^{n,m} = B_\infty^{n,m}$ and $C_r^{n,m} = C_\infty^{n,m}$ for $r \ge R$, compatibly with α and β, so

$$0 \to {}_WE_\infty^{n,m}(F^pK) \to {}_WE_\infty^{n,m}(K) \to {}_WE_\infty^{n,m}(K/F^pK) \to 0$$

is exact and the three filtrations agree on ${}_WE_\infty$.

Step 6: from the limit page to the complex. By theorem 4.2.6 and the naturality recorded in its proof, $\alpha_\infty^{n,m}$ is gr_n^W of the map $H^m(F^pK) \to H^m(K)$ induced by the inclusion, for the filtrations $W_nH^m(-) = \mathrm{im}\big(H^m(W_n-) \to H^m(-)\big)$, both finite in each degree. Apply theorem 2.1.15 to the map of short exact sequences

$$\begin{array}{ccccccccc}
0 & \to & W_{n-1}H^m(F^pK) & \to & W_nH^m(F^pK) & \to & \mathrm{gr}_n^W H^m(F^pK) & \to & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \to & W_{n-1}H^m(K) & \to & W_nH^m(K) & \to & \mathrm{gr}_n^W H^m(K) & \to & 0
\end{array}$$

The sequence $0 \to \ker(W_{n-1}) \to \ker(W_n) \to \ker(\mathrm{gr}_n^W)$ is exact, the third kernel vanishes by Step 5, and $W_nH^m(F^pK) = 0$ for n small; so induction upwards on n gives $\ker\big(W_nH^m(F^pK) \to W_nH^m(K)\big) = 0$ for every n, and since the filtration is exhaustive in each degree, $H^m(F^pK) \to H^m(K)$ is a monomorphism for every p and m.

Strictness of d follows at once. Let $z \in d(K^{m-1}) \cap F^pK^m$, say $z = dx$ with $x \in K^{m-1}$. Then z is a cocycle of $F^pK^\bullet$ whose class maps to $[z] = 0$ in $H^m(K)$, so by injectivity $z = dy$ for some $y \in F^pK^{m-1}$. Hence $d(K^{m-1}) \cap F^pK^m = d(F^pK^{m-1})$, which is strict compatibility with F in the sense of definition 4.6.1.

Step 7: degeneration of the F-sequence. The filtration F is decreasing and biregular and $K^\bullet$ is bounded below, so after a shift of the index corollary 4.2.7 applies, and by corollary 4.2.8(2)

$$E_r^{p,q}(K,F) \cong \frac{Z_r^{p,q}}{Z_{r-1}^{p+1,q-1} + d\, Z_{r-1}^{p-r+1,\,q+r-2}}, \qquad Z_r^{p,q} = \{\, x \in F^pK^{p+q} : dx \in F^{p+r}K^{p+q+1} \,\}$$

Write $Z_\infty^{p,q} = \{x \in F^pK^{p+q} : dx = 0\}$. For $r \ge 1$ we claim $Z_r^{p,q} = Z_\infty^{p,q} + Z_{r-1}^{p+1,q-1}$. Given $x \in Z_r^{p,q}$, strictness at level $p+r$ gives $dx \in d(K^{p+q}) \cap F^{p+r}K^{p+q+1} = d(F^{p+r}K^{p+q})$, say $dx = dy$ with $y \in F^{p+r}K^{p+q}$. Then $x - y \in Z_\infty^{p,q}$, while $y \in F^{p+1}K^{p+q}$ has $dy \in F^{p+r}K^{p+q+1} = F^{(p+1)+(r-1)}K^{p+q+1}$, so $y \in Z_{r-1}^{p+1,q-1}$, which is a summand of the denominator. Hence every class of $E_r^{p,q}(K,F)$ is represented by a cocycle of $K^\bullet$, on which d vanishes, so $d_r = 0$ for every $r \ge 1$ and the spectral sequence of $(K^\bullet, F)$ degenerates at E_1. This proves (3), completing the proof.

The proof never manipulated the objects Z_r of definition 4.2.3 on the W-side, and it is worth saying what those would show. By proposition 4.2.4, with $Z_r^{n,m} = \{x \in W_nK^m : dx \in W_{n-r}K^{m+1}\}$ and $D_r^{n,m} = Z_{r-1}^{n-1,m} + d\,Z_{r-1}^{n+r-1,m-1}$, one has ${}_WE_r^{n,m} = Z_r^{n,m}/D_r^{n,m}$, and F_d^p is the image of $F^pK^m \cap Z_r^{n,m}$, while $F_{d^*}^p$ is the image of $Z_r^{n,m} \cap \left(F^pK^m + \widehat{D}_r^{n,m}\right)$, where $\widehat{D}$ is built from the objects $\{y \in W_aK^b : dy \in W_{a-s}K^{b+1} + F^pK^{b+1}\}$ in place of $Z_s^{a,b}$. The gap between the two is exactly the failure of the boundary condition to be detectable modulo F^p, and closing it by hand is the several-page computation the argument above replaces with a long exact sequence.

Example 4.6.6: Sharpness of the Hypothesis

Take $\mathcal{A}$ the category of $\mathbb{Q}$-vector spaces and let $K^\bullet$ be $\mathbb{Q}e$ in degree 0 and $\mathbb{Q}f$ in degree 1, with $de = f$. Filter it by

$$W_{-1}K = 0, \quad W_0K^0 = 0,\ W_0K^1 = \mathbb{Q}f, \quad W_1K = K$$

$$F^0K = K, \quad F^1K^0 = 0,\ F^1K^1 = \mathbb{Q}f, \quad F^2K = 0$$

Both are biregular filtrations by subcomplexes. The graded pieces are $\mathrm{gr}_0^W K = \mathbb{Q}f$ in degree 1 and $\mathrm{gr}_1^W K = \mathbb{Q}e$ in degree 0, each with zero differential, so the differential of every $\mathrm{gr}_n^W K^\bullet$ is trivially strict for F: the hypothesis of theorem 4.6.5(2) holds with $r_0 = 0$ and no more.

It holds with no more. The differential $d_1\colon {}_WE_1^{1,0} = \mathbb{Q}e \to {}_WE_1^{0,1} = \mathbb{Q}f$ is the connecting map of $0 \to W_0K \to K \to \mathrm{gr}_1^W K \to 0$, hence $e \mapsto f$, an isomorphism; its source has $F^1 = 0$ and its target has F^1 equal to everything, so d_1 is not strict. Likewise $H^\bullet(K) = 0$ while $\mathrm{gr}_F^0 K = \mathbb{Q}e$ in degree 0 and $\mathrm{gr}_F^1 K = \mathbb{Q}f$ in degree 1, so the spectral sequence of (K, F) has $E_1^{0,0} = \mathbb{Q}e$ and $E_1^{1,0} = \mathbb{Q}f$ with $E_\infty = 0$ and does not degenerate at E_1; equivalently $d(K^0) \cap F^1K^1 = \mathbb{Q}f$ while $d(F^1K^0) = 0$, so d is not strict for F. Conclusion (2) of theorem 4.6.5 with $r_0 = 0$ is therefore exactly right: here ${}_WE_2 = 0$, so the three filtrations do agree on ${}_WE_1$ and ${}_WE_2$, and nothing beyond that is true.

The example is not pathological, and a consumer should read it as a warning about the shape of the input. The strictness of d_0 is a statement about the graded pieces of W alone and can never see how those pieces are glued; the strictness of d_1 is a statement about the gluing, and it has to be supplied. In mixed Hodge theory it is supplied by weights: the terms of ${}_WE_1$ are cohomologies of smooth projective varieties, d_1 is a morphism of pure Hodge structures of a single weight, and such a morphism is strict, while d_r for $r \geq 2$ lowers the weight and vanishes. With all d_r strict, part (3) applies and the Hodge filtration degenerates at E_1.

The last of the three exercises below is used often enough elsewhere to be worth stating as a result in its own right.

Lemma 4.6.7: Degeneration at E_1 and Strictness

Let $(K^\bullet, F)$ be a complex with F a biregular decreasing filtration by subcomplexes. The spectral sequence of $(K^\bullet, F)$ degenerates at E_1 if and only if the differential of $K^\bullet$ is strictly compatible with F.

***Proof*:**
This is exercise 4.6.1 (3), proved there in both directions. No finiteness or field hypothesis is needed: biregularity alone makes the correction argument in the reverse direction terminate.

Exercise 4.6.1

1. Exhibit filtered vector spaces and strict morphisms f and g with $g \circ f$ not strict, confirming the last claim of proposition 4.6.2.
2. Give a morphism of filtered complexes that is a quasi-isomorphism but not a filtered quasi-isomorphism.
3. Show that the spectral sequence of a filtered complex (K, F) degenerates at E_1 if and only if the differential of K is strictly compatible with F.